

November 1, 2024

Environmental Approvals Branch
Manitoba Environment and Climate Change
14 Fultz Boulevard, Box 35
Winnipeg, Manitoba R3Y 0L6

Attention: Agnes Wittmann
Director

**Re: Orphaned and Abandoned Mines Program – Ruttan Mine Site
Summary of Site Activities – Final Rev 0**

Dear Agnes Wittmann:

On behalf of the Orphaned and Abandoned Mines (OAM) Program of Manitoba Environment and Climate Change, Kontzamanis Graumann Smith MacMillan Inc. (KGS Group) has prepared this Summary of Site Activities describing the components and activities for on-going environmental management of the former Ruttan Mine to support closing the existing Environment Act License No. 858 VO.

This report provides the following:

- Site Description and Operational History
- Closure and Remediation Activities/Infrastructure
- On-going Water Monitoring/Management
- Proposed Future Changes in Water Monitoring/Management

1.0 SITE DESCRIPTION

The Ruttan Mine Site is a closed mine located within UTM NAD 83, Zone 14V, Northing 6259596, Easting 459508, approximately 23 kilometers east of the Town of Leaf Rapids, Manitoba. Access to the site is via all-weather road Provincial Road (PR) 493, which transitions into Mine Road and leads directly into the site.

The main waterbodies on site are Ruttan Lake, Open Pit, Mill Pond, and Brehaut Lake, which are described as follows (see Figure 1).

- Ruttan Lake was a natural lake prior to mine construction but has since been contaminated with Acid Mine Drainage (AMD) from mine tailings. Ruttan Lake is a submerged tailings storage area which requires water cover over the tailings to minimize the production of AMD water. The lake presently receives runoff from the surrounding undisturbed areas (west and south) and from overland flow that has come into contact with potential acid generating rock sources that are located around the Ruttan Lake, including relatively insignificant amount of waste rock used to build haul road (an approximately 2.5 hectare area). Ruttan Lake discharges into the Open Pit.
- The Open Pit is a constructed waterbody that resulted from flooding the open pit mine and receives AMD impacted water from Ruttan Lake and any uncapped areas. The pit serves as a storage reservoir for water to be treated by the Water Treatment Plant (WTP).
- Mill Pond was a natural pond prior to mine construction and during operation was used to store clean water for milling purposes. Currently, the pond serves as a polishing pond for treated effluent from the WTP. Water discharged from the north end of Mill Pond flows north through a discharge ditch/creek, under Highway 493 through two (2) culverts and eventually reaches Brehaut Lake.
- Brehaut Lake was a natural lake prior to mine construction and was used as a settling pond for effluent from the water treatment system used during mining operations. Brehaut Lake also contains some submerged tailings, therefore it remains within the development limits. The lake is the most downstream water body on-site and therefore the overflow weir located north of the lake is the designated compliance point for all water discharged from the site.

2.0 SITE OPERATIONAL HISTORY

The Ruttan Mine was a base metal copper/zinc deposit initially mined as an open pit and eventually converted to an underground operation (Government of Manitoba, 2020). The property was owned and operated by Sherritt Gordon Mines from 1973 until 1987, at which time Hudson Bay Mining and Smelting (HBM&S) took over ownership. The mine, along with the mill, operated for approximately 30 years until closure in 2002. The area impacted by mining activities covers approximately 12 km². HBM&S completed all of their closure obligations from 2003 to 2005. The closure activities included the removal of most of the buildings, dismantling and removing of any infrastructure owned by HBM&S, the capping of shafts and ventilation raises, and the removal of the headframe. Any concrete foundations were knocked down to ground level and basements and trenches were backfilled. HBM&S also constructed diversion dams and channels to direct clean water away from the contaminated areas of the site (HBM&S, 2002).

After operations ceased, the Ruttan Mine Site contained over 50 million tonnes of mine rock (waste rock) that was produced during the development of the Open Pit, and over 40 million tonnes of tailings. Approximately 6.8 million tonnes of tailings were used as backfill on site during decommissioning activities. At the time of closure, the only known and proven technology applicable to the Ruttan area to prevent sulphide oxidation was to place the material of concern under a water cover; however, due to the cost to relocate the rock and exposed tailings,

this option was not considered feasible. Therefore, the best available technology for treating the AMD water was to raise the pH high enough to cause dissolved metals to form precipitates, followed by removal of these as a sludge (HBM&S, 2002). This practice is in operation currently at the Ruttan Mine Site.

Since 2005, the Province of Manitoba has assumed all property interest at the Ruttan Mine Site and is responsible for performing all ongoing rehabilitation work in connection with the mine closure.

3.0 REMEDIATION AND CLOSURE ACTIVITIES

Since obtaining responsibility for site remediation in 2005, the Province of Manitoba initiated site rehabilitation activities by completing a project to revegetate the area of the former Ruttan Mill Site (an approximate area of 19.8 hectares). Between 2006 and 2009, the Province retained KGS Group to provide engineering services involving physical inspections of several key features of the site such as dams, tailings areas, and diversion channels. Water level surveys within the Open Pit and a full Lidar survey of the site was also completed. In 2009, KGS Group developed a conceptual design study for the on-site water treatment.

As part of the HBM&S Original Closure Plan, on-site AMD was directed to the Open Pit using the large (34,000,000 m³) volume of the pit and underground workings as storage for all on-site AMD water, anticipated to be for many years into the future. However, KGS Group's survey showed that the open pit would reach its maximum storage capacity by 2014, versus the HBM&S forecast of approximately 2040.

In October 2010, KGS Group was retained again to provide engineering services in support of mine closure activities. Initial activities were carried out in 2011 and included the following.

- Collection and analysis of water samples from various bodies of water at the site.
- Collection and analysis of samples of tailings and waste rock at the site.
- Completion of a geophysical survey and geotechnical investigations such as test holes and test pits at the TMA dikes.

The information from the site investigations were used to develop a revised closure plan that aimed to reduce risk to the surrounding environment from metals contamination at the site and minimize future operation and maintenance costs for the Province of Manitoba. Major components of the revised closure plan included the following items.

- Grading of the TMA to direct clean surface runoff towards the south.
- Stabilizing and grading of the downstream slope of West Dam to a slope of 4 horizontal to 1 vertical.
- Capping of the TMA including exterior slopes of containment dikes with a low permeability clay cover to allow off-site discharge of run-off.
- Raising the water level of Ruttan Lake by constructing an outlet control structure (North Control Dam) and raising the existing Ruttan Dam to provide a water cover over the tailing washout area adjacent to Ruttan Lake.

- Building a water treatment facility for the site, based on a low-density sludge lime treatment plant where lime is used to raise the pH of the raw water to promote the heavy metals in solution to precipitate out as sludge in the settling ponds.

Since 2014, remediation components from the revised closure plan have been completed by several different contractors including the following.

- Construction of a water treatment plant that was completed by Tervita in 2014 and has been operated by a private contractor since that time (see Section 3.1.3).
- Construction of two settling ponds (SP1 and SP2) to receive treated effluent discharged from the water treatment plant. The maximum capacity of SP1 is approximately 500,000 m³ and SP2 has a maximum capacity of approximately 250,000 m³.
- Installation of a 12.47 kV transformer at the entrance of the Ruttan Mine Site, to provide 120 to 347 V power to the site.
- Construction of an overhead 12.47 kV distribution line from the transformer to the water treatment plant.
- Construction of the North Control Dam at the north end of Ruttan Lake (as described in Section 3.1.2).
- Construction of the site access road and excavation of a channel south of the settling ponds.
- Construction of the East Haul Road.
- Upgrade of Ruttan Dam west of Ruttan Lake (as described in Section 3.1.2).
- Major works related to TMA capping, West Dam regrading, and seepage cut-off installation (as described in Section 3.1.1).
- Installation of a vinyl sheet pile (VSP) along the West Dam (as described in Section 3.1.1.2).
- Regrading and capping of tailings and tailings containment dikes.
- Installation of the Seepage Control System (SCS) Drain along the West Dam (as described in Section 3.1.1.2).

In 2020, Strilkiwski Contracting Ltd. (SCL) was retained by the Province to complete the remedial works for the TMA and West Dam. The TMA capping completion included capping of all TMA cells and the West Dam, repair of the VSP wall, improvement of the North Control Dam discharge ditch, construction of a clay berm at the site access road, and improvements to the east haul road. The remedial work was completed in July 2022 (KGS Group, 2024a).

3.1 Current Remedial Infrastructure

3.1.1 TAILINGS MANAGEMENT AREA

Before site remediation was completed, all of the runoff from rainfall and snow melt at the Ruttan Mine Site interacted with tailings and waste rock and was considered AMD with an estimated average annual volume of runoff between 3.0 and 3.3 million m³. Of this total, approximately 2.0 to 2.2 million m³ flowed through Ruttan Lake, and 1.0 to 1.1 m³ originated from the immediate area of the Open Pit. Completion of the clay cap on the tailings in the Tailings Management Area diverted approximately 0.5 million m³ of runoff from the Open Pit directly off-site, reducing the average volume of runoff to the Open Pit from Ruttan Lake to approximately 1.5 to 1.8 million m³ per year, and the total average volume of runoff to the Open Pit from the entire site to approximately 2.5 to 2.9 million m³ per year. The major benefit of the cap on the TMA is that Ruttan Lake will no

longer receive AMD and eventually the water in Ruttan Lake can be diverted off-site without treatment. When this point is reached, the total volume of AMD water that requires treatment will average approximately 1.0 to 1.1 million m³ per year.

3.1.1.1 TMA Cells

There are three cells within the TMA. Cell 1 is the eastern cell and contains mainly coarse tailings that are contained by perimeter rockfill berms on the north, east and south sides. The tailings are capped with clay and vegetated. Surface water runoff from TMA Cell 1 is discharged off site to the south through a constructed outlet weir lined with rockfill. Cell 2 is located to the west of TMA Cell 1 and contains mainly coarse tailings that are contained by perimeter rockfill berms along the north and south sides and the West Dam adjacent to Ruttan Lake. The tailings are capped with clay and vegetated. Surface water runoff from TMA Cell 2 is directed to TMA Cell 3 through rockfill lined ditches and swales. Cell 3 is located south of TMA Cell 2 and contains fine grained tailings that are contained by perimeter rockfill berms on the north, east and south sides and the West Dam adjacent to Ruttan Lake. The tailings are capped with clay and vegetated. Surface water runoff from TMA Cell 3 is discharged to the south through a constructed rockfill lined outlet weir (KGS Group, 2024a). Cell 3 is a low-lying area that generally had a soft subgrade that required stabilization by placement of 600 mm thick rockfill on a geotextile/geogrid to facilitate construction of the clay cap.

3.1.1.2 West Dam and Seepage Collection System

The West Dam is located on the west side of TMA Cell 2 and Cell 3 and along the east shore of Ruttan Lake. The slope of the West Dam was graded to a slope of 4 horizontal to 1 vertical and capped with clay and vegetated. A seepage cut-off wall, to prevent AMD seepage into Ruttan Lake, consisting of one row of VSP was installed through the tailings down to bedrock along the toe of the West Dam slope. The dam also includes a toe berm used as a lower access road.

A SCS drain was also constructed at the upstream side of the VSP wall and the lower access road to intercept and collect AMD seepage coming through the West Dam tailings. The SCS was completed in 2022 with the drain routed north to divert the AMD seepage water to the Open Pit.

3.1.2 WATER MANAGEMENT INFRASTRUCTURE

Management of water levels is an important aspect of site management. The main bodies of contaminated water are Ruttan Lake and the Open Pit, both of which contain large volumes of AMD water that must be managed properly to prevent contaminated water from entering the environment. There are four main dams that control the flow of contaminated and uncontaminated water across and around the site.

- **North Control Dam (NCD):** Located on the northeast end of Ruttan Lake to control and regulate the water level of Ruttan Lake. The dam was constructed in 2014 and is composed of zoned earthfill and contains a 400 mm diameter pipe at invert elevation of 285.5 m and valve chamber with a 450 mm gate valve to control discharge of water from Ruttan Lake towards the Open Pit. The original gate chamber design had a gate valve on the upstream and downstream ends, however in 2020 both valves froze and broke during winter and only the upstream valve was replaced. The NCD is equipped with an overflow weir that contains VSP with an invert elevation of 288.0 m which is the design operating level of Ruttan Lake. When the Ruttan Lake water level is higher than the design operating level, contaminated AMD water spills over the weir and

the water is directed towards the Open Pit through multiple culverts beneath a haul road immediately downstream of the North Control Dam.

- **Ruttan Dam:** Located on the northwest side of Ruttan Lake. Its purpose is to prevent AMD water from entering Brehaut Lake so that the AMD water can be collected in the Open Pit. Three HDPE pipes (600 mm diameter) are installed at invert elevation of 288.0 m to accommodate the future discharge of clean, untreated water from Ruttan Lake into Brehaut Lake once the water in Ruttan Lake is acceptable for off-site discharge. The culverts are currently plugged using HDPE caps that will be removed only once the water quality is suitable. A drainage creek exists at the downstream side of the Ruttan Dam which drains to the south end to Brehaut Lake. Ruttan Dam was originally constructed during the operation of the mine, however, to raise the Ruttan Lake water levels to submerge tailings the crest of the dam was raised in 2013 and the drainage pipes installed.
- **South Diversion Dam:** Located at the southeast end of Ruttan Lake. The dam was constructed to block clean surface water runoff from the area south of the dam from entering Ruttan Lake thereby reducing the volume of water requiring treatment. Water that ponds upstream of the South Diversion Dam is directed by a constructed channel to the south end of Brehaut Lake. The dam was originally constructed during the mine closure by HBM&S and there was no work on the dam included in the more recent site remediation works.
- **Brehaut Dam:** Located at the north end of Brehaut Lake. It was originally constructed at the outlet from Brehaut Lake for mining operations to ensure adequate water cover over lime sludge that resulted from treatment of mine effluent at the south end of the lake. Brehaut Dam is the compliance point where all water discharged from the site must meet water quality standards. There was no work completed at the dam included in site remediation works.

The objective of the water management strategy is to minimize the volume of AMD water requiring treatment. Since the main source of AMD water is Ruttan Lake, the overall goal is to improve the quality of the water in Ruttan Lake until it is acceptable to be discharged directly to Brehaut Lake without treatment. Studies have shown that when AMD water freezes, the ice is relatively clean, and the metals concentrate in the water below the ice. In the winter, from early February to late March, the valve in the NCD is opened to allow the concentrated water below the ice to enter the Open Pit, while the relatively clean ice is retained in Ruttan Lake. In this way, the time for Ruttan Lake to achieve acceptable water quality standards is reduced. However, it is important that the water level in Ruttan Lake be maintained at 287.5 m or higher to maintain a minimum of 1.5 m water cover over the submerged tailings delta.

3.1.3 WATER TREATMENT INFRASTRUCTURE

A major component of the overall site remediation project was the construction and on-going operation of a WTP to treat the AMD from the Open Pit. In 2013 and 2014, the main WTP was constructed in the former hoist building east of the Open Pit. The WTP is a low-density sludge (LDS) system using lime to treat AMD by raising the pH to a target of 10 which removes the metals dissolved in the water into a solid state. The system is composed of a treatment plant inside a metal clad building, two pumping stations (PS-1 and PS-2), a clean water reservoir and 2 settling ponds. The main features of the water treatment system are shown in Appendix A and the WTP As-Builts are included in Appendix B.

The treatment plant includes an electrical room, office, lunchroom, washroom with sewage pump and holding tank, water sample testing station, and plant area that processes lime and clean water into lime slurry to treat the contaminated water from the Open Pit. The equipment consists of a slaker, mixing tanks with agitators, and reactor tanks and a lime storage silo. Seven progressive cavity pumps are used to move the lime slurry throughout the plant. Pumping station PS-1 is located west of the WTP on the east side of the Open Pit and consists of four submersible pumps and a structure to store electrical equipment. Pumping station PS-2 is located on the west side of the clean water reservoir and consists of a structure to store the clean water pump and electrical equipment.

Clean water is pumped from PS-2 to the water treatment plant and is used for process water during the lime slaking process, and for flushing of plant equipment. Quick lime from truck shipments is added to the slaker or to the lime storage silo. The slaker mixes the clean water and lime to create a lime slurry. The lime slurry is then pumped into the mixing tanks. An agitator within the mixing tanks keeps the solids in suspension while a pump transfers lime slurry into the reactor to treat the contaminated water. The contaminated water is pumped out of the Open Pit from PS-1 into the reactors. The reactors mix the contaminated water and lime slurry, until it reaches a pH of approximately 10 units. The treated water discharges by gravity out of the water treatment plant into an excavated channel, which leads to the settling ponds.

The WTP was designed to treat 20,000 m³ of acidic water in a 24-hour period over an annual operating summer/fall season of 150 days, for a total treatment capacity of 3 million m³. Modifications to the plant since commissioning in 2014 have increased the daily treatment capacity to approximately 30,000 m³/day and resulted in several changes to the configuration of the equipment and piping. Appendix A contains a set of as-built drawings that were updated in 2023 to show the existing water treatment system.

There is also a portable WTP that was installed during late summer of 2017 due to the extreme volumes of AMD and limited storage capacity for the AMD upstream of the Open Pit. The portable WTP is located just south of SP1 and operates in the same manner as the main WTP with a capacity to treat 20,000 m³ of AMD per day.

In addition, a BNR Reactor (BNR) treatment system was installed and has operated at the site since 2021. The Reactor is owned by a third party and is not included as infrastructure of the Ruttan mine. The BNR involves a cylindrical reactor where lime is mixed with the AMD before being discharged to the sludge removal system. Treatment capacity of the BNR is unknown.

3.1.4 SETTLING PONDS

As part of the water treatment system for the Ruttan Mine Site, two settling ponds (SP1 and SP2) contained by clay dikes were developed north of the Open Pit. Water treated at the WTP is discharged by gravity to the two-stage Settling Ponds to allow the low-density lime sludge to settle. In normal operation, the effluent is discharged to SP1, then SP2, and then Mill Pond (for final polishing). Flow from SP1 is routed into SP2 through an overflow weir. SP2 releases treated water over a weir into Mill Pond for polishing which is then discharged off-site over a third weir at the north end of Mill Pond into the drainage swale to Brehaut Lake, which is the location of the compliance point (KGS Group, 2024a). The drainage swale crosses under Highway 493 through a set of culverts that are the responsibility of Manitoba Transportation and Infrastructure.

As the lime sludge is deposited and builds up, the overall retention capacity of the ponds is reduced. Therefore, sludge removal from the settling ponds is required every 3 to 5 years, depending on the volume of water treated and the resulting amount of sludge produced. The sludge removed from the settling ponds is pumped into the Open Pit for storage.

3.2 Water Treatment Operations

The WTP has been operated annually during the summer months since 2014 (KGS, 2024b). The water treatment system is not designed to operate during winter months, and the operating season typically runs from late May to mid-October (100 to 120 days per year). The operation and maintenance of the WTP is handled by experienced third party operators, including startup in the spring and winterizing in the fall. The General Plant Operation Manual (Appendix C) describes the operating procedures and the operation and maintenance requirements for the WTP. A detailed Operation and Maintenance Manual is maintained at the WTP.

At present, all water treatment activities at the Ruttan Mine Site are sub-contracted to Mine Water Recovery (MWR). Water treatment activities include operation of both WTPs and the BNR Reactor with a target total treatment volume of 5.5 million m³ (per operational season), operation of the valve in the North Control Dam to regulate water flow from Ruttan Lake to the Open Pit, and weekly compliance sampling at Brehaut Lake.

Sludge management is sub-contracted to North Fringe of Saskatchewan to remove sludge from SP1, SP2, Mill Pond, and the bypass channel and pump the sludge into the Open Pit, with a target sludge removal volume of 400,000 m³ (per operational season).

The total volume of AMD treated, total volume of lime used for treatment, and total volume of sludge removed for 2021 – 2023 is summarized in Table 1.

TABLE 1: SUMMARY OF 2021 – 2023 WATER TREATMENT ACTIVITIES

	2021	2022	2023
AMD Volume Treated (m³)⁽¹⁾	5,693,538	5,568,802	5,515,981
Lime Consumption (tonnes)	8,249.29	7,088.26	4,928.00
Sludge Removed (m³)⁽²⁾	273,202	421,844	345,992

Notes:

1. Total volumes include water treated by the WTP, portable WTP, and the BNR Reactor.
2. Sludge is removed from SP1, SP2, Mill Pond, and bypass channel.

3.3 Environmental Water Quality Monitoring

In 2021, KGS Group was contracted by the OAM Program of Manitoba Environment and Climate Change for the overall environmental management of the Ruttan Mine Site. The scope of work for the environmental

monitoring at the Mine Site, as provided in the Adaptive Management Plan (KGS Group, 2024a), included the following.

- Conduct annual water quality monitoring of five (5) select on-site lakes, outfalls, weirs, and culvert sampling locations during the spring/summer/fall season.
- Prepare an annual water quality report following fall seasonal monitoring to include tabulated results, discussion of results relative to historical data, recommendations for future monitoring, and GPS coordinates for each sampling location.

The surface water monitoring locations specified in the AMP for environmental monitoring are shown on Figure 2. The GPS coordinates, location descriptions, and sampling frequency of each location is summarized in Table 2.

TABLE 2: SURFACE WATER SAMPLING LOCATIONS

Category	Location ID	Sample ID	Sampling Frequency	Description	Coordinates (m) ⁽¹⁾
On-Site Surface Water	TMA Cell 1 Outfall	TMA1-WQ-01	Yearly	Outlet swale at south toe of TMA Cell 1.	N 6257839.26 E 460903.19
	TMA Cell 3 Outfall	TMA3-WQ-01	Yearly	Outlet swale at south toe of TMA Cell 3.	N 6257131.12 E 460516.31
	Mill Pond Exit Weir	MLP-WQ-01	Yearly	Mill Pond discharge weir at north end of Mill Pond.	N 6260658.99 E 460751.50
	Ruttan Lake	RL-WQ-01	Yearly	Upstream of the North Control Dam.	N 6258951.76 E 459747.93
	PR493 Culvert	PR493-WQ-01	Yearly	Located approximately 1 km north (downstream) of Mill Pond Weir at a culvert crossing of PR 493.	N 6261680.97 E 461040.03
Compliance Point	Brehaut Lake Outfall	BHL-WQ-01	Weekly (by water treatment operator)	Dam structure at north point of Brehaut Lake.	N 6263862.52 E 459557.16

Notes:

1. All coordinates within UTM Zone 14V.

The 2023 sampling and monitoring event at Ruttan Mine Site was the first year that water samples were collected as part of the implementation of the AMP. The Environmental monitoring and sampling program is planned to be completed in spring 2024 and will include the same water sampling locations.

MWR is currently contracted to complete weekly compliance sampling at the Brehaut Lake Outfall. A summary of the compliance water sampling from 2020 – 2023 is summarized in Table 3.

TABLE 3: SUMMARY OF COMPLIANCE SAMPLING

	2020	2021	2022	2023
Number of Samples Taken	17	19	19	20
Start and End Dates	June 22 – October 11	June 1 – October 3	June 3 – October 2	May 23 – October 1
Results in Compliance with MDMER 2002 (amended 2021) and CEC Order No. 858VO	Yes (criteria was MMER at this time)	Yes (criteria was MMER at this time)	Yes	Yes

3.3.1 SITE CONDITION ASSESSMENT

The overall environmental management of the Ruttan Mine Site also involves an annual site condition assessment which includes the scope of work listed below:

- Site Condition Assessment
 - Conduct one site visit every year starting in 2023 to inspect the overall site conditions, engineered covers, drainage pathways, and dikes in accordance with the National Orphaned/Abandoned Mines Initiative (NOAMI) Site Characterization Field Form.
 - Prepare a summary report to include a site plan, photographic documentation of the site conditions, completed NOAMI checklist, and recommendations for remedial works.

The purpose of the site assessment is to ensure that all remediation components are in working order and to identify any deficiencies at the site that require attention. If deficiencies or repairs are identified, recommendations will be integrated into the summary report with a Class C cost estimate.

4.0 CHANGES IN WATER MONITORING/MANAGEMENT

Adaptive management of the water at the Ruttan Mine Site involves annual sampling of the water quality in Ruttan Lake. The water in Ruttan Lake is sampled and analyzed for the parameters specified in the federal Metal and Diamond Mine Effluent Regulations (MDMER) which are periodically updated. When the water quality in Ruttan Lake is in compliance with the requirements of the MDMER (approximately 5 to 10 years after completion of the TMA Cap in 2022), the invert of the weir at the NCD will be raised to 288.2 m and the caps on the pipes in Ruttan Dam will be removed.

Compliance is considered to be achieved when the concentrations of all relevant parameters in the MDMER comply with the standards specified in the MDMER for 2 consecutive years. Sampling and analysis of the water quality in Ruttan Lake will be conducted annually to ensure that the water remains suitable for discharge.

Proposed future changes in the frequency of water treatment and water monitoring will be based on water quality improvements observed in the site water bodies from the on-going mitigation measures implemented on

site. Over time, the amount of AMD contaminated runoff that flows into Ruttan Lake will decrease and water quality will improve such that it is suitable for off-site discharge through the Ruttan Dam into Brehaut Lake. Once water from Ruttan Lake is discharged to Brehaut Lake, and not the Open Pit, the demand for water treatment would decrease and annual target values for water treatment and sludge removal would be adjusted.

5.0 REFERENCES

Government of Manitoba. 2020. Stage 2 – Request for Proposals for Engineering and Architectural Services. Mining Reclamation – Construction Management. December 18, 2020.

Hudson Bay Mining & Smelting Co., Limited. 2002. Ruttan Mine Property Decommissioning Plan. Final Version. January 2002.

KGS Group. 2024a. Ruttan Mine Site Adaptive Management Plan. Final Version. March 25, 2024.

KGS Group. 2024b. OAM Subprogram – Ruttan Mine Site 2023 Water Treatment Annual Report. Final Version. April 3, 2024.

6.0 STATEMENT OF LIMITATIONS AND CONDITIONS

Limitations

This report has been prepared for Manitoba Environment and Climate Change in accordance with the agreement between KGS Group and Manitoba Environment and Climate Change (the “Agreement”). This report represents KGS Group’s professional judgment and exercising due care consistent with the preparation of similar reports. The information, data, recommendations and conclusions in this report are subject to the constraints and limitations in the Agreement and the qualifications in this report. This report must be read as a whole, and sections or parts should not be read out of context.

This report is based on information made available to KGS Group by Manitoba Environment and Climate Change. Unless stated otherwise, KGS Group has not verified the accuracy, completeness or validity of such information, makes no representation regarding its accuracy and hereby disclaims any liability in connection therewith. KGS Group shall not be responsible for conditions/issues it was not authorized or able to investigate or which were beyond the scope of its work. The information and conclusions provided in this report apply only as they existed at the time of KGS Group’s work.

Third Party Use of Report

Any use a third party makes of this report or any reliance on or decisions made based on it, are the responsibility of such third parties. KGS Group accepts no responsibility for damages, if any, suffered by any third party as a result of decisions made or actions undertaken based on this report.

Geo-Environmental Statement of Limitations

KGS Group prepared the geo-environmental conclusions and recommendations for this report in a professional manner using the degree of skill and care exercised for similar projects under similar conditions by reputable and competent environmental consultants. The information contained in this report is based on the information that was made available to KGS Group during the investigation and upon the services described, which were performed within the time and budgetary requirements of Manitoba Environment and Climate Change. As this report is based on the available information, some of its conclusions could be different if the information upon which it is based is determined to be false, inaccurate or contradicted by additional information. KGS Group makes no representation concerning the legal significance of its findings or the value of the property investigated.

7.0 CLOSING

The past and current remediation and closure activities that have taken place at the Ruttan Mine Site have reduced the amount of AMD impacted water and improved the overall water quality of the water bodies on site. Continuing to implement the on-going mitigation measures and water management will further reduce the potential to impact the surrounding environment. KGS Group trusts that the above meets your requirements. If you have any questions regarding this submission, please contact the undersigned.

Prepared By:



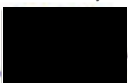
Rachel Dyck, B.Env.Sc., C.E.T.
Environmental Scientist-Technologist

Reviewed By:



Shaun Moffatt, M.Sc.
Senior Environmental Scientist

Approved By:



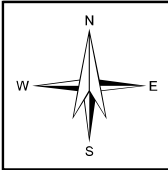
Dami Adedapo, Ph.D., P.Eng.
Principal

RED/jr

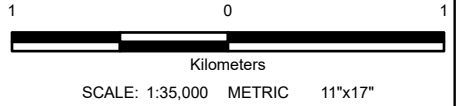
Attachments

cc: Morgan Hallet
Cory Graham

FIGURES



NOTES:
1. Imagery Source: Ruttan LiDAR imagery, dated as 2022.
2. All units are metric and in metres unless otherwise specified.
Transverse Mercator Projection, NAD 1983 CSRS, Zone 14.
Elevations are in metres above sea level (MSL).



➡	0	24/10/31	ISSUED WITH FINAL REPORT	RED	SM
	NO.	YYMM/DD	DESCRIPTION	ISSUED BY	CHECK BY
REVISIONS / ISSUE					
					
OAM SUBPROGRAM - RUTTAN MINE SITE SUMMARY OF SITE ACTIVITIES					
SITE LOCATION MAP					
OCTOBER 2024			FIGURE 01	REV:	0



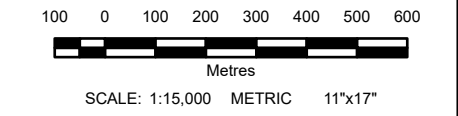
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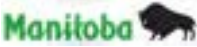
- Capped Mine Opening
- Water sampling location
- Site Access Road
- Tailings Management Area (TMA)

NOTES:

1. Imagery Source: Ruttan LiDAR imagery, dated as 2022.

2. All units are metric and in metres unless otherwise specified. Transverse Mercator Projection, NAD 1983 CSRS, Zone 14. Elevations are in metres above sea level (MSL).

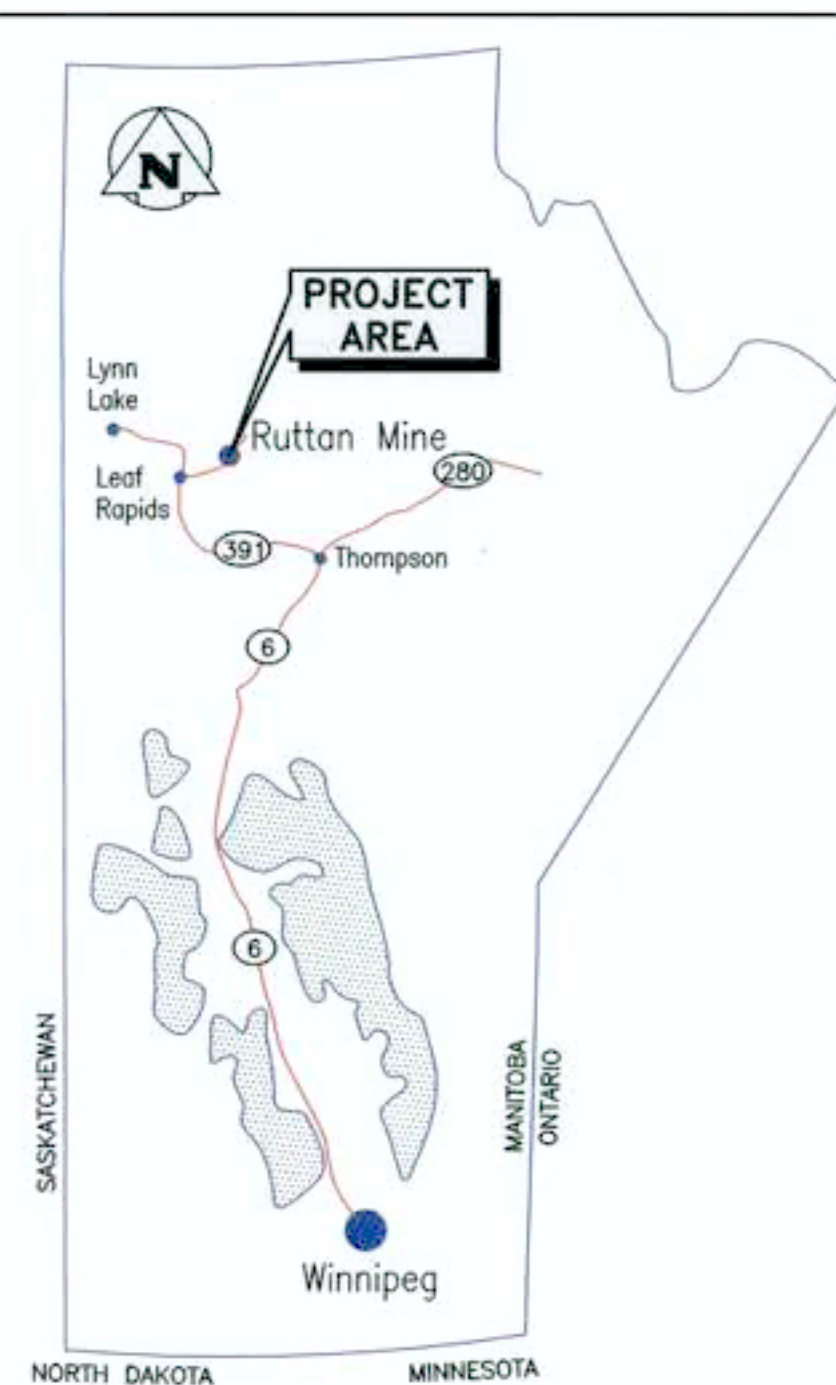
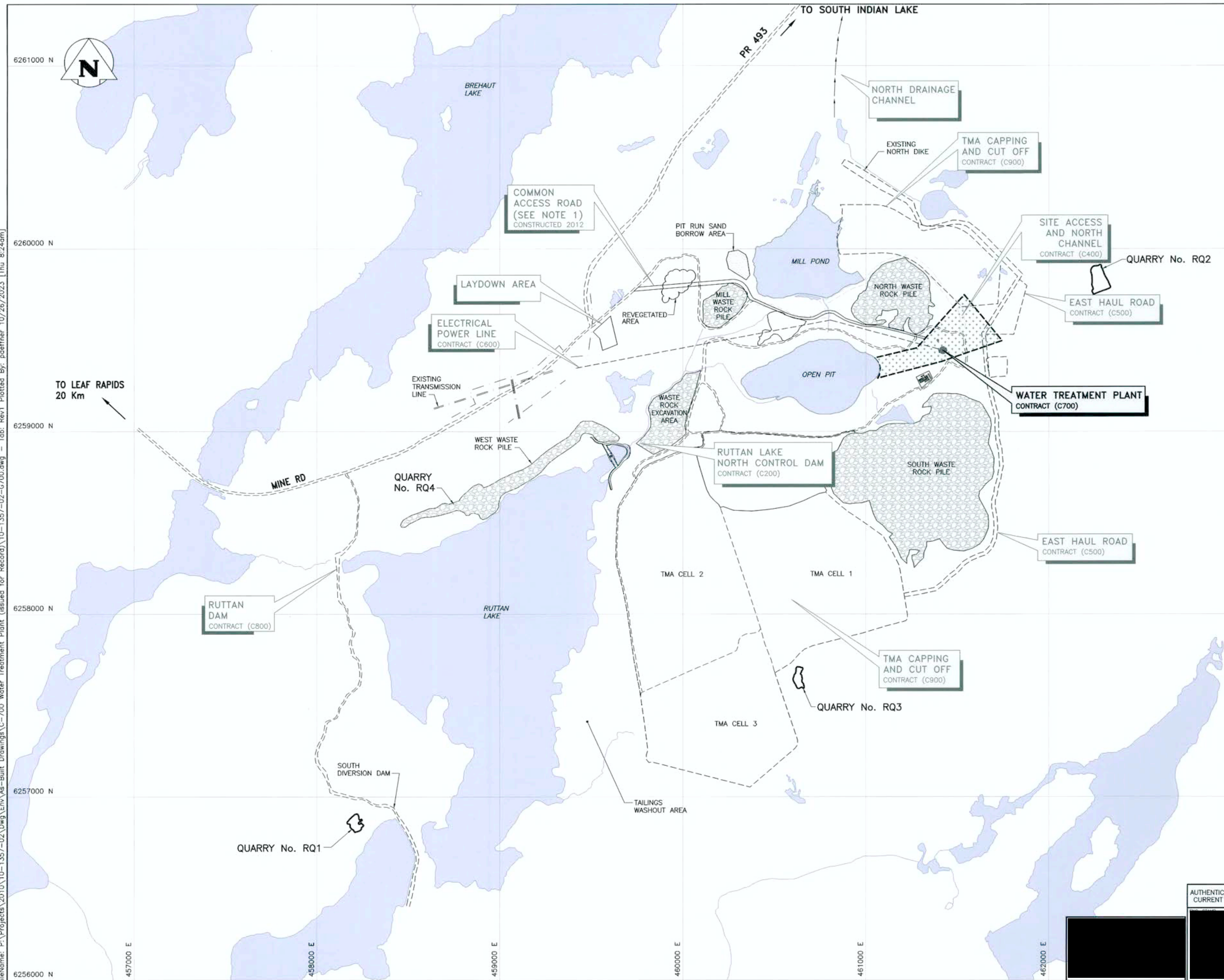


0	24/10/31	ISSUED WITH FINAL REPORT		RED	SM
NO.	YYMMDD	DESCRIPTION		ISSUED BY	CHECK BY
REVISIONS / ISSUE					
					
OAM SUBPROGRAM - RUTTAN MINE SITE SUMMARY OF SITE ACTIVITIES					
SITE OVERVIEW & SAMPLE LOCATIONS					
OCTOBER 2024		FIGURE 02		REV:	0

APPENDIX A

Water Treatment Plant Drawings

File Name: P:\Projects\2010\10-1357-02\Drawings\Built Drawings\700 Water Treatment Plant (issued for Record)\10-1357-02-G700.dwg - Tab: Rev1 Plotted By: pdeffner 10/26/2023 [Thu 8:24am]



KEY MAP

LEGEND:

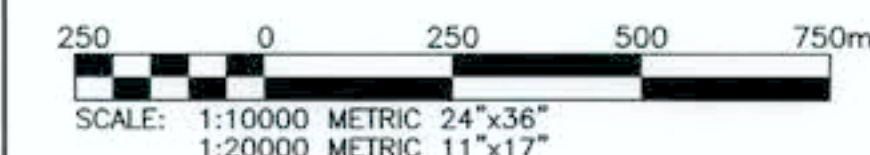
- ROADS
- BUILDINGS
- LIMITS OF WORK AREA THIS CONTRACT

6259900 N UTM GRID, NAD 83, METRIC

NOTES:

- COMMON ACCESS ROAD CONTRACTOR IS RESPONSIBLE FOR GENERAL MAINTENANCE AND COMPLETION OF CLAY SEAL WORK IN 2013.

RECORD DRAWING



1	23/10/31	ISSUED RECORD DRAWING	JBS
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	JBS
NO.	YY/MM/DD	DESCRIPTION	BY

REVISIONS / ISSUE



PROJECT: RUTTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION: GEOTECHNICAL
PROJECT AREA MAP

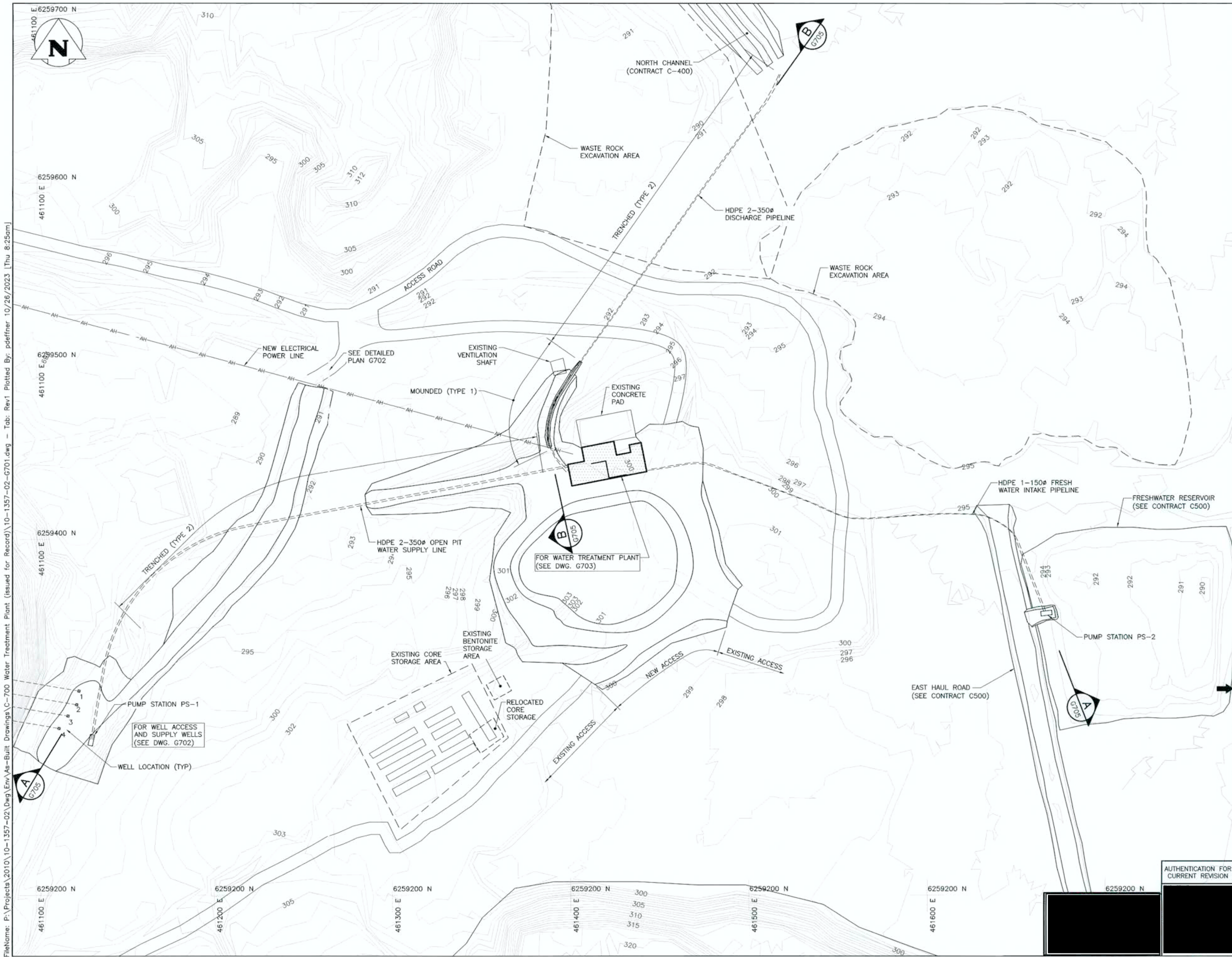
AUTHENTICATION FOR
CURRENT REVISION

KGS
GROUP
CONSULTING
ENGINEERS

DESIGN BY:	RGN	DATE (YY/MM/DD):	12/06/21
DESIGN CHECK:	JBS	DATE:	12/10/17
DRAWN BY:	PwD	DATE:	12/06/21
DWG CHECK:	jS	DATE:	12/10/17

DWG. NO. 10-1357-02 G700 REV: 1

File Name: E:\Projects\2010\10-1357-02.Dwg, Env\As-Built Drawings\C-700 Water Treatment Plant (issued for Record)\10-1357-02-G701.dwg - Tab: Rev1 Plotted By: pdeffner 10/26/2023 [Thu 8:25am]



LEGEND:

CP-1 CONTROL POINT

285 TOPOGRAPHIC CONTOUR (m) GEODETIC

6259900 N UTM GRID, NAD 83, METRIC

REFERENCE:

CONTOURS HAVE BEEN DERIVED FROM LIDAR DATA CAPTURED ON SEPTEMBER 17, 2009.

RECORD DRAWING

20 0 20 40 60 80m

SCALE: 1:1000 METRIC 24"x36"
1:2000 METRIC 11"x17"

1	23/10/31	ISSUED RECORD DRAWING	JBS
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	JBS
NO.	YY/MM/DD	DESCRIPTION	BY

REVISIONS / ISSUE

CLIENT:

Manitoba
MANITOBA MINES BRANCH

PROJECT:

RUTTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:

GEOTECHNICAL
GENERAL ARRANGEMENT

AUTHENTICATION FOR
CURRENT REVISION

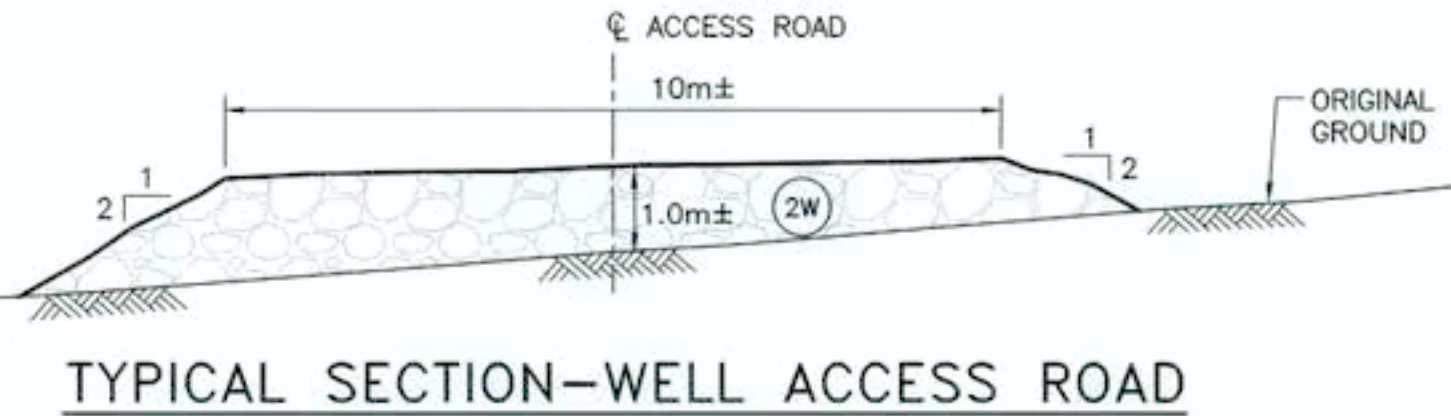
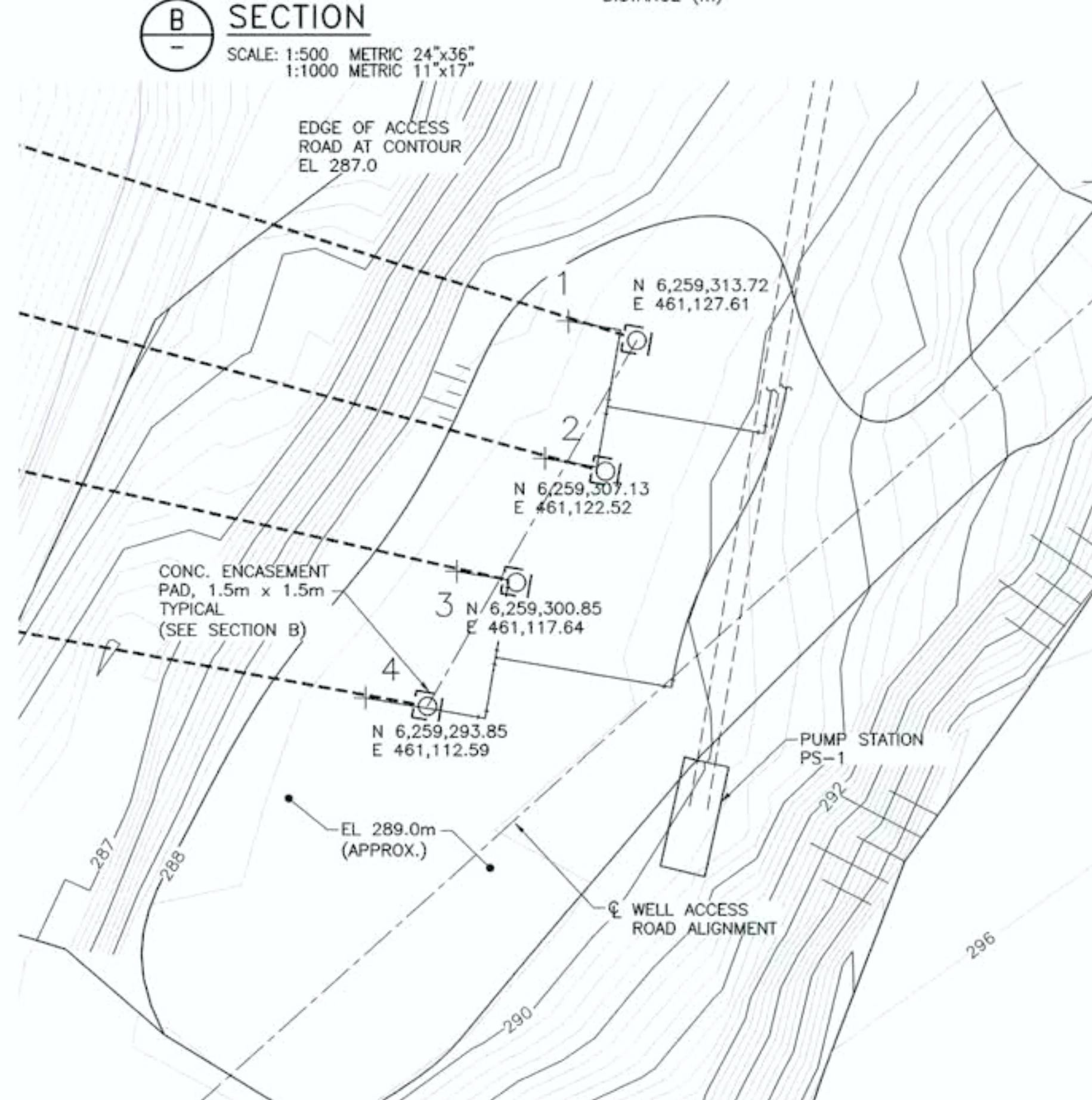
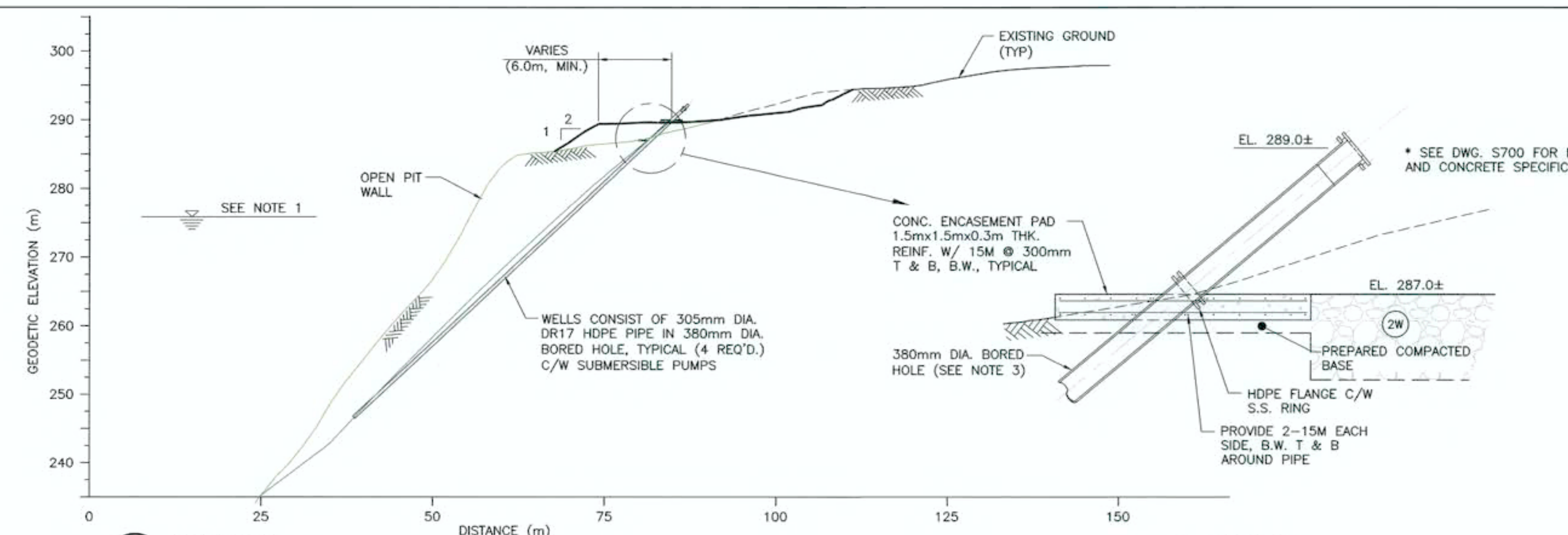
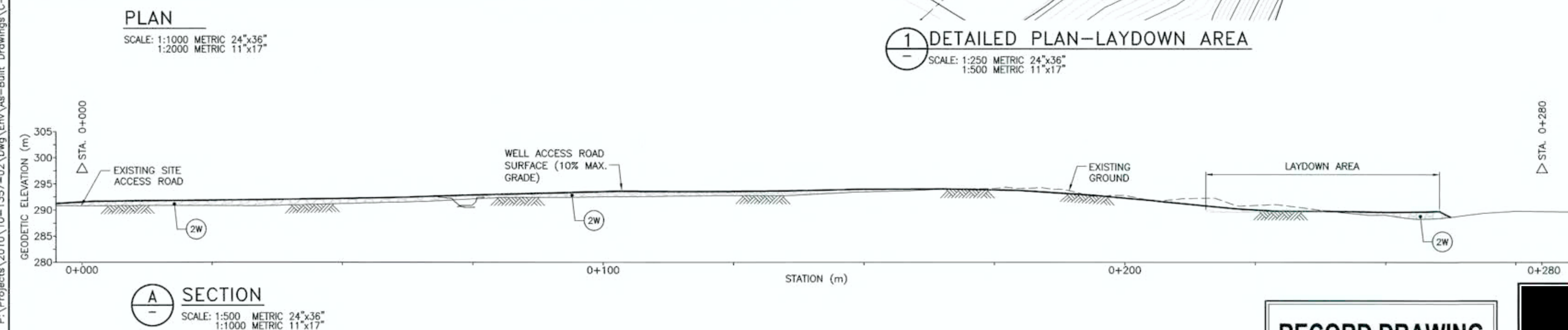
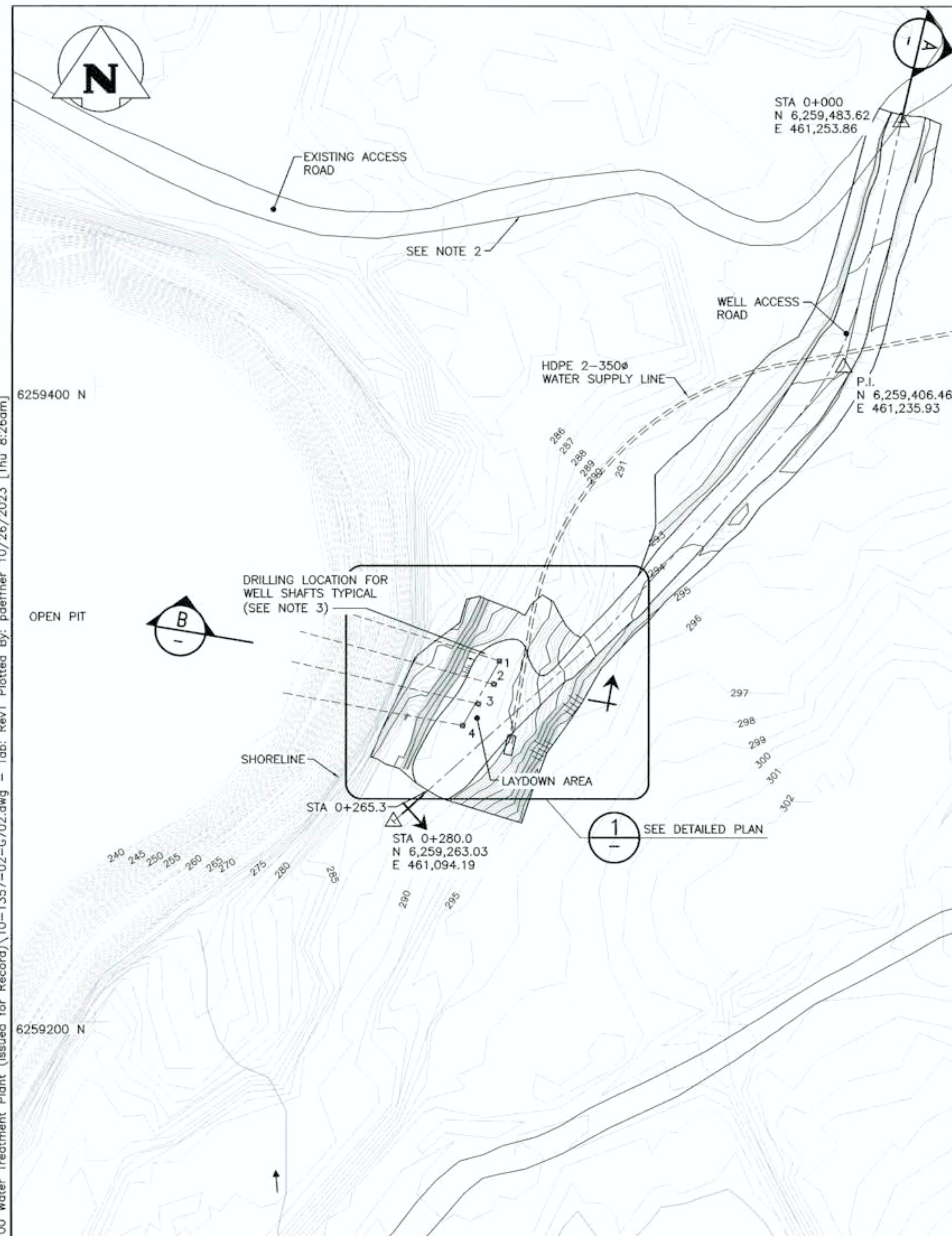
KGS
GROUP
CONSULTING
ENGINEERS

DESIGN BY:	RGD	DATE (YY/MM/DD):	12/07/05
DESIGN CHECK:	jbs	DATE:	12/10/17
DRAWN BY:	PwD/ESm	DATE:	12/07/05
DWG CHECK:	js	DATE:	12/10/17

DWG. NO. 10-1357-02 G701

REV: 1

File Name: P:\Projects\2010-1357-02-Dwg-Env\As-Built Drawings\C-700 Water Treatment Plant (Issued for Record)\10-1357-02-G702.dwg - Tab: Rev1 Plotted By: pdfntr 10/26/2023 [Thu 8:26am]



LEGEND:

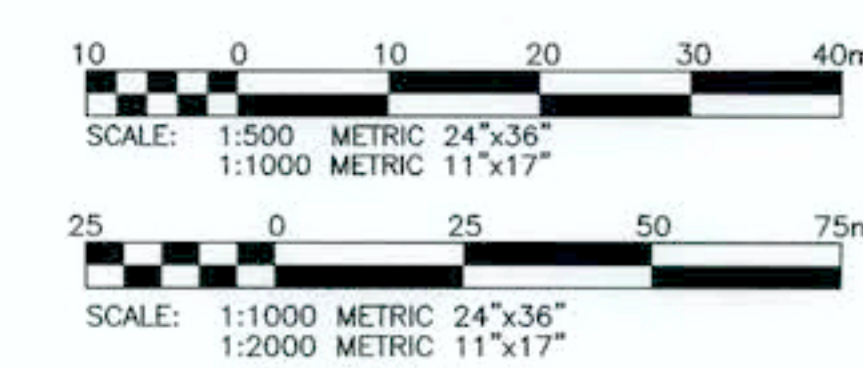
WELL

NOTES:

- OPEN PIT WATER LEVEL (AUGUST, 2012) AT EL. 276.0, APPROX., AND IS ANTICIPATED TO RISE WITH TIME.
- WELL ACCESS ROAD TIED IN SMOOTHLY TO EXISTING ROAD.
- HORIZONTAL ORIENTATION AND MINIMUM DIP OF 45° FROM HORIZONTAL FIELD FINALIZED AS DIRECTED BY THE CONSULTANT.

REFERENCE:

CONTOURS HAVE BEEN DERIVED FROM LIDAR DATA CAPTURED ON SEPTEMBER 17, 2009.



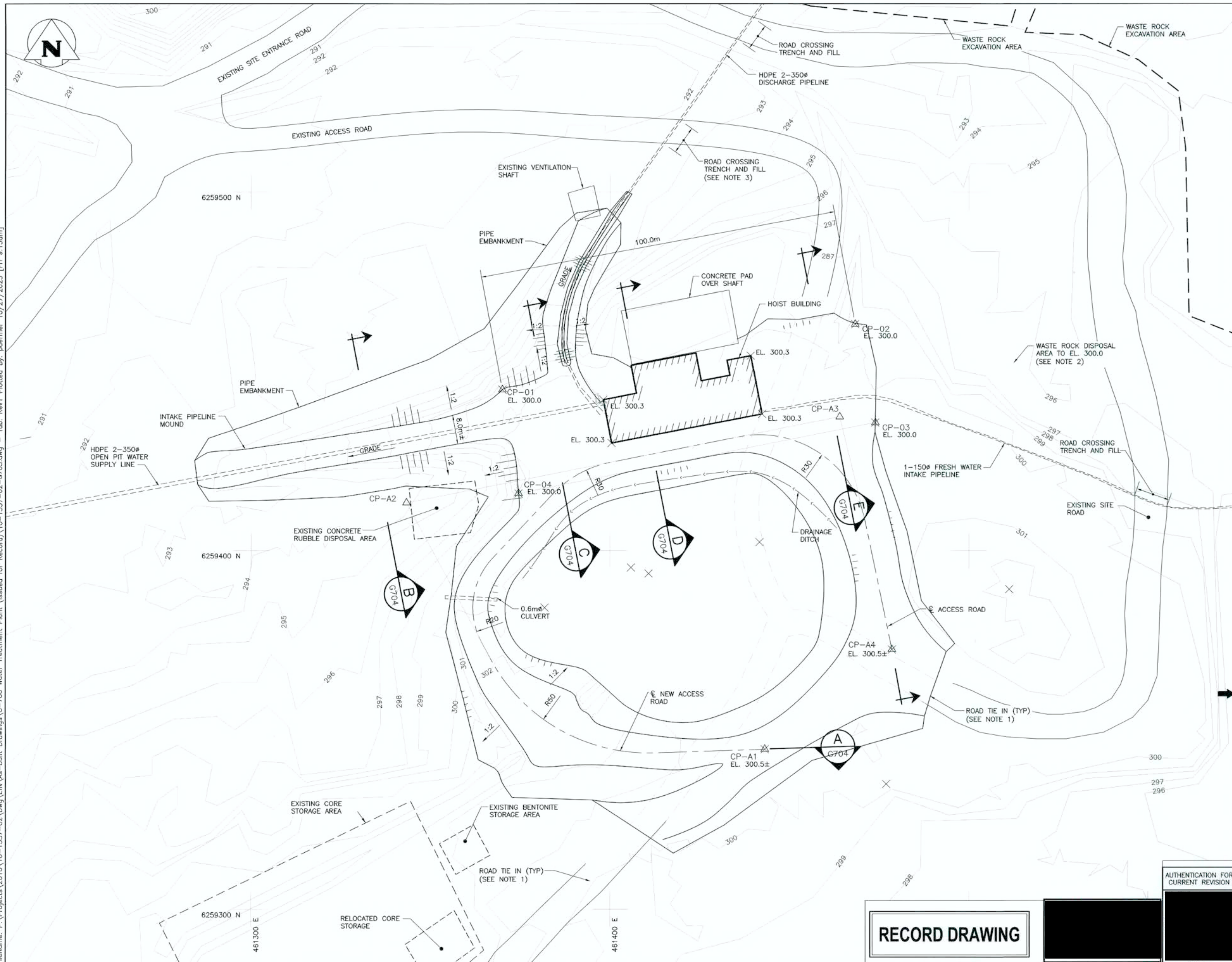
1	23/10/31	ISSUED RECORD DRAWING	
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
GEOTECHNICAL WELL ACCESS PLAN, SECTIONS AND DETAILS			
AUTHENTICATION FOR CURRENT REVISION		DESIGN BY: RGN	DATE (YY/MM/DD): 12/08/20
		DESIGN CHECK: JBS	DATE: 12/10/17
		DRAWN BY: ESm	DATE: 12/08/20
		DWG CHECK: JS	DATE: 12/10/17
DWG. NO.		10-1357-02	G702
REV:			1

RECORD DRAWING

APPENDIX A

Water Treatment Plant Drawings

File Name: P:\Projects\2010\10-1357-02\Drawings\As-Built Drawings\C-700 Water Treatment Plant (Issued for Record)\10-1357-02-G703.dwg - Tab: Rev1 Plotted By: pdfntr 10/27/2023 [Fri 9:15am]



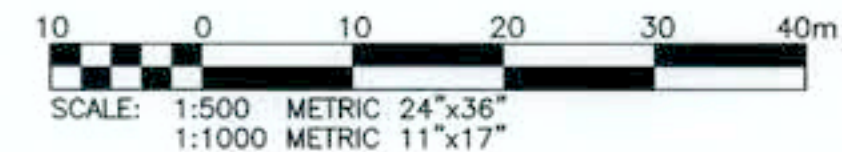
- LEGEND:**
- CP-R1 CONTROL POINT
 - EL. 300.0 PARKING GRADE ELEVATION
 - 295 TOPOGRAPHIC CONTOUR (GEODETIC, METRES)

ACCESS ROAD LAYOUT TABLE		
CP #	EASTING	NORTHING
CP-A1	461,409.44	6,259,322.53
CP-A2	461,343.30	6,259,413.46
CP-A3	461,464.00	6,259,437.36
CP-A4	461,482.47	6,259,371.79

PARKING LAYOUT TABLE		
CP #	EASTING	NORTHING
CP-01	461,370.02	6,259,444.91
CP-02	461,460.84	6,259,461.01
CP-03	461,466.48	6,259,433.59
CP-04	461,374.52	6,259,416.07

- NOTES:**
- HOIST BUILDING ROAD ACCESS TIE IN SMOOTHLY TO EXISTING SITE ROAD.
 - EXTENT OF DISPOSAL AREA IS DEPENDENT ON THE AMOUNT OF WASTE ROCK AND EXPANDED TO THE EAST AS DIRECTED BY THE CONSULTANT.
 - ROAD CROSSINGS REQUIRED THE RAISING OF THE ROAD SURFACE TO MAINTAIN GRADES OF PIPES.
 - RECORD DRAWINGS AS OF 2012, INITIAL CONTRACT COMPLETION, SUBSEQUENTLY REFINEMENTS ADDED TO THE PLAN.

REFERENCE:
CONTOURS HAVE BEEN DERIVED FROM LIDAR DATA CAPTURED ON SEPTEMBER 17, 2009.



1	23/10/31	ISSUED RECORD DRAWING	JBS
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	JBS
NO.	YY/MM/DD	DESCRIPTION	BY

REVISIONS / ISSUE	
CLIENT:	

PROJECT:
RUTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

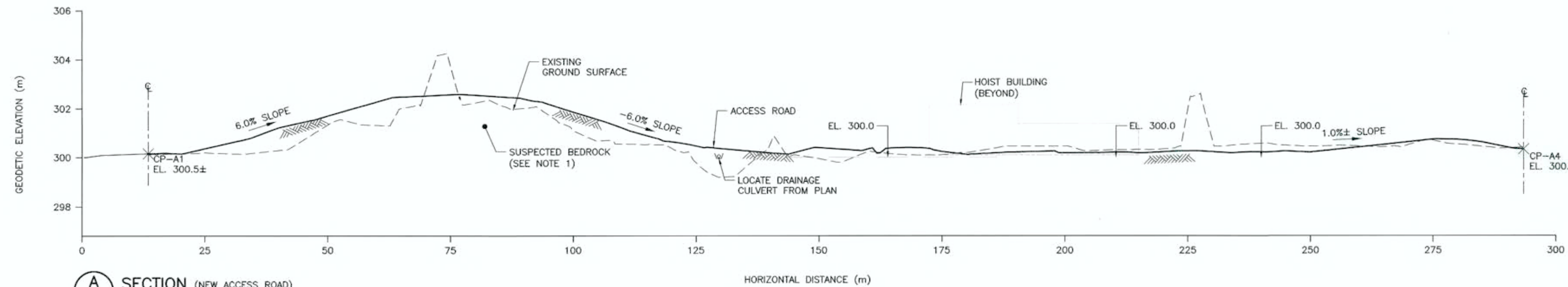
DWG. DESCRIPTION:
GEOTECHNICAL
SITE GRADING PLAN

AUTHENTICATION FOR CURRENT REVISION	DESIGN BY:	RGN	DATE (YY/MM/DD):	12/06/26
	DESIGN CHECK:	JBS	DATE:	12/10/17
	DRAWN BY:	PwD/SDW	DATE:	12/06/26
	DWG CHECK:	JBS	DATE:	12/10/17

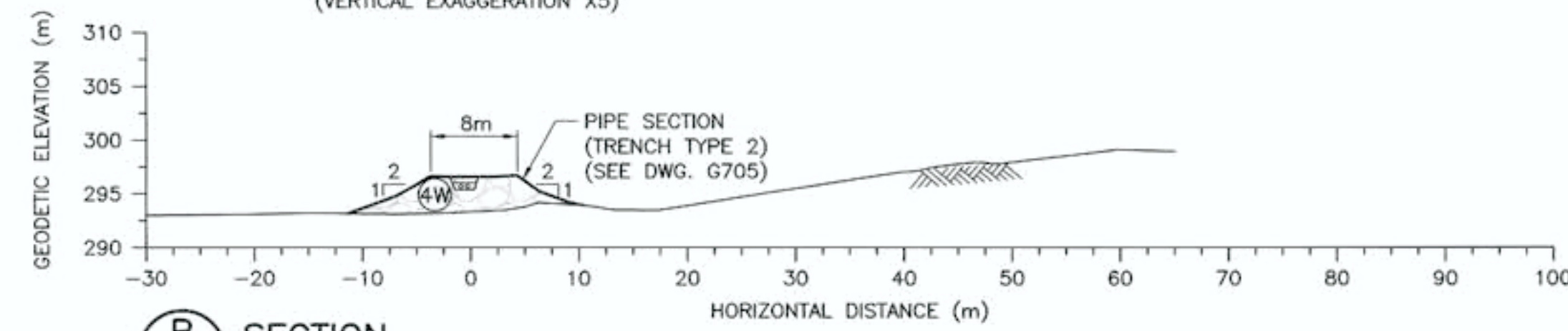
DWG. NO. 10-1357-02 G703 REV: 1

RECORD DRAWING

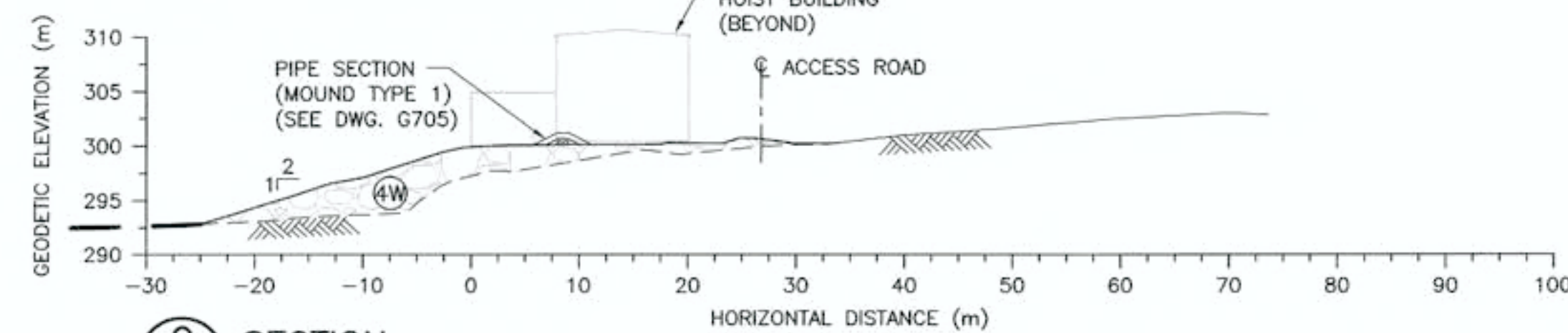
File Name: P:\Projects\2010\10-1357-02\Drawings\As-Built Drawings\C-700 Water Treatment Plant (Issued for Record)\10-1357-02-G704.dwg - Tab: Rev1 Plotted By: pdeffner 10/26/2023 [Thu 8:38am]



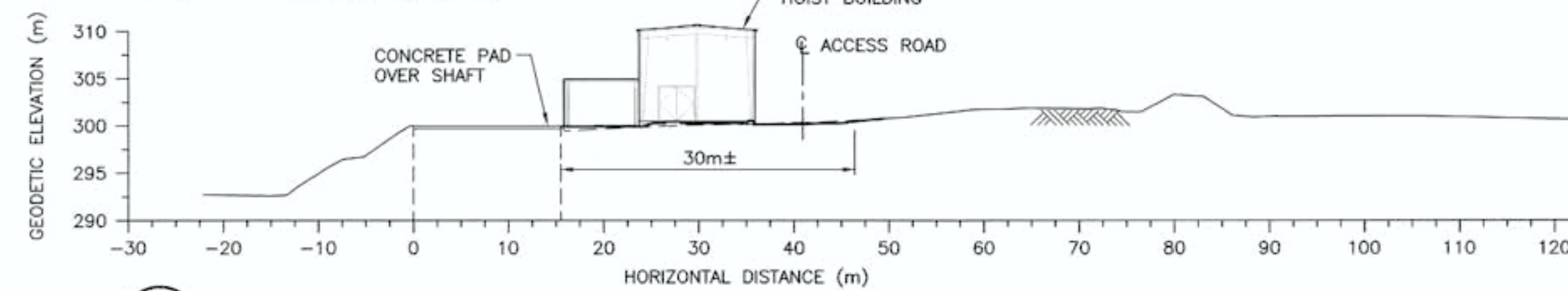
A SECTION (NEW ACCESS ROAD)
G703 SCALE: 1:500 METRIC 24"x36"
1:1000 METRIC 11"x17"
(VERTICAL EXAGGERATION X5)



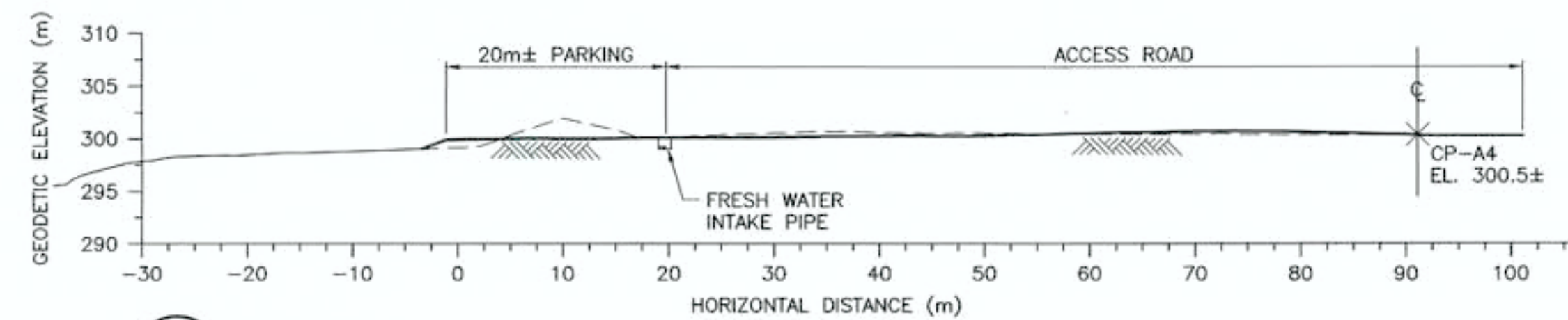
B SECTION
G703 SCALE: 1:1000 METRIC 24"x36"
1:2000 METRIC 11"x17"



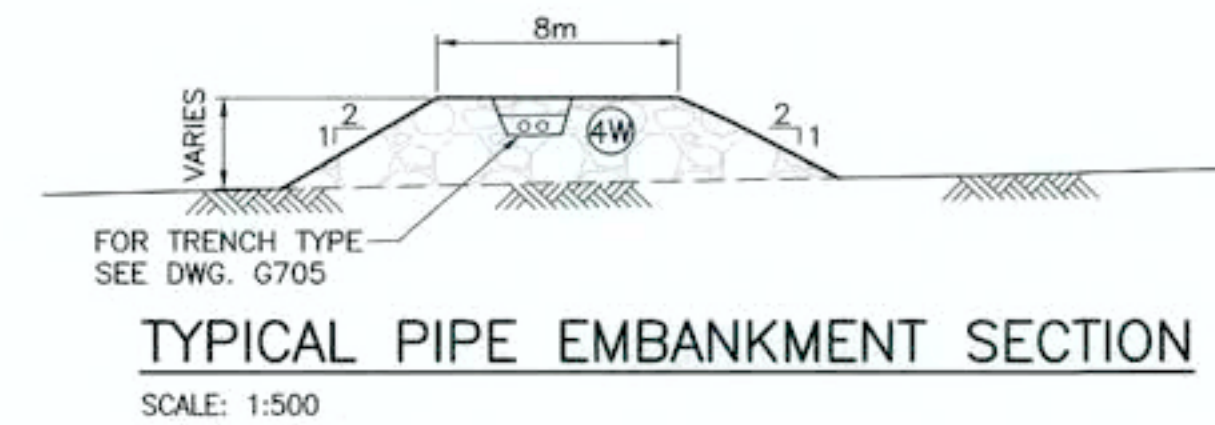
C SECTION
G703 SCALE: 1:1000 METRIC 24"x36"
1:2000 METRIC 11"x17"



D SECTION
G703 SCALE: 1:1000 METRIC 24"x36"
1:2000 METRIC 11"x17"



E SECTION
G703 SCALE: 1:1000 METRIC 24"x36"
1:2000 METRIC 11"x17"



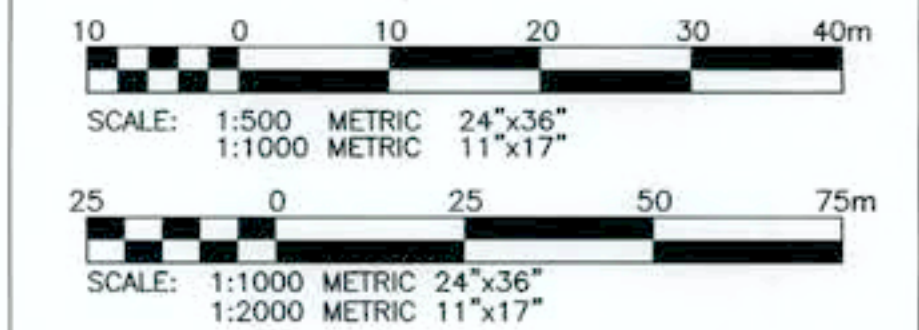
TYPICAL PIPE EMBANKMENT SECTION
SCALE: 1:500

LEGEND:

- RT - ROAD TOPPING
- 1 - IMPERVIOUS CLAY
- 2C - TYPE 2C AS SPECIFIED
- 2W - TYPE 2W AS SPECIFIED
- 3C - TYPE 3C AS SPECIFIED
- 3W - TYPE 3W AS SPECIFIED
- 4W - TYPE 4W AS SPECIFIED
- 5 - SAND

NOTES:

- ACCESS ROAD GRADE IN AREA OF POSSIBLE BEDROCK NO STEEPER THAN 6.0% AS DIRECTED BY THE CONSULTANT.



1	23/10/31	ISSUED RECORD DRAWING	JBS
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	JBS
NO.	YY/MM/DD	DESCRIPTION	BY

REVISIONS / ISSUE



PROJECT:
RUTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:
GEOTECHNICAL
SITE GRADING SECTIONS AND DETAILS

AUTHENTICATION FOR
CURRENT REVISION

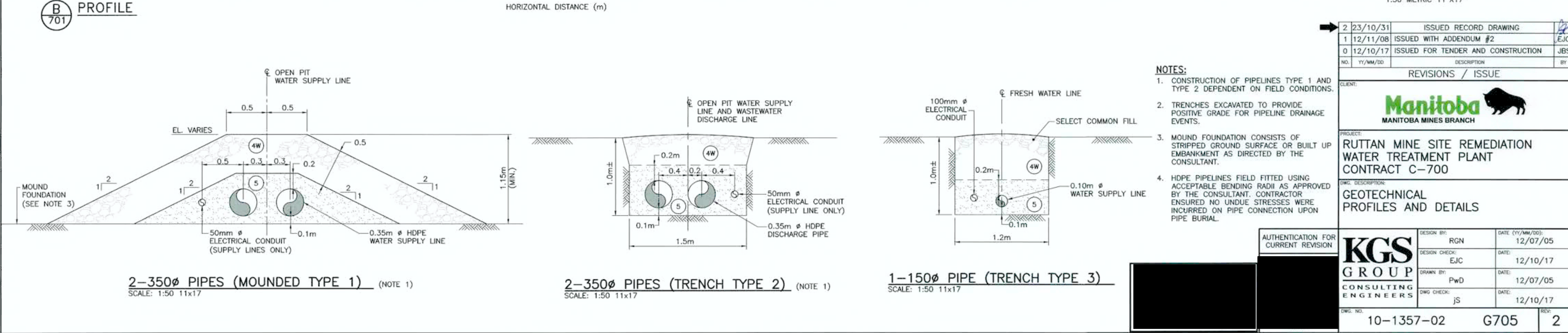
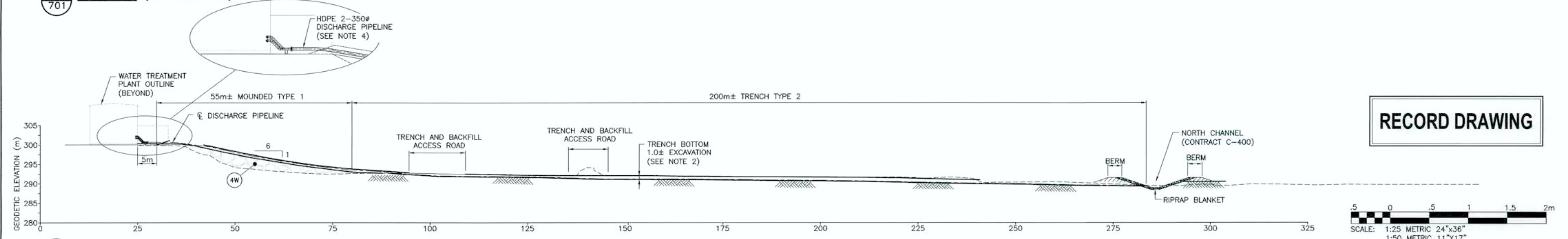
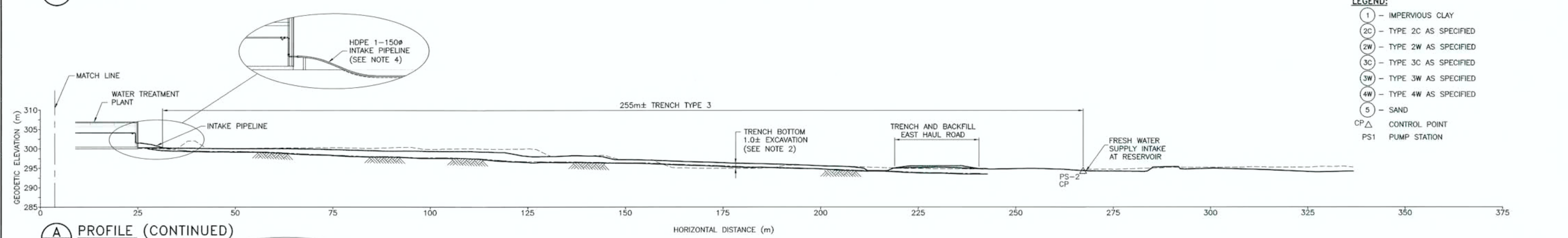
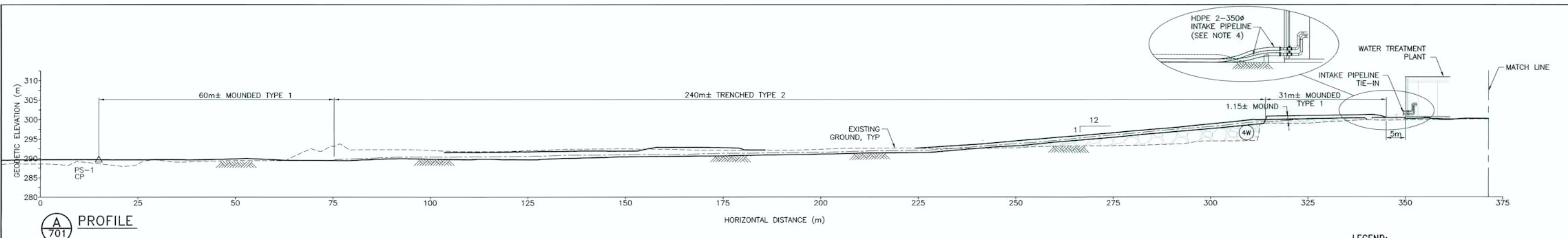
KGS
GROUP
CONSULTING
ENGINEERS

DESIGN BY:	RGH	DATE: (YY/MM/DD):	12/06/26
DESIGN CHECK:	JBS	DATE:	12/10/17
DRAWN BY:	SDW	DATE:	12/06/26
DWG CHECK:	JS	DATE:	12/10/17

DWG. NO. 10-1357-02 G704 REV. 1

RECORD DRAWING

File Name: P:\Projects\2010\10-1357-02\Drawings\A-Built Drawings\C-700 Water Treatment Plant (Issued for Record)\10-1357-02-G705.dwg - Tab: Rev2 Plotted By: pdefiner 10/26/2023 [Thu 8:39am]



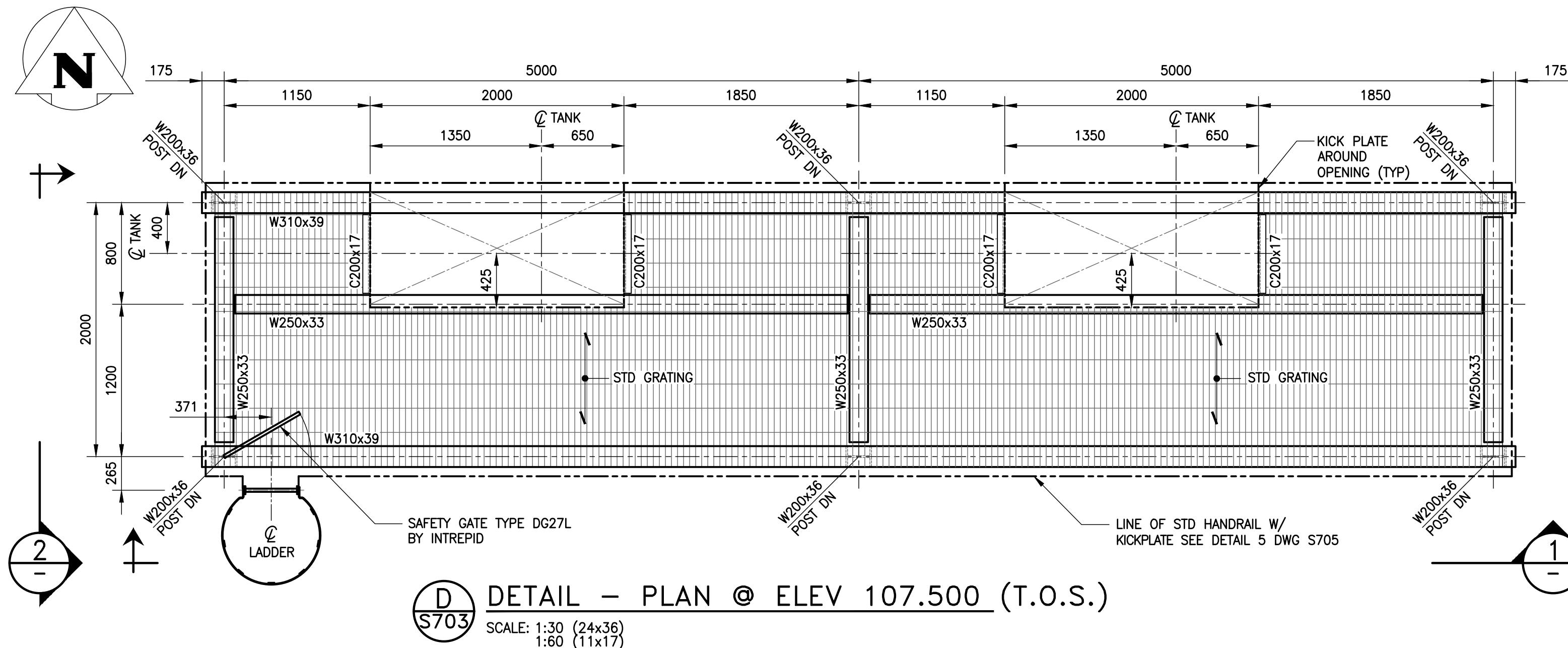
RECORD DRAWING

2	23/10/31	ISSUED RECORD DRAWING	JJC
1	12/11/08	ISSUED WITH ADDENDUM #2	EJC
0	12/10/17	ISSUED FOR TENDER AND CONSTRUCTION	JBS
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
GEOTECHNICAL PROFILES AND DETAILS			
AUTHENTICATION FOR CURRENT REVISION			
DESIGN BY:		RGD	DATE: 12/07/05
DESIGN CHECK:		EJC	DATE: 12/10/17
DRAWN BY:		PwD	DATE: 12/07/05
DWG. CHECK:		JS	DATE: 12/10/17
DWG. NO. 10-1357-02 G705			
REV. 2			

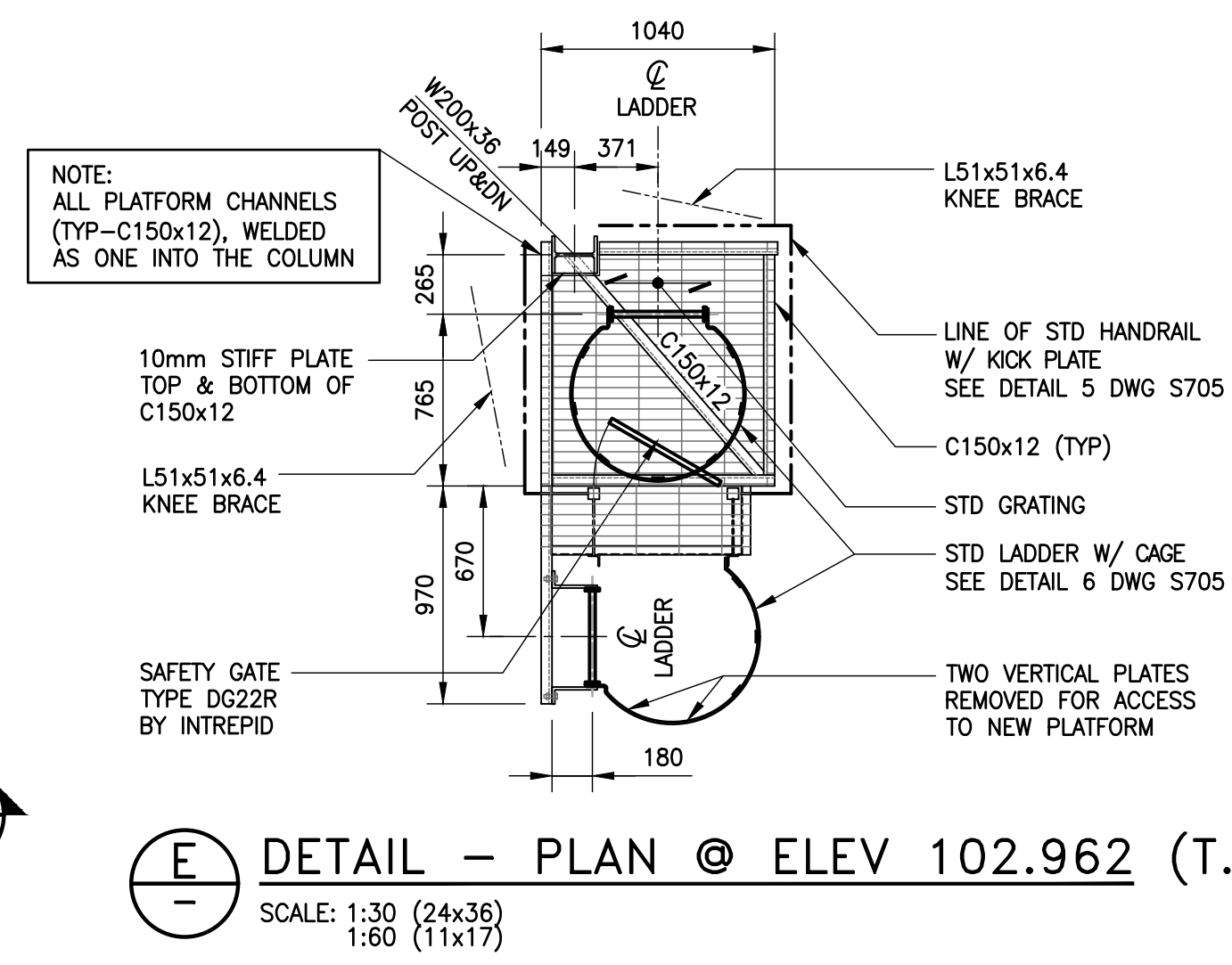
APPENDIX A

Water Treatment Plant Drawings

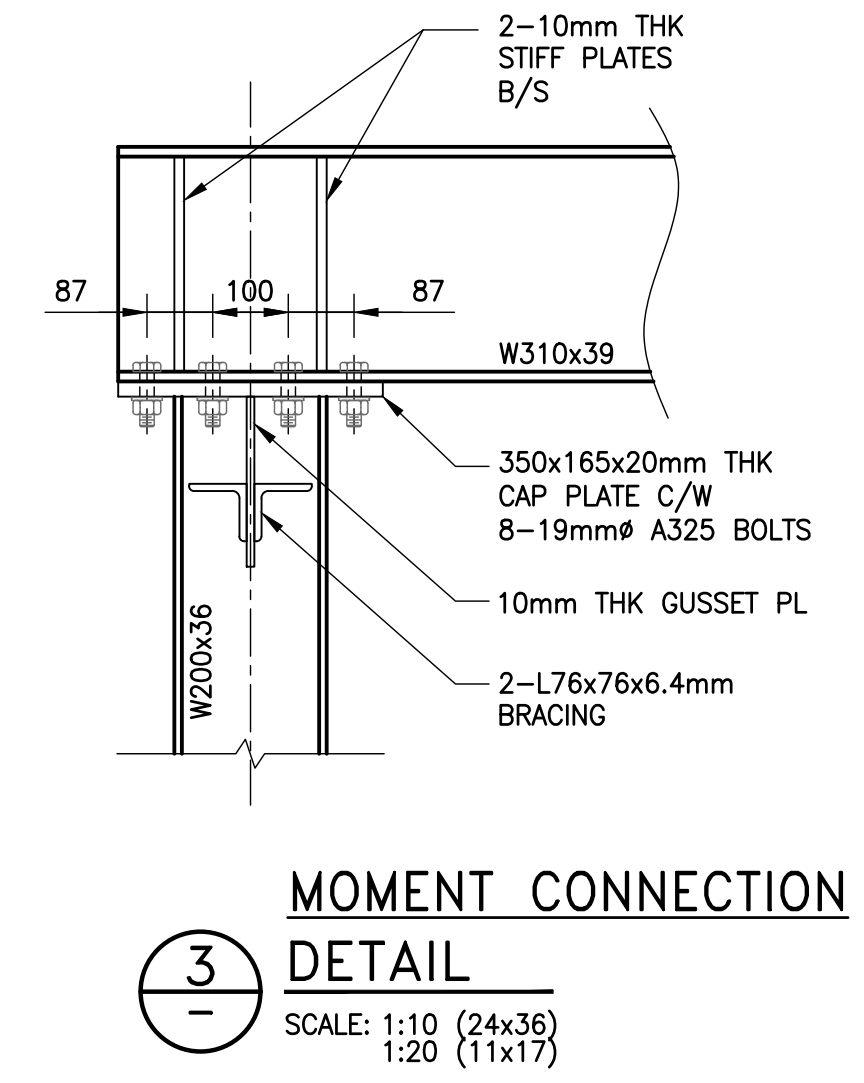
File Name: U:\FMS\21-1357-001\21-1357-001_S704.dwg - Tab: S704 Plotted By: emaling 23/10/25 [Wed 7:04pm]
24x36 PLOT SCALE: 1"=1'-0"



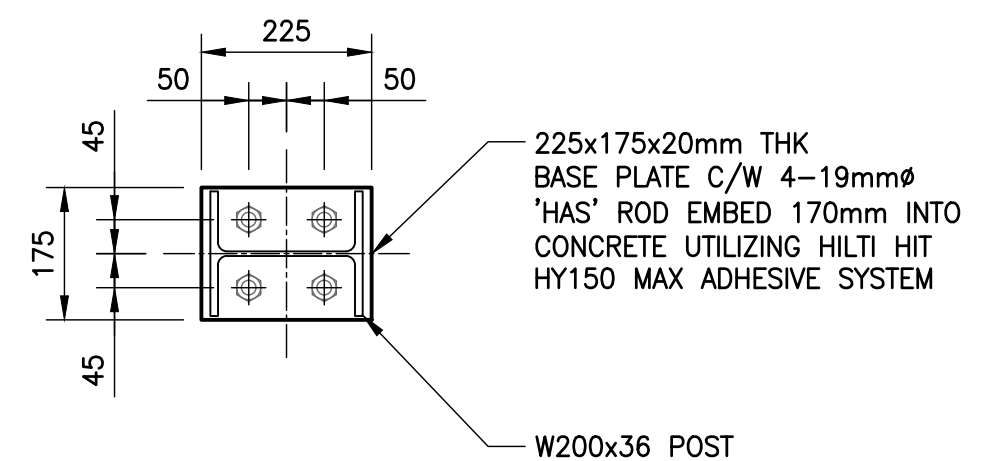
D DETAIL - PLAN @ ELEV 107.500 (T.O.S.)
SCALE: 1:30 (24x36)
1:60 (11x17)



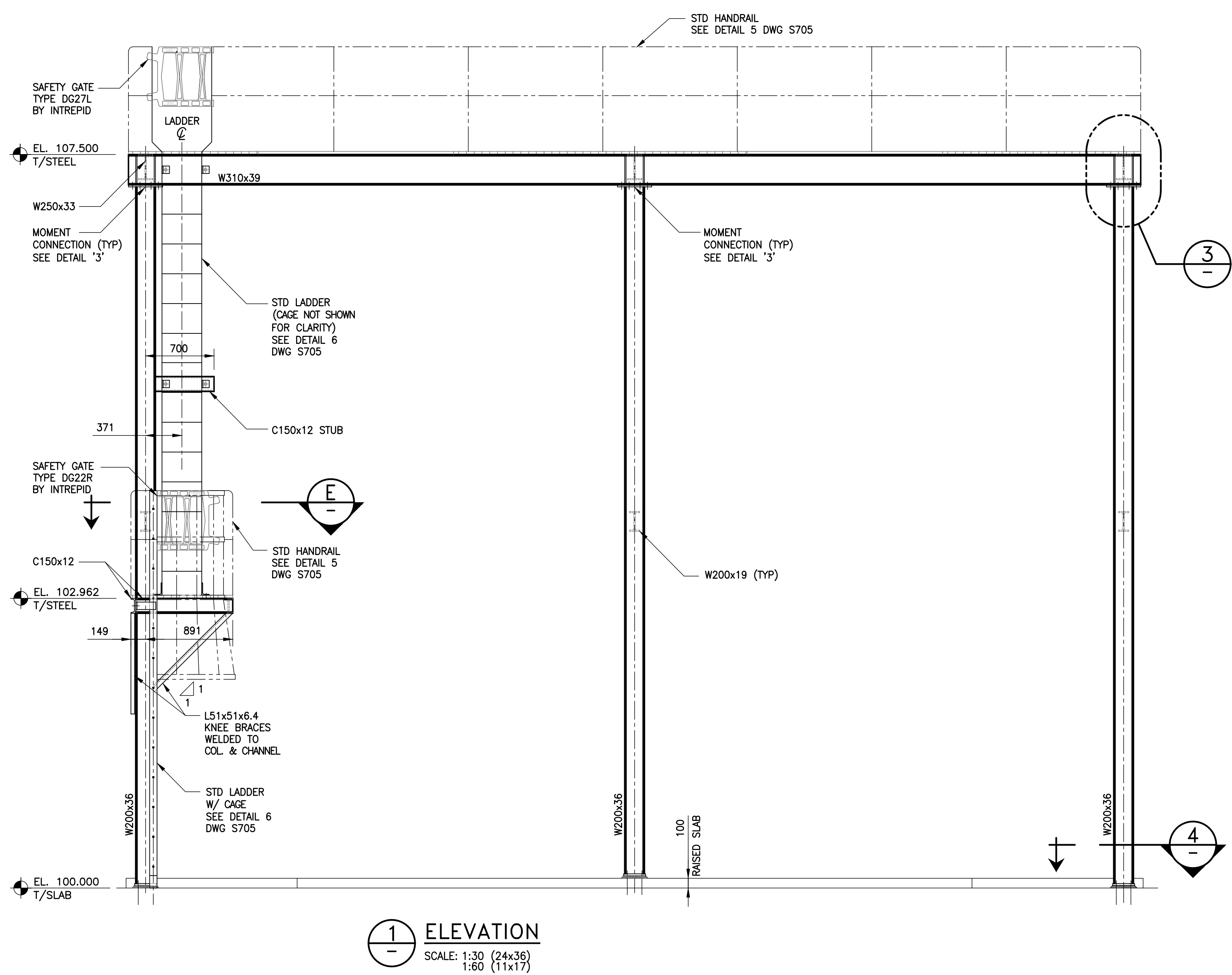
E DETAIL - PLAN @ ELEV 102.962 (T.O.S.)
SCALE: 1:30 (24x36)
1:60 (11x17)



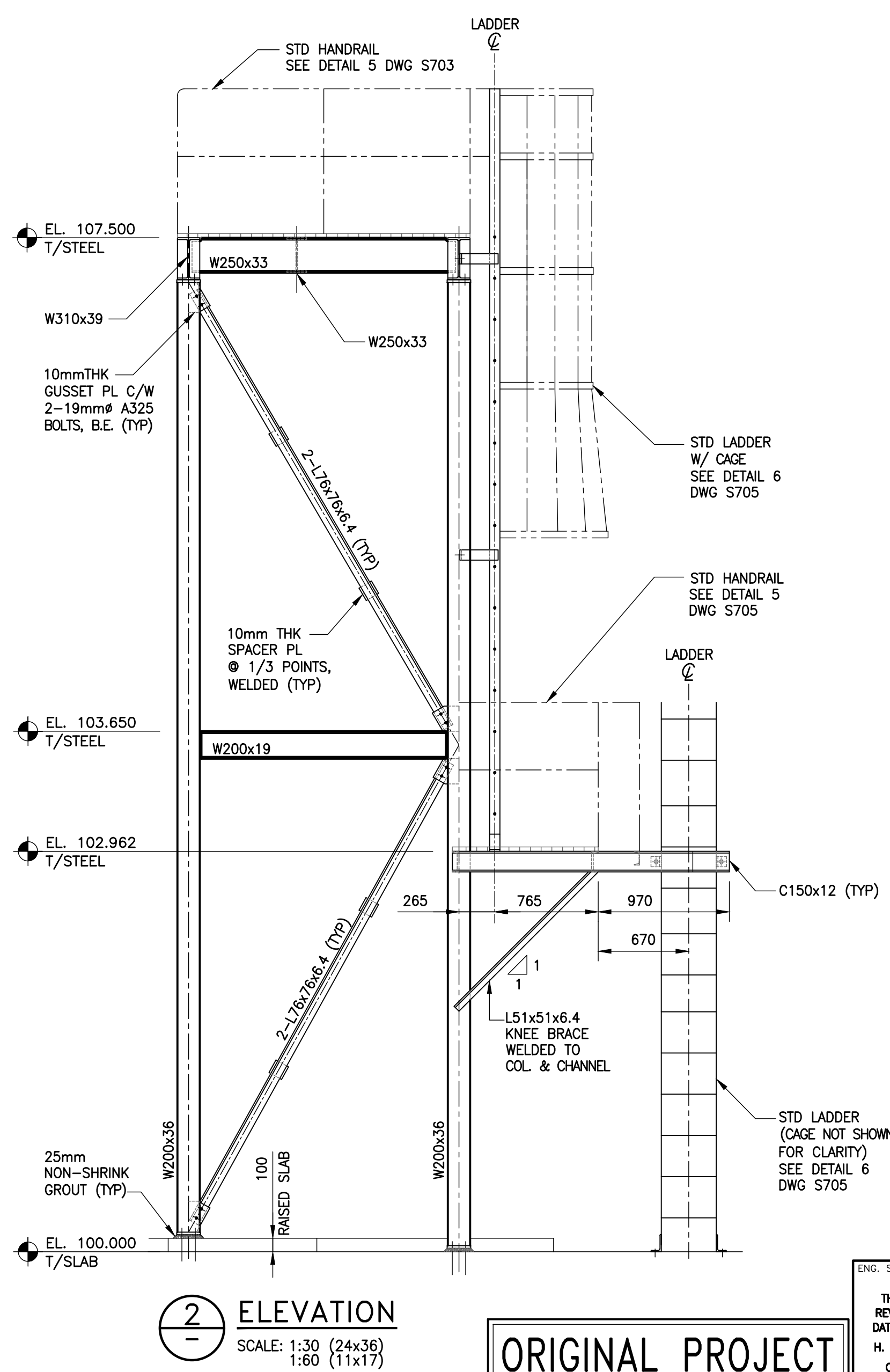
3 MOMENT CONNECTION
DETAIL
SCALE: 1:10 (24x36)
1:20 (11x17)



4 SECTION
SCALE: 1:10 (24x36)
1:20 (11x17)



1 ELEVATION
SCALE: 1:30 (24x36)
1:60 (11x17)



2 ELEVATION
SCALE: 1:30 (24x36)
1:60 (11x17)

ORIGINAL PROJECT
10-1357-02

ENG. STAMP
THE ORIGINAL ISSUE
REV. 0 WAS STAMPED,
DATED AND SIGNED BY:
H. R. FOERSTER P.Eng
ON 2012/10/12

AS-BUILT
AS-BUILT DRAWING REVISIONS ARE
BASED ON FIELD OBSERVATIONS BY
KGS GROUP ON:
2023/07/11
THE DESIGN OF THESE CHANGES HAVE
NOT BEEN VERIFIED BY KGS GROUP.

0 23/10/26 ISSUED AS-BUILT		
NO.	YY/MM/DD	DESCRIPTION
REVISIONS / ISSUE		
CLIENT:		
Manitoba MANITOBA MINES BRANCH		
PROJECT:		
RUTAN MINE SITE REMEDIAL WATER TREATMENT PLANT CONTRACT C-700		
DWG. DESCRIPTION:		
STRUCTURAL BUILDING - STEEL WORKS SECTIONS & DETAILS		
DESIGN BY:	DATE (YY/MM/DD):	
HRF	12/06/08	
DESIGN CHECK:	DATE:	
RUL	12/10/17	
DRAWN BY:	DATE:	
EDC	12/06/08	
DWG. CHECK:	DATE:	
HRF	12/10/12	
DWG. NO.	REV.	
21-1357-001 S704	0	



SCALE: 1:20 (24x36)
1:40 (11x17)



SCALE: 1:15 (24x36)
1:30 (11x17)



ENG. STAMP

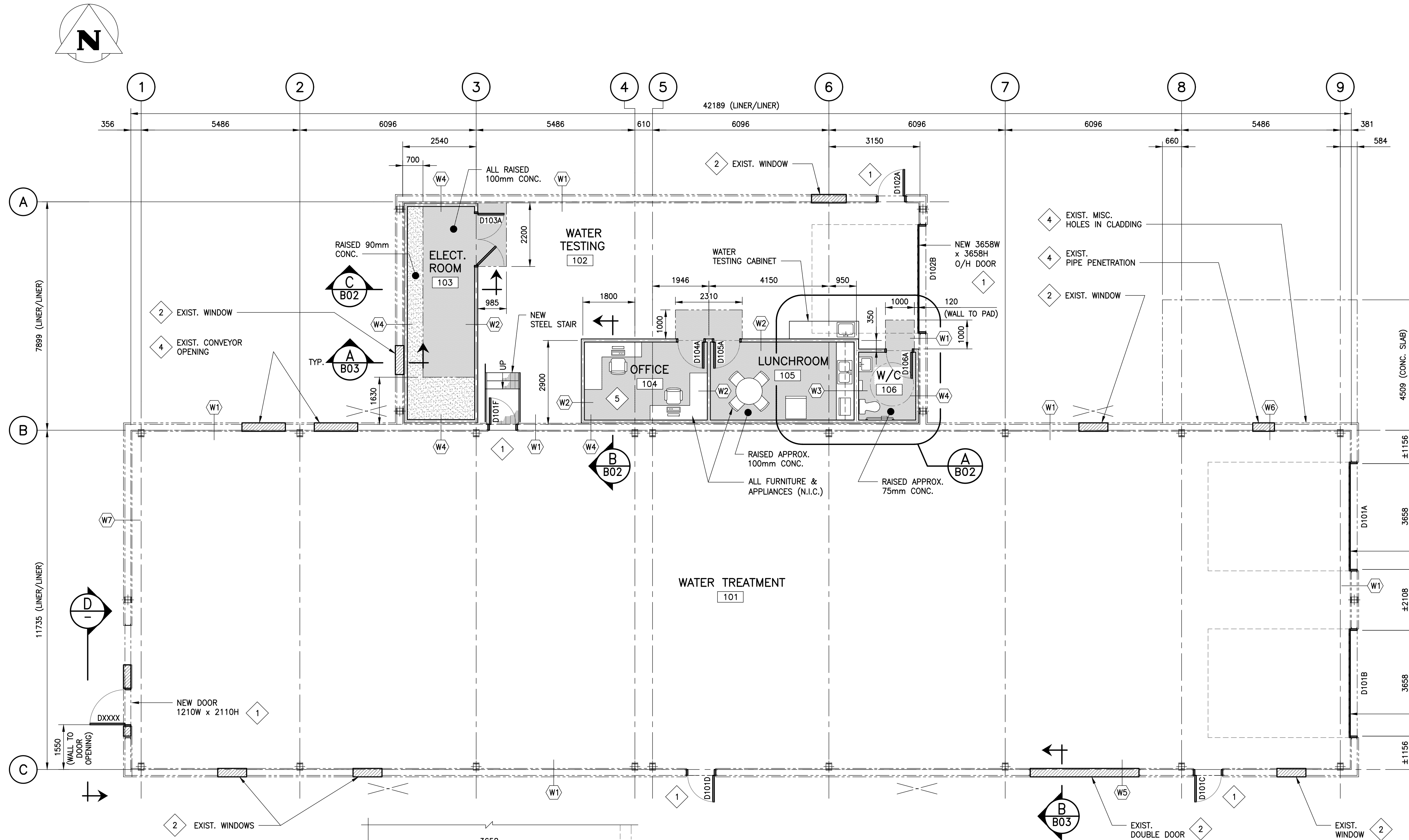
THE ORIGINAL ISSUE
REV. 0 WAS STAMPED,
DATED AND SIGNED BY:
H. R. FOERSTER P.Eng
ON 2012/10/12

AS-BUILT

AS-BUILT DRAWING REVISIONS ARE
BASED ON FIELD OBSERVATIONS BY
KGS GROUP ON:
2023/07/11

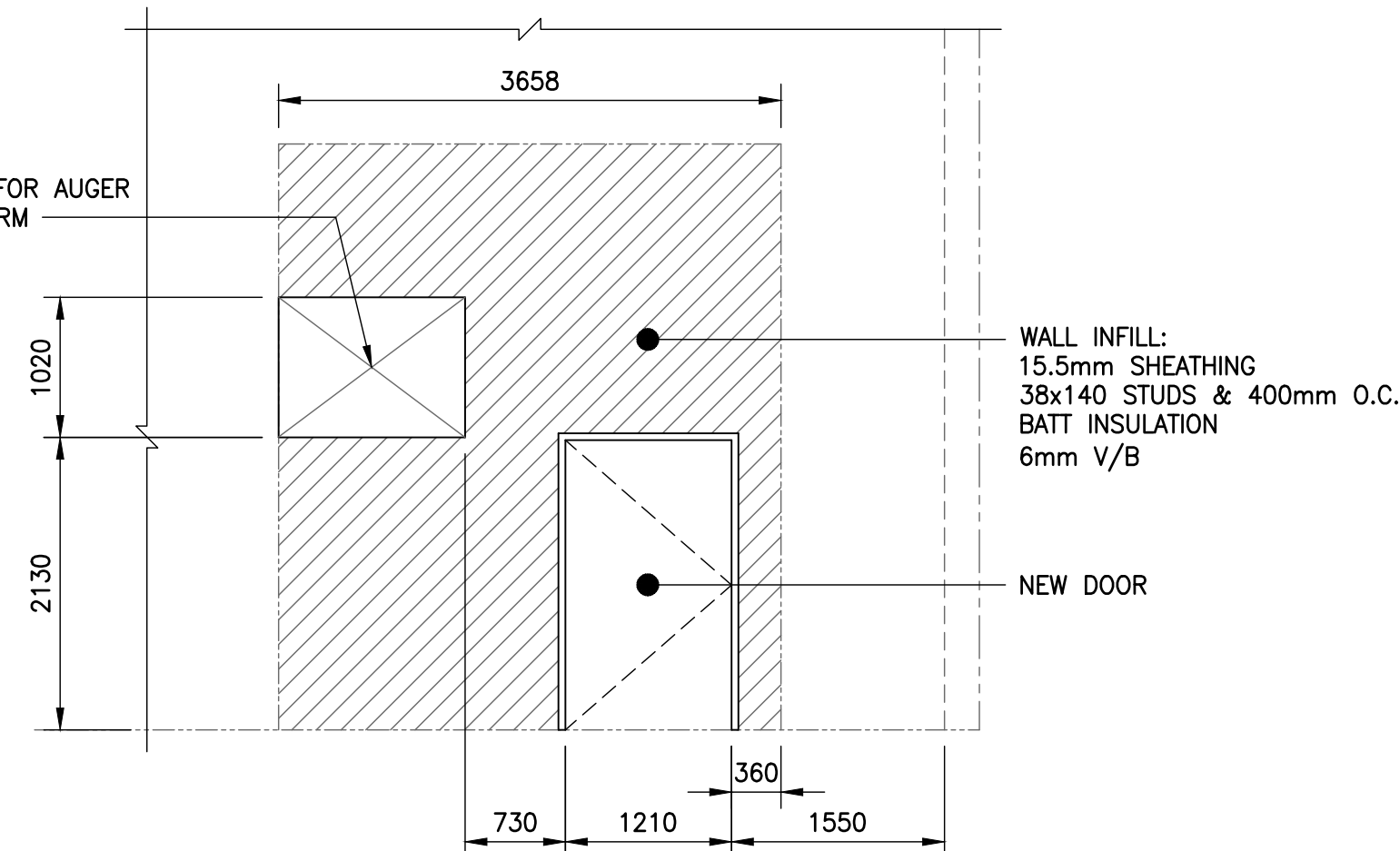
THE DESIGN OF THESE CHANGES HAVE
NOT BEEN VERIFIED BY KGS GROUP.

File Name: U:\PMS\21-1357-001\21-1357-001_S706.dwg - Tab: S706 Plotted By: emalong 23/10/25 [Wed 7:06pm]
24x36/PLOT SCALE: 1"=1'-11"



FLOOR PLAN
SCALE: 1:75 (24x36)
1:150 (11x17)

OPENING FOR AUGER
& PLATFORM



D SECTION
SCALE: 1:50 (24x36)
1:100 (11x17)

BUILDING CODE ANALYSIS

BUILDING DESCRIPTION (GENERAL):

CHANGE IN OCCUPANCY FROM A MINE PROCESSING FACILITY
TO A SEASONAL (SUMMER) WATER TREATMENT PLANT.

BUILDING CLASSIFICATION (NBC 2010)

OCCUPANCY CLASSIFICATION(S) GROUP F, DIVISION 3
N.B.C.C. ARTICLE NO. (GOVERNING) 3.2.2.83

BUILDING AREA (MAX. HORIZ.) 668 SQ. M

NUMBER OF STOREYS 1

BASEMENT NO

SPRINKLER NO

CONSTRUCTION COMBUSTIBLE/NONCOMBUSTIBLE

TOTAL OCCUPANT LOAD (ACTUAL MAXIMUM) 4

SCOPE OF WORK:

THE PROJECT IS TO INCLUDE, BUT IS NOT LIMITED TO:

- 1 REMOVE EXISTING DOOR & FRAME. INSTALL NEW DOOR.
(SEE DOOR SCHEDULE)
- 2 REMOVE EXISTING DOOR OR WINDOW & FRAME. FILL IN
OPENING WEATHERTIGHT. (SEE SECTIONS)
- 3 FRAME NEW OPENING AND INSTALL NEW DOOR.
(SEE DOOR SCHEDULE)
- 4 FILL IN OPENING WEATHERTIGHT.
(SEE SECTIONS ON DWG B03)
- 5 CONSTRUCT NEW:
- ELECTRICAL ROOM
- OFFICE
- LUNCHROOM INCLUDING CABINETS
- WASHROOM
- TESTING CABINET AND SINK

ORIGINAL PROJECT
10-1357-02

WALL TYPES:

W1 EXISTING EXTERIOR WALL - TYPICAL UNLESS NOTED

- EXTERIOR CLADDING
- STRUCTURAL GIRTS
- BATT INSULATION
- INTERIOR CLADDING

W2 INTERIOR PARTITION WALL

- 12 PLYWOOD G1S (PAINTED)
- 38x89 STUDS @ 400 O/C
- 12 PLYWOOD G1S (PAINTED)

W3 PLUMBING WALL

- 12 PLYWOOD G1S (PAINTED)
- 38x140 STUDS @ 400mm O/C
- 12 PLYWOOD G1S (PAINTED)

W4 NEW SUPPORT WALL

- EXISTING WALL CONSTRUCTION (SEE W1)
- ±25 AIR SPACE
- 38x89 STUDS @ 400mm O/C
- 12 PLYWOOD G1S (PAINTED)

W5 DOOR & WINDOW INFILL

- EXTERIOR CLADDING TO MATCH EXIST.
- NEW STRUCTURAL INFILL GIRTS

W6 SMALL PENETRATION INFILL

- EXTERIOR CLADDING TO MATCH EXIST.

W7 RECONSTRUCTED EXTERIOR WALL

- NEW EXTERIOR CLADDING
- STRUCTURAL GIRTS

GENERAL NOTES:

1. SEE MECHANICAL & ELECTRICAL DRAWINGS FOR
REQUIRED OPENINGS.
2. ALL EXPOSED PLYWOOD TO BE G1S. SEE STRUCT.
3. REFER TO DRAWING B03 FOR WALL FRAMING DETAILS.

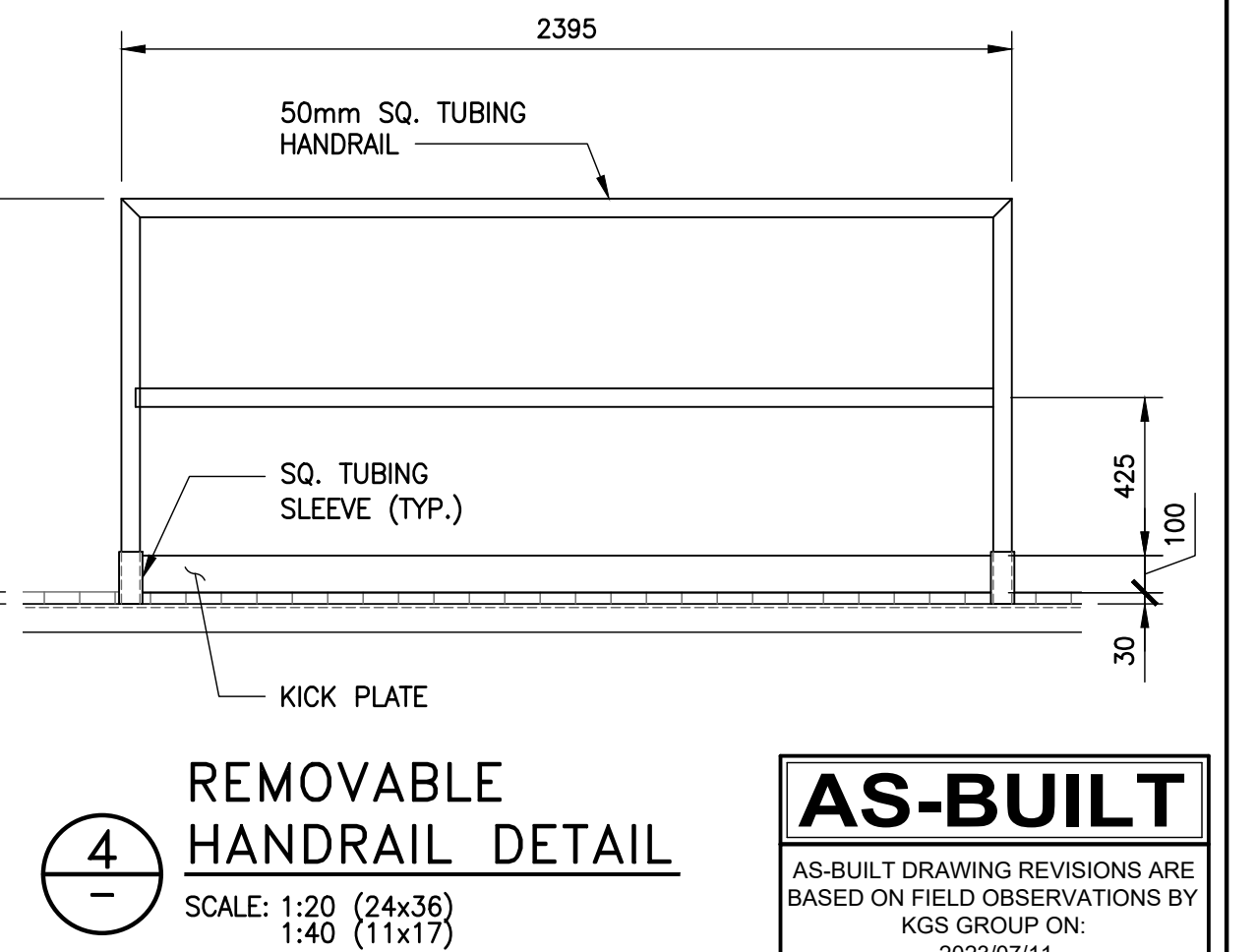
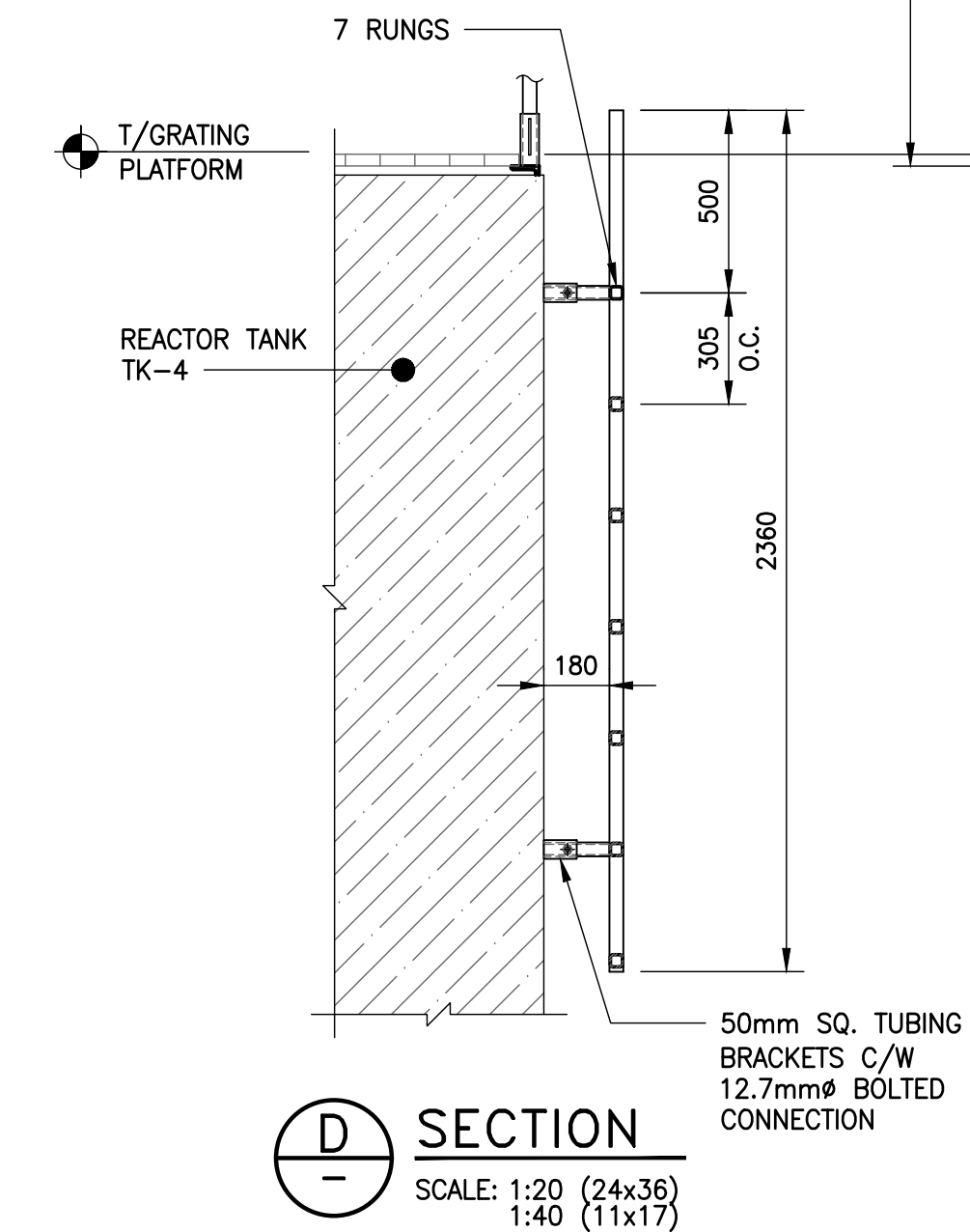
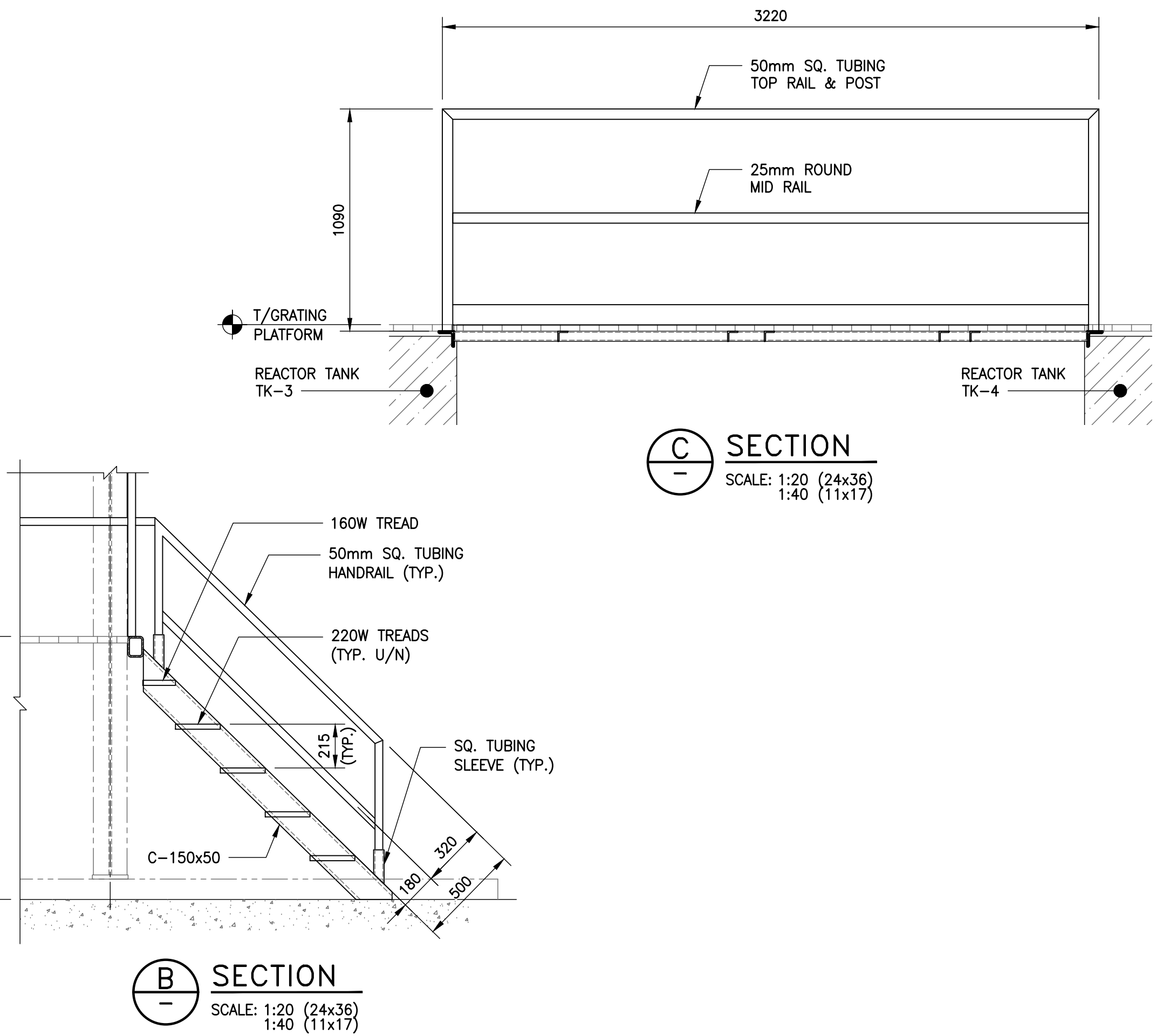
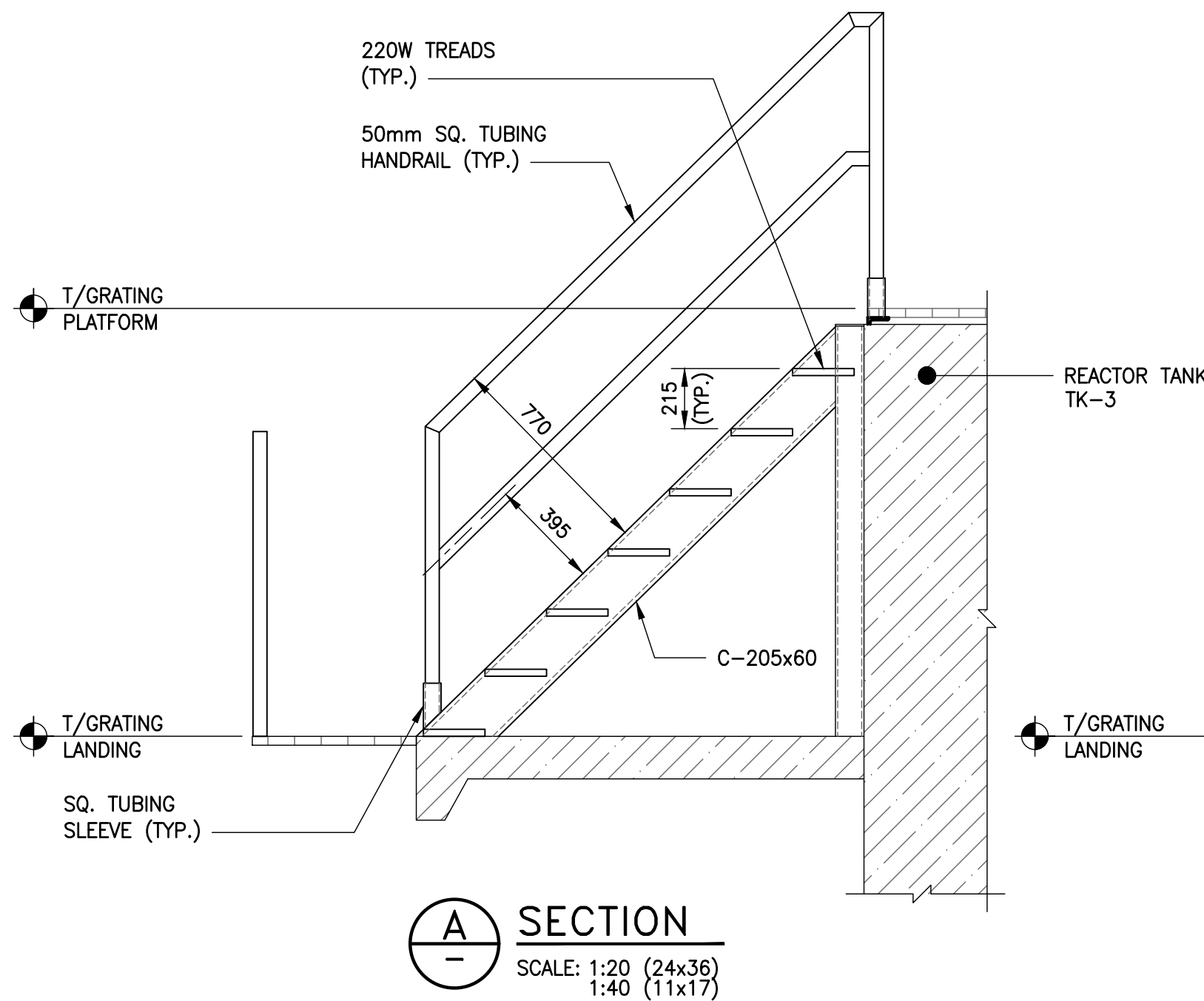
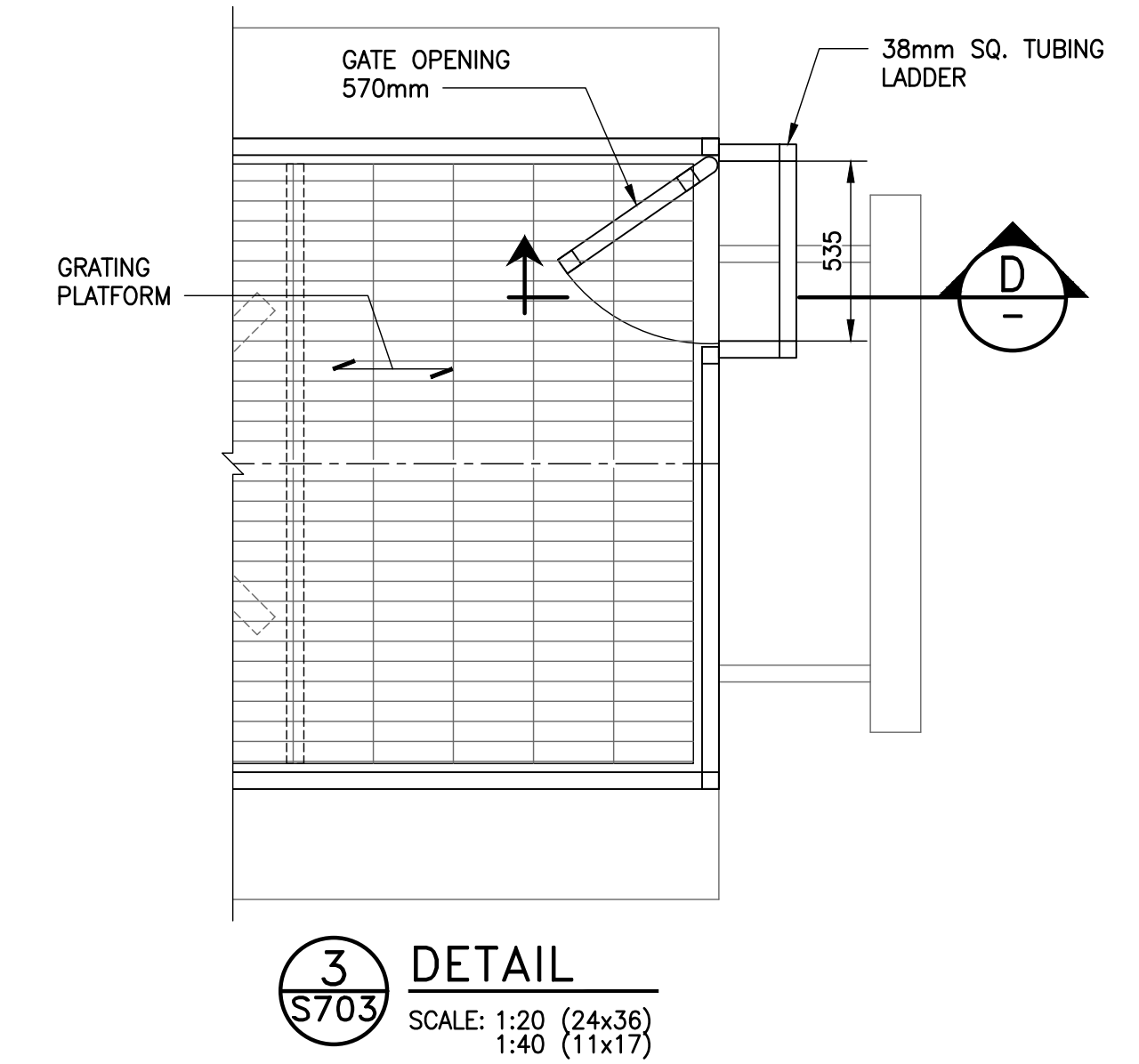
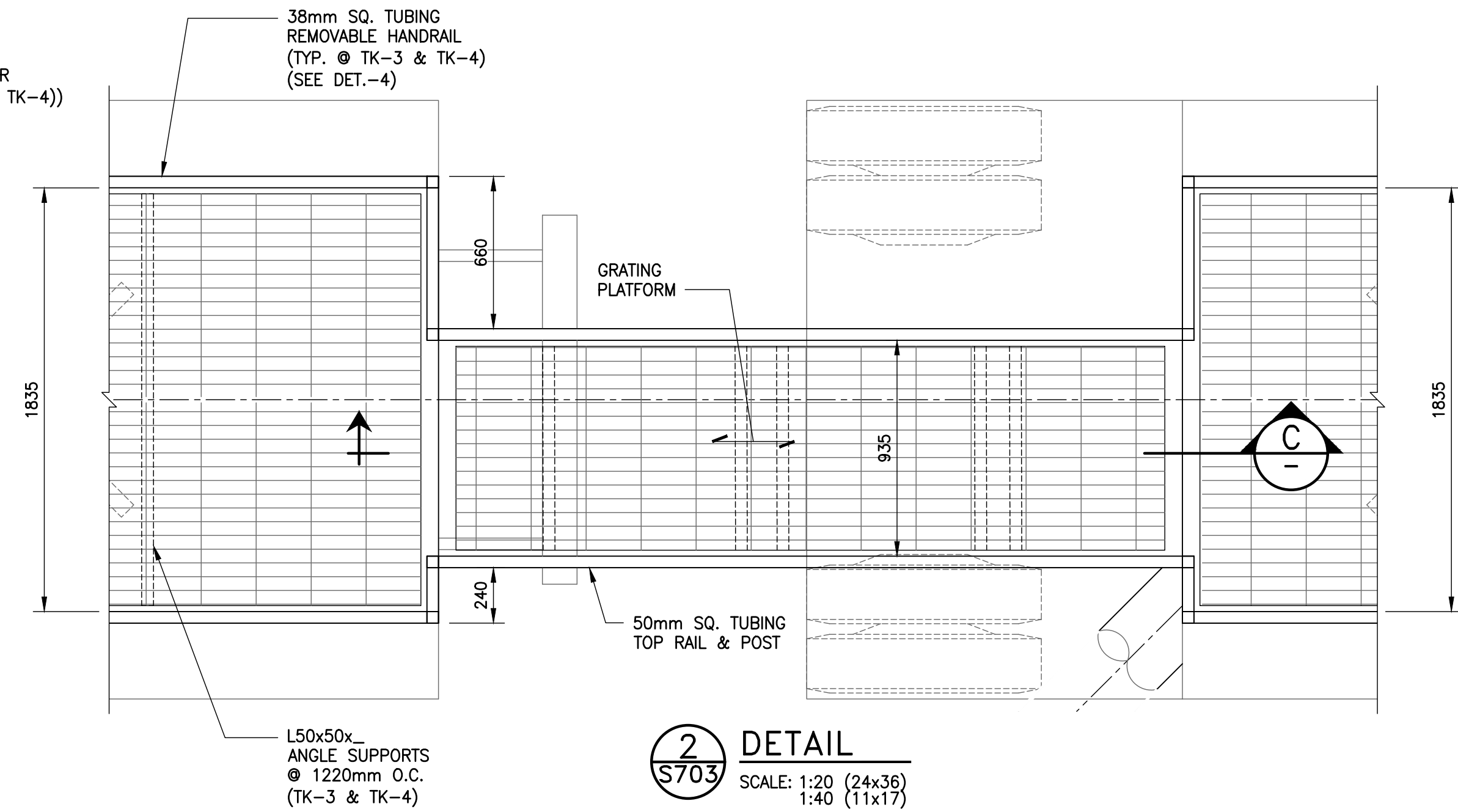
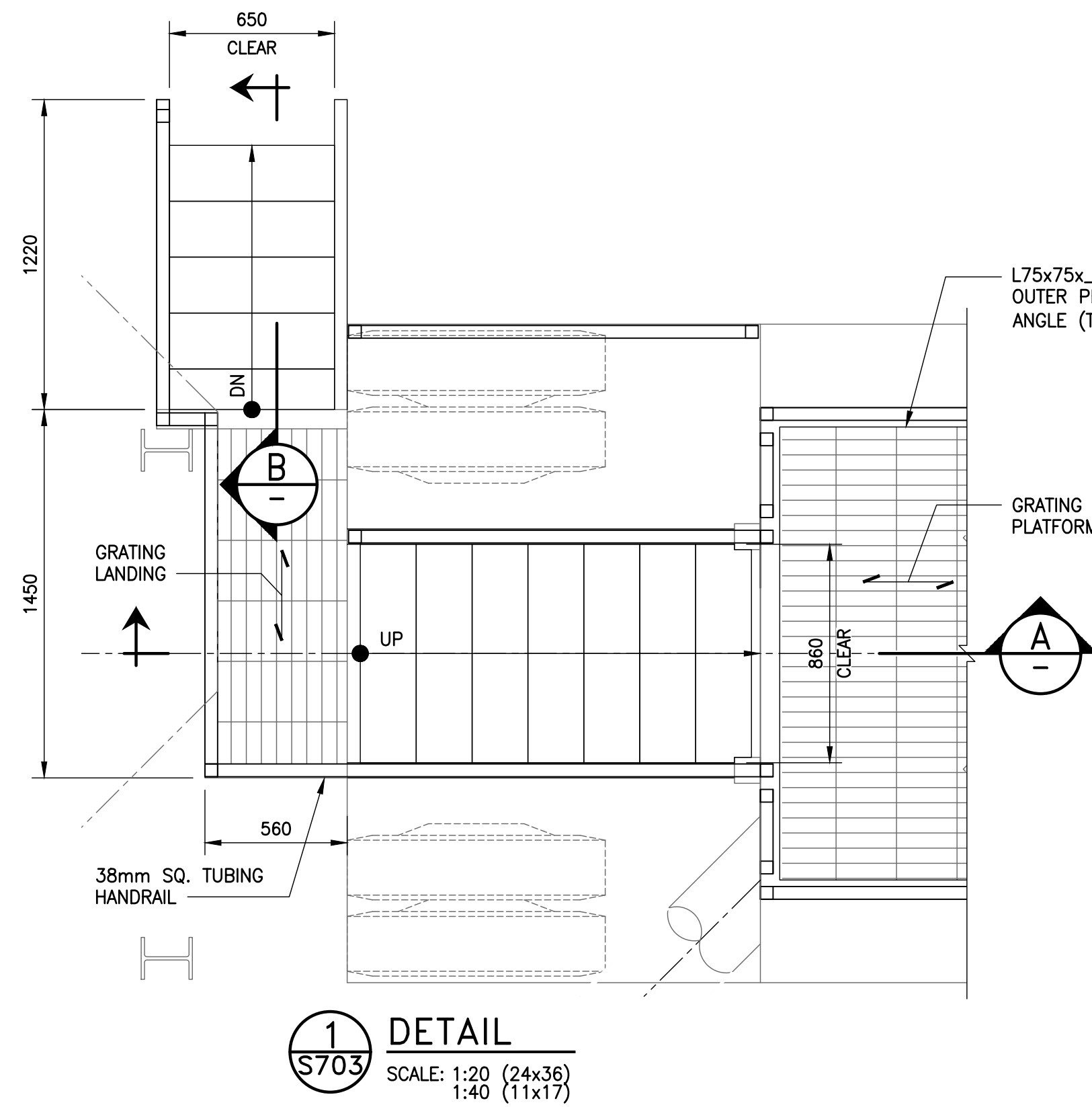
NEW 3658W
x 4267H
O/H DOOR

NEW 3658W
x 4267H
O/H DOOR

AS-BUILT
AS-BUILT DRAWING REVISIONS ARE
BASED ON FIELD OBSERVATIONS BY
KGS GROUP ON:
2023/07/11
THE DESIGN OF THESE CHANGES HAVE
NOT BEEN VERIFIED BY KGS GROUP.

0 23/10/26 ISSUED AS-BUILT			
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT: Manitoba MANITOBA MINES BRANCH			
PROJECT: RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION: STRUCTURAL BUILDING - FLOOR PLAN			
KGS GROUP	DESIGN BY:	HF	DATE: 12/04/18
	DESIGN CHECK:	RJL	DATE: 12/10/17
	DRAWN BY:	MBB	DATE: 12/05/03
	DWG CHECK:	HF	DATE: 12/10/12
DWG. NO. 21-1357-001 S706			REV. 0

File Name: U:\FMS\21-1357-001\21-1357-001_S711.dwg - Tab: S711 Plotted By: emalong 23/10/25 [Wed 7:08pm]
24x36/PLOT SCALE: 1"=1'-0"



AS-BUILT

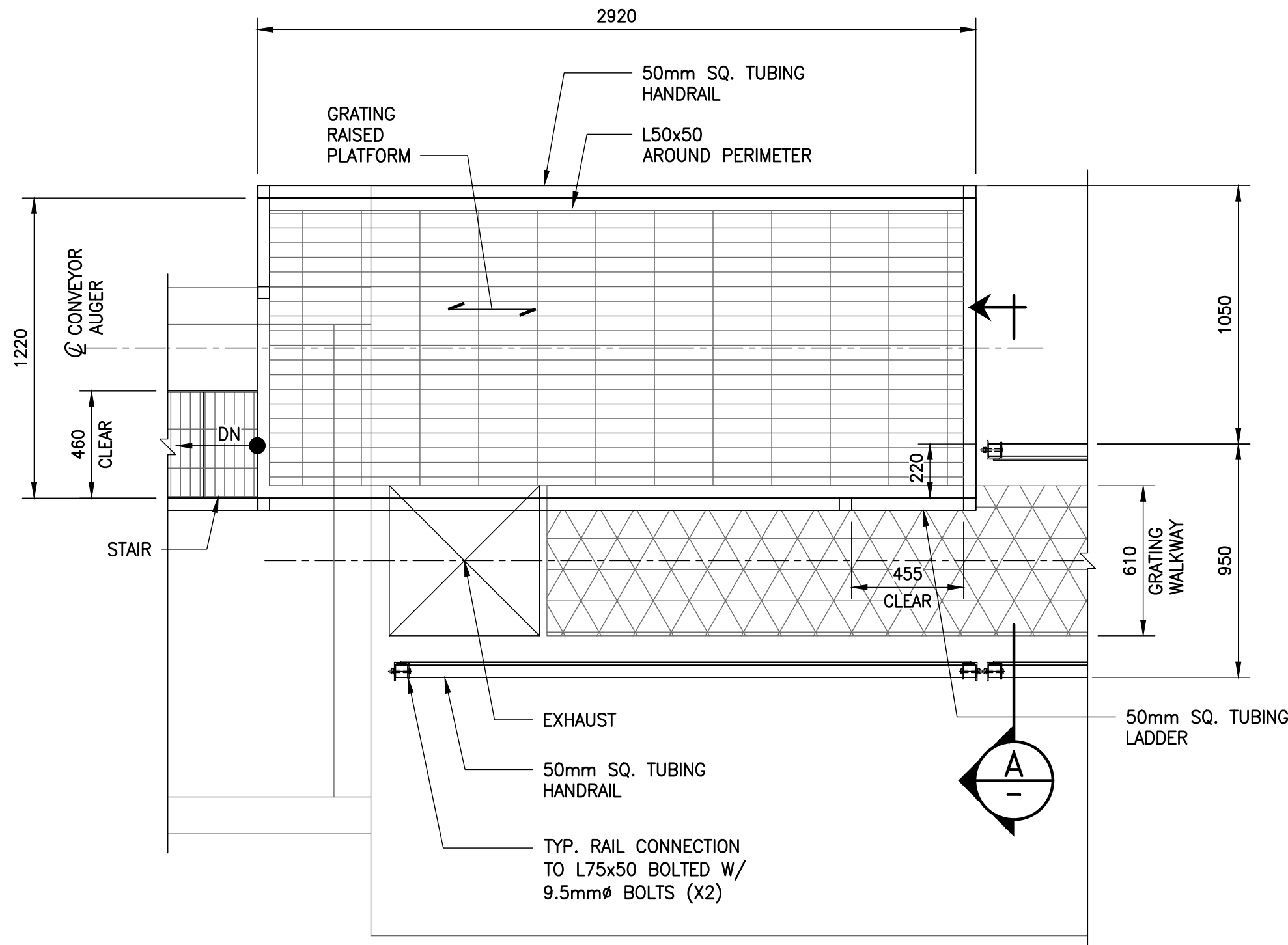
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THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

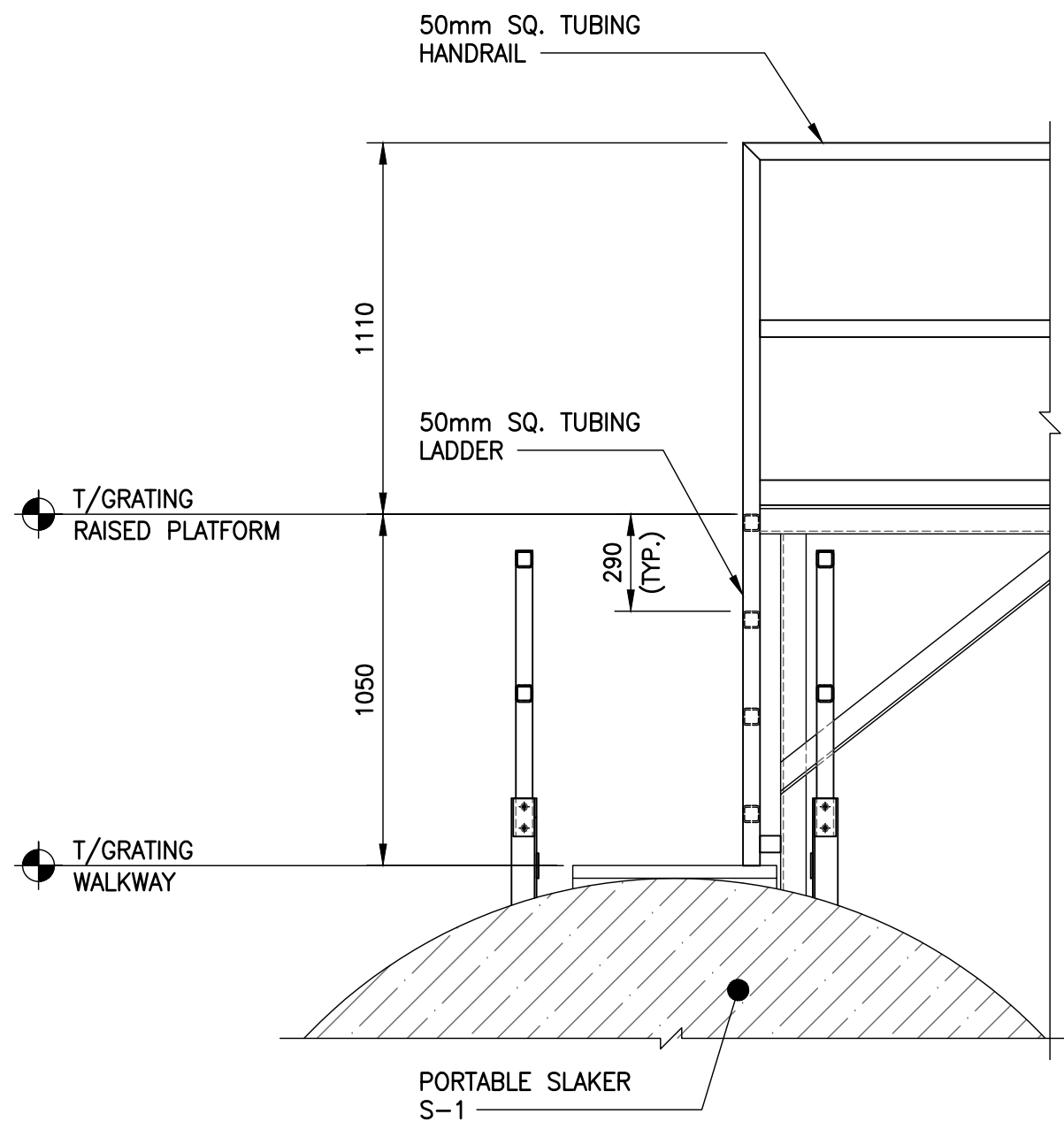
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NO.	YY/MM/DD	DESCRIPTION	BY	
REVISIONS / ISSUE				
CLIENT:				
Manitoba MANITOBA MINES BRANCH				
PROJECT:				
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
STRUCTURAL SECTIONS & DETAILS				
KGS GROUP		DESIGN BY:	DATE: (YY/MM/DD):	
		DESIGN CHECK:	DATE:	
		DRAWN BY: EM	DATE: 23/08/08	
		DWG CHECK: GE	DATE: 23/09/20	
DWG. NO. 21-1357-001 S711				REV. 0

ORIGINAL PROJECT
10-1357-02

File Name: U:\FMS\21-1357-001\21-1357-001_S712.dwg - Tab: S712 Plotted By: emalong 23/10/25 [Wed 7:08pm]
24x36/PLOT SCALE: 1"=1'



1 DETAIL
S703 SCALE: 1:20 (24x36)
1:40 (11x17)



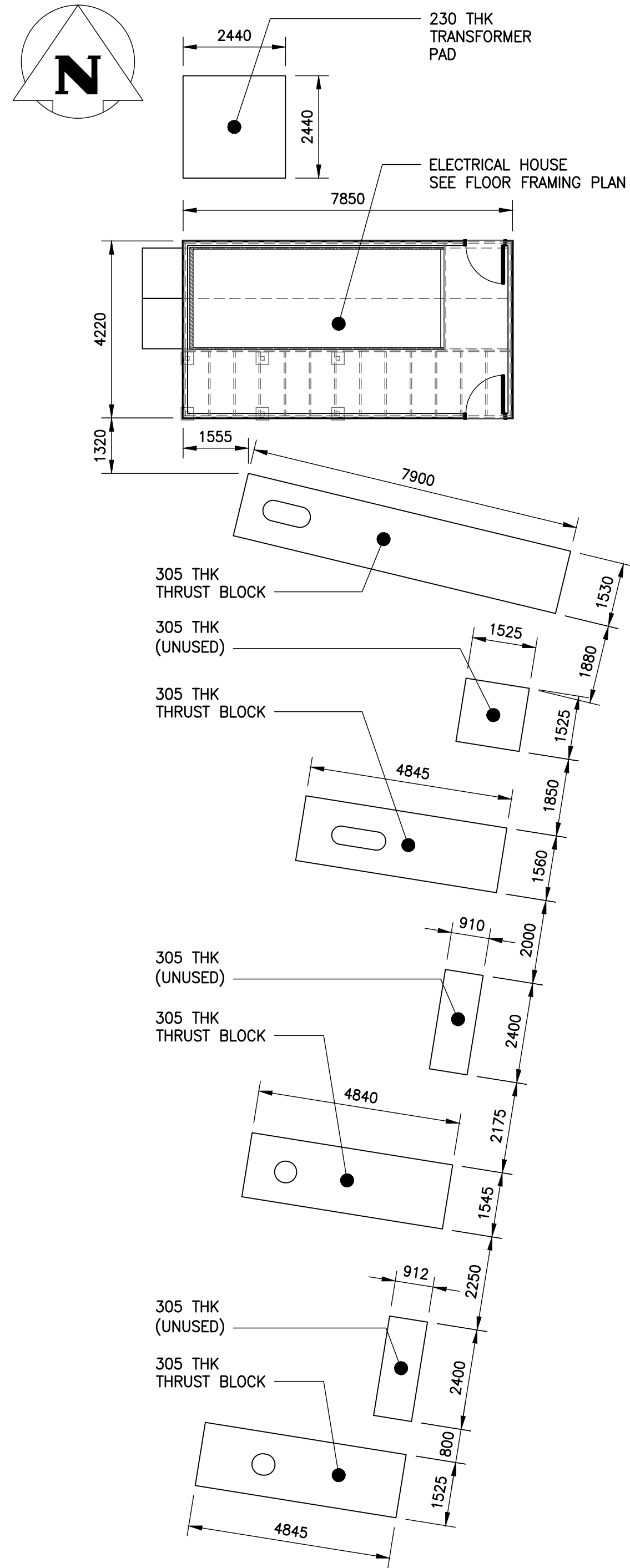
A SECTION
SCALE: 1:20 (24x36)
1:40 (11x17)

AS-BUILT
AS-BUILT DRAWING REVISIONS ARE
BASED ON FIELD OBSERVATIONS BY
KGS GROUP ON:
2023/07/11
THE DESIGN OF THESE CHANGES HAVE
NOT BEEN VERIFIED BY KGS GROUP.

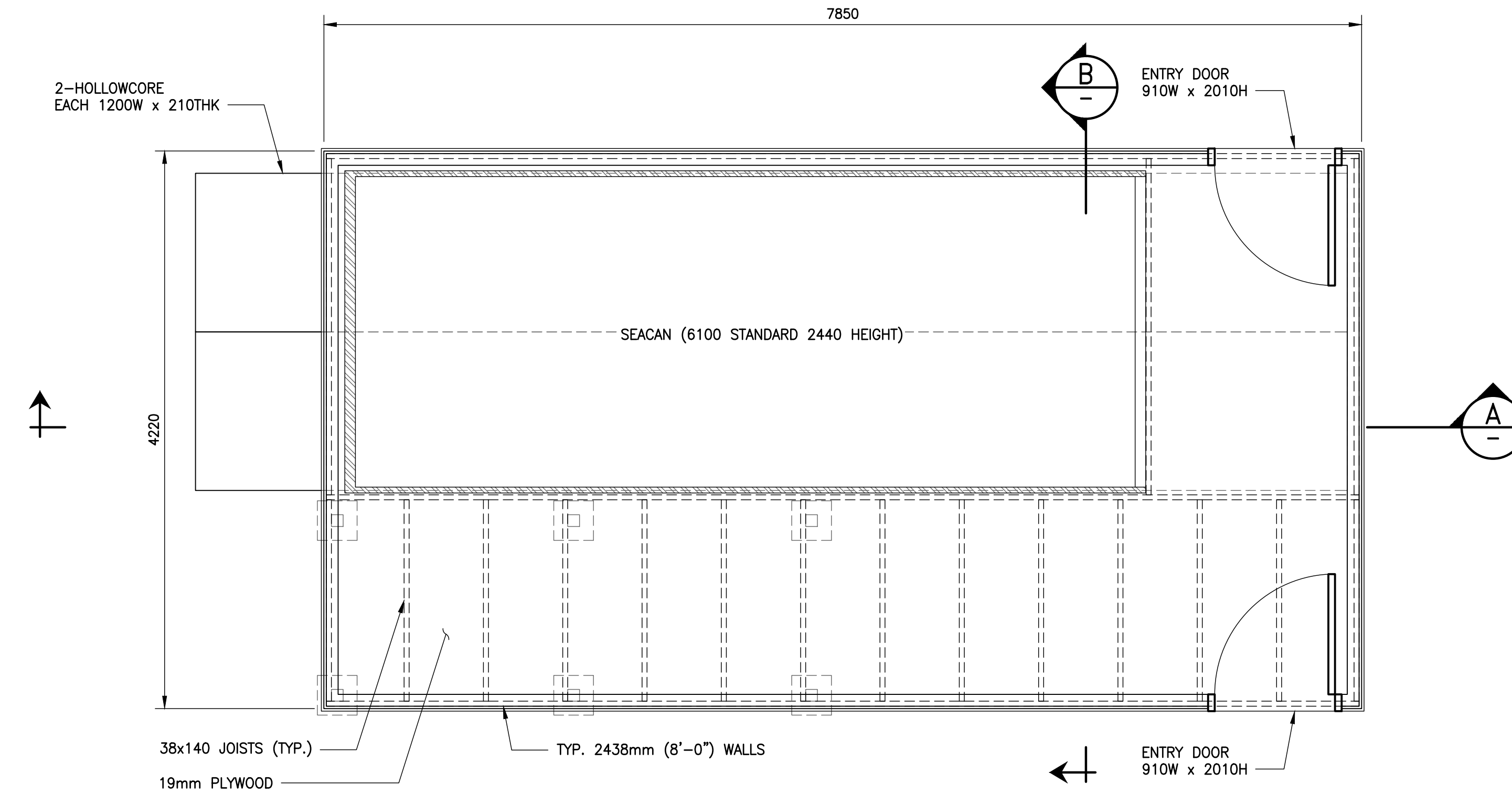
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NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
STRUCTURAL SECTIONS & DETAILS			
KGS GROUP	DESIGN BY:	DATE: (YY/MM/DD):	
	DESIGN CHECK:	DATE:	
	DRAWN BY: EM	DATE: 23/08/10	
	DWG CHECK: GE	DATE: 23/09/20	
DWG. NO. 21-1357-001 S712			REV: 0

ORIGINAL PROJECT
10-1357-02

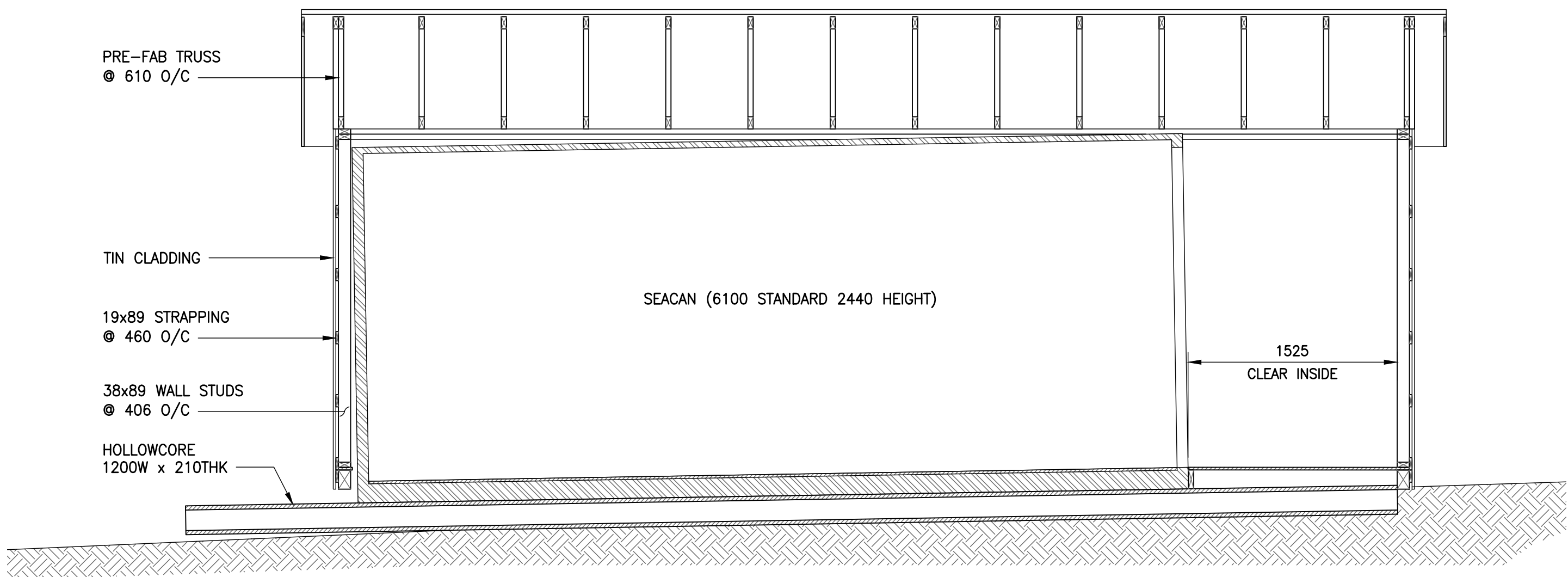
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24x36/PLOT SCALE: 1"=1'



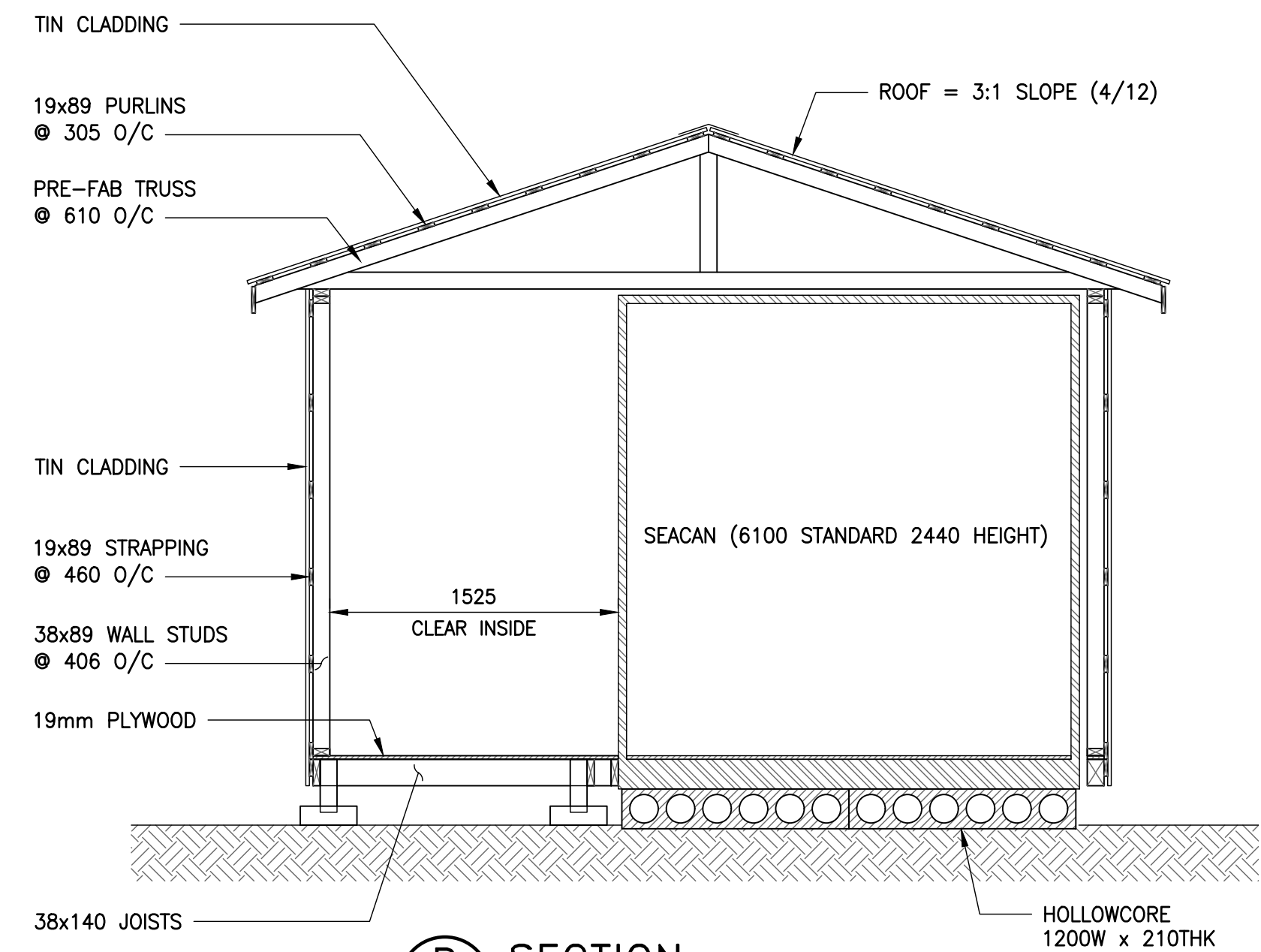
PLAN - PUMP STATION 1
SCALE: 1:100 (24x36)
1:200 (11x17)



1 FLOOR FRAMING PLAN
SCALE: 1:30 (24x36)
1:60 (11x17)



A SECTION
SCALE: 1:30 (24x36)
1:60 (11x17)



B SECTION
SCALE: 1:30 (24x36)
1:60 (11x17)

AS-BUILT

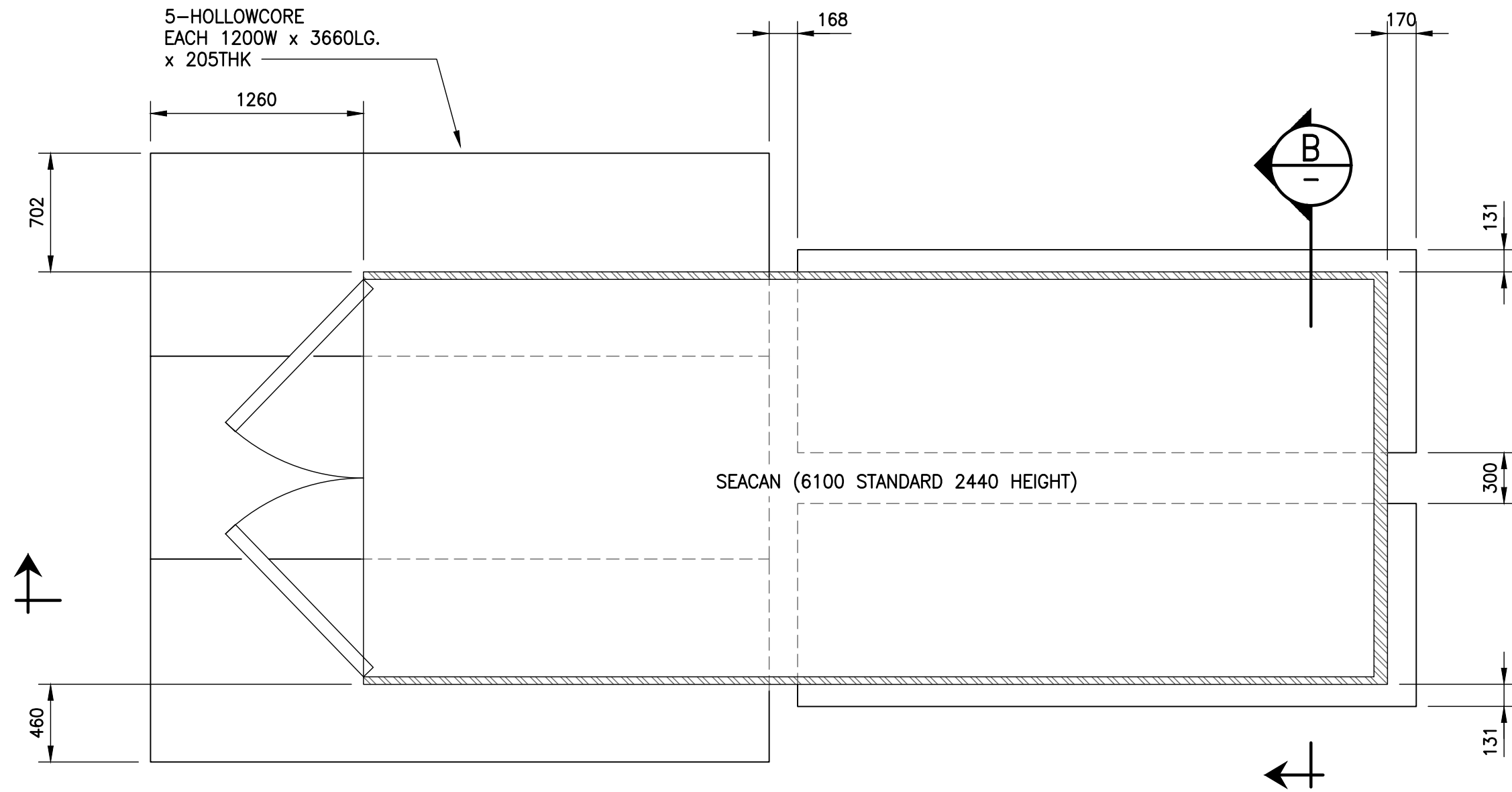
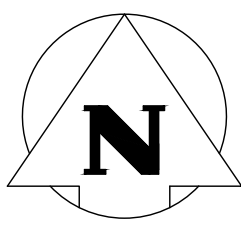
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KGS GROUP ON:
2023/07/11
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0	23/10/26	ISSUED AS-BUILT	
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
STRUCTURAL PUMP STATION 1 PLAN & SECTIONS			
KGS GROUP	DESIGN BY:	DATE (YY/MM/DD):	
	DESIGN CHECK:	DATE:	
	DRAWN BY: EM	DATE: 23/08/11	
	DWG CHECK: GE	DATE: 23/09/20	
DWG. NO. 21-1357-001 S713			REV. 0

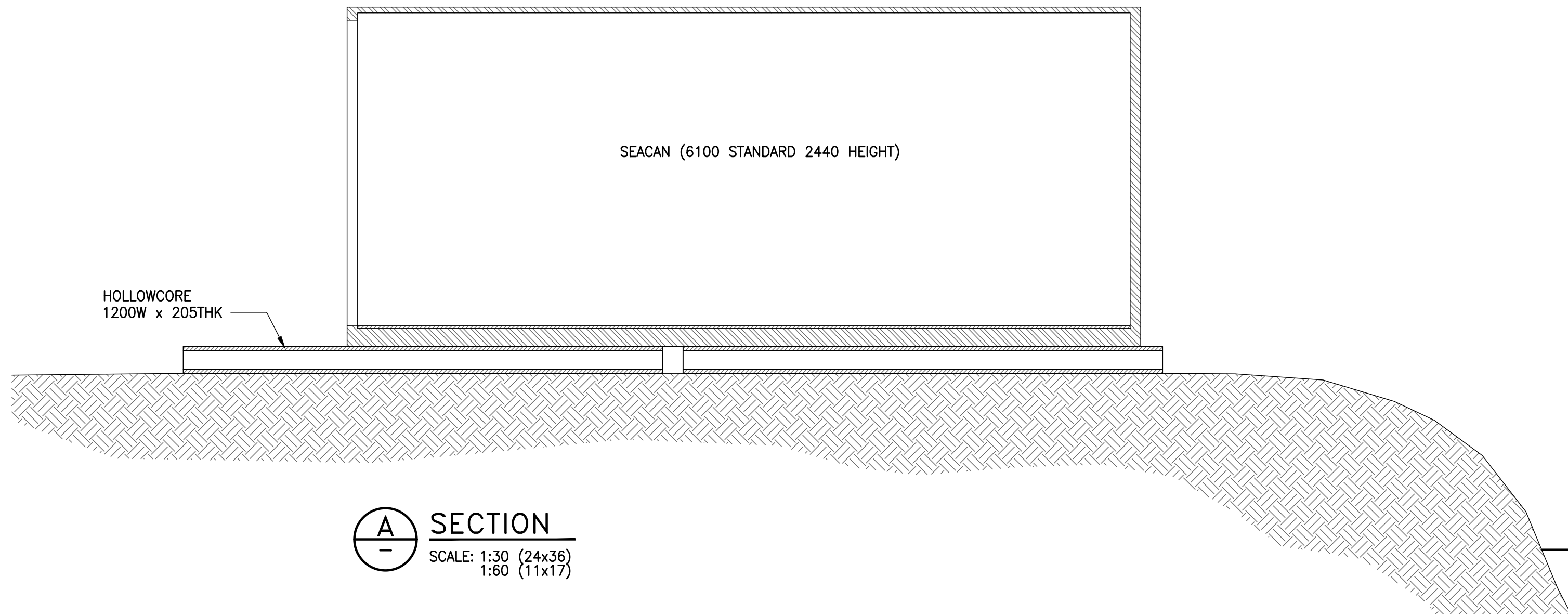
ORIGINAL PROJECT
10-1357-02

ENG. STAMP

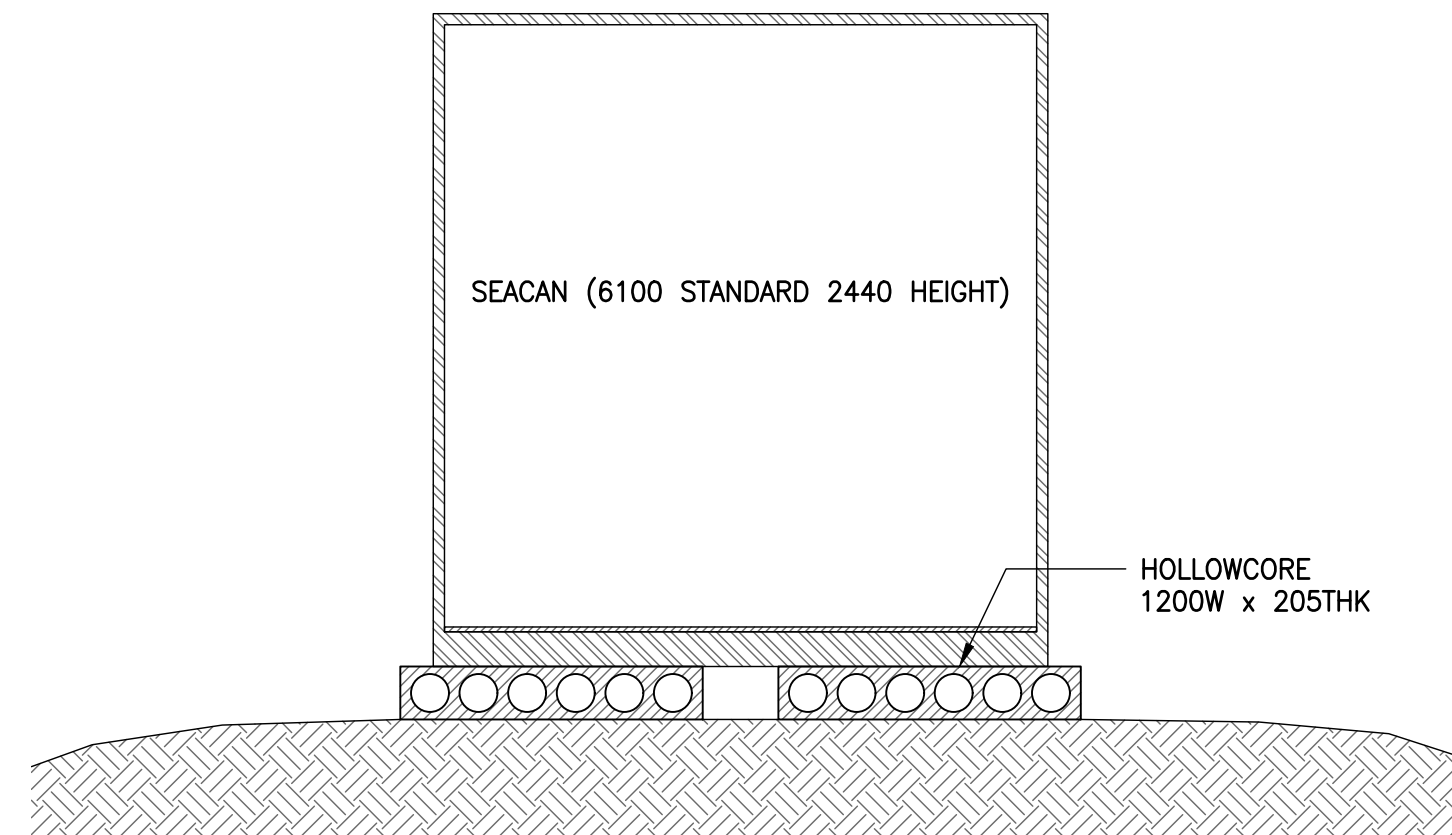
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24x36 / PLOT SCALE: 1"=1'



1 PLAN - PUMP STATION 2
SCALE: 1:30 (24x36)
1:60 (11x17)



A SECTION
SCALE: 1:30 (24x36)
1:60 (11x17)



B SECTION
SCALE: 1:30 (24x36)
1:60 (11x17)

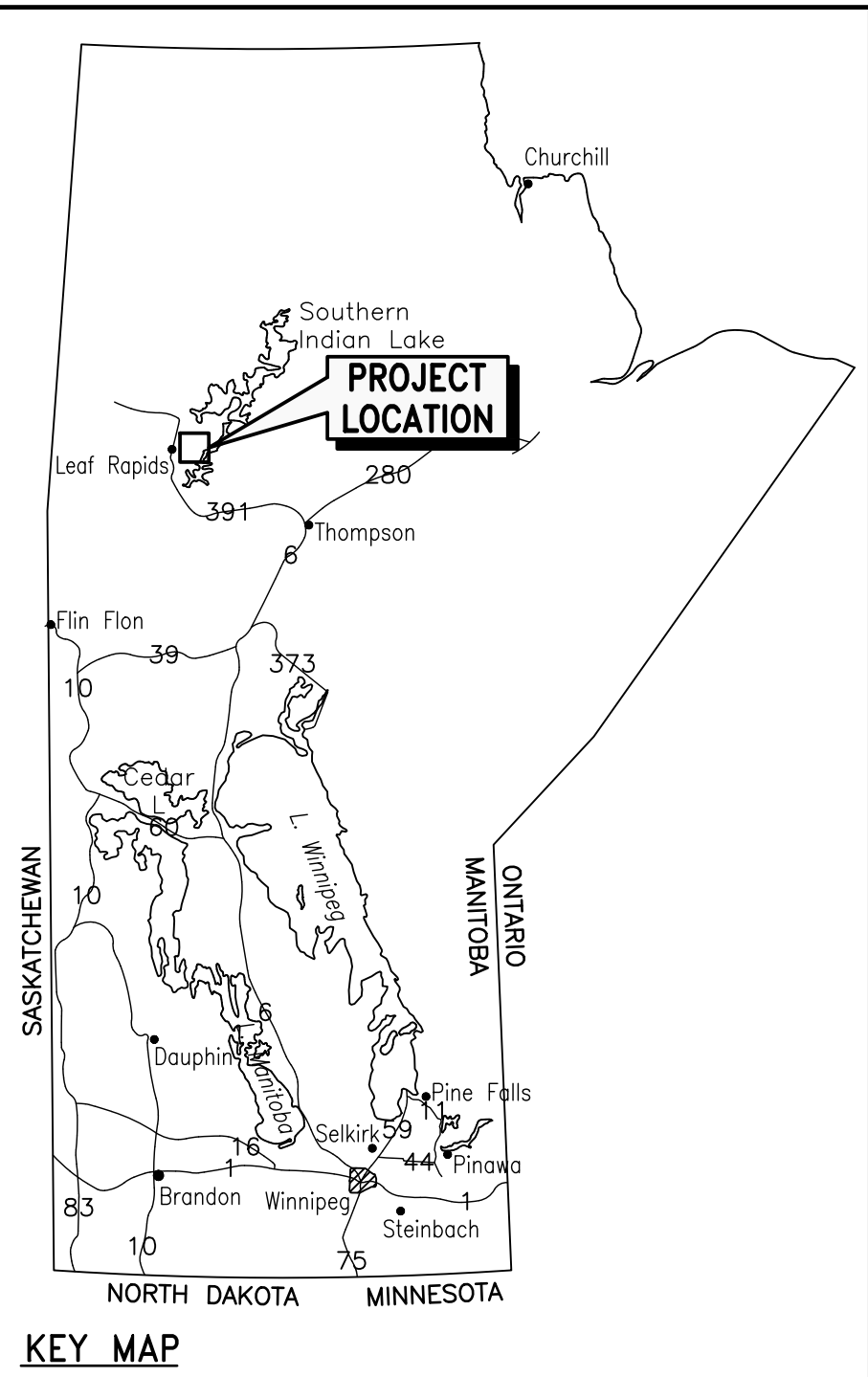
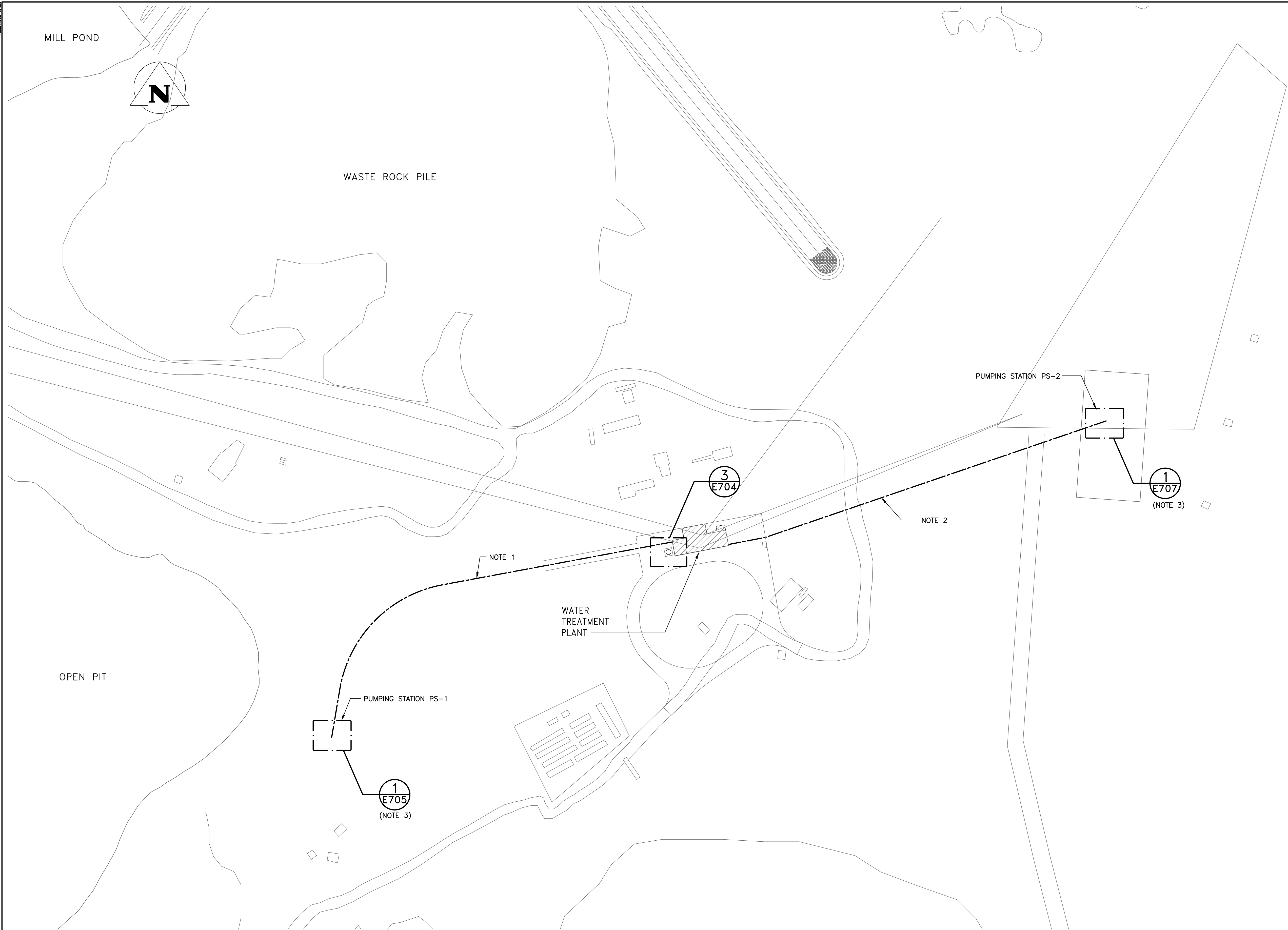
ORIGINAL PROJECT
10-1357-02

ENG. STAMP

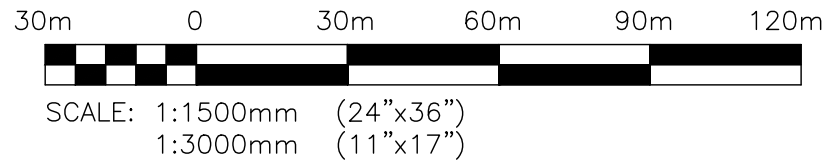
AS-BUILT
AS-BUILT DRAWING REVISIONS ARE
BASED ON FIELD OBSERVATIONS BY
KGS GROUP ON:
2023/07/11
THE DESIGN OF THESE CHANGES HAVE
NOT BEEN VERIFIED BY KGS GROUP.

0	23/10/26	ISSUED AS-BUILT	
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
STRUCTURAL PUMP STATION 2 PLAN & SECTIONS			
KGS GROUP	DESIGN BY:	DATE: (YY/MM/DD):	
	DESIGN CHECK:	DATE:	
	DRAWN BY: EM	DATE: 23/08/14	
	DWG CHECK: GE	DATE: 23/09/20	
DWG. NO. 21-1357-001 S714			REV: 0

File Name: \\kgs-file\file\Projects\2021\21-1357-001\Drawings\Elec\21-1357-001-E701.dwg - Tab: E701 Plotted By: DShabataev 23/10/26 [Thu 8:16am]
24"x36" PLOT SCALE: 1"=100'



- NOTES:**
1. REFER TO DRAWING G705 FOR TYPE 1 AND TYPE 2 TRENCH DETAILS.
 2. REFER TO DRAWING G705 FOR TYPE 3 TRENCH DETAILS.
 3. LOCATION OF PUMPING STATIONS ARE APPROXIMATE. REFER TO DRAWING G701 FOR ACCURATE LOCATIONS.



AS-BUILT

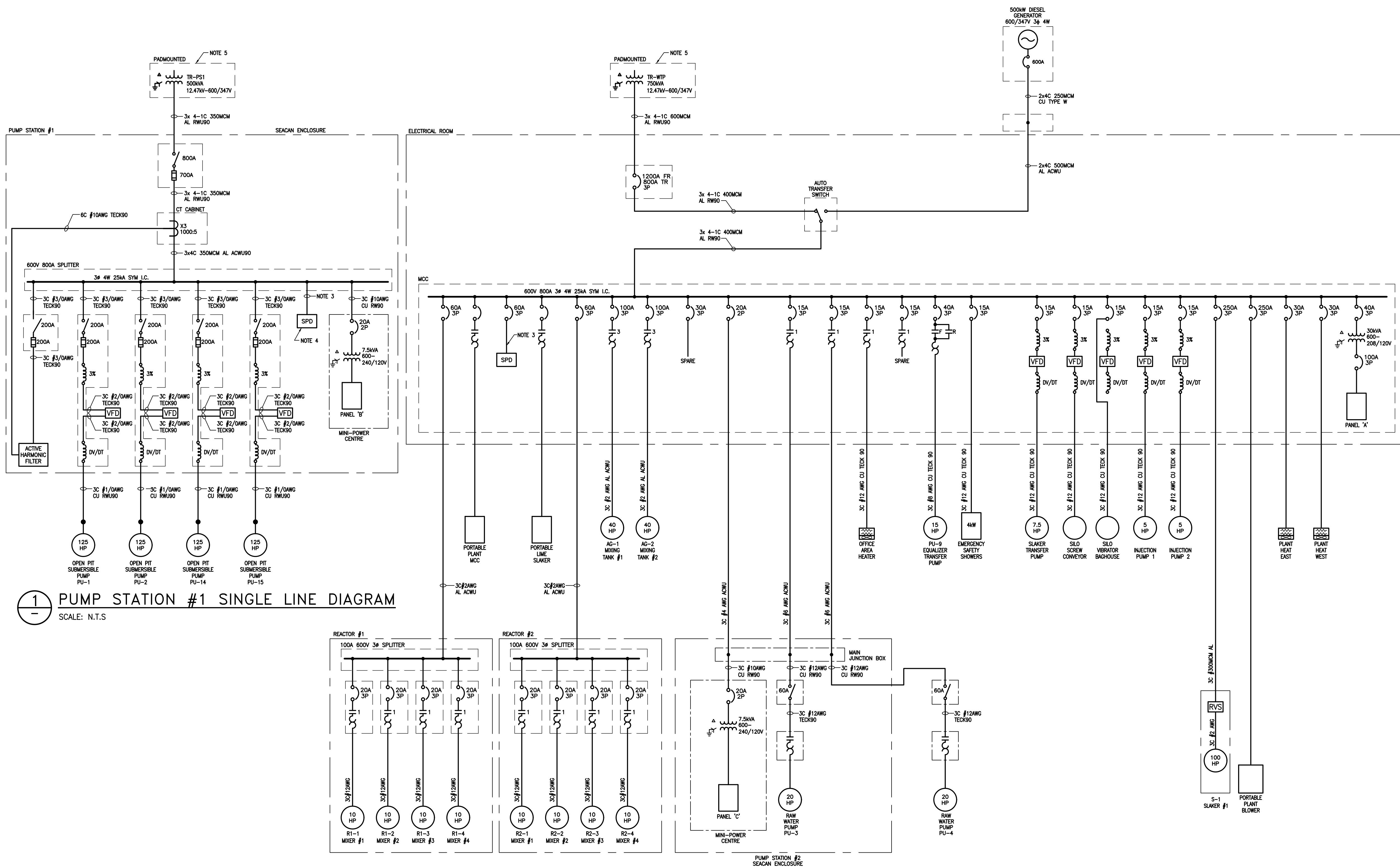
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THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

0 23/10/26 ISSUED AS-BUILT			
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL ELECTRICAL CONDUIT SITE PLAN			
	DESIGN BY:	CLS	DATE: (YY/MM/DD): 12/01/04
	DESIGN CHECK:	JAB	DATE: 12/10/17
	DRAWN BY:	GCN	DATE: 12/07/09
	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO. 21-1357-001 E701			REV. 0

ORIGINAL PROJECT
10-1357-02

ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17



1 PUMP STATION #1 SINGLE LINE DIAGRAM
SCALE: N.T.S

2 WATER TREATMENT PLANT SINGLE LINE DIAGRAM
SCALE: N.T.S

NOTES:

1. GRAY ITEMS INDICATES EQUIPMENT PROVIDED BY OTHERS.
2. PROVIDE RAYCHEM GELCAP RE-ENTERABLE WATER TIGHT SPLICE KITS FOR CONNECTIONS WITHIN PUMP WELL HEAD.
3. CONDUCTOR LENGTH BETWEEN BREAKER AND SPD SHALL BE AS SHORT AS POSSIBLE (300mm MAX) AND SHALL BE TWISTED TOGETHER.
4. SURGE PROTECTION DEVICE SHALL BE RATED 120kA SURGE CURRENT PER PHASE, 60kA PER MODE, COMPLETE WITH COMPONENT LEVEL FUSING. TOTAL PROTECTION SOLUTIONS MODEL TK-ST120-3Y600.
5. TRANSFORMER AND PRIMARY CONDUCTORS INSTALLED BY CONTRACT C-600 AND ARE NOT PART OF SCOPE FOR CONTRACT C-700.

LEGEND:

- STANDBY GENERATOR
- VARIABLE FREQUENCY DRIVE
- REDUCED VOLTAGE STARTER
- SURGE PROTECTION DEVICE
- REACTOR/LONG LEAD FILTER
- PLUG AND RECEPTACLE CONNECTION
- SPLICE CONNECTION

AS-BUILT

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NO. YY/MM/DD DESCRIPTION BY

REVISIONS / ISSUE

CLIENT:



PROJECT:

RUTTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:

ELECTRICAL
SINGLE LINE DIAGRAMS

DESIGN BY: CLS DATE: 11/11/07

DESIGN CHECK: JAB DATE: 12/10/17

DRAWN BY: JPL DATE: 11/11/07

DWG CHECK: JAB DATE: 12/10/17

DWG. NO. 21-1357-001 E702

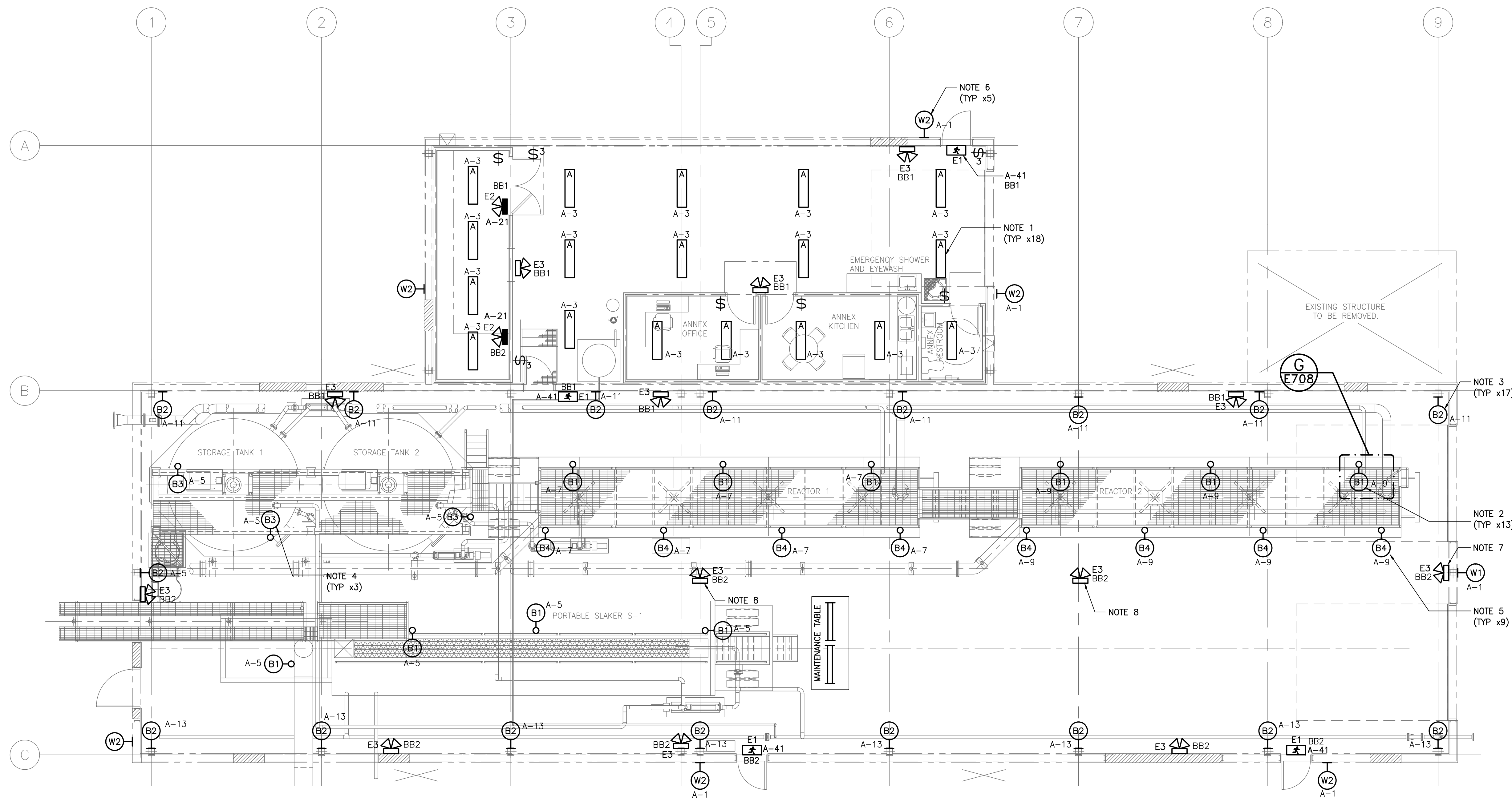
REV. 0

ORIGINAL PROJECT
10-1357-02

ENG. STAMP

ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

File Name: R:\Projects\2021\21-1357-001\Drawings\Elec\21-1357-001_E703.dwg - Tab: E703 Plotted By: DShabanev 23/10/26 [Thu 8:16am]
24"x36" PLOT SCALE: 1:1



NOTES:

1. ALL TYPE 'A' FIXTURES TO BE SUSPENDED FROM CEILING AT ELEVATION 3.0m AFF.
2. ALL TYPE 'B1' FIXTURES TO BE STANCHION MOUNTED TO EQUIPMENT HANDRAIL AT ELEVATION 2.4m ABOVE PLATFORM WHICH HANDRAIL SERVES.
3. ALL TYPE 'B2' FIXTURES TO BE WALL MOUNTED AT ELEVATION 2.5m AFF.
4. ALL TYPE 'B3' FIXTURES TO BE STANCHION MOUNTED TO HANDRAIL ABOVE MIXING TANKS AT ELEVATION 1.8m ABOVE PLATFORM.
5. ALL TYPE 'B4' FIXTURES TO BE STANCHION MOUNTED TO EQUIPMENT HANDRAIL AT ELEVATION 0.6m ABOVE PLATFORM WHICH HANDRAIL SERVES AND ORIENTED AWAY FROM PLATFORM TO ILLUMINATE FLOOR LEVEL.
6. ALL TYPE 'W1' FIXTURES TO BE MOUNTED 4.5m ABOVE FINISHED GRADE AND ALL TYPE 'W2' FIXTURES TO BE MOUNTED 2.75m ABOVE FINISHED GRADE.
7. ALL EMERGENCY LIGHT FIXTURES AND REMOTE HEADS TO BE MOUNTED 2.1m AFF UNLESS INDICATED OTHERWISE.
8. MOUNT REMOTE HEAD TO UNDERSIDE OF STRUCTURAL STEEL.

LEGEND:

- FLUORESCENT FIXTURE
- DENOTES:
A = FIXTURE DESIGNATION
LP-1 = PANELBOARD AND CIRCUIT DESIGNATION
- WALL MOUNTED LIGHTING FIXTURE
- POLE OR STANCHION MOUNT LIGHTING FIXTURE
- REMOTE EMERGENCY DUAL HEAD
- DENOTES BATTERY UNIT SUPPLY
- EMERGENCY LIGHTING BATTERY UNIT
- DENOTES BATTERY UNIT
- TOGGLE SWITCH
- DENOTES 3-WAY
- PICTOGRAM EXIT LITE

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AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

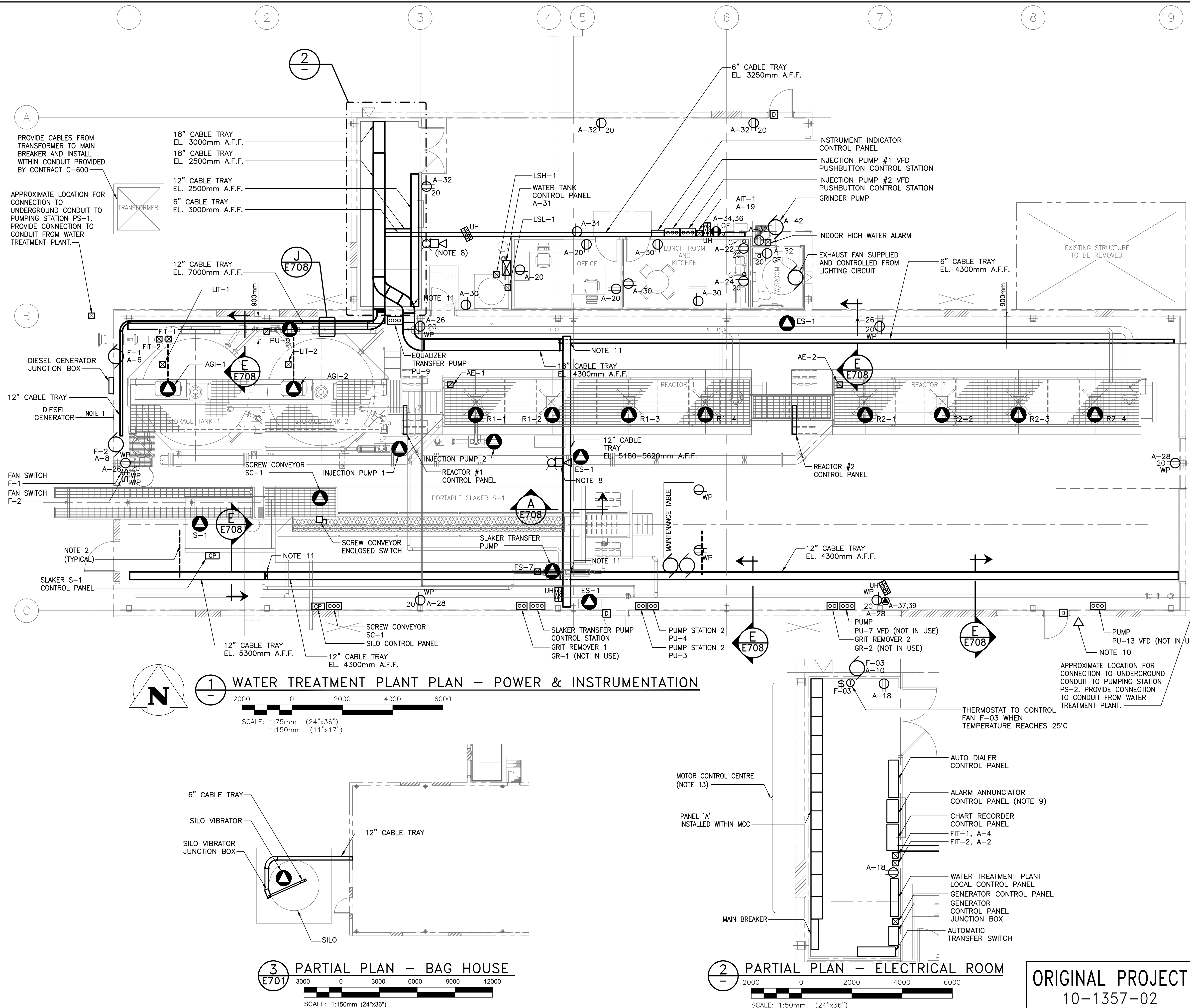
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NO.	YY/MM/DD	DESCRIPTION
REVISIONS / ISSUE		
CLIENT:		
MANITOBA MINES BRANCH		
PROJECT:		
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700		
DWG. DESCRIPTION:		
ELECTRICAL WATER TREATMENT PLANT LIGHTING PLAN		
	DESIGN BY:	CLS
	DATE:	12/04/09
	DESIGN CHECK:	JAB
	DATE:	12/10/17
	DRAWN BY:	JPL
	DATE:	12/04/9
	DWG CHECK:	JAB
	DATE:	12/10/17
DWG. NO.		21-1357-001 E703
REV.		0

ENG. STAMP

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SEALED BY:
J.A. BOUCHARD
2012/10/17

ORIGINAL PROJECT
10-1357-02

File Name: R:\Projects\2021\21-1357-001\Drawings\Elec\21-1357-001_E704.dwg - Tab: E704 Plotted By: DShabataev 23/10/26 [Thu 8:16am]
24"x36" PLOT SCALE: 1:1



- NOTES:**
1. PROVIDE A MINIMUM OF 1.5m CLEARANCE AROUND THE GENSET.
 2. POWER TO THE MOTOR BY STRAPPING THE POWER CABLE TO UNISTRUT FROM THE CABLE TRAY TO THE MOTOR.
 3. (SPARE NOTE).
 4. (SPARE NOTE).
 5. (SPARE NOTE).
 6. (SPARE NOTE).
 7. (SPARE NOTE).
 8. GENERAL ALARM COMBINATION AUDIBLE/VISUAL SIGNAL. (FEDERAL SIGNAL - AV1-LED) MOUNTED TO UNDERSIDE OF LOWER CEILING.
 9. RTK-725 PROGRAMMABLE ALARM ANNUNCIATOR.
 10. OEM SATELLITE TERMINAL (AD-1) ANTENNA ON THE SOUTH EXTERIOR OF THE BUILDING 8m AFG.
 11. TRANSITION BETWEEN CABLE TRAY ELEVATIONS USING A CABLE TRAY SLOPED AT 45°.
 12. (SPARE NOTE).
 13. MOTOR CONTROL CENTRE MOUNTED ON 90mm HIGH HOUSEKEEPING PAD.

- LEGEND:**
- 3 PHASE LOAD
 - SINGLE PHASE LOAD
 - FIELD DEVICE
 - UNISTRUT LOCATION (NOTE 2)
 - DUPLEX RECEPTACLE
 - DENOTES:
 - 0 - MOUNTED ABOVE COUNTER/TABLE
 - GF1 - GROUND FAULT INTERRUPT
 - 20 - 20A RECEPTACLE
 - SPLIT WIRED DUPLEX RECEPTACLE
 - SINGLE RECEPTACLE
 - PUSHBUTTON CONTROL STATION
 - HORN/STROBE
 - CABLE TRAY
 - DOOR STRIKE
 - THERMOSTAT
 - TOGGLE SWITCH
 - DENOTES:
 - WP - WEATHERPROOF COVER
 - ANTENNA

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0 23/10/26 ISSUED AS-BUILT		
NO.	YY/MM/DD	DESCRIPTION
REVISIONS / ISSUE		
CLIENT:		
Manitoba MANITOBA MINES BRANCH		
PROJECT:		
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700		
DWG. DESCRIPTION:		
ELECTRICAL WATER TREATMENT PLANT POWER AND INSTRUMENTATION PLAN		
KGS GROUP	DESIGN BY:	CLS
	DESIGN CHECK:	JAB
	DRAWN BY:	JPL
	DWG CHECK:	JAB
DATE: (YY/MM/DD):		12/04/09
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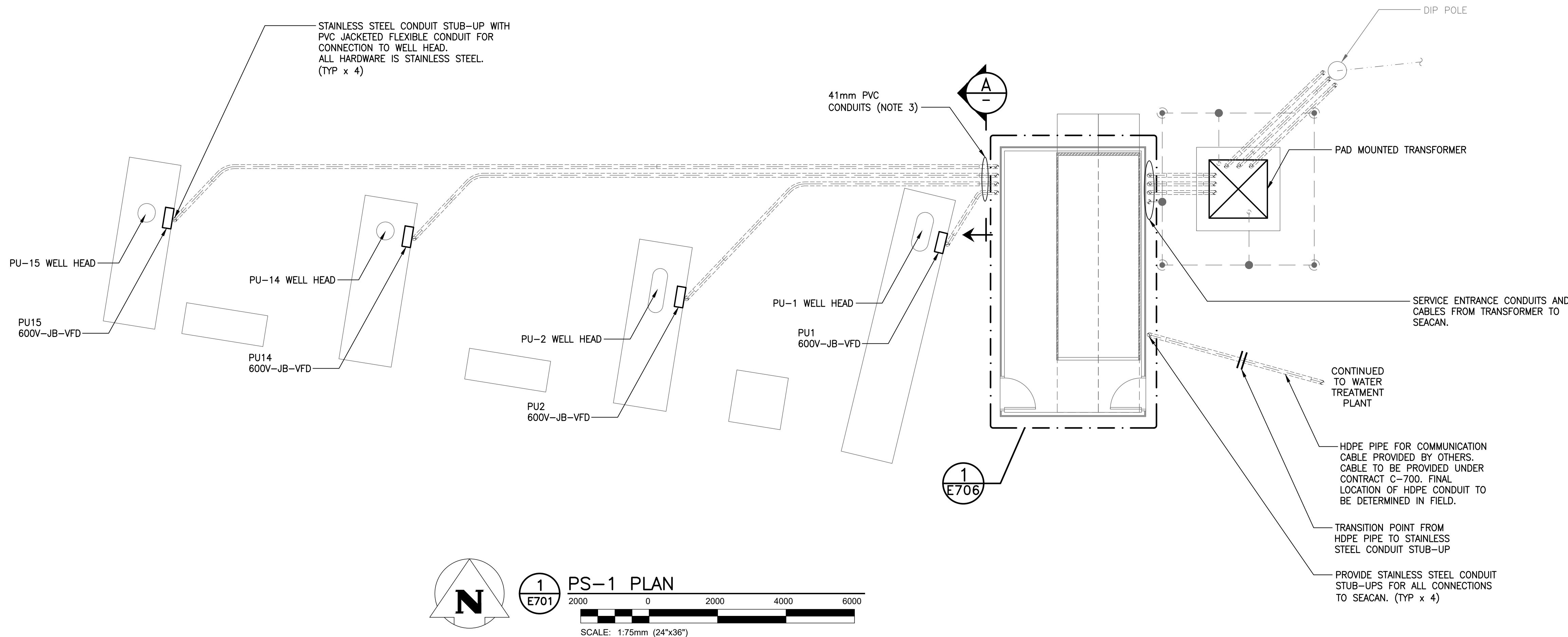
ORIGINAL PROJECT
10-1357-02

ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

3 PARTIAL PLAN - BAG HOUSE
E701
SCALE: 1:150mm (24"x36")

2 PARTIAL PLAN - ELECTRICAL ROOM
SCALE: 1:150mm (24"x36")

File Name: R:\Projects\2021\21-1357-001\Drawings\Elec\21-1357-001_E705.dwg - Tab: E705 Plotted By: DShabataev 23/10/26 [Thu 8:16am]
24"x36" PLOT SCALE: 1:1

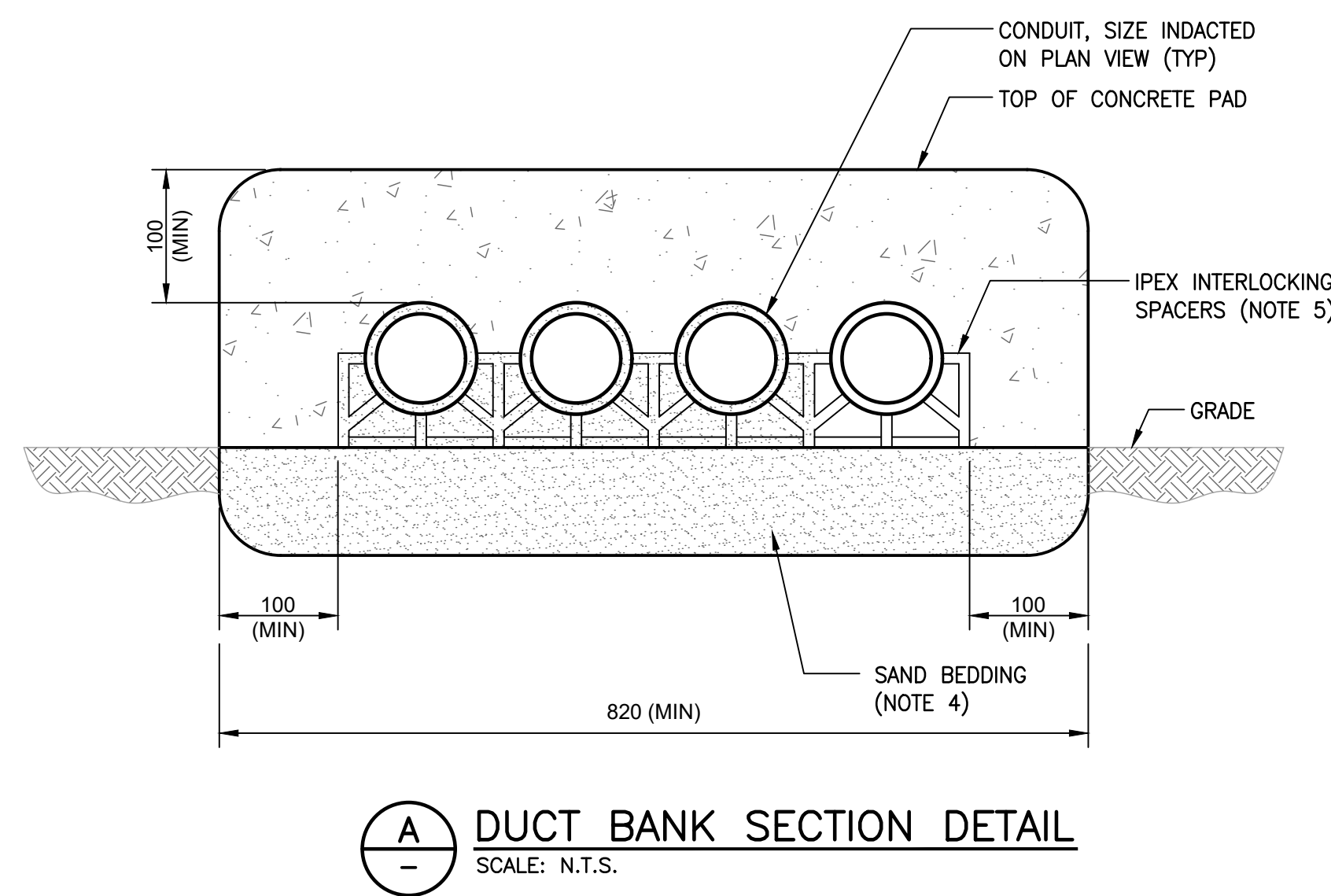


NOTES:

1. GRAY ITEMS INDICATE WORK BY OTHERS.
2. RAYCHEM GELCAP RE-ENTERABLE WATER TIGHT SPLICE KITS FOR CONNECTIONS WITHIN PUMP WELL HEAD.
3. PVC CONDUITS WITHIN CONCRETE PAD AND TRANSITION TO STAINLESS STEEL CONDUIT WHERE CONDUITS STUB UP FROM CONCRETE PAD.
4. NO MINIMUM THICKNESS FOR SAND BASE. SAND BASE IS TO PROVIDE A LEVEL SURFACE FOR CONDUITS AND SPACERS OVER UNEVEN GRANITE TRENCH.
5. IPEX INTERLOCKING SPACER ASSEMBLY CONSISTS OF THREE SPACERS WIDE SPACED MAXIMUM 2.1m APART THROUGHOUT THE LENGTH OF TRENCH. NO REBAR REQUIRED.

LEGEND

- OVERHEAD POWER LINE
- UNDERGROUND CABLES IN CONDUIT
- UNDERGROUND #2/0 GREEN INSULATED COPPER GROUND CONDUCTOR
- COPPER CLAD GROUNDING ROD
- WELDED GROUND CONNECTION



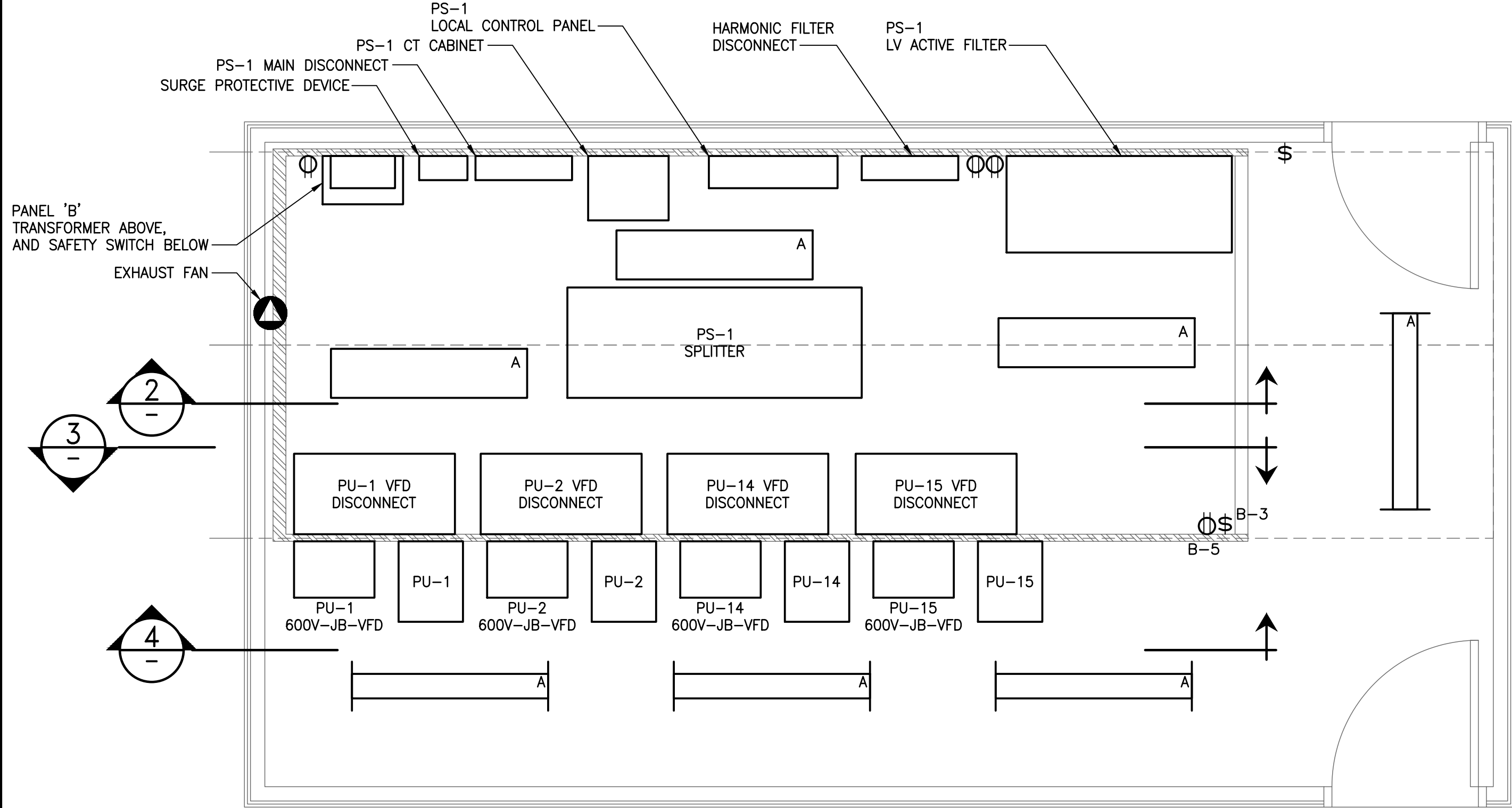
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NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT: Manitoba MANITOBA MINES BRANCH			
PROJECT: RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION: ELECTRICAL PUMPING STATION PS-1 PLAN			
KGS GROUP	DESIGN BY:	MAM	DATE: 12/06/13
	DESIGN CHECK:	JAB	DATE: 12/10/17
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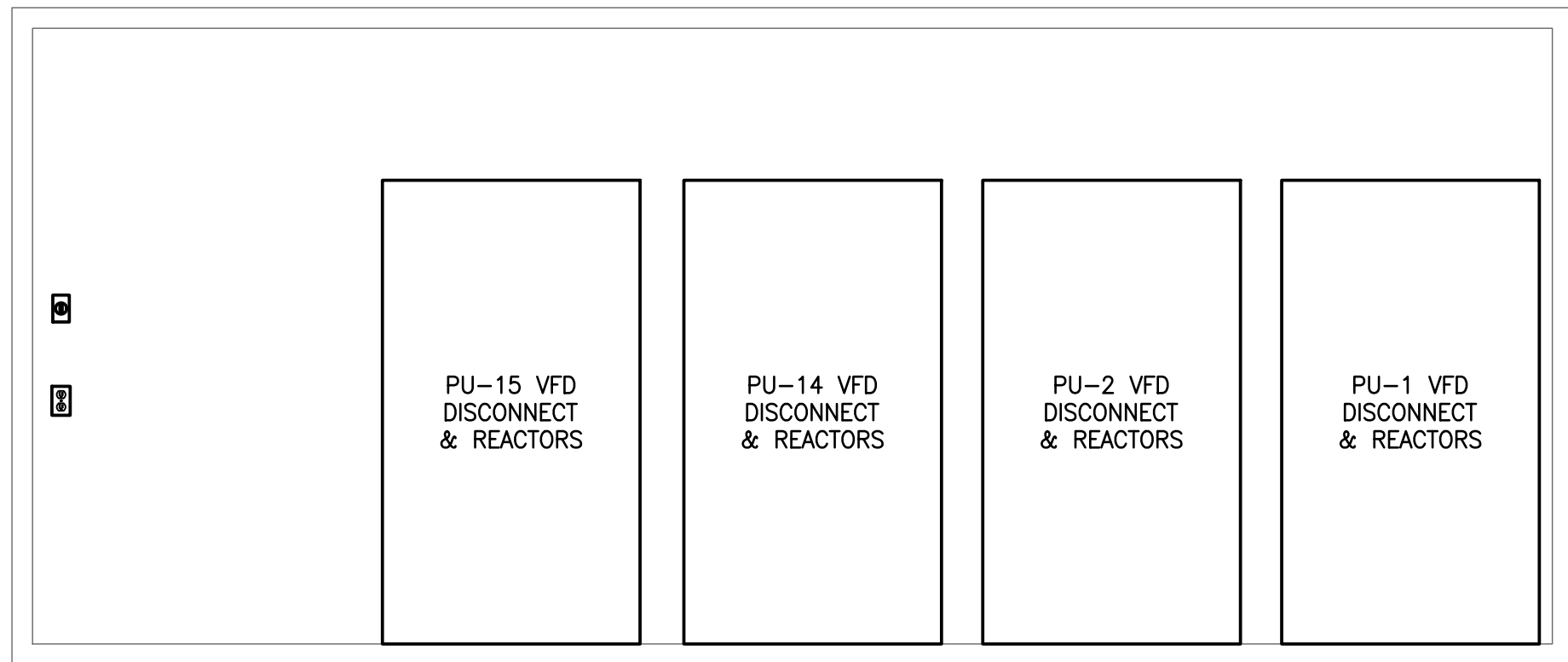
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10-1357-02

ENG. STAMP
ORIGINAL DRAWING SEALED BY:
J.A. BOUCHARD
2012/10/17

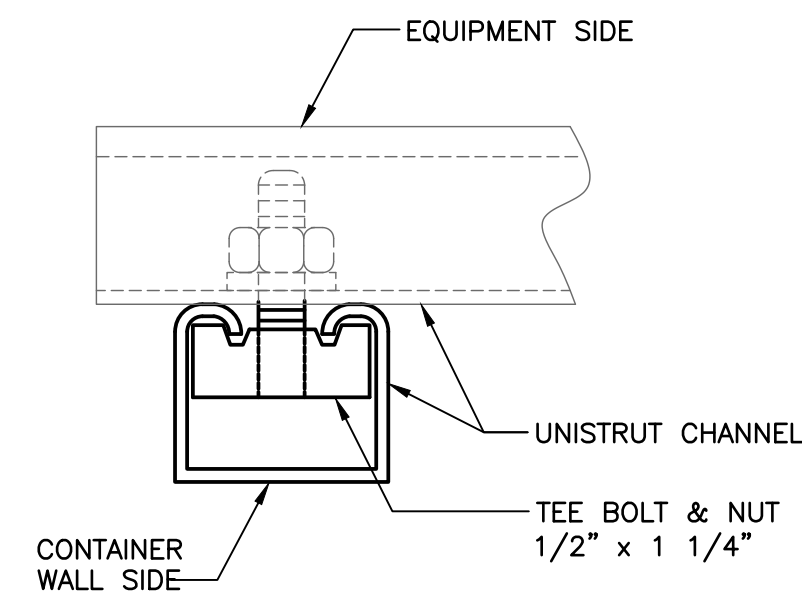
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24"x36" PLOT SCALE: 1:1



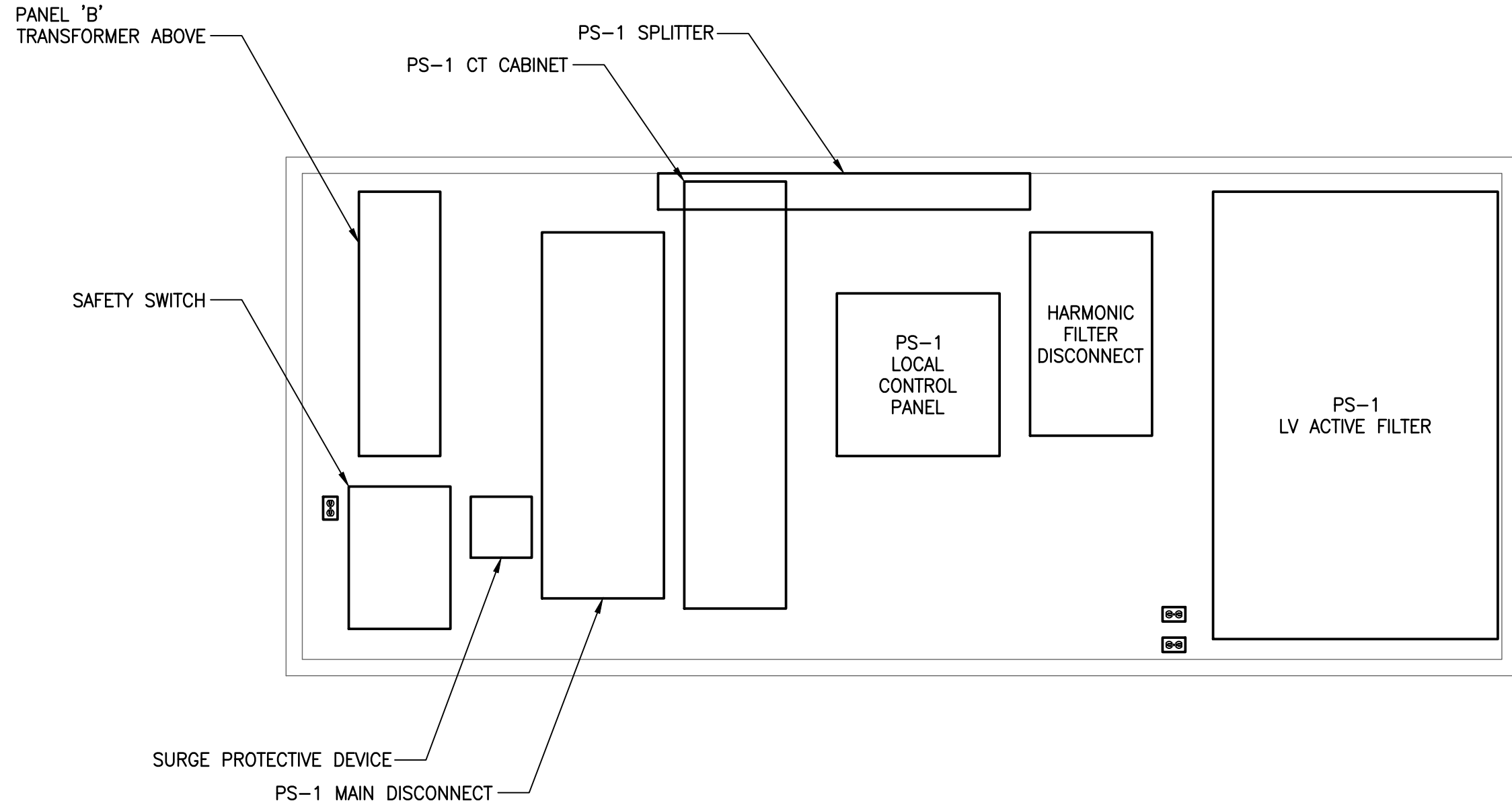
1 ELECTRICAL EQUIPMENT LAYOUT
SCALE: 1:25



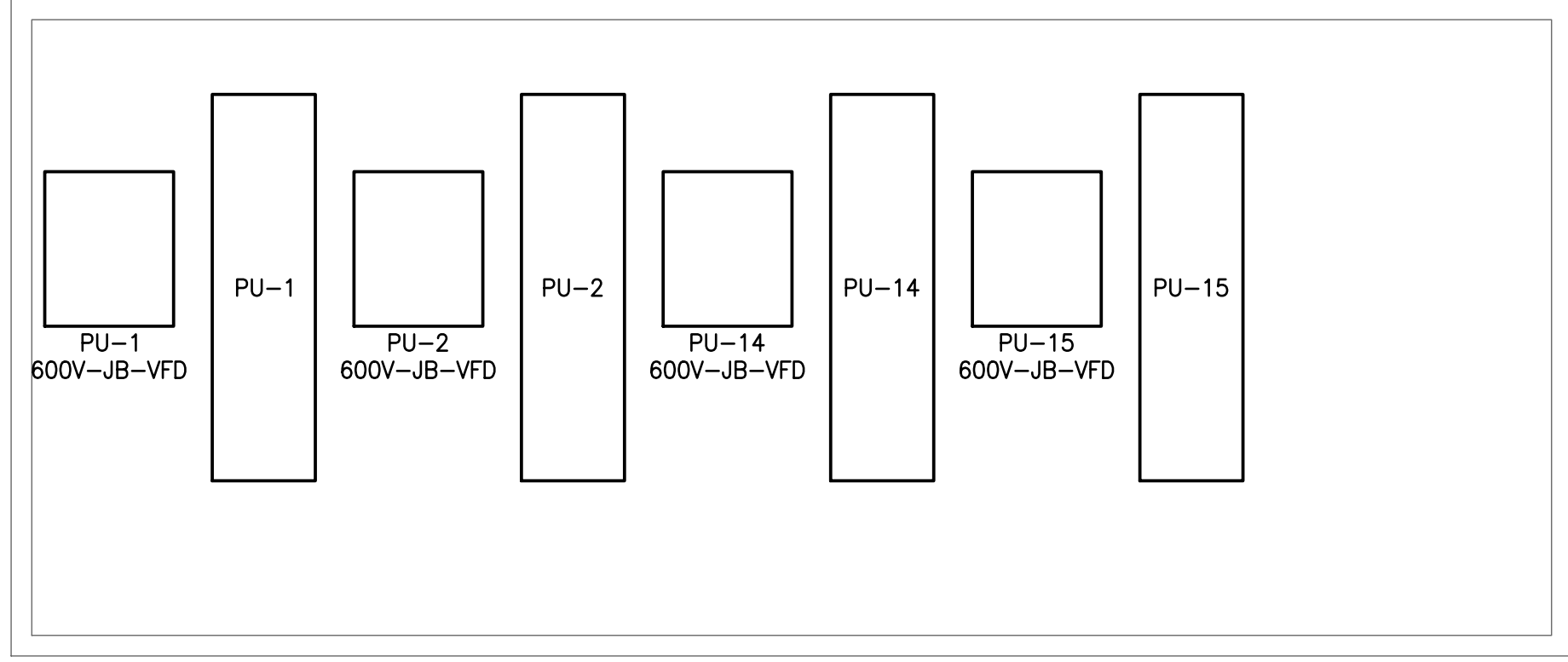
3 EQUIPMENT ELEVATION
SCALE: 1:25



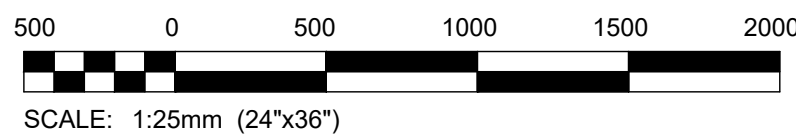
5 TYPICAL BOLT ASSEMBLY
SCALE: N.T.S.



2 EQUIPMENT ELEVATION
SCALE: 1:25



4 EQUIPMENT ELEVATION
SCALE: 1:25



ORIGINAL PROJECT
10-1357-02

ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

- NOTES:**
- WELD STUD BOLTS TO SHIPPING CONTAINER SIDE WALL TO MOUNT STRUT CHANNELS.
 - LOCATE CHANNELS AS NECESSARY TO SUPPORT EQUIPMENT.
 - PREPARE AND PRIME INSIDE OF SEACAN AND PAINT WHITE.
 - ALL PENETRATIONS SHALL BE SEALED WITH A WATER RATED SEALANT -40 SILICON.

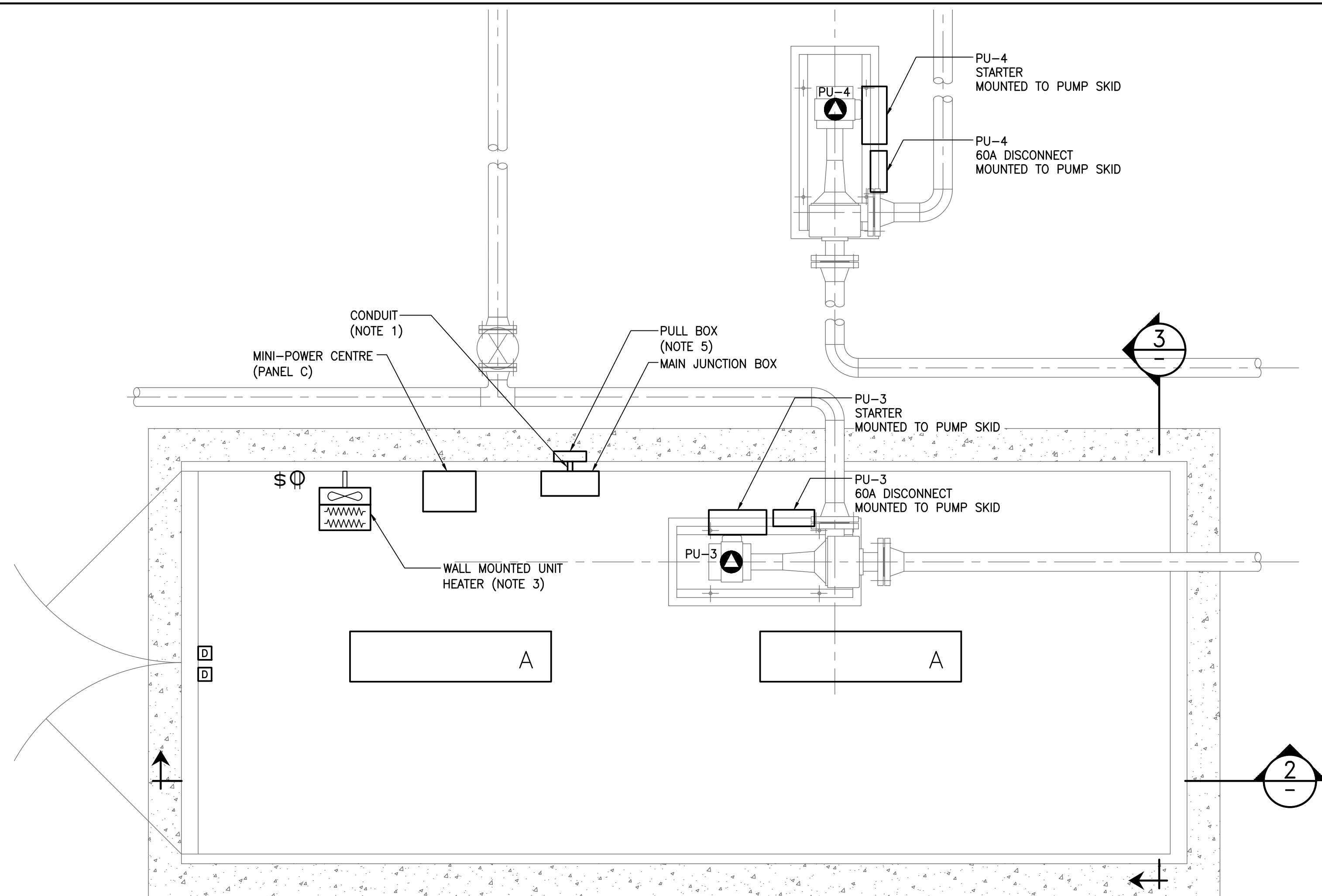
LEGEND:

	A	SURFACE MOUNT FLUORESCENT FIXTURE
	D	DOOR STRIKE
	T	THERMOSTAT
	⊕	DUPLEX RECEPTACLE
	\$	WALL SWITCH
	⊕	3 PHASE LOAD

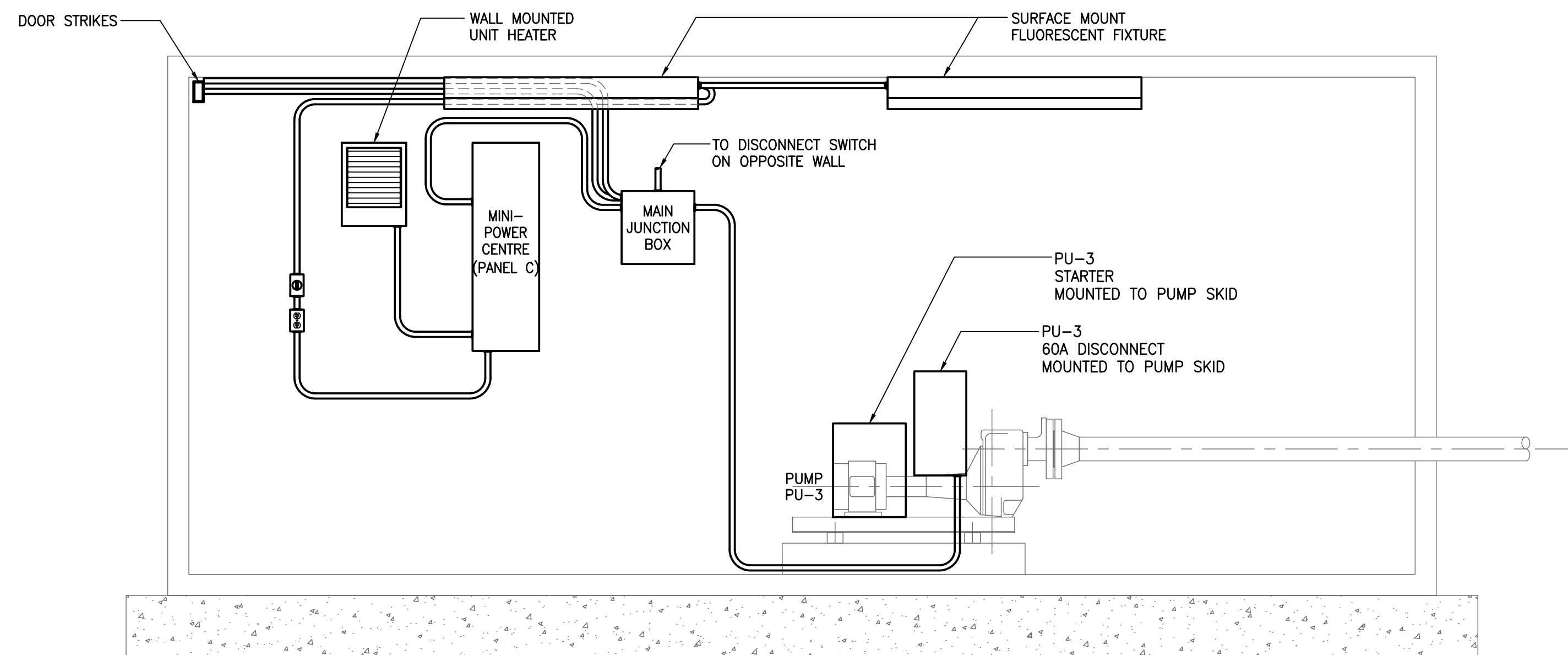
AS-BUILT
AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

0 23/10/26 ISSUED AS-BUILT			
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL PUMP STATION 1 - EQUIPMENT ENCLOSURE PLAN AND SECTIONS			
	DESIGN BY:	DAK	DATE: 12/03/01
	DESIGN CHECK:	JAB	DATE: 12/10/17
	DRAWN BY:	JPL	DATE: 12/03/01
	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO. 21-1357-001 E706			REV. 0

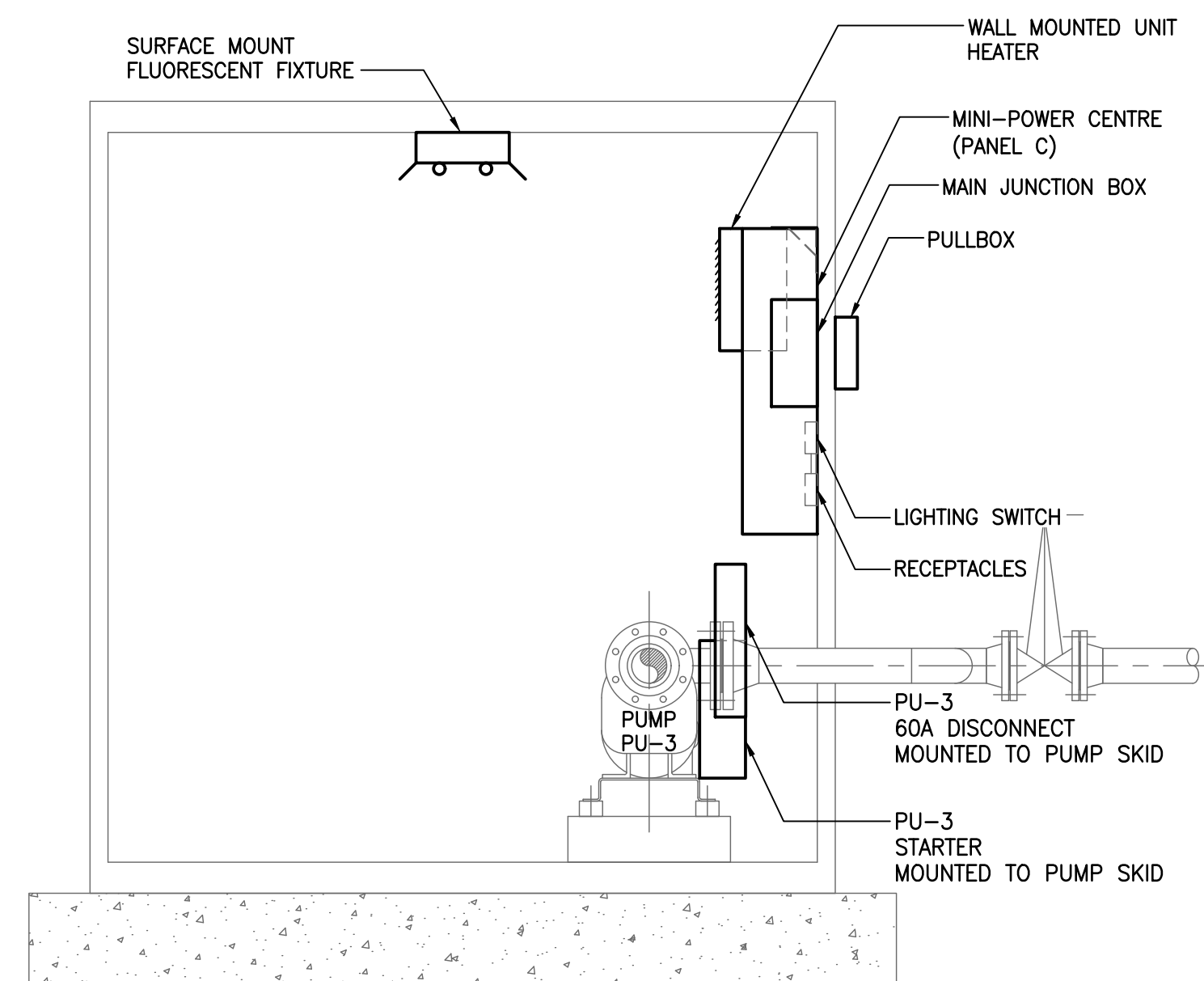
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24"x36" PLOT SCALE: 1:1



1 EQUIPMENT ENCLOSURE PLAN
SCALE: 1:20mm (24"x36")
1:40mm (11"x17")



2 EQUIPMENT ENCLOSURE LONGITUDINAL SECTION
SCALE: 1:20mm (24"x36")
1:40mm (11"x17")



3 EQUIPMENT ENCLOSURE SECTION
SCALE: 1:20mm (24"x36")
1:40mm (11"x17")

- NOTES:**
- ALL PENETRATIONS SEALED WITH WATER PROOF SEALANT -40 SILICON.
 - PLYWOOD BACKBOARD TO MOUNT ALL ELECTRICAL EQUIPMENT TO WALLS.
 - UNIT HEATER DIMPLEX #DGWH4031 OR APPROVED EQUIVALENT.
 - ALL CONDUIT WITHIN EQUIPMENT ENCLOSURE SHALL BE RGS CONDUIT.
 - PULL BOX LOCATION TO BE DETERMINED BY LOCATION OF UNDERGROUND CONDUIT FROM WATER TREATMENT PLANT. REFER TO DRAWING E701 FOR UNDERGROUND CONDUIT ROUTE. PROVIDE CONNECTION TO UNDERGROUND CONDUIT FROM PULL BOX

- LEGEND:**
- SURFACE MOUNT FLUORESCENT FIXTURE
 - DOOR STRIKE
 - DUPLEX RECEPTACLE
 - WALL SWITCH
 - 30 MOTOR LOAD
 - 240V RECEPTACLE (TYPE 6-20R)
 - UNIT HEATER

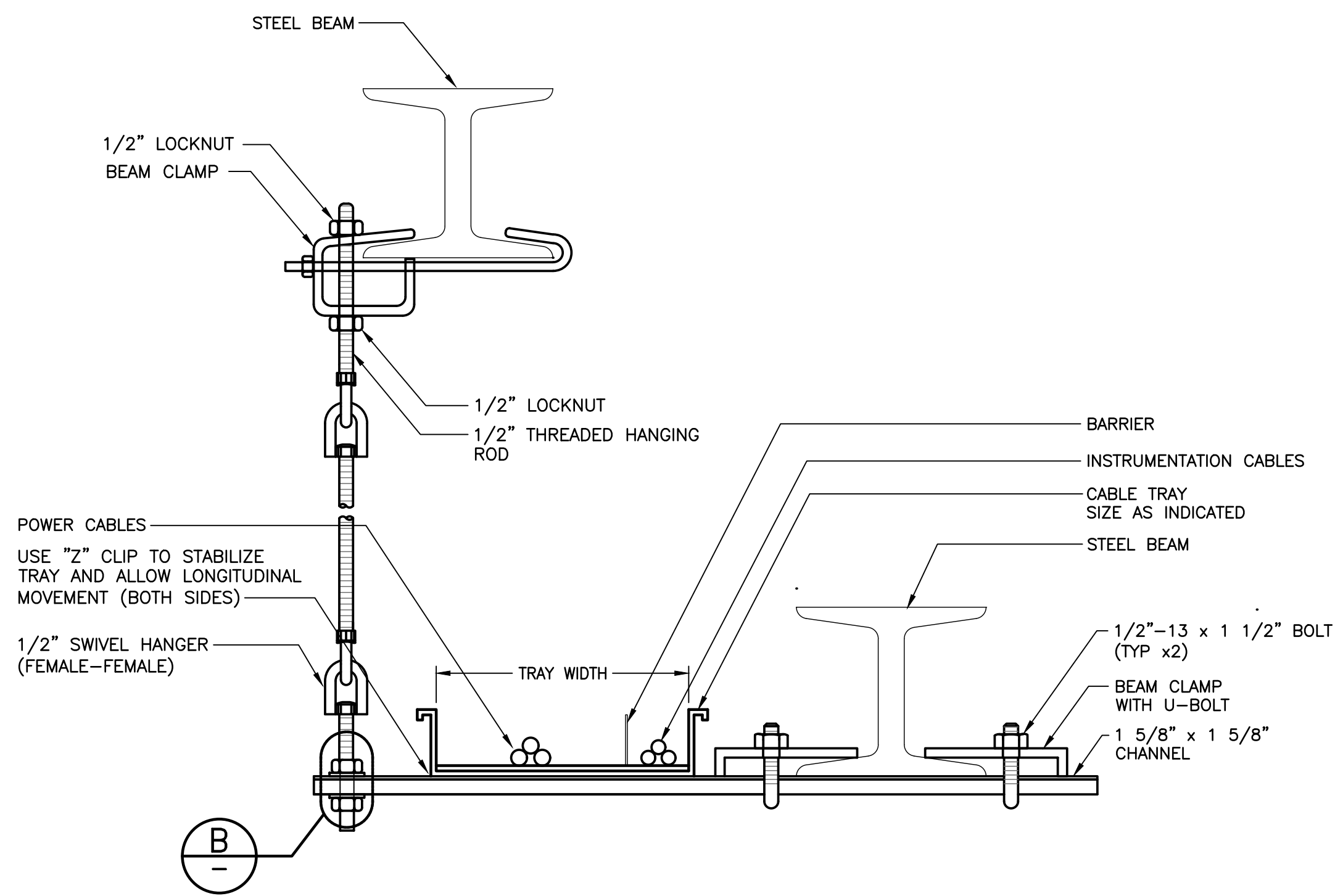
AS-BUILT
AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
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0	23/10/26	ISSUED AS-BUILT		
NO.	YY/MM/DD	DESCRIPTION	BY	
REVISIONS / ISSUE				
CLIENT:				
PROJECT:				
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
ELECTRICAL PUMP STATION 2 - EQUIPMENT ENCLOSURE PLAN & SECTIONS				
	DESIGN BY:	DAK	DATE: (YY/MM/DD):	12/03/30
	DESIGN CHECK:	JAB	DATE:	12/10/17
	DRAWN BY:	JPL	DATE:	12/03/30
	DWG CHECK:	JAB	DATE:	12/10/17
DWG. NO.	21-1357-001 E707			REV: 0

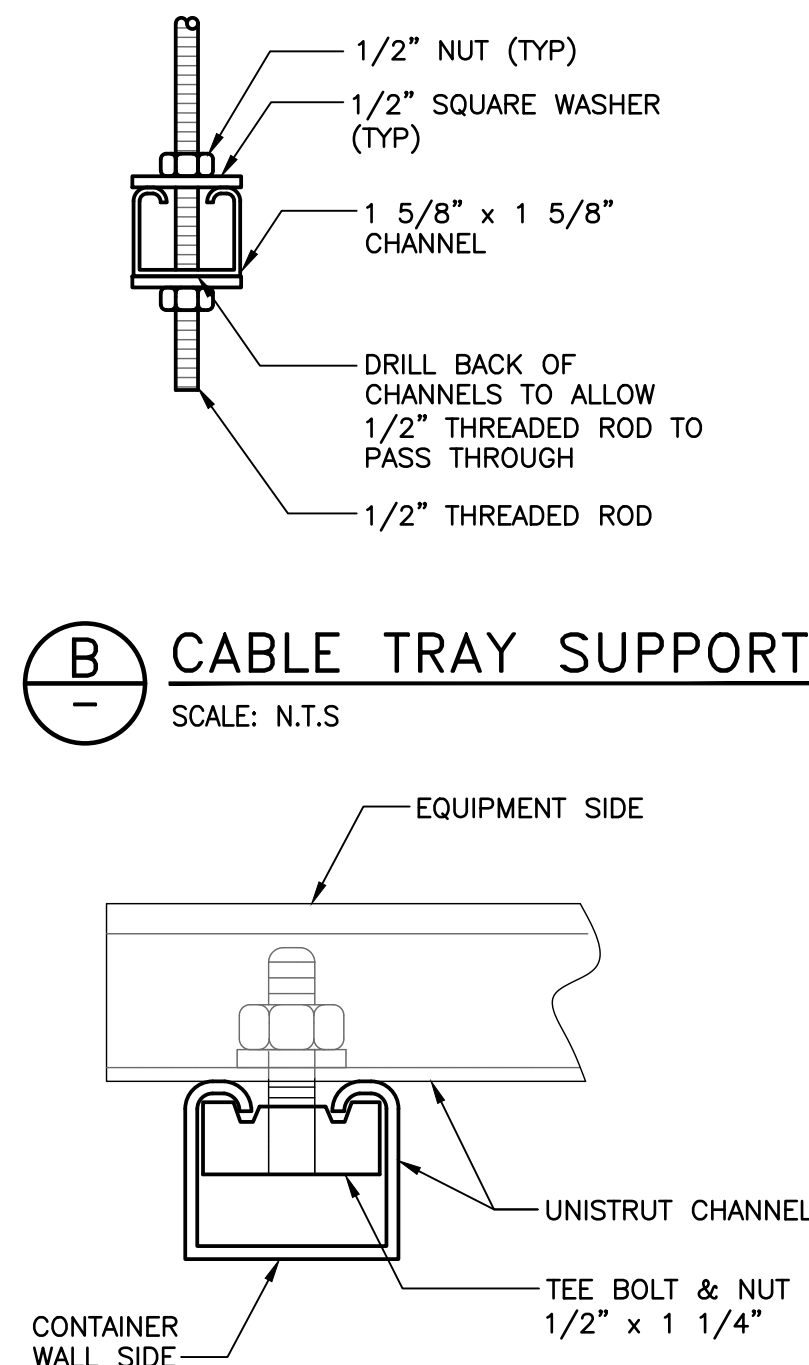
ORIGINAL PROJECT
10-1357-02

ENG. STAMP
ORIGINAL DRAWING SEALED BY:
J.A. BOUCHARD
2012/10/17

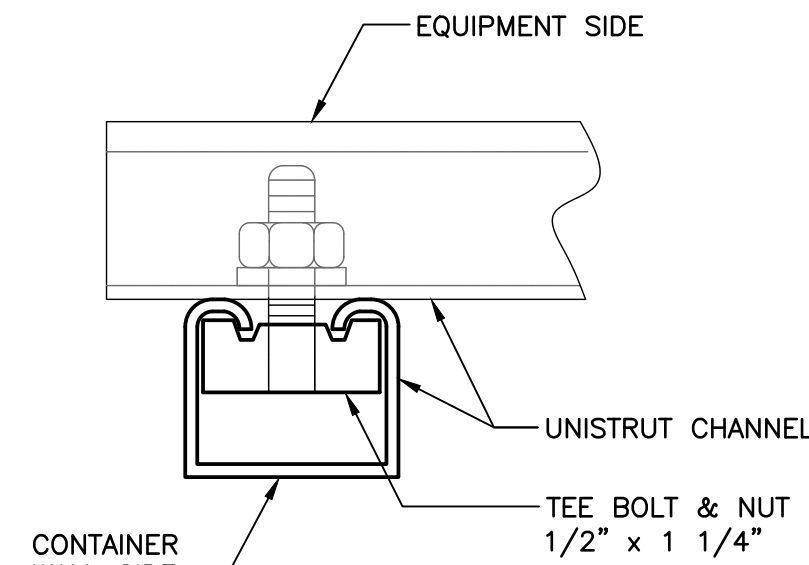
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24"x36" Plot Scale: 1:1



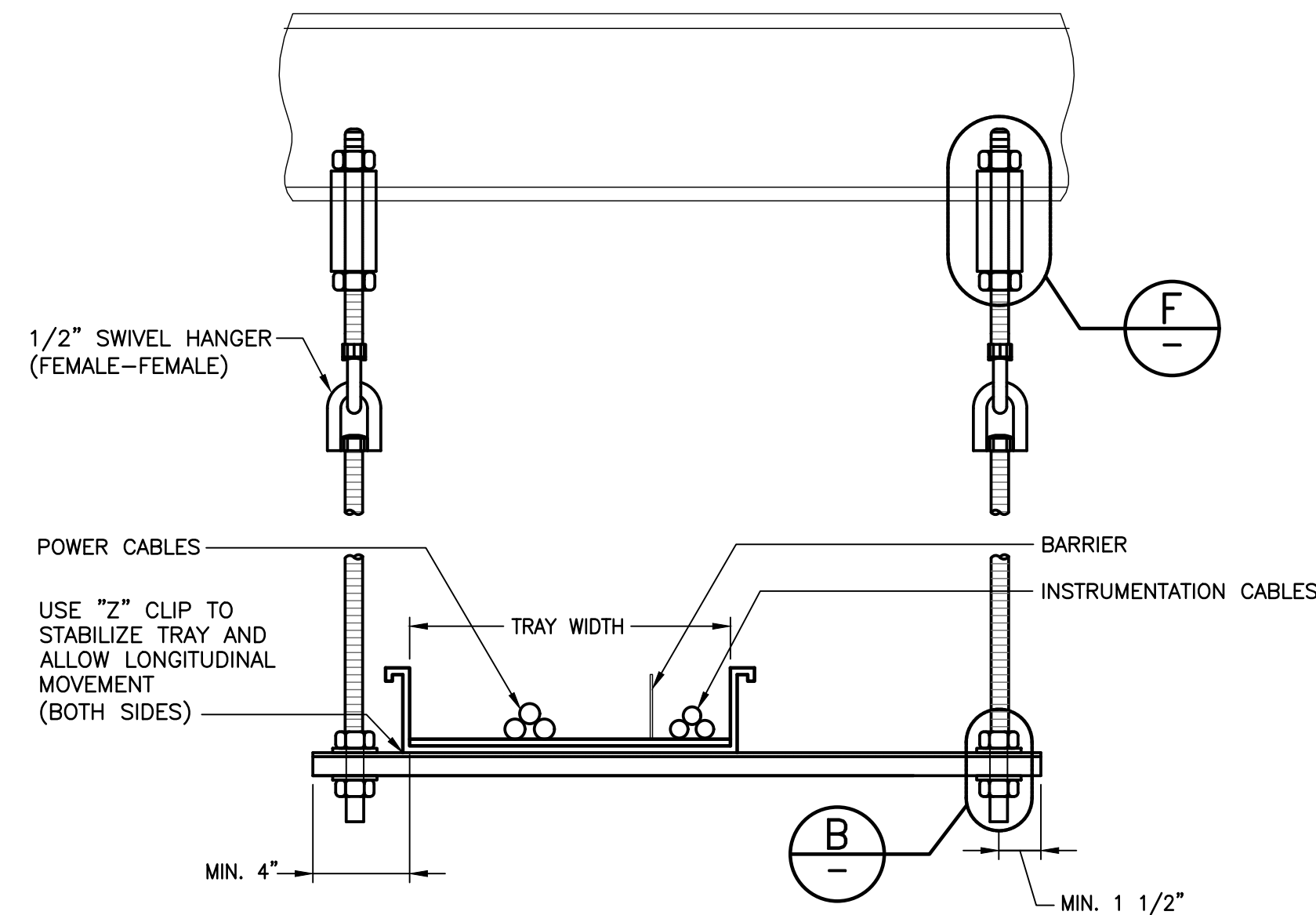
A CABLE TRAY TRAPEZE MOUNTING DETAIL
E704 SCALE: N.T.S.



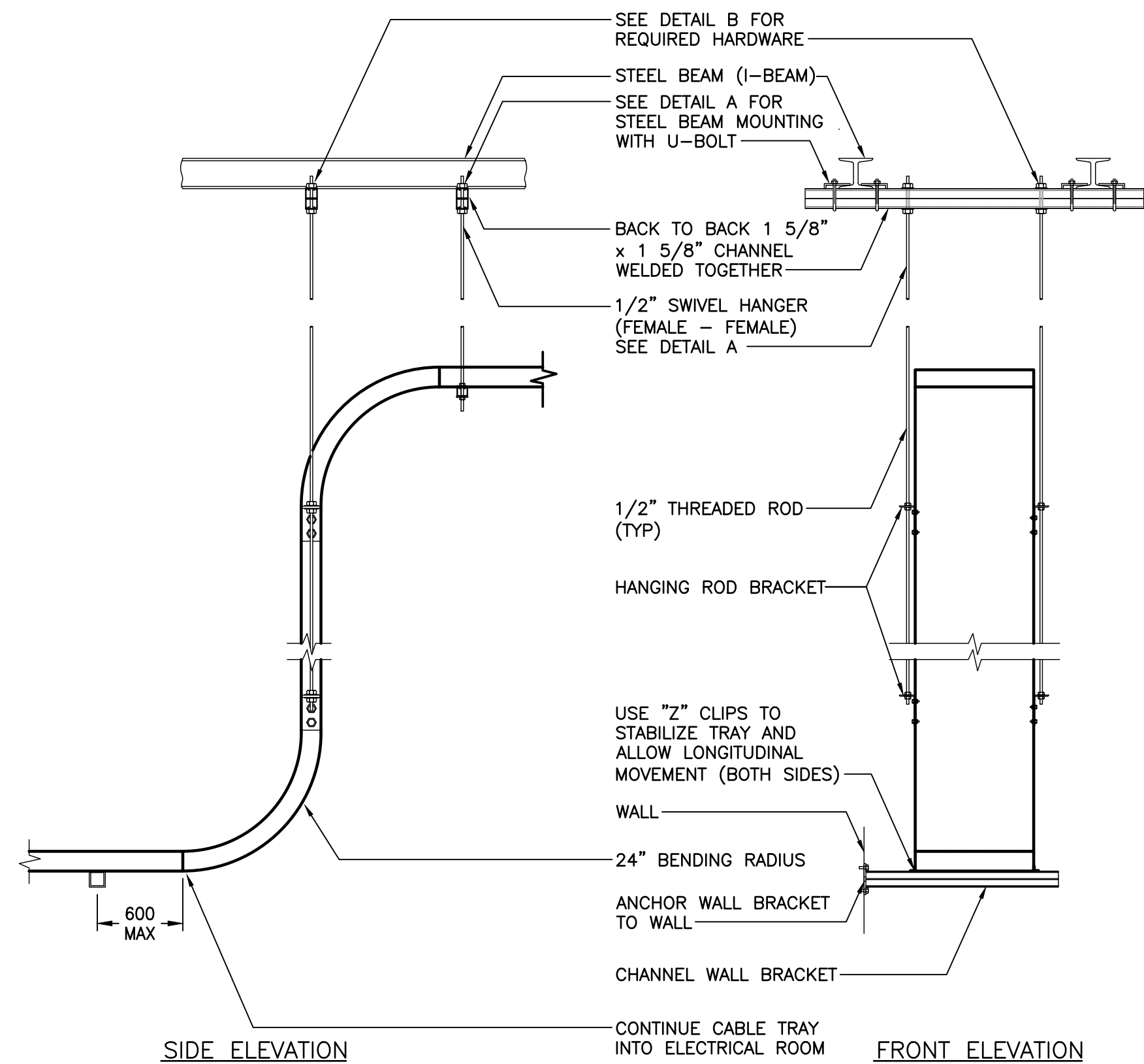
B CABLE TRAY SUPPORT
SCALE: N.T.S.



C TYPICAL BOLT ASSEMBLY
E707 SCALE: N.T.S.



E CABLE TRAY TRAPEZE MOUNTING DETAIL
E704 SCALE: N.T.S.



J CABLE TRAY SUPPORT DETAIL
E704 N.T.S.

NOTES:

1. PRE-DRILL HOLE AND USE TWO 0.25"-20 SELF TAPPING SHEET METAL SCREWS TO SECURE THE LIMIT SWITCH TO THE SEA CAN. ADJUST TOP PUSH ROD AND THE POSITION OF THE SWITCH SUCH THAT THE CONTACT OPENS AND CLOSSES. ADJUST THE TOP PUSH ROD SUCH THAT IT DOES NOT REACH THE END OF TRAVEL.

AS-BUILT
AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
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NO.	YY/MM/DD	DESCRIPTION	BY	
REVISIONS / ISSUE				
CLIENT:				
Manitoba MANITOBA MINES BRANCH				
PROJECT:				
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
ELECTRICAL DETAILS				
DESIGN BY:		CLS	DATE: (YY/MM/DD):	12/06/15
DESIGN CHECK:		JAB	DATE:	12/10/17
DRAWN BY:		JPL	DATE:	12/06/15
DWG CHECK:		JAB	DATE:	12/10/17
DWG. NO.				REV
21-1357-001 E708				0

ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

ORIGINAL PROJECT
10-1357-02

File Name: R:\Projects\2021\21-1357-001\Dwg\Elec\21-1357-001_E709.dwg - Tab: E709 Plotted By: DShabataev 23/10/26 [Thu 8:17am]
24"x36" PLOT SCALE: 1:1

SUPPLY: 208V 3Ø 4W MAINS AMPACITY: 100 INCOMING: ☐ BREAKER ☒ LUGS MAIN BREAKER REQUIRED: ☐ YES ☒ NO MAIN BREAKER AMPS: BUS MATERIAL: ☒ COPPER ☐ ALUMINUM BUS AND BREAKER RATING: 10KA			NAME PLATE: PANEL ' A '			LOCATION: WATER TREATMENT PLANT – ELECTRICAL RM MOUNTING: ☒ SURFACE ☐ FLUSH MANUFACTURER: MODEL NO: BREAKER: ☐ PLUG-IN ☒ BOLT-ON SPD: ☐ YES ☒ NO SERVICE ENTRANCE RATED: ☐ YES ☒ NO								
REV	CCT	DESCRIPTION	VA	P	BKR	A	B	C	BKR	P	VA	DESCRIPTION	CCT	REV
0	1	EXTERIOR LIGHTING	680	1	20	•			15	1	50	FLOWMETER FIT–02	2	0
0	3	FLUORESCENT LIGHTING	1152	1	20		•		15	1	50	FLOWMETER FIT–01	4	0
0	5	TANK LIGHTING & SLAKER #1 LIGHTING	1175	1	20	•		•	15	1	696	EXHAUST FAN F–01	6	0
0	7	REACTOR TK–3 LIGHTING	1225	1	20		•		15	1	696	EXHAUST FAN F–02	8	0
0	9	REACTOR TK–4 LIGHTING	1225	1	20		•		15	1	696	EXHAUST FAN F–03	10	0
0	11	NORTH WALL LIGHTING	1050	1	20			•	20	1	–	MCC HEATER	12	0
0	13	SOUTH WALL LIGHTING	1225	1	20	•			–	–	–		14	
0	15	SLAKER #2 LIGHTING	1225	1	20		•		–	–	–		16	
0	17	CONTROL PANEL	150	1	15			•	15	1	600	ELECTRICAL ROOM RECEPTACLES	18	0
0	19		–	–	–			•	15	1	600	OFFICE RECEPTACLES	20	0
0	21	EMERGENCY LIGHTING	200	1	15	•		•	20	1	600	KITCHEN COUNTER RECEPTACLE	22	0
0	23	ALARM ANNUNCIATOR RTK–725 (UA–1)	100	1	15	•		•	20	1	600	KITCHEN COUNTER RECEPTACLE	24	0
0	25	MEMOGRAPH M RSG40 (UR–1)	50	1	15	•		•	20	1	600	PROCESS AREA RECEPTACLES	26	0
0	27	LIQUILINE CM444 (AIT–1) PH TRANSMITTER	50	1	15	•		•	20	1	600	PROCESS AREA RECEPTACLES	28	0
0	29	SATELLITE PHONE AND AUTO DIALER RECEPTACLE	100	1	15	•		•	15	1	600	KITCHEN RECEPTACLES	30	0
0	31	WATER TANK LEVEL CONTROLS	50	1	15	•		•	20	1	600	CONVENIENCE RECEPTACLES	32	0
0	33	GENERATOR SILO SUB PANEL	–	2	30		•	•	20	1	600	TEST SINK COUNTER RECEPTACLE	34	0
0	35		–	–	–		•	•	20	1	600	HOT WATER TANK	36	0
0	37		–	–	–		•	•	–	–	–		38	
0	39	208V RECEPTACLE	–	2	15		•	•	–	–	–		40	
0	41	EXIT LIGHTS – LOCKED IN ON POSITION	15	1	15			•	30	1	2000	GRINDER PUMP	42	0
			LOAD 'A'		LOAD 'B'		LOAD 'C'							
			0		250		7,686							

SUPPLY: 208V 3Ø 4W MAINS AMPACITY: 100 INCOMING: ☐ BREAKER ☒ LUGS MAIN BREAKER REQUIRED: ☐ YES ☒ NO MAIN BREAKER AMPS: BUS MATERIAL: ☒ COPPER ☐ ALUMINUM BUS AND BREAKER RATING: 10kA		NAME PLATE: PANEL ' B '		LOCATION: PUMP STATION #1 MOUNTING: ☒ SURFACE ☐ FLUSH MANUFACTURER: MODEL NO: BREAKER: ☐ PLUG-IN ☒ BOLT-ON SPD: ☐ YES ☒ NO SERVICE ENTRANCE RATED: ☐ YES ☒ NO									
REV\CCT		VA	P	BKR	A	B	C	BKR	P	VA		CCT\REV	
0	1	300	1	15	•			15	1	—	SPARE	2	0
0	3	600	1	15		•		15	1	150	CONTROL PANEL	4	0
0	5	200	1	15			•	15	1	—	SPARE	6	0
	7	—	—	—		•		—	—	—		8	
	9	—	—	—			•	—	—	—		10	
	11	—	—	—				•	—	—		12	
	13	—	—	—					—	—		14	
	15	—	—	—		•		—	—	—		16	
	17	—	—	—			•	—	—	—		18	
		LOAD 'A'		LOAD 'B'		LOAD 'C'							
		300		750		750							

SUPPLY: 240V 1Ø 3W MAINS AMPACITY: 100 INCOMING: ☐ BREAKER ☒ LUGS MAIN BREAKER REQUIRED: ☐ YES ☒ NO MAIN BREAKER AMPS: BUS MATERIAL: ☒ COPPER ☐ ALUMINUM BUS AND BREAKER RATING: 10KA				NAME PLATE: PANEL ' C '				LOCATION: PUMP STATION #2 MOUNTING: ☒ SURFACE ☐ FLUSH MANUFACTURER: MODEL NO: BREAKER: ☐ PLUG-IN ☒ BOLT-ON SPD: ☐ YES ☒ NO SERVICE ENTRANCE RATED: ☐ YES ☒ NO							
REV	CCT			VA	P	BKR	L1	L2	BKR	P	VA			CCT	REV
0	1	LIGHTS		100	1	15	•							2	
0	3	RECEPTACLE		600	1	15		•		30	2	4800	UNIT HEATER RECEPTACLE	4	0
	5			-	-	-	•		-	-	-			6	
	7			-	-	-		•	-	-	-			8	
	9			-	-	-	•		-	-	-			10	
	11			-	-	-		•	-	-	-			12	
	13			-	-	-			-	-	-			14	
	15			-	-	-	•		-	-	-			16	
	17			-	-	-		•	-	-	-			18	
				LOAD 'L1'		LOAD 'L2'									
				2,500		3,000									

LUMINAIRE SCHEDULE				
TYPE	DESCRIPTION	VOLTAGE	LAMPS	DESIGN BASICS
A	4' FLUORESCENT STRIP C/W REFLECTOR WITH WHITE ENAMEL FINISH, CHAIN HANGER KIT AND ELECTRONIC PROGRAMMED START BALLAST	120V	2-32W T8 4100°K	LITHONIA: L 232 120 GEB10PS HC36
B1	STANCHION MOUNT FIXTURE, 25 DEGREE ANGLE, ALUMINUM ENCLOSURE, TYPE 5 DISTRIBUTION, DOME REFLECTOR C/W 8' SWIVEL POLE SUPPORT.	120V	1-175W MH	CROUSE-HINDS: VMVMP175GP/FA/RD70/120 SWIVEL POLE: S3-2400-T-HG
B2	WALL MOUNT FIXTURE, ALUMINUM ENCLOSURE, TYPE 5 DISTRIBUTION, 30 DEGREE DOME REFLECTOR, 3/4 INCH THROUGH FEED HUBS.	120V	1-175W MH	CROUSE-HINDS: VMVM2TW175GP/FA/RA70/120
B3	STANCHION MOUNT FIXTURE, 25 DEGREE ANGLE, ALUMINUM ENCLOSURE, TYPE 5 DISTRIBUTION, DOME REFLECTOR C/W 6' SWIVEL POLE SUPPORT.	120V	1-100W MH	CROUSE-HINDS: VMVMP100GP/FA/RD70/120 SWIVEL POLE: R1-40
B4	STANCHION MOUNT FIXTURE, 25 DEGREE ANGLE, ALUMINUM ENCLOSURE, TYPE 5 DISTRIBUTION, DOME REFLECTOR	120V	1-175W MH	CROUSE-HINDS: VMVMP175GP/FA/RD70/120
W1	WALL PACK C/W LAMP, PHOTOCELL CONTROL, PULSE START BALLAST AND BRONZE FINISH.	120V	1-400W PSMH	KEENE: 553 400 PS L 2
W2	WALL PACK C/W LAMP, PHOTOCELL CONTROL, PULSE START BALLAST AND BRONZE FINISH.	120V	1-70W PSMH	KEENE: 553 070 PS L 1

EMERGENCY LIGHTING EQUIPMENT SCHEDULE				
TYPE	DESCRIPTION	VOLTAGE	LAMPS	DESIGN BASICS
E1	PICTOGRAM EXIT SIGN AVAILABLE IN CEILING MOUNT AND WALL MOUNT, SINGLE SIDED NEMA 4X POLYMERIC GASKETED ENCLOSURE WITH POLYCARBONATE FACE PLATE, MEETS CSA 22.2 NO. 141 FOR PICTOGRAM EXIT SIGNS, SUITABLE FOR MINUS 40° OPERATION C/W 2 PICTOGRAM FILMS PER FACE	120/12V	WHITE LED	LUMACELL: LN-1-W-U-CW
E2	DUAL HEAD EMERGENCY LIGHT WITH BATTERY PACK. MINIMUM 250W FOR 1/2 HOUR. ALL CHARGING SYSTEMS TO HAVE SELF TEST AND INDICATING LIGHTS FOR TROUBLE AND ALARM, FIBERGLASS NEMA 4X HOUSING, SOLID-STATE PULSE TYPE CHARGER C/W HEATER AND THERMOSTAT.	120/12V	2-12W MR16	LUMACELL: RG12S-250-NX-2M12W-AT-HTR
E3	REMOTE DUAL HEAD EMERGENCY LIGHT, ADJUSTABLE, ALUMINUM BACK PLATE WITH CLEAR POLYCARBONATE COVER, NEMA 4X.	12V	2-12W MR16	LUMACELL: MQM2NX-12V12W

NOTES:
1. -

AS-BUILT

AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

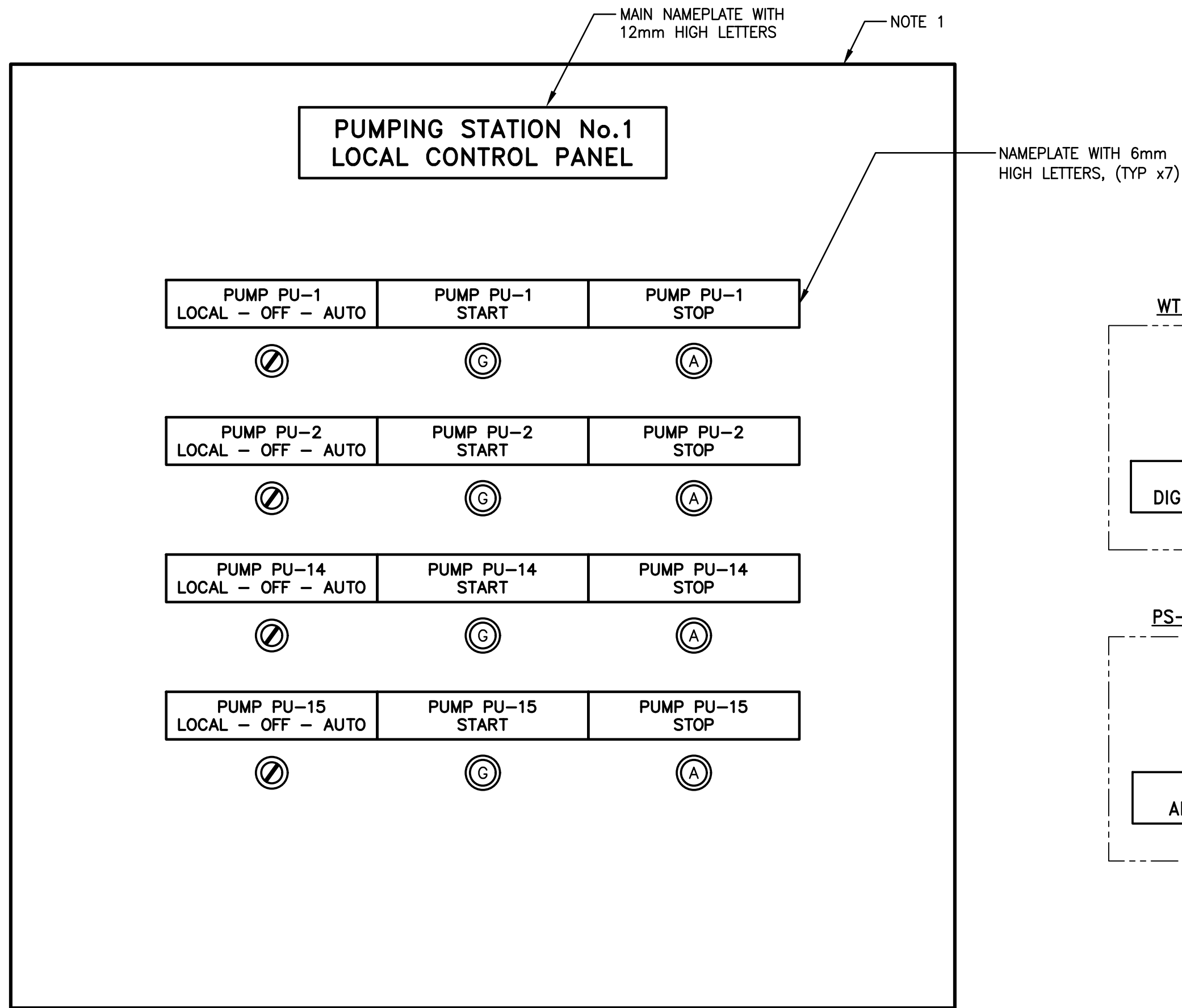
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NO.	YY/MM/DD	DESCRIPTION
REVISIONS / ISSUE		
CLIENT:		
		
PROJECT:		
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700		
DWG. DESCRIPTION:		
ELECTRICAL LUMINAIRE AND PANEL SCHEDULES		
	DESIGN BY:	CLS
	DESIGN CHECK:	JAB
	DRAWN BY:	GCN
	DWG CHECK:	JAB
DATE: (YY/MM/DD):		12/04/27
DATE:		12/10/17
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DATE:		12/10/17
DWG. NO.		21-1357-001 E709
REV.		0

ENG. STAMP

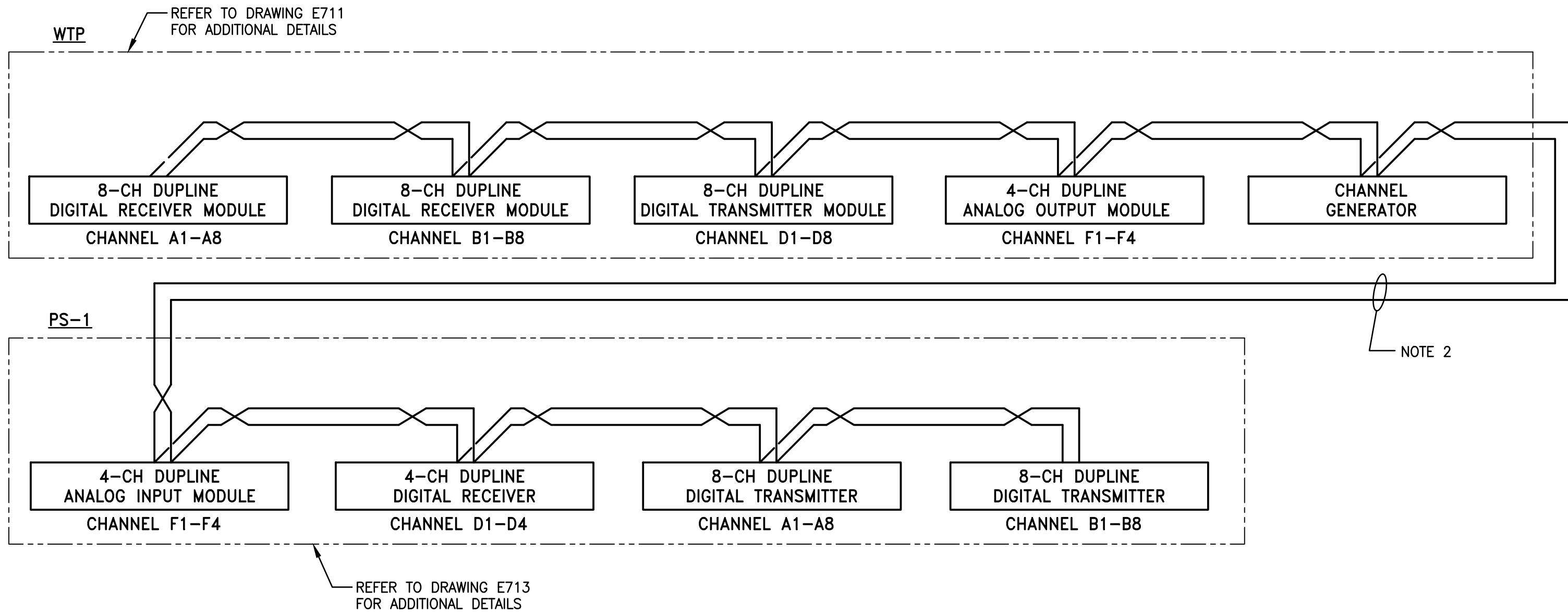
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

ORIGINAL PROJECT
10-1357-02

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24"x36" PLOT SCALE: 1:1



1 WTP PUMP CONTROL PANEL – DOOR LAYOUT
SCALE: N.T.S



2 DUPLINE 2-WIRE BUS INTERCONNECTION DIAGRAM
SCALE: N.T.S

NOTES:

1. SIZE OF PANEL TO SUIT ENCLOSED EQUIPMENT.
2. PROVIDE A 1 PR #16AWG SHIELDED, TWISTED PAIR, ARMoured CABLE FROM WATER TREATMENT PLANT TO PUMPING STATION PS-1, RUN IN UNDERGROUND CONDUIT PROVIDED BY OTHERS.

LEGEND:

- Ⓢ ON-OFF SELECTOR SWITCH
- Ⓢ PUSHBUTTON
- Ⓢ POTENTIOMETER
- Ⓢ PILOT LIGHT
- DENOTES:
G – GREEN
A – AMBER
- Ⓢ EMERGENCY STOP PUSH BUTTON

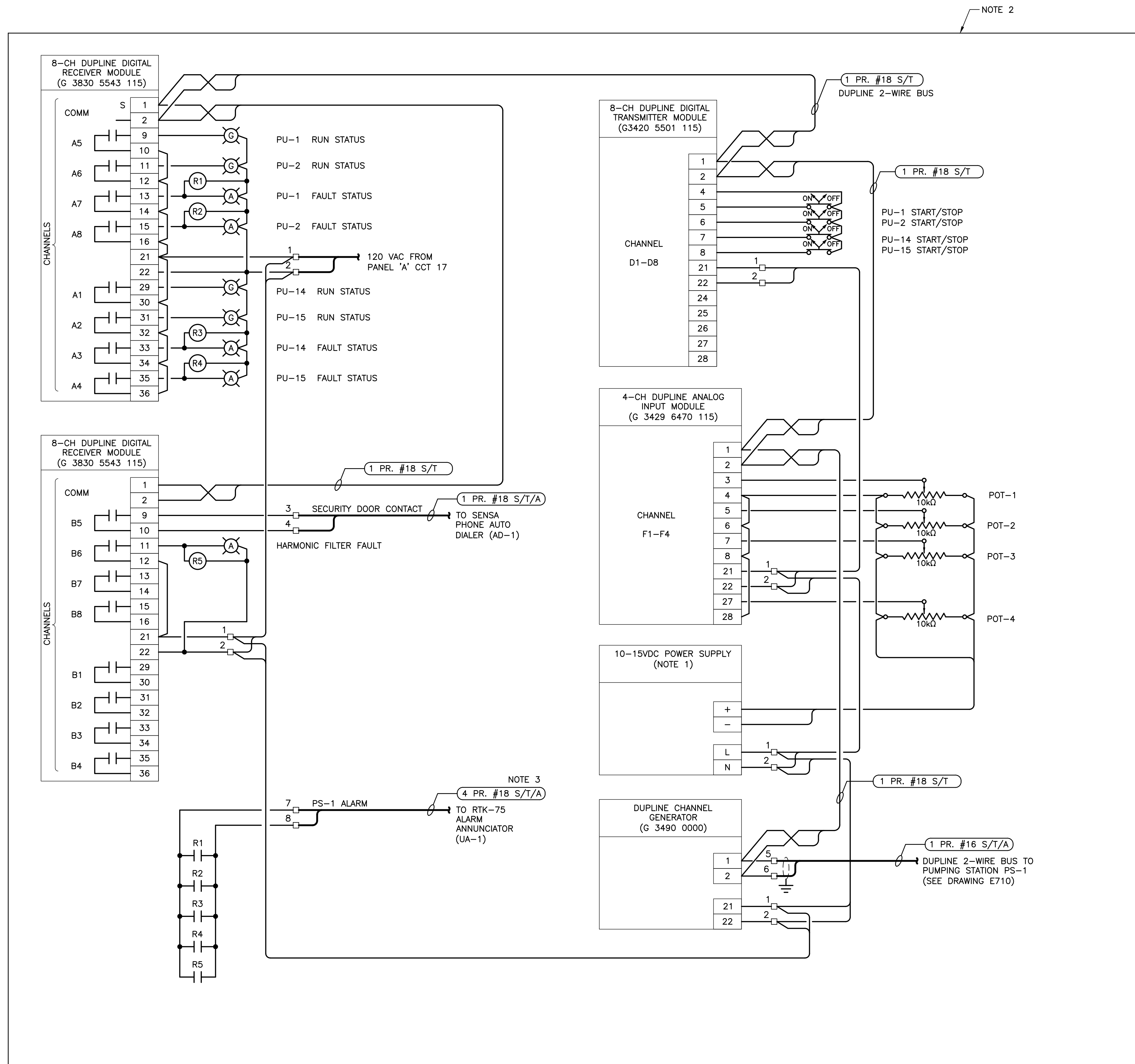
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REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL CONTROL PANEL LAYOUT AND BLOCK DIAGRAMS			
KGS GROUP	DESIGN BY:	CLS	DATE: (YY/MM/DD): 12/03/01
	DESIGN CHECK:	JAB	DATE: 12/10/17
	DRAWN BY:	JPL	DATE: 12/03/01
	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO. 21-1357-001 E710			REV. 0

ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

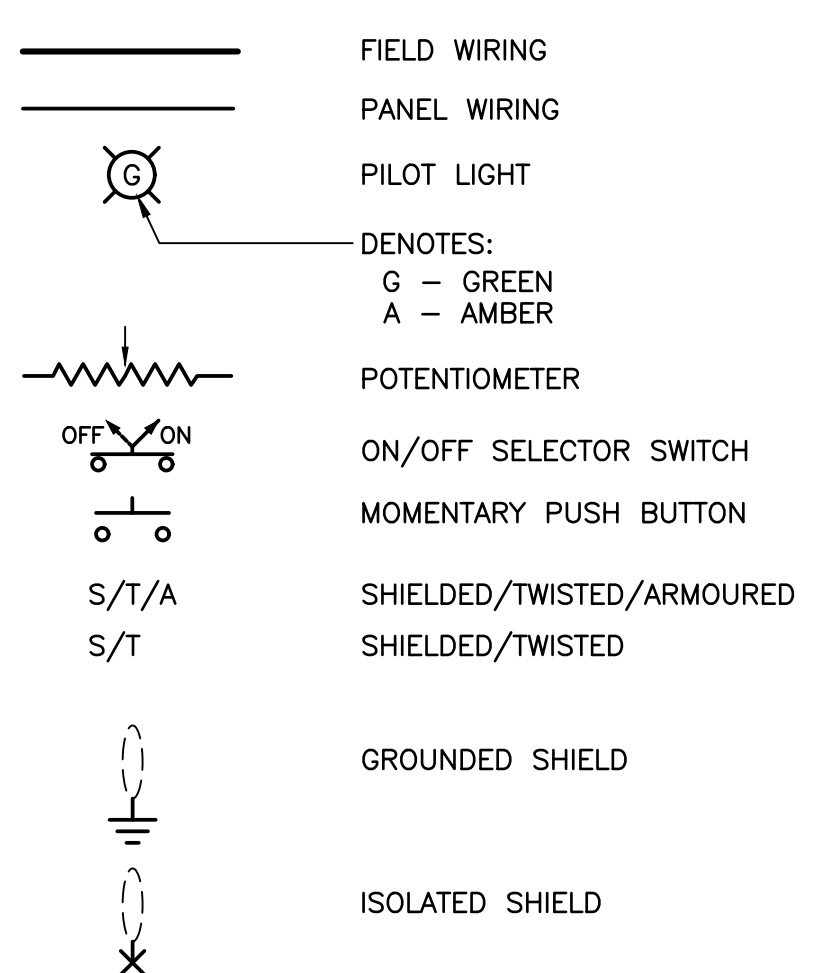
ORIGINAL PROJECT
10-1357-02



NOTES:

1. 10-15VDC PHOENIX CONTACT POWER SUPPLY CATALOG NUMBER MINI-PS-100-240AC-15DC12. OUTPUT VOLTAGE TO BE ADJUSTED TO 10VDC.
2. SIZE ENCLOSURE TO SUIT ENCLOSED EQUIPMENT. PROVIDE ROOM FOR AN ADDITIONAL 10% FREQUENCY RUMBLE, AND TWICE AS MANY THERMAL BLOCKS AS SHOWN. ENCLOSURE TO BE COMPLETE WITH ENVIRONMENTAL CONTROLS TO MAINTAIN ENCLOSURE TEMPERATURE BETWEEN 0°C AND 35°C.
3. LEAVE SPARE CONDUCTORS COILED IN ENCLOSURE AND LABEL AS SPARE.

LEGEND:



AS-BUILT

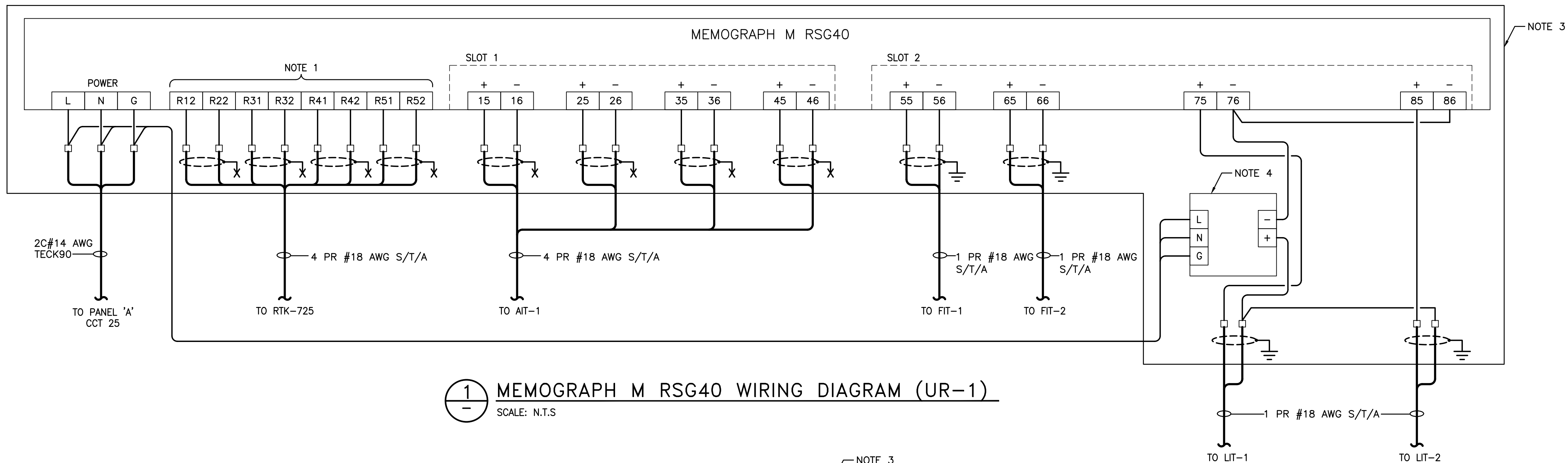
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KGS GROUP ON:
2023/07/11

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	23/10/26		ISSUED AS-BUILT	
	NO.	YY/MM/DD	DESCRIPTION	
BY				
REVISIONS / ISSUE				
CLIENT:				
 MANITOBA MINES BRANCH				
PROJECT:				
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
ELECTRICAL WTP CONTROL PANEL WIRING DIAGRAM				
	DESIGN BY:		DATE (YY/MM/DD):	
	CLS		11/11/07	
	DESIGN CHECK:		DATE:	
	JAB		12/10/17	
	DRAWN BY:		DATE:	
	JPL		11/11/07	
DWG CHECK:		DATE:		
JAB		12/10/17		
DWG. NO.				REV:
21-1357-001 E711				0

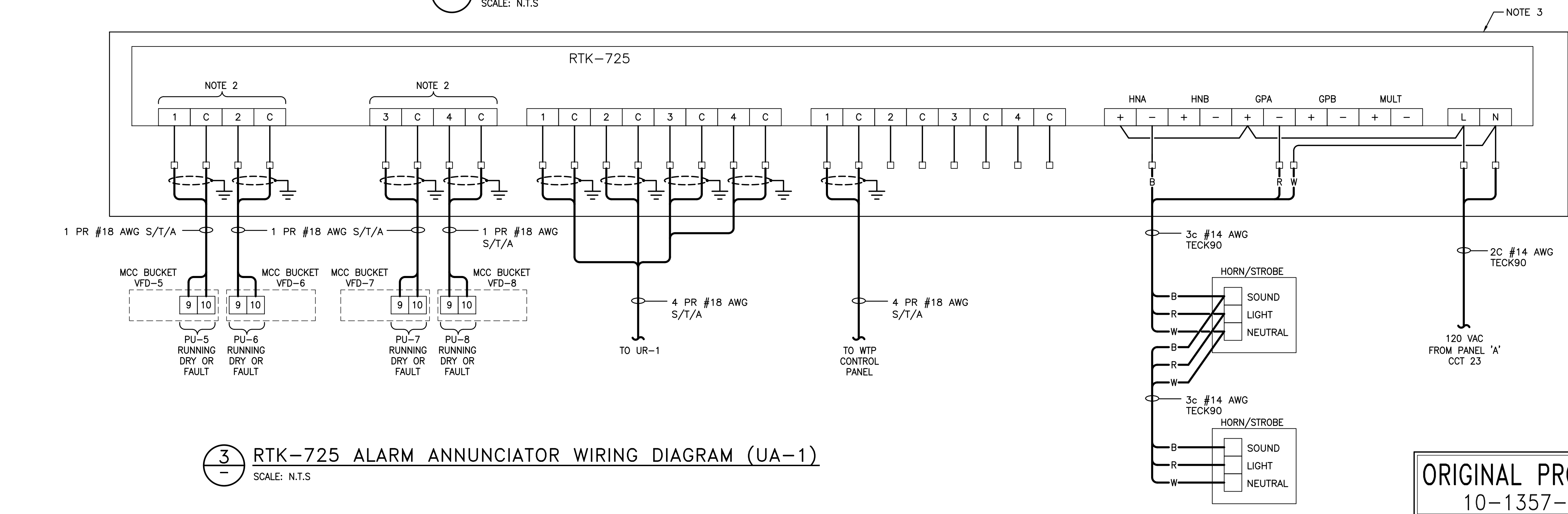
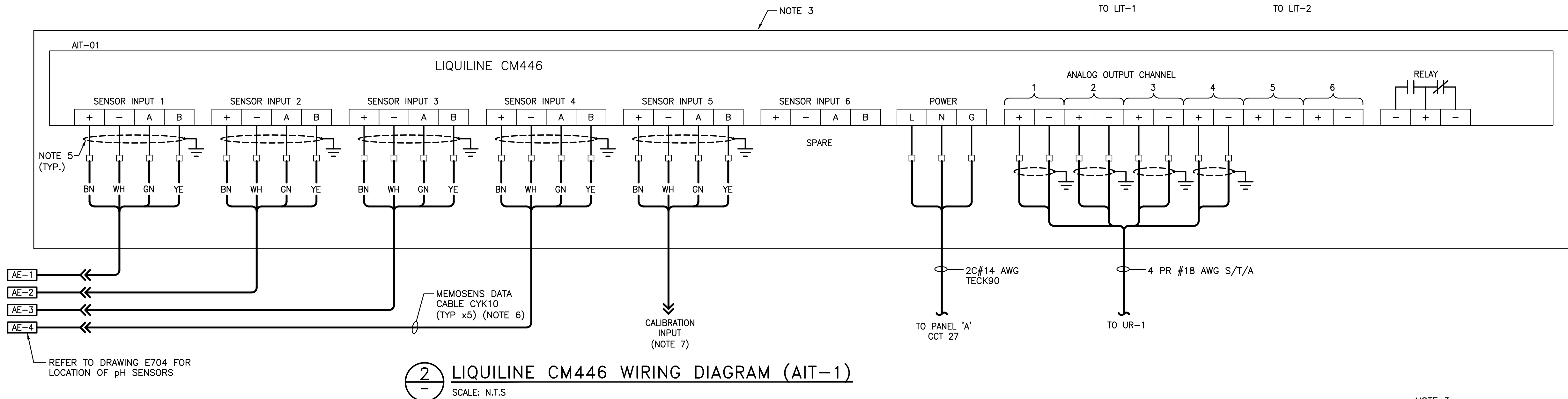
ORIGINAL PROJECT
10-1357-02

File Name: R:\Projects\2021\21-1357-001\Dwg\Elec\21-1357-001_E712.dwg - Tab: E712 Plotted By: DShabataev 23/10/26 [Thu 8:17am]
24x36 Plot Scale: 1:1



- NOTES:**
- MEMOGRAPH RELAY 2 AND 3 OUTPUTS TO CLOSE WHEN MIXING TANK 1 AND 2 REACHES 10% OR 90% LEVEL. RELAYS 4 AND 5 TO CLOSE WHEN pH AT OUTLET OF REACTORS IS LESS THAN 7 OR GREATER THAN 10.
 - RTK-725 INPUT OPTION AD3 TIME DELAY INPUTS TO BE SET TO 5 SECONDS.
 - SIZE ENCLOSURE TO SUIT ENCLOSED EQUIPMENT. ENCLOSURE TO BE COMPLETE WITH ENVIRONMENTAL CONTROLS TO MAINTAIN ENCLOSURE TEMPERATURE BETWEEN 0°C AND 35°C.
 - SINGLE CHANNEL, 12W, 24VDC POWER SUPPLY. (PHEONIX CONTACT-STEP-PS/1AC/24DC/0.5)
 - FOLLOW MANUFACTURES RECOMMENDED GROUNDING PRACTICES.
 - CONTRACTOR SHALL DETERMINE LENGTHS OF MEMOSENS DATA CABLES IN FIELD PRIOR TO ORDERING CABLES.
 - CALIBRATION INPUT IS INTENDED TO PERFORM PERIODIC CALIBRATION CHECKS OF pH SENSORS AND CABLE LENGTH MUST BE SUFFICIENT TO REACH NEAREST SINK.
 - LEAVE SPARE CONDUCTORS COILED IN ENCLOSURE AND LABEL AS SPARE.

- LEGEND:**
- FIELD WIRING
 - PANEL WIRING
 - MEMOSENS PLUG CONNECTION
 - S/T/A SHIELDED/TWISTED/ARMoured
 - GROUNDING SHIELD
 - ISOLATED GROUND



AS-BUILT

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THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

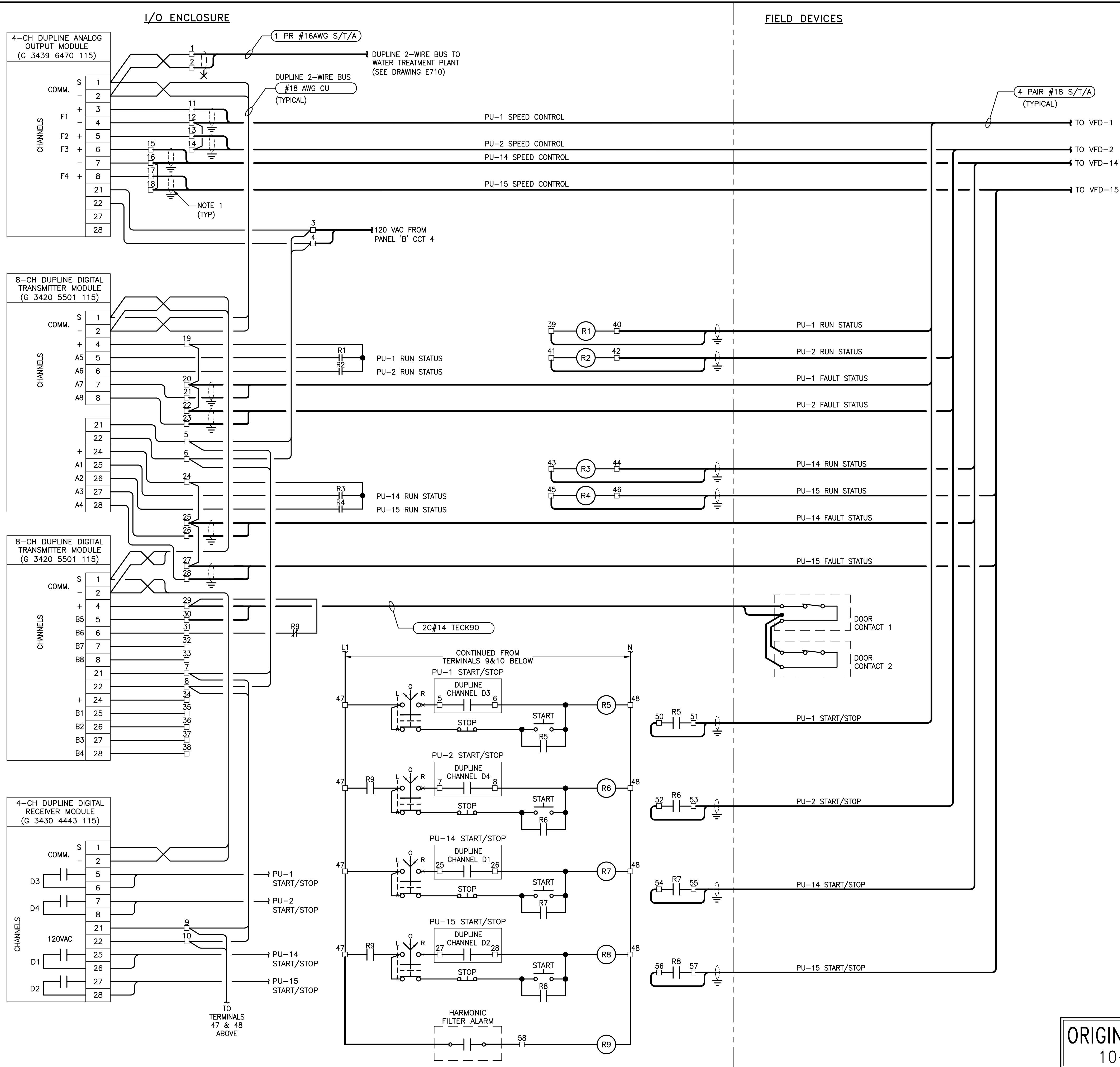
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NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL WTP WIRING DIAGRAM			
KGS GROUP	DESIGN BY:	CLS	DATE: 11/11/07
	DESIGN CHECK:	JAB	DATE: 12/10/17
	DRAWN BY:	JPL	DATE: 11/11/07
	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO.		REV.	
21-1357-001 E712		0	

ENG. STAMP

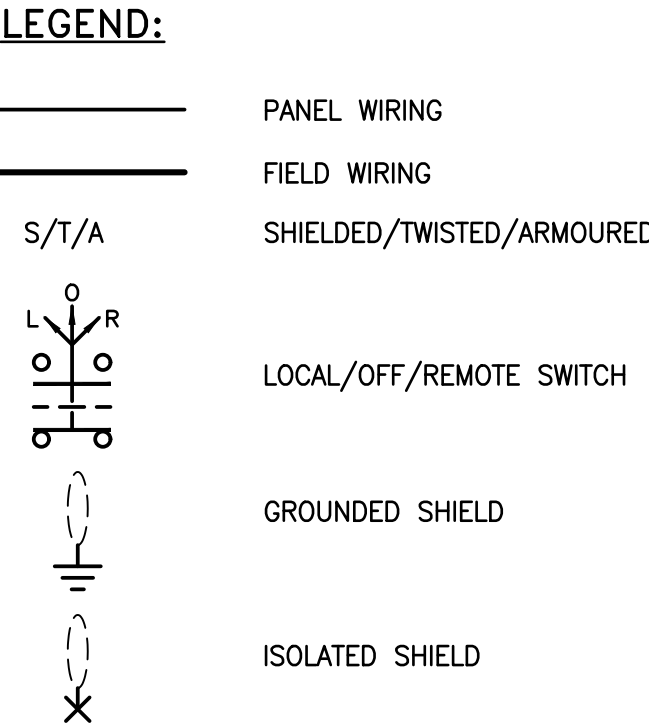
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

ORIGINAL PROJECT
10-1357-02

File Name: R:\Projects\2021\21-1357-001\Drawg\Elec\21-1357-001_E713.dwg - Tab: E713 Plotted By: DShabataev 23/10/26 [Thu 8:17am]
24"x36" PLOT SCALE: 1:1



- NOTES:**
- GROUND CABLE SHIELDING AT ONE END ONLY.
 - PROVIDE ENCLOSURE ENVIRONMENTAL CONTROLS TO MAINTAIN ENCLOSURE TEMPERATURE BETWEEN 0°C AND 35°C.
 - PS-1 CONTROL PANEL ENCLOSURE SHALL BE SIZED TO SUIT ALL COMPONENTS WITH SUFFICIENT SPACE FOR TWO ADDITIONAL DUPLINE MODULES AND ASSOCIATED TERMINAL BLOCKS. WALL MOUNTABLE ENCLOSURE TO BE RATED TYPE 12 OR IP52.



AS-BUILT

AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11

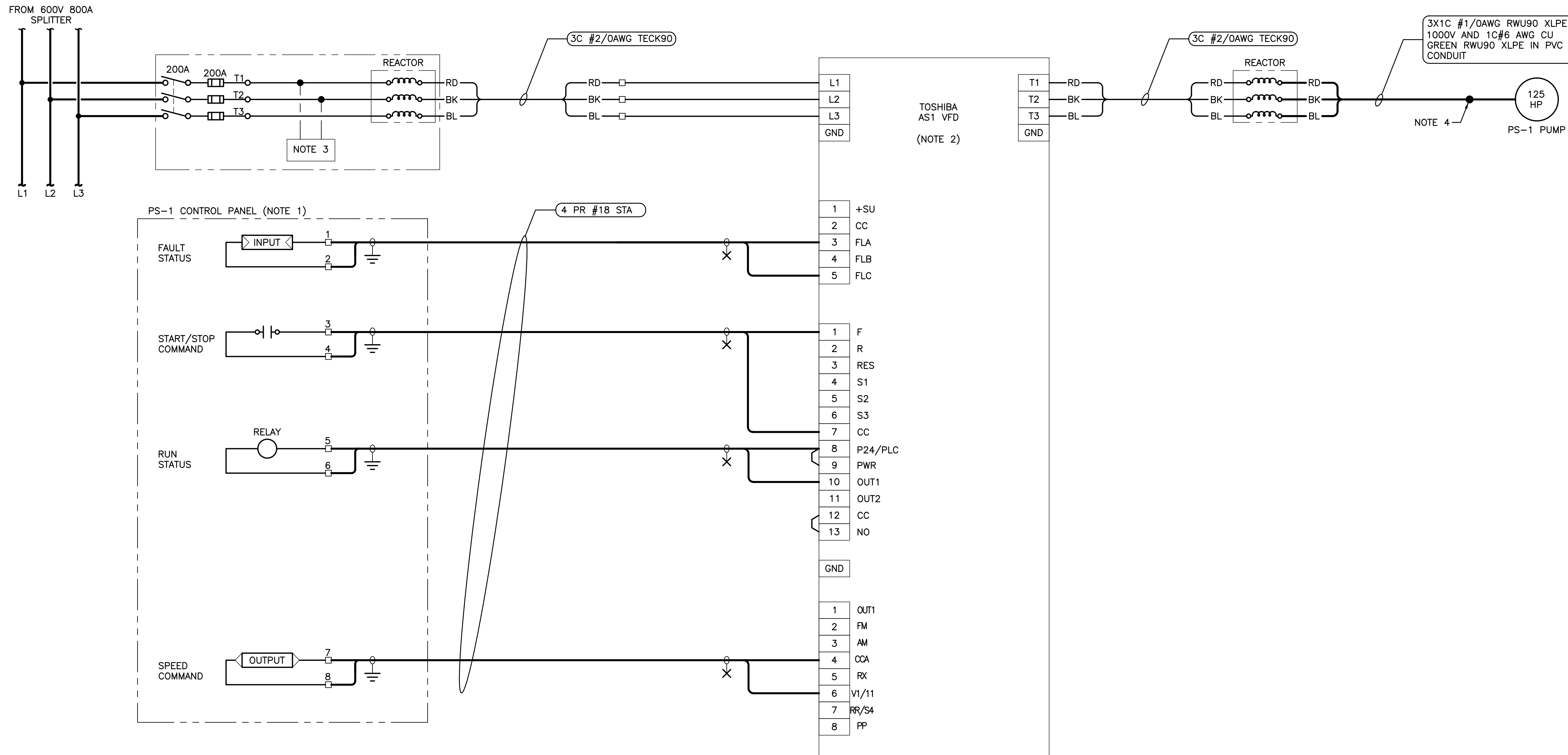
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NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH			
PROJECT:			
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL PS-1 CONTROL PANEL WIRING DIAGRAM			
KGS GROUP	DESIGN BY:	CLS	DATE: 11/11/07
	DESIGN CHECK:	JAB	DATE: 12/10/17
	DRAWN BY:	JPL	DATE: 11/11/07
	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO.			REV
21-1357-001 E713			0

ORIGINAL PROJECT
10-1357-02

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ORIGINAL DRAWING
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J.A. BOUCHARD
2012/10/17

File Name: R:\Projects\2021\21-1357-001\Dwg\Elec\21-1357-001_E714.dwg - Tab: E714 Plotted By: DShabataev 23/10/26 [Thu 8:17am]
24x36/PLOT SCALE: 1:1



NOTES:

1. WIRING WITHIN PS-1 CONTROL PANEL IS TYPICAL FOR ALL PS-1 PUMPS. REFER TO VFD TERMINATION LEGEND (DETAIL 2) FOR TERMINAL BLOCK NUMBERS CORRESPONDING TO VFD NUMBER. REFER TO DRAWING E713 FOR DETAILED INFORMATION ON DUPLINE CHANNELS, RELAYS AND START/STOP COMMAND CONTACTS.
2. VFD AND LINE & LOAD REACTORS TO BE MOUNTED WITHIN A TYPE 12 OR IP52 RATED ENCLOSURE.
3. PROVIDE ENCLOSURE ENVIRONMENTAL CONTROLS TO MAINTAIN ENCLOSURE TEMPERATURE BETWEEN 0°C AND 35°C.
4. PROVIDE RAYCHEM GELCAP STUB SPLICE COVER OVER BOLTED SPLICE CONNECTION TO MOTOR CONDUCTORS. PERMANENT COMPRESSION CONNECTION TO MOTOR CONDUCTORS INSULATED WITH ELECTRICAL TAPE IS NOT ACCEPTABLE.

LEGEND:

	DUPLINE INPUT
	DUPLINE OUTPUT
	GROUNDIED SHIELD
	ISOLATED SHIELD
	SHIELDED/TWISTED/ARMoured
	FIELD WIRING
	PANEL WIRING

1 TYPICAL VFD WIRING DIAGRAM
SCALE:

TYPICAL WIRING DIAGRAM TERMINAL NUMBER	VFD #1	VFD #2	VFD #14	VFD #15
1	20	22	25	27
2	21	23	26	28
3	50	52	54	56
4	51	53	55	57
5	39	41	43	45
6	40	42	44	46
7	11	13	15	17
8	12	14	16	18
RUN STATUS - RELAY	R1	R2	R3	R4
START/STOP COMMAND RELAY	R5	R6	R7	R8

2 VFD TERMINATION LEGEND
SCALE:

AS-BUILT

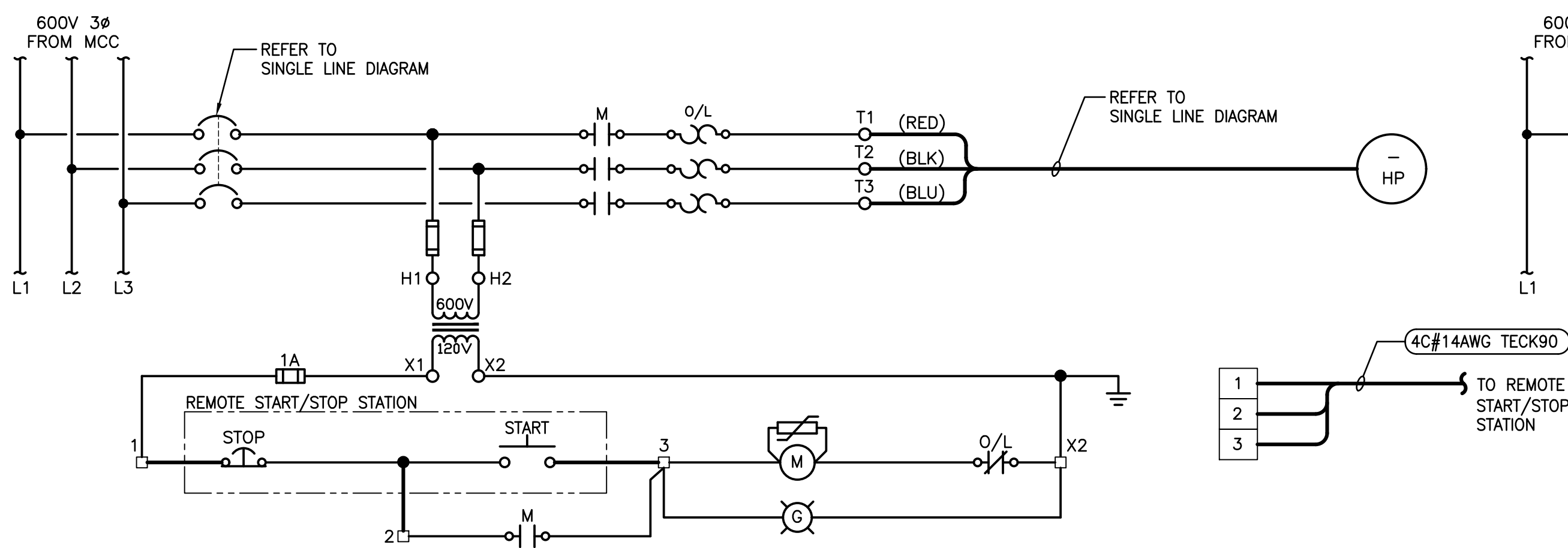
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NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
DWG. DESCRIPTION:			
ELECTRICAL VFD WIRING DIAGRAM			
	DESIGN BY:	CLS	DATE: 11/11/07
	DESIGN CHECK:	JAB	DATE: 12/10/17
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	DWG CHECK:	JAB	DATE: 12/10/17
DWG. NO.	21-1357-001 E714		REV: 0

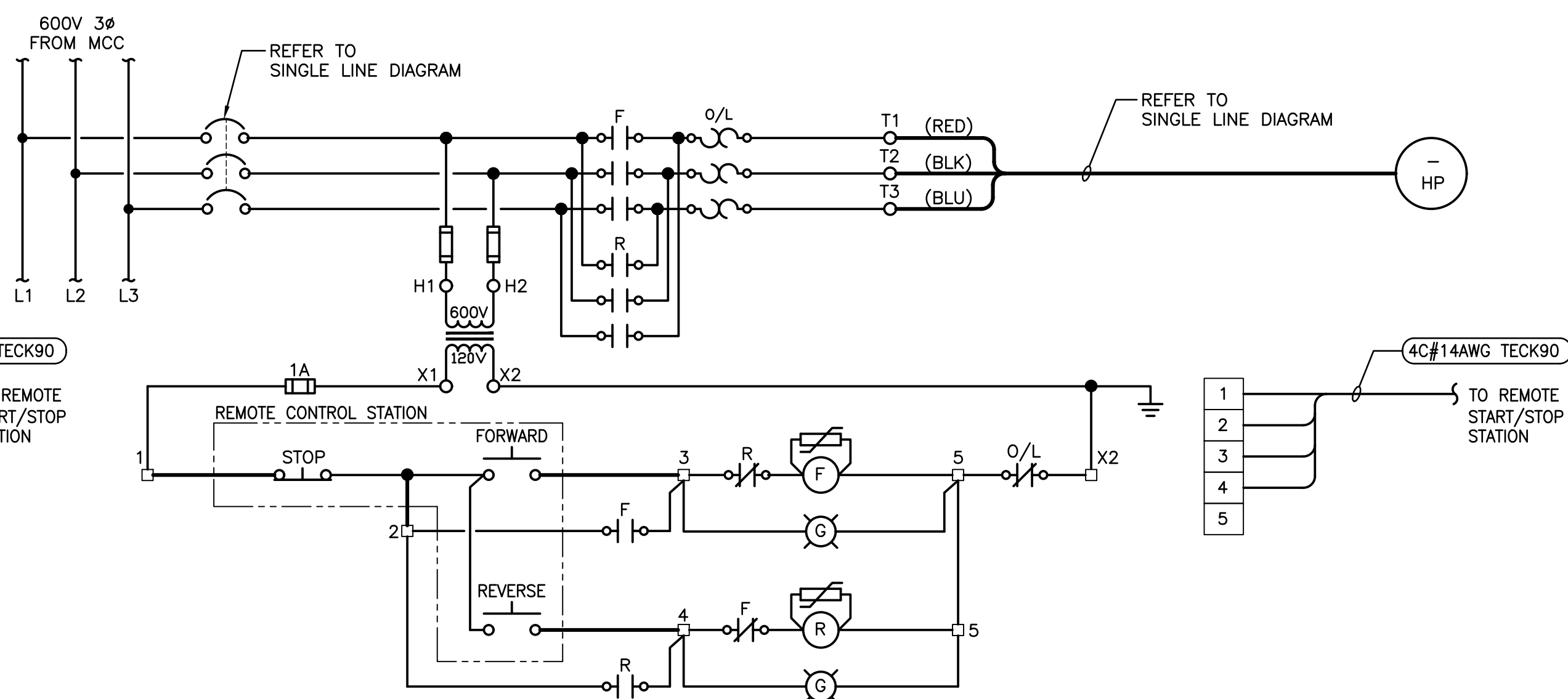
ORIGINAL PROJECT
10-1357-02

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ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

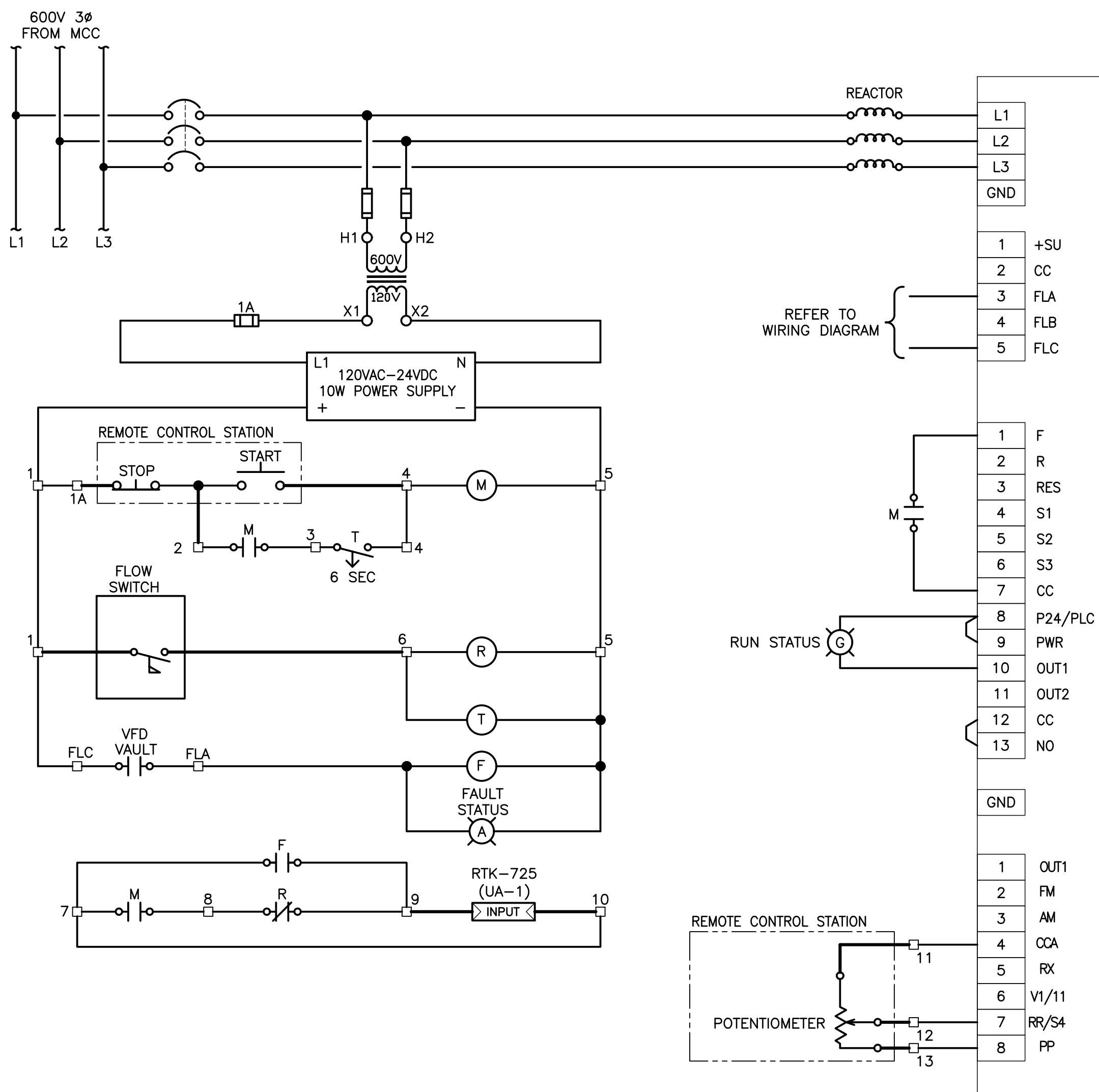
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24x36 PLOT SCALE: 1:1



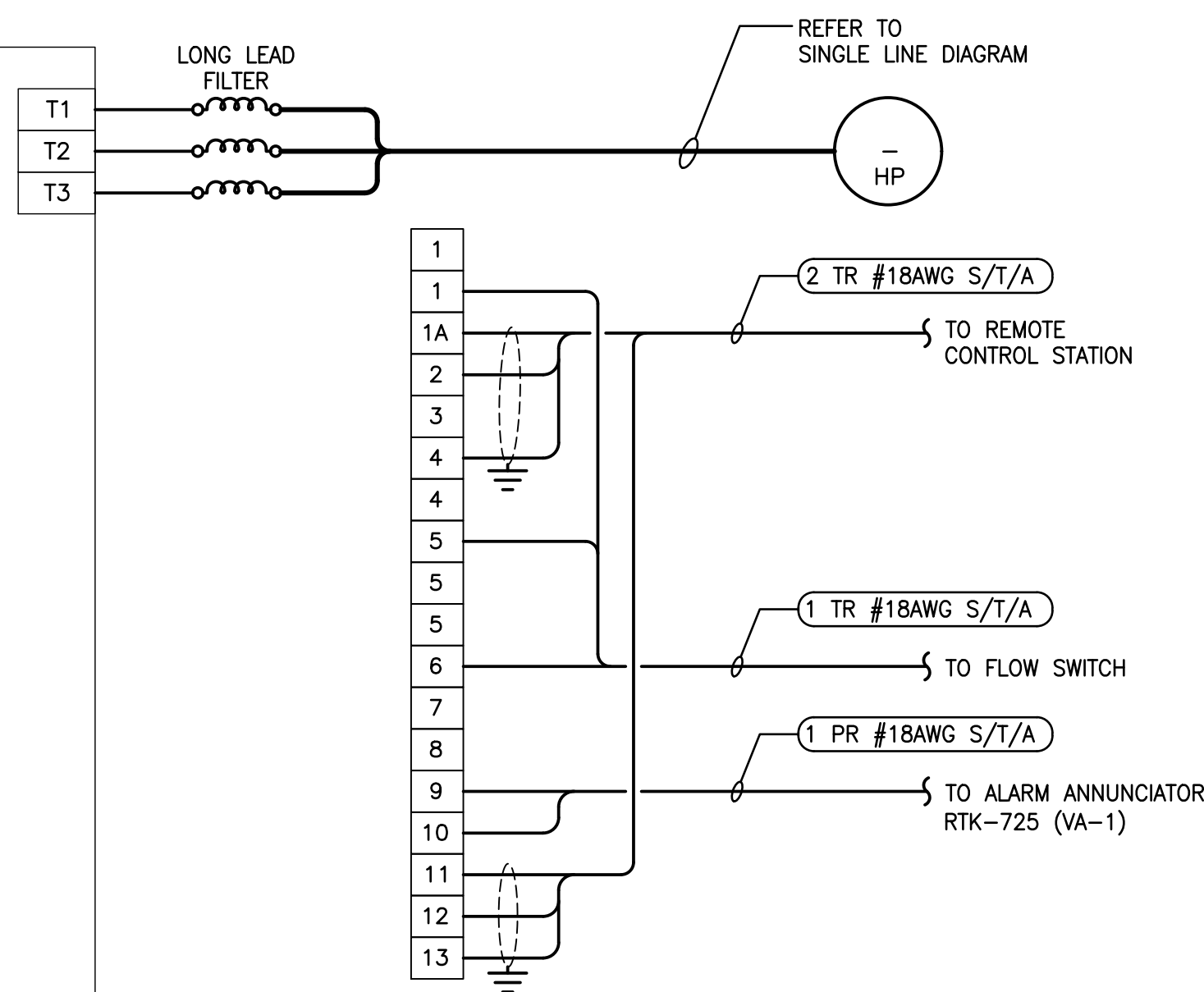
1 GR-1,2 FULL VOLTAGE STARTER WIRING SCHEMATIC (NOTE1)
SCALE: N.T.S



2 TYPICAL FULL VOLTAGE REVERSING STARTER WIRING SCHEMATIC (NOTE 2)
SCALE: N.T.S



3 PU-6,8,12 VFD WIRING DIAGRAM (NOTE 3)
SCALE: N.T.S



NOTES:

- GRIT REMOVER STARTERS ARE EITHER SPARE OR HAVE BEEN REPURPOSED.
- PROGRESSIVE CAVITY PUMP FIBERGLASS CONTROL STATION C/W GREEN FORWARD, GREEN REVERSE AND RED STOP 22.5mm PLASTIC PUSHBUTTONS.
ENCLOSURE: CROUSE HINDS - FFB23
FORWARD PUSHBUTTON: AB - 800FP-F309
REVERSE PUSHBUTTON: AB - 800FP-F310
STOP PUSHBUTTON: AB - 800FP-F402
- REFER TO DWG E702 FOR PUMP STARTERS STILL IN SERVICE.
PLASTIC BUTTONS.
ENCLOSURE: CROUSE HINDS - FFB22
START PUSHBUTTON: AB - 800FP-F301
STOP PUSHBUTTON: AB - 800FP-F402
POTENTIOMETER: AB - 800FP-POT6

LEGEND:

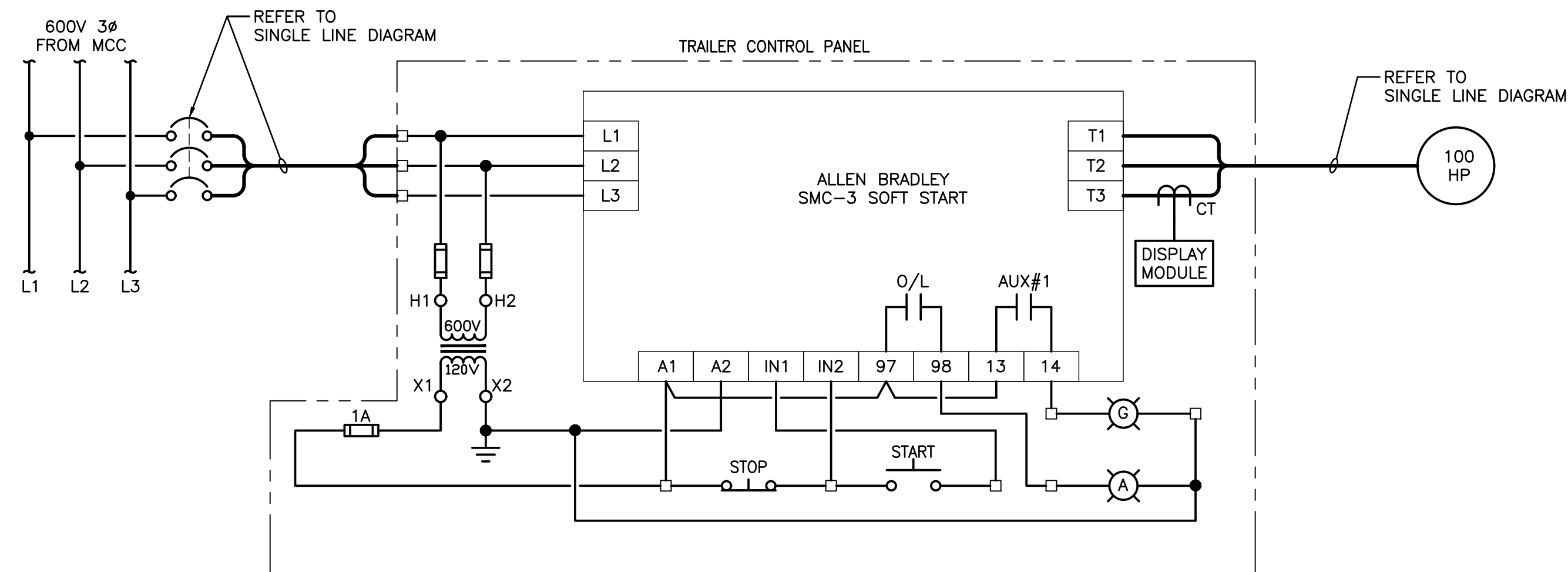
- FIELD WIRING
- PANEL WIRING
- GROUNDING SHIELD
- ISOLATED SHIELD
- S/T/A SHIELDED/ISOLATED/ARMoured

AS-BUILT
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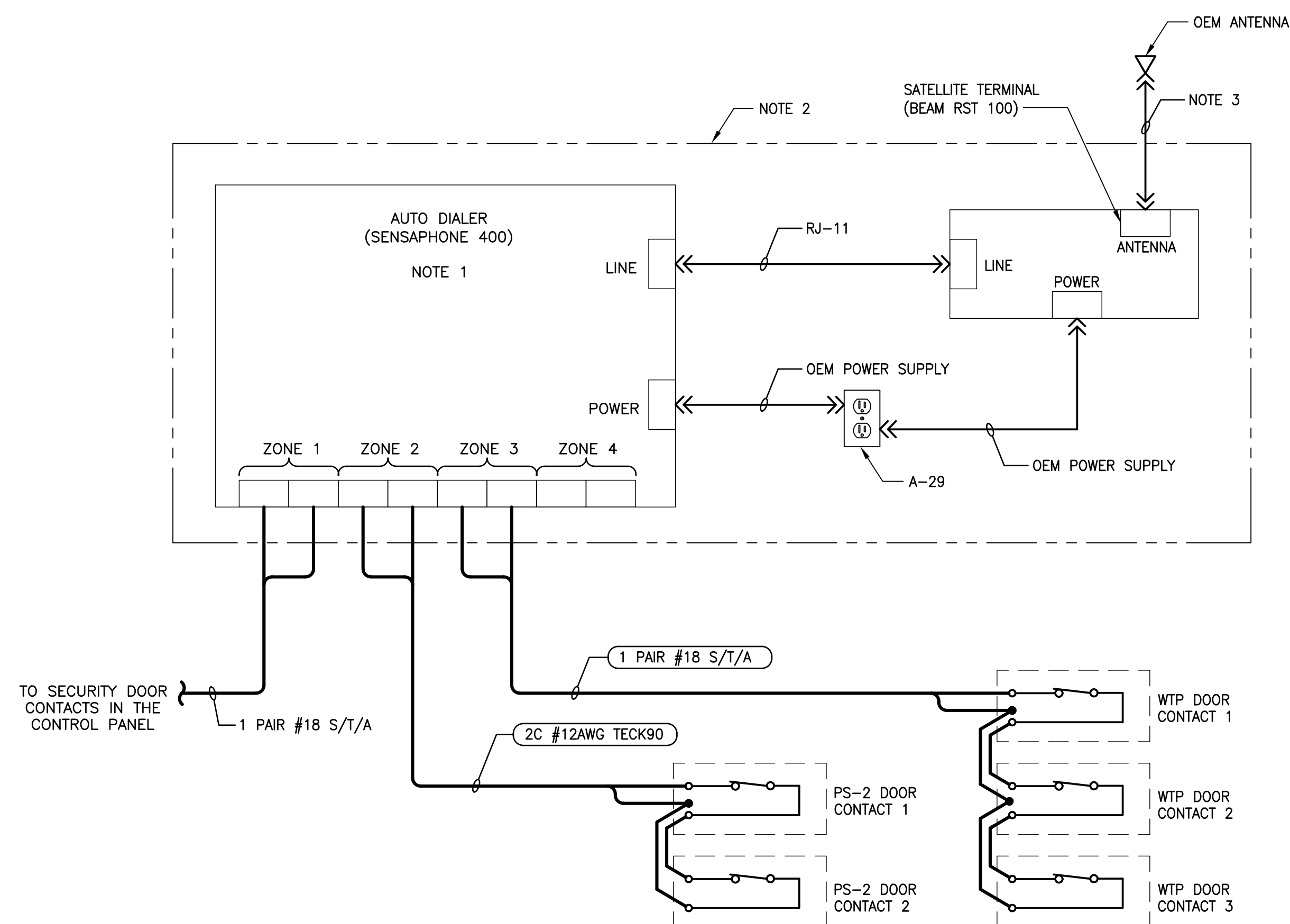
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NO.	YY/MM/DD	DESCRIPTION	BY	
REVISIONS / ISSUE				
CLIENT:				
Manitoba MANITOBA MINES BRANCH				
PROJECT:				
RUTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
ELECTRICAL WIRING DIAGRAM				
DESIGN BY:		CLS	DATE: (YY/MM/DD):	11/11/07
DESIGN CHECK:		JAB	DATE:	12/10/17
DRAWN BY:		JPL	DATE:	11/11/07
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DWG. NO.				REV
21-1357-001 E715				0

ORIGINAL PROJECT
10-1357-02

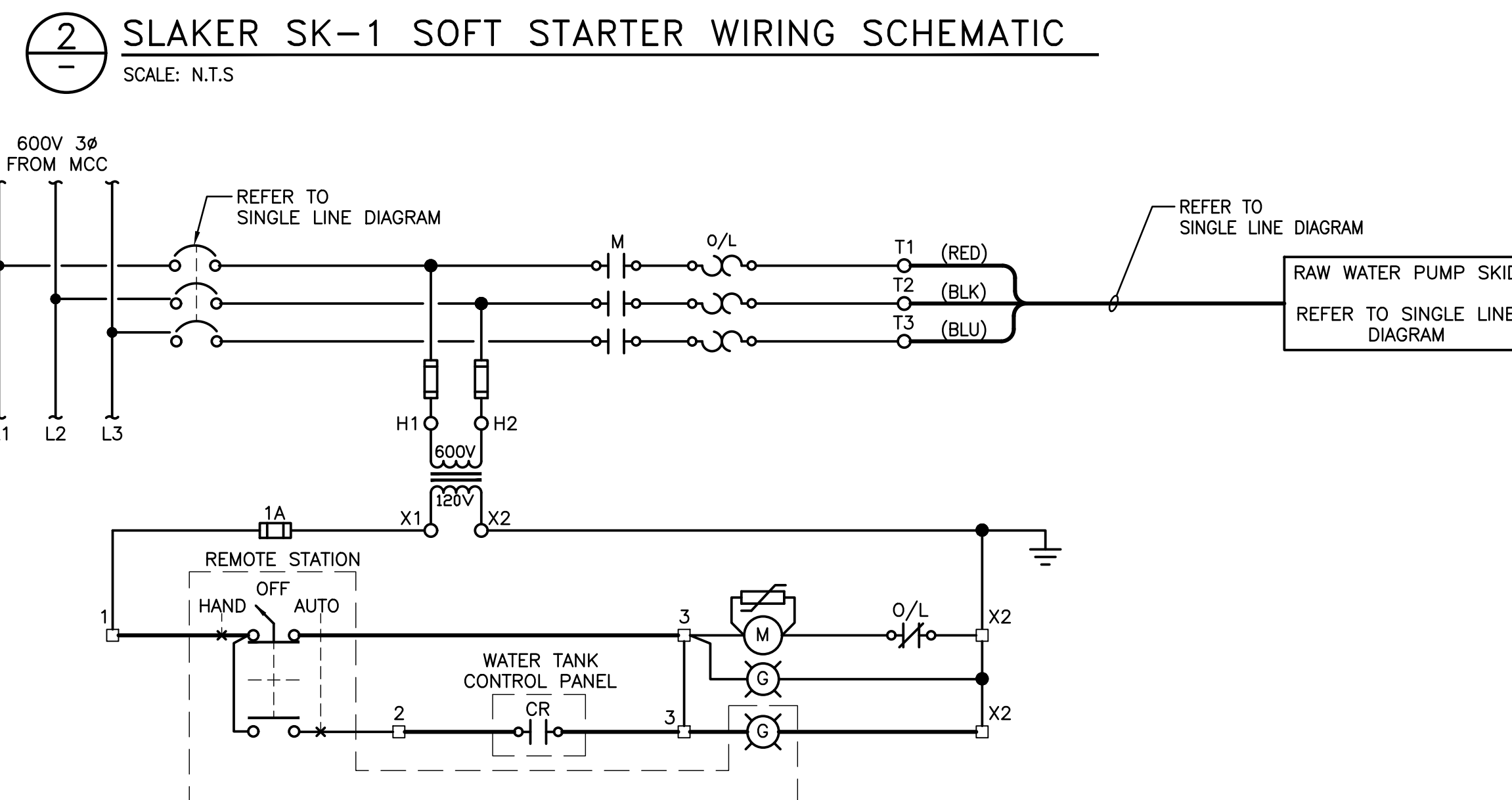
ENG. STAMP
ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17



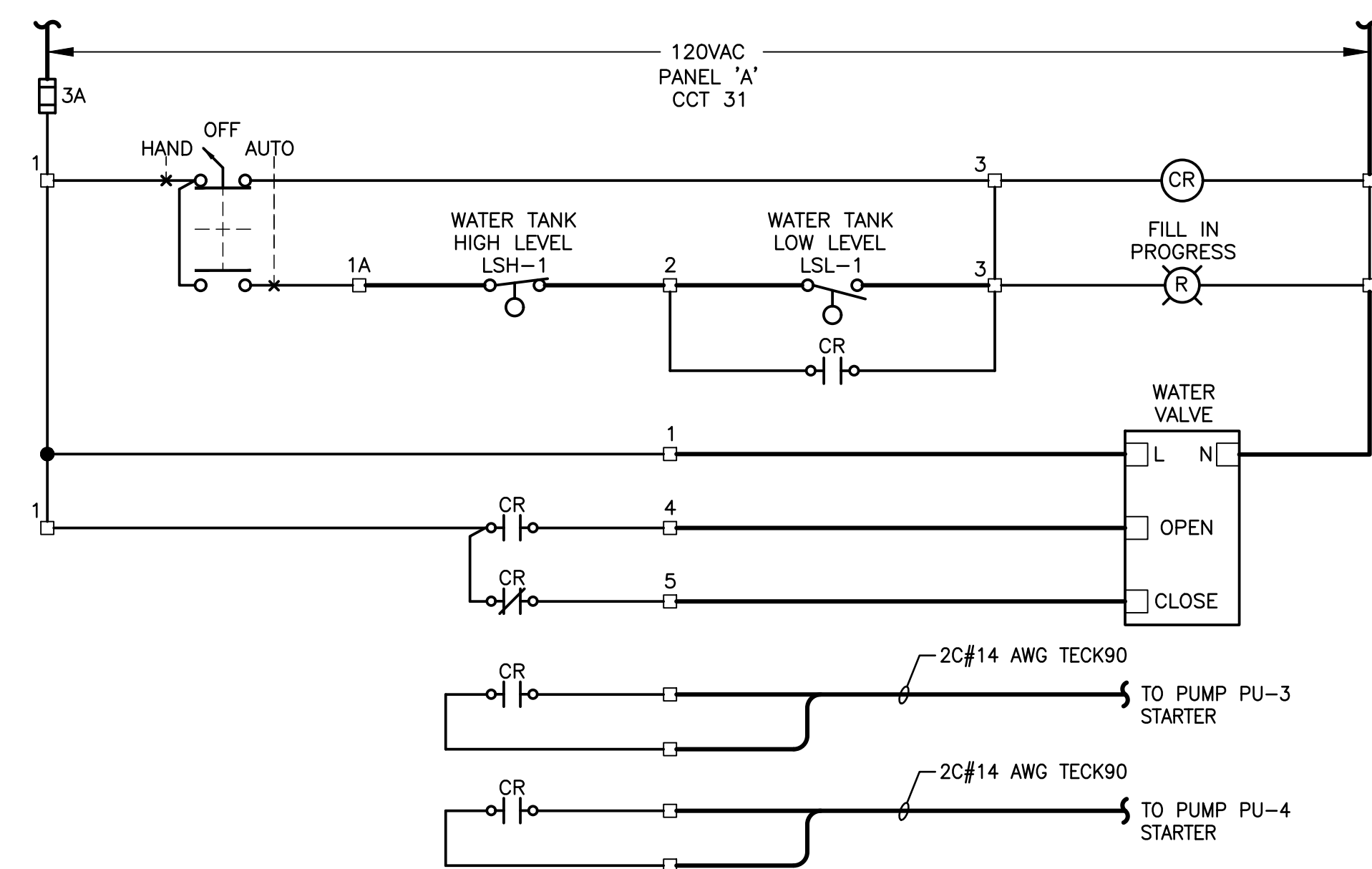
- # 1 TYPICAL FULL VOLTAGE STARTER WIRING SCHEMATIC SCALE: N.T.S



3 AUTO DIALER WIRING SCHEMATIC (AD-1)



4 — RAW WATER PUMPS PU-3 & PU-4 STARTER WIRING SCHEMATICS SCALE: N.T.S



5 — WATER TANK CONTROL PANEL WIRING SCHEMATIC SCALE: N.T.S

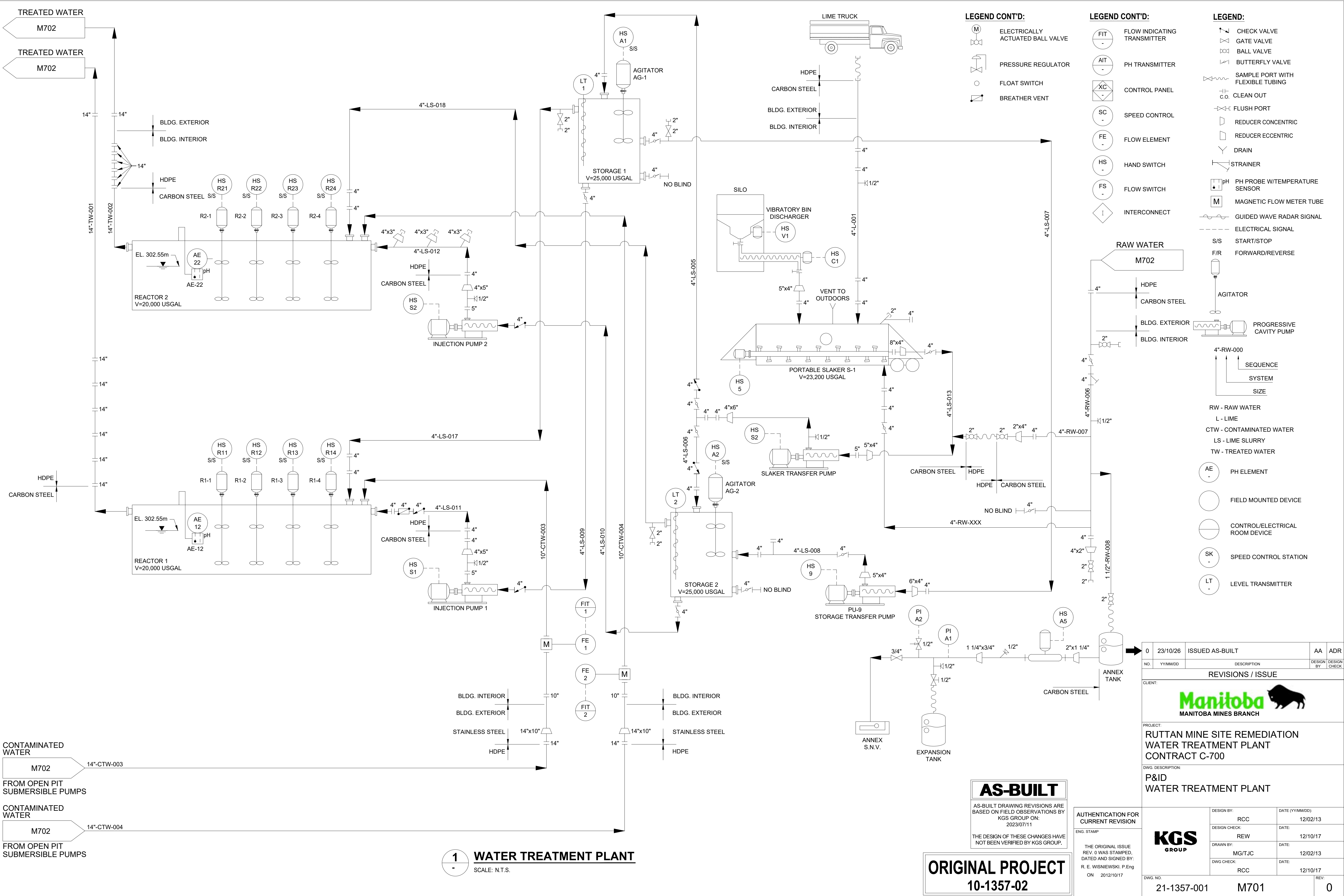
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ENG. STAMP

ORIGINAL DRAWING
SEALED BY:
J.A. BOUCHARD
2012/10/17

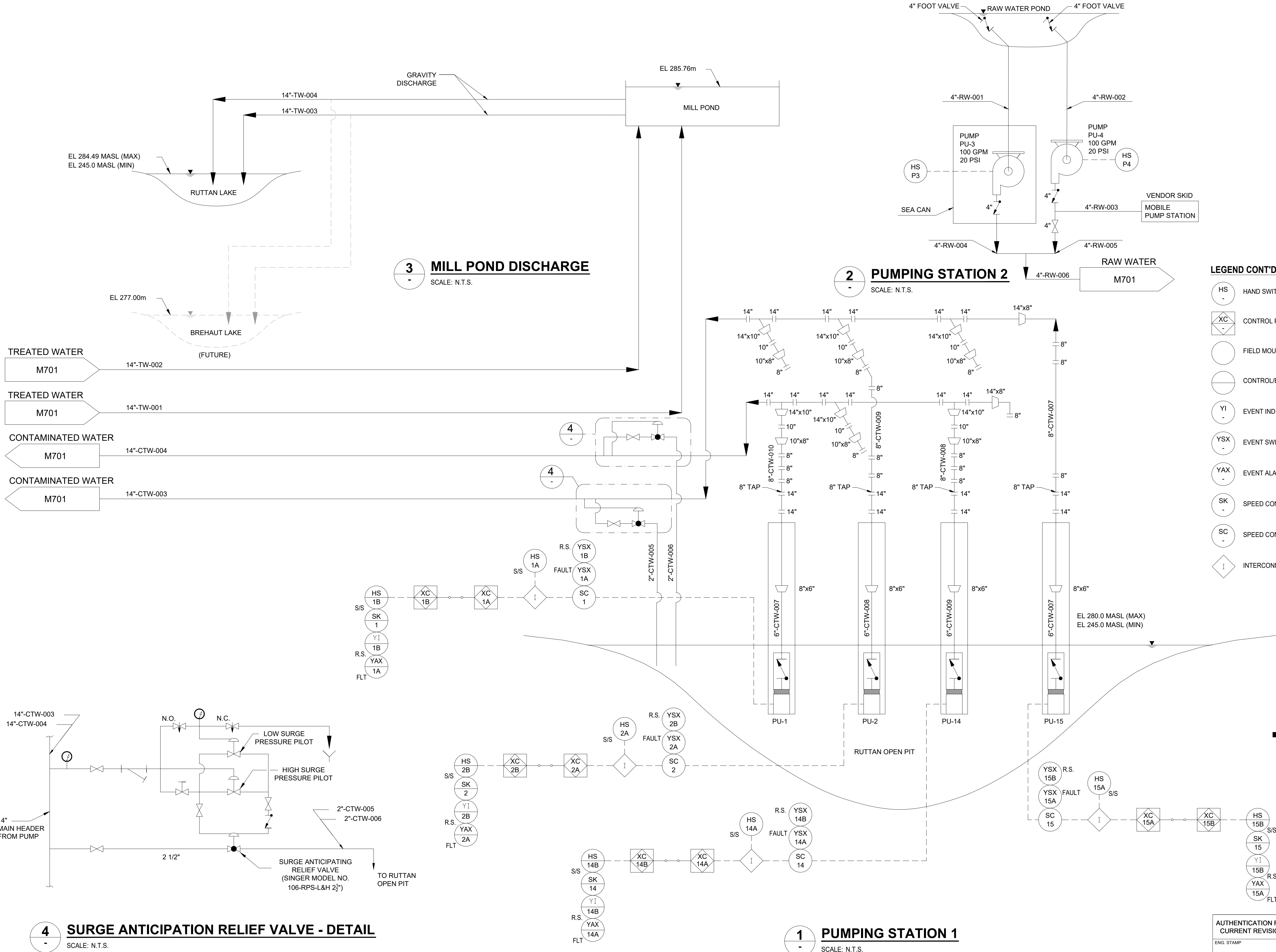
AS-BUILT AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11 THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.			
0 23/10/26		ISSUED AS-BUILT	
NO.	YY/MM/DD	DESCRIPTION	BY
REVISIONS / ISSUE			
CLIENT:			
Manitoba MANITOBA MINES BRANCH 			
PROJECT:			
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700			
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	DWG. NO.		
21-1357-001 E716			REV: 0

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24"x36" PLOT SCALE: 1:1 (METRIC)



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NO.	YYMMDD	DESCRIPTION	DESIGN BY	DESIGN CHECK
REVISIONS / ISSUE				
CLIENT:				
PROJECT:				
RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700				
DWG. DESCRIPTION:				
P&ID WATER TREATMENT PLANT				
DESIGN BY:		RCC	DATE (YYMMDD):	
DESIGN CHECK:		REW	DATE:	
DRAWN BY:		MG/TJC	DATE:	
DWG. CHECK:		RCC	DATE:	
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21-1357-001			M701	
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24"x36" PLOT SCALE: 1:1 (METRIC)



LEGEND:

- CHECK VALVE
- GATE VALVE
- NEEDLE VALVE
- CONTROL VALVE
- SURGE ANTICIPATING RELIEF GLOBE VALVE
- PRESSURE PILOT
- GLOBE VALVE
- SAMPLE PORT WITH FLEXIBLE TUBING
- CLEAN OUT
- FLUSH PORT
- PRESSURE GAUGE
- REDUCER
- DRAIN
- ELECTRICAL SIGNAL
- COMMUNICATION LINE

LEGEND CONT'D:

- HS - HAND SWITCH
- XC - CONTROL PANEL
- FIELD MOUNTED DEVICE
- CONTROL/ELECTRICAL ROOM DEVICE
- YI - EVENT INDICATOR
- YSX - EVENT SWITCH
- YAX - EVENT ALARM
- SK - SPEED CONTROL STATION
- SC - SPEED CONTROL
- I - INTERCONNECT
- CENTRIFUGAL PUMP
- SUBMERSIBLE PUMP W/INTEGRAL CHECK VALVE
- SEQUENCE
- SYSTEM
- SIZE
- RW - RAW WATER
- L - LIME
- CTW - CONTAMINATED WATER
- LS - LIME SLURRY
- TW - TREATED WATER

AS-BUILT

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THE DESIGN OF THESE CHANGES HAVE NOT BEEN VERIFIED BY KGS GROUP.

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NO.	YYMMDD	DESCRIPTION	DESIGN BY	DESIGN CHECK

REVISIONS / ISSUE

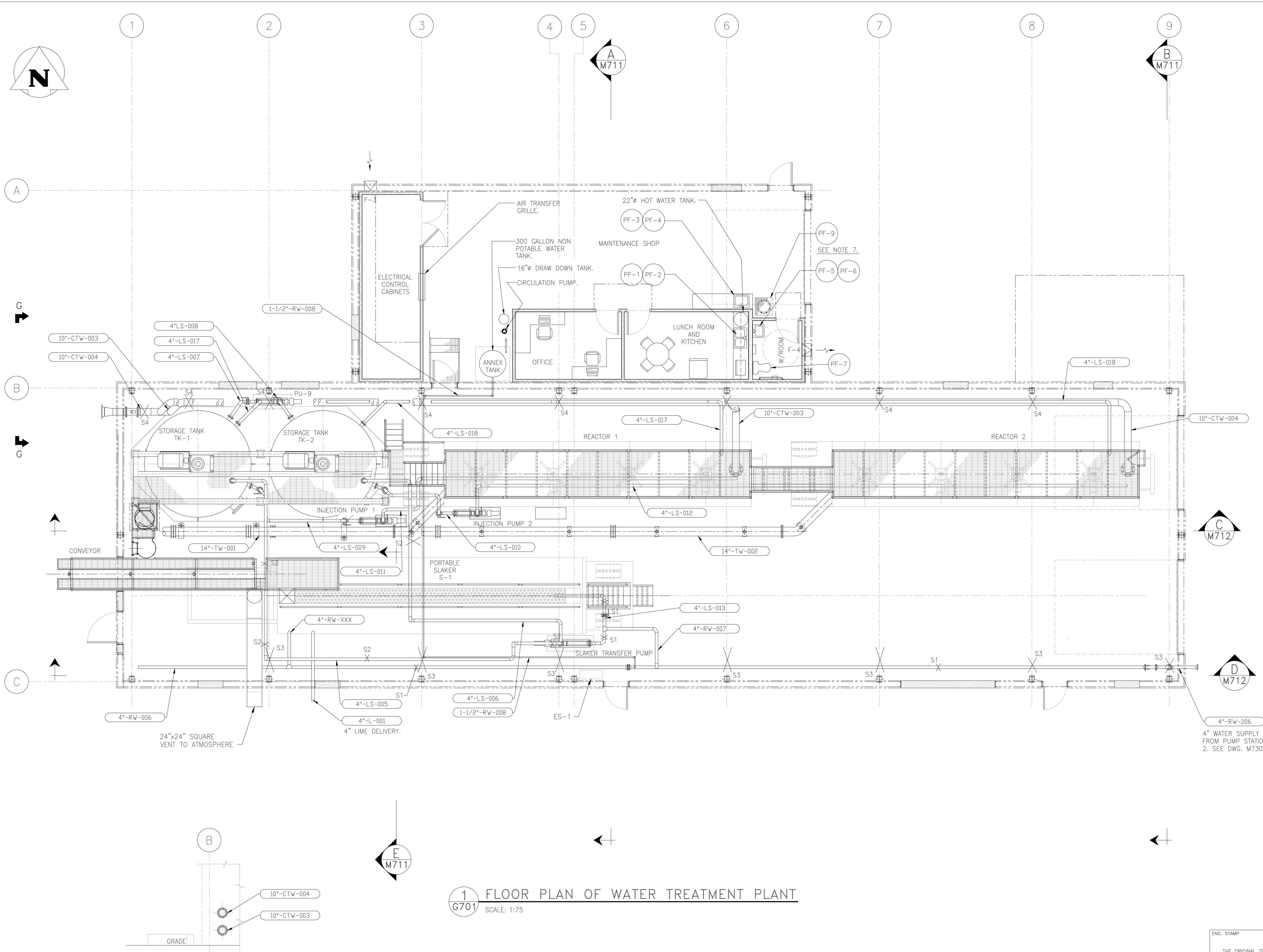


PROJECT: RUTTAN MINE SITE REMEDIATION WATER TREATMENT PLANT CONTRACT C-700

DWG. DESCRIPTION: P&ID PUMPING STATIONS

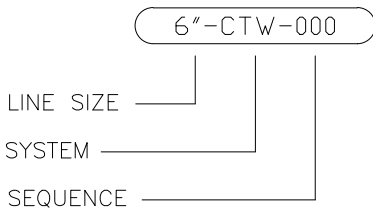
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	DESIGN CHECK:	DATE:
	DRAWN BY: MG/TJC	DATE: 12/02/13
	DWG CHECK:	DATE:
DWG. NO: 21-1357-001	M702	REV: 0

ORIGINAL PROJECT
10-1357-02



LEGEND:

PIPE LINE IDENTIFICATION:



SYSTEM	RW	RAW WATER
	L	LIME
	CTW	CONTAMINATED WATER
	LS	LIME SLURRY
	TW	TREATED WATER

PIPE SUPPORT IDENTIFICATION:

REFER TO KGS GROUP PIPE SUPPORT DESIGN STANDARDS.

S1 - SEE PS-2.05. STANCHION PIPE ANCHOR.

S2 - SEE PS-2.09. PIPE HANGER ASSEMBLY.

S3 - SEE PS-2.11. AUXILIARY STEEL SUPPORT.

S4 - SEE PS-3.02. SUPPORT ARRANGEMENT AT SHOP COLUMNS.

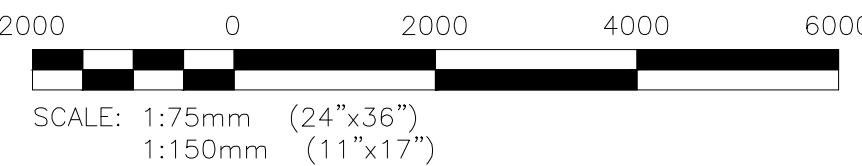
SEE ALSO PS-2.10 FOR U-BOLT GUIDE FOR PIPE 18" AND SMALLER.

PLUMBING FIXTURES:

(PF-X) DENOTES PLUMBING FIXTURES. SEE BILL OF MATERIAL DWG. M713 FOR DESCRIPTION.

NOTES:

- ALL DIMENSIONS ARE IN MILLIMETERS.
- ALL EXHAUST VENTS SHALL BE INSTALLED TO GRAVITY DRAIN BACK TO THE EQUIPMENT.
- ALL PIPE SUPPORT LOCATIONS ARE INDICATED THUS
- PIPE SUPPORTS LOCATIONS INDICATED ARE MINIMAL AND FURTHER REQUIREMENTS WILL BE AT DISCRETION OF ENGINEER IN THE FIELD.
- BILL OF MATERIAL FOR THIS DRAWING SEE: DWG. M713.
- PIPING SCHEDULE FOR THIS DRAWING SEE: DWG. M714.
- SUMP REQUIRED IN CONCRETE TO HOUSE GRINDER PUMP, MINIMUM 800 x 800 x 1500 DEEP. HOLDING TANK BY OTHERS.
- CONTRACTOR TO SUPPLY, INSTALL, TEST AND COMMISSION ALL EQUIPMENT AS PER THE DRAWINGS AND SPECIFICATIONS.
- CONTRACTOR TO SUPPLY AND INSTALL ALL SAFETY LABELS AND FLOW DIRECTION WARNINGS FOR PIPING AND EQUIPMENT.



AS-BUILT

AS-BUILT DRAWING REVISIONS ARE BASED ON FIELD OBSERVATIONS BY KGS GROUP ON: 2023/07/11
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NO.	YY/MM/DD	DESCRIPTION	BY
0	23/10/26	ISSUED AS-BUILT	AA

CLIENT:

PROJECT:
RUTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:
MECHANICAL
WATER TREATMENT PLANT
GENERAL ARRANGEMENT

	DESIGN BY:	RC/JHH	DATE (YY/MM/DD):	11/12/02
	DESIGN CHECK:	REW	DATE:	12/10/17
	DRAWN BY:	JHH	DATE:	11/12/02
	DWG. CHECK:	RCC	DATE:	12/10/17

DWG. NO.	REV.
21-1357-001 M710	0

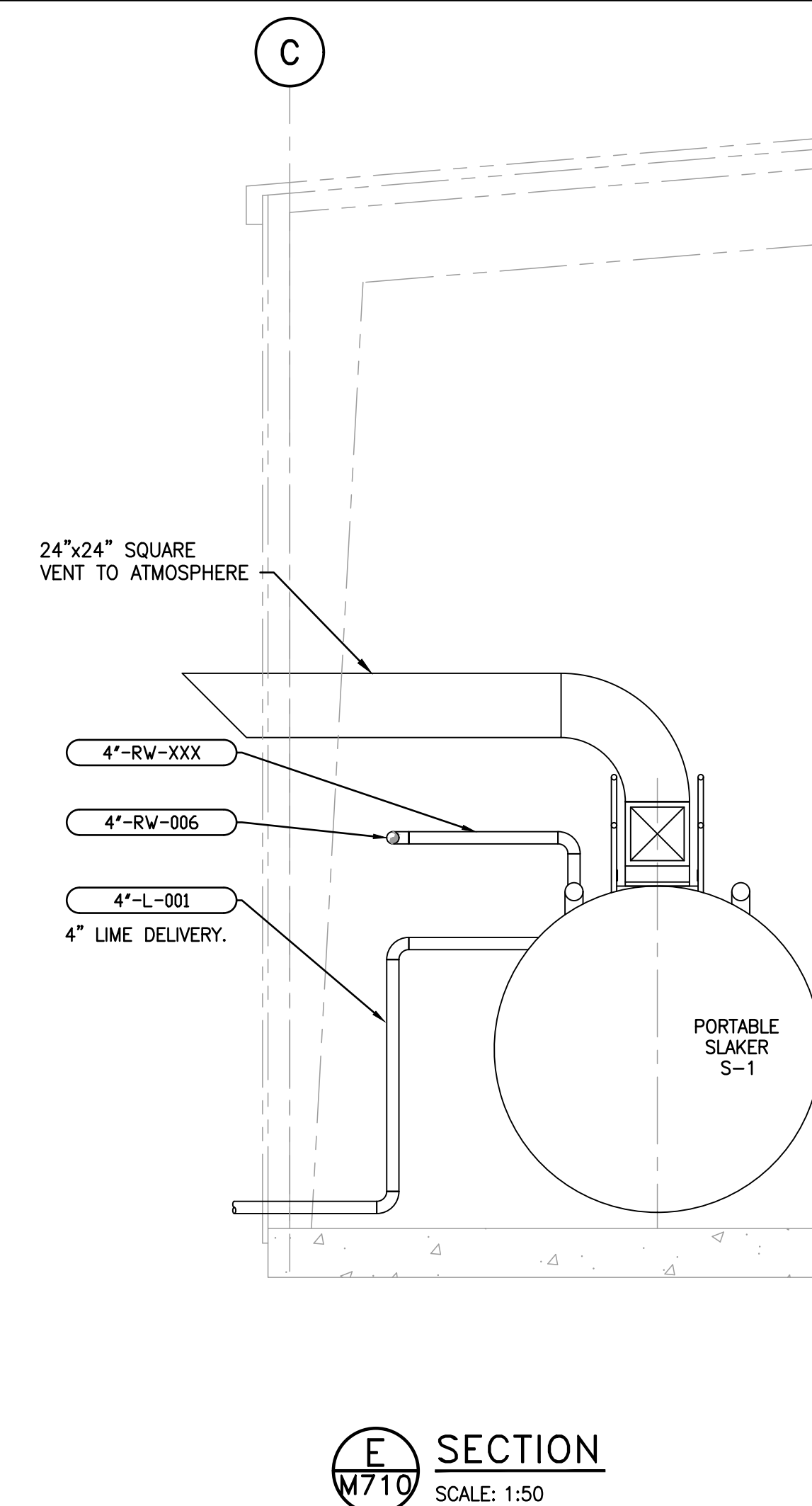
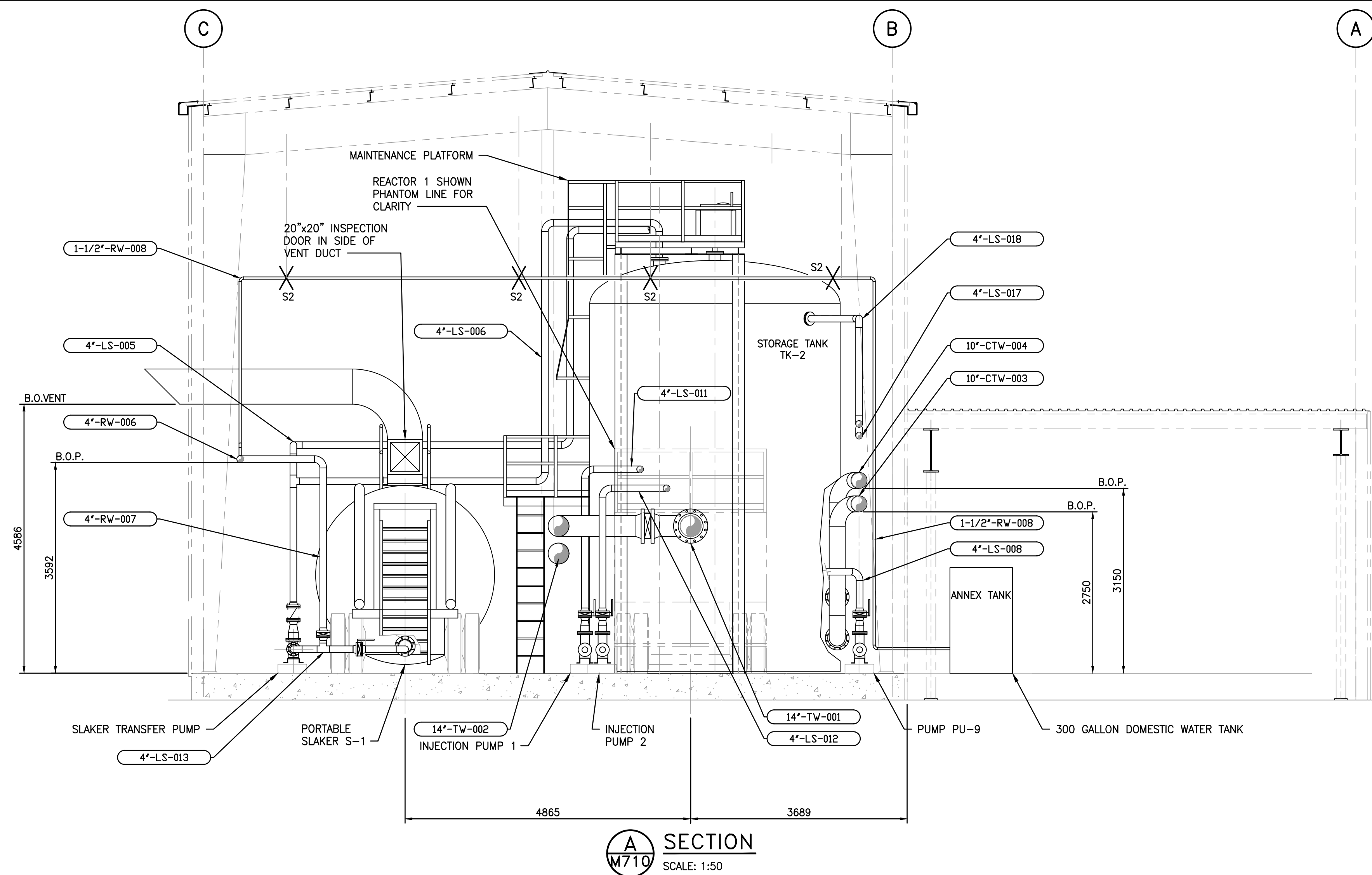
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R. E. WISNIEWSKI, P.Eng
ON 2012/10/17

ORIGINAL PROJECT
10-1357-02

1 FLOOR PLAN OF WATER TREATMENT PLANT
G701 SCALE: 1:75

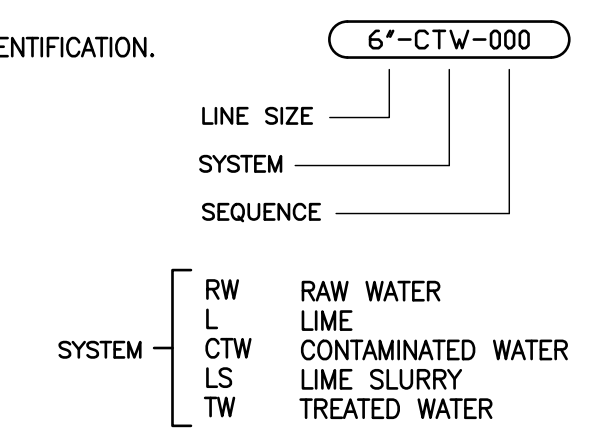
SECTION G-G
SCALE: 1:75

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24"x36" PLOT SCALE: 1:1



LEGEND:

PIPE LINE IDENTIFICATION.



PIPE SUPPORT IDENTIFICATION:

REFER TO KGS GROUP PIPE SUPPORT DESIGN STANDARDS.

S1 - SEE PS-2.05. STANCHION PIPE ANCHOR.

S2 - SEE PS-2.09. PIPE HANGER ASSEMBLY.

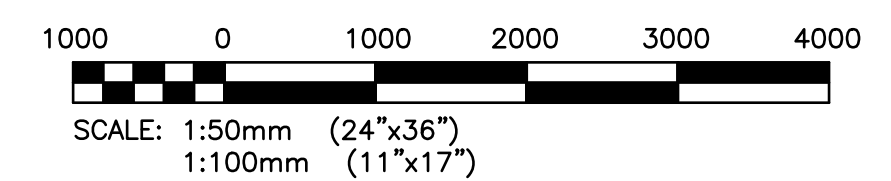
S3 - SEE PS-2.11. AUXILIARY STEEL SUPPORT.

S4 - SEE PS-3.02. SUPPORT ARRANGEMENT AT SHOP COLUMNS.

SEE ALSO PS-2.10 FOR U-BOLT GUIDE FOR PIPE 18\"/>

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. ALL EXHAUST VENTS SHALL BE INSTALLED TO GRAVITY DRAIN BACK TO THE EQUIPMENT.
3. ALL PIPE SUPPORT LOCATIONS ARE INDICATED THUS
4. PIPE SUPPORTS LOCATIONS INDICATED ARE MINIMAL AND FURTHER REQUIREMENTS WILL BE AT DISCRETION OF ENGINEER IN THE FIELD.
5. BILL OF MATERIAL FOR THIS DRAWING SEE: DWG. M713.
6. PIPING SCHEDULE FOR THIS DRAWING SEE: DWG. M714.
7. SUMP REQUIRED IN CONCRETE TO HOUSE GRINDER PUMP, MINIMUM 800 x 800 x 1500 DEEP. HOLDING TANK BY OTHERS.
8. CONTRACTOR TO SUPPLY, INSTALL, TEST AND COMMISSION ALL EQUIPMENT AS PER THE DRAWINGS AND SPECIFICATIONS.
9. CONTRACTOR TO SUPPLY AND INSTALL ALL SAFETY LABELS AND FLOW DIRECTION WARNINGS FOR PIPING AND EQUIPMENT.
10. REFER TO DRAWING M715 FOR GRIT REMOVER INLET DETAIL.



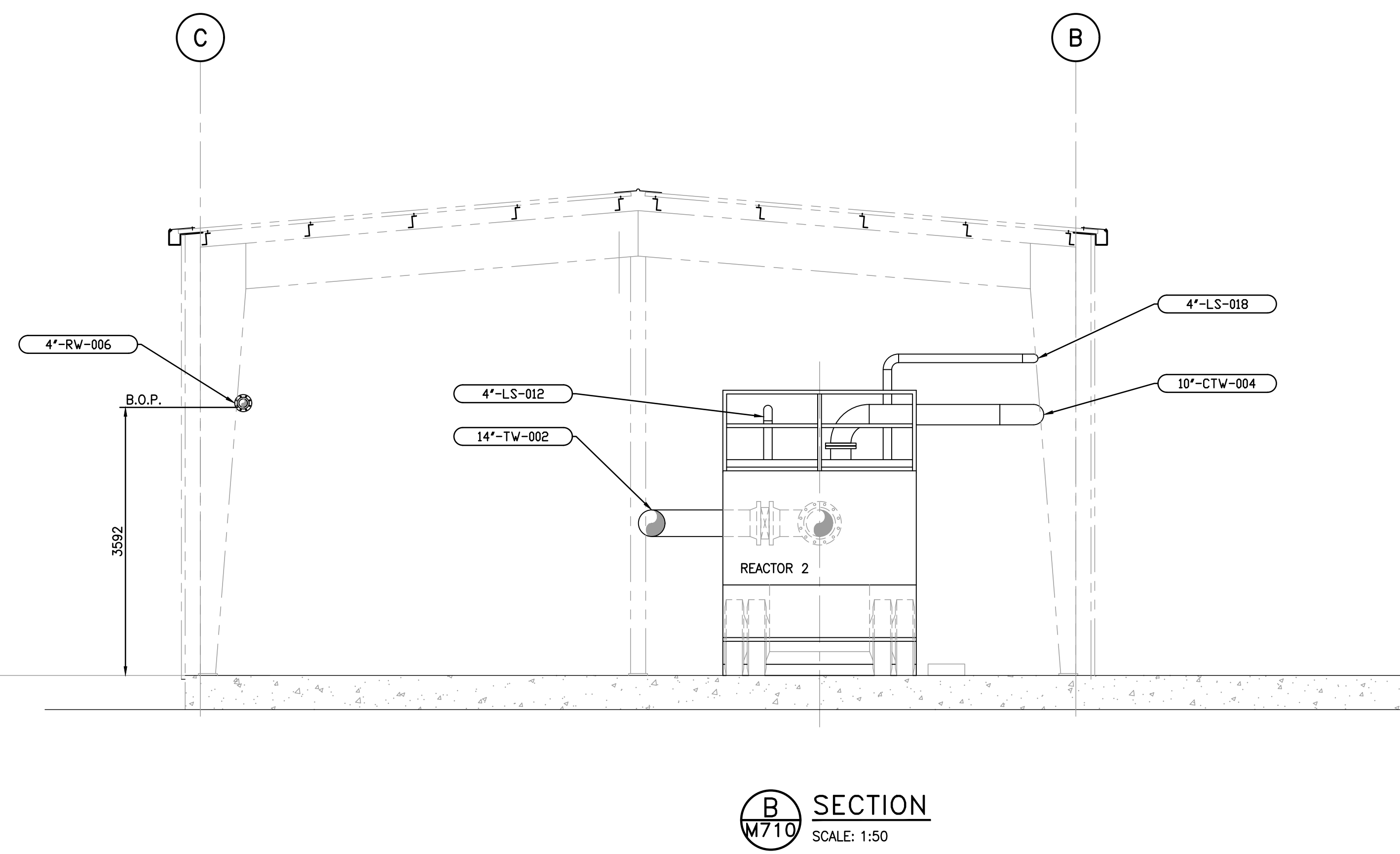
AS-BUILT

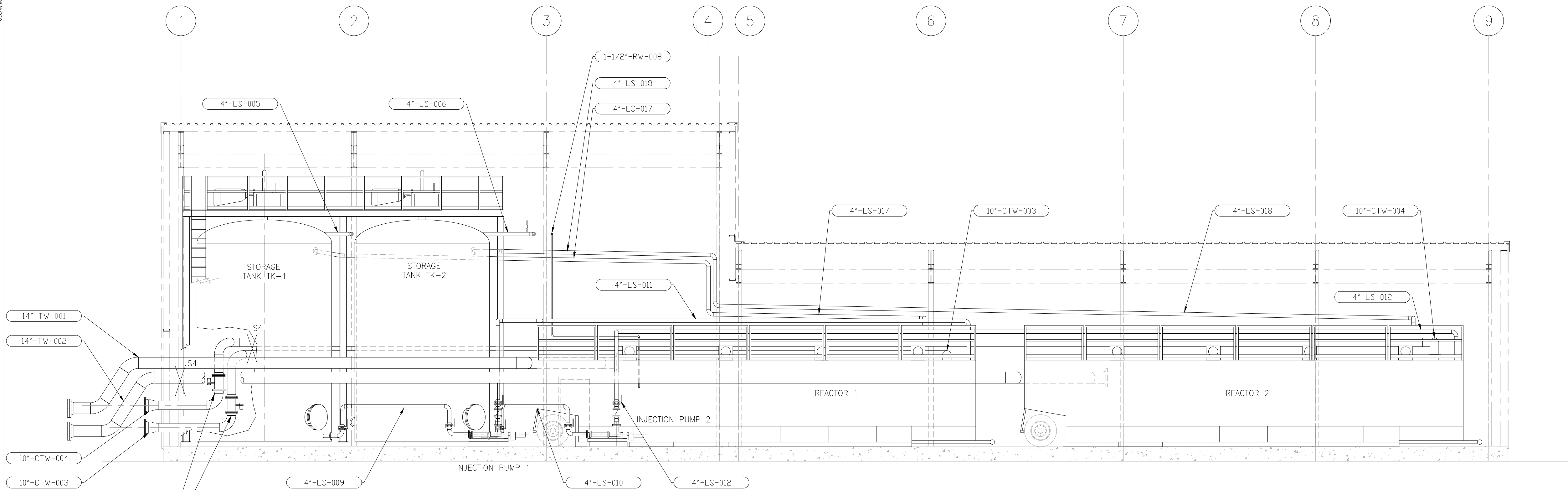
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0 23/10/26 ISSUED AS-BUILT		AA
NO.	YY/MM/DD	DESCRIPTION
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MECHANICAL WATER TREATMENT PLANT SECTIONS		
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	RC/JHH	11/12/02
	DESIGN CHECK:	DATE:
	REW	12/10/17
DRAWN BY:	JHH	DATE: 11/12/02
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RCC	DATE:	12/10/17
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21-1357-001 M711		0

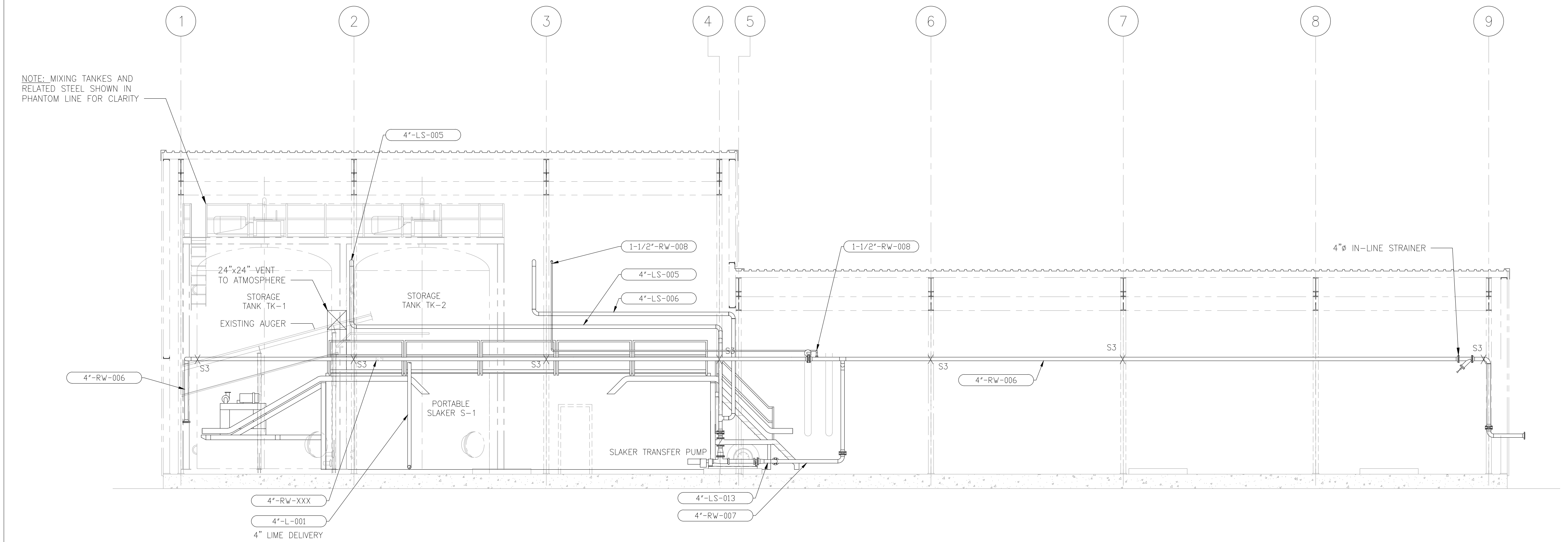
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10-1357-02

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DATED AND SIGNED BY:
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ON 2012/10/17





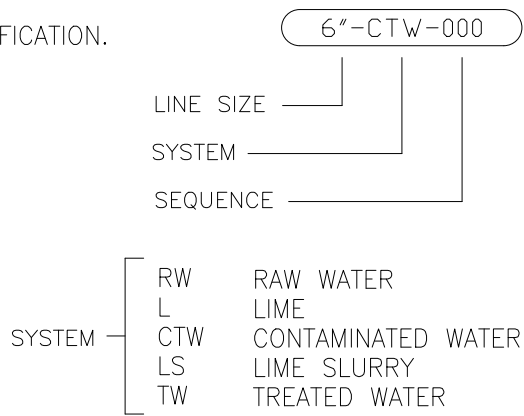
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M710 SCALE: 1:75



D SECTION
M710 SCALE: 1:75

LEGEND:

PIPE LINE IDENTIFICATION.



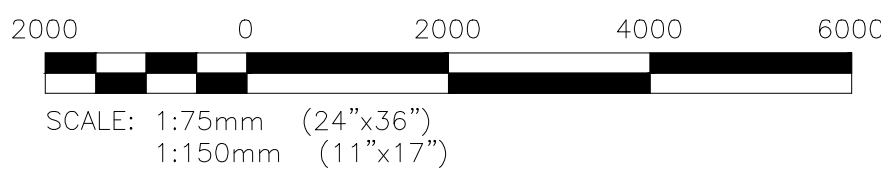
PIPE SUPPORT IDENTIFICATION:

REFER TO KGS GROUP PIPE SUPPORT DESIGN STANDARDS.

- S1 - SEE PS-2.05. STANCHION PIPE ANCHOR.
 - S2 - SEE PS-2.09. PIPE HANGER ASSEMBLY.
 - S3 - SEE PS-2.11. AUXILIARY STEEL SUPPORT.
 - S4 - SEE PS-3.02. SUPPORT ARRANGEMENT AT SHOP COLUMNS.
- SEE ALSO PS-2.10 FOR U-BOLT GUIDE FOR PIPE 18"Ø AND SMALLER.

NOTES:

- 1. ALL DIMENSIONS ARE IN MILLIMETERS.
- 2. ALL EXHAUST VENTS SHALL BE INSTALLED TO GRAVITY DRAIN BACK TO THE EQUIPMENT.
- 3. ALL PIPE SUPPORT LOCATIONS ARE INDICATED THUS
- 4. PIPE SUPPORTS LOCATIONS INDICATED ARE MINIMAL AND FURTHER REQUIREMENTS WILL BE AT DISCRETION OF ENGINEER IN THE FIELD.
- 5. BILL OF MATERIAL FOR THIS DRAWING SEE: DWG. M713.
- 6. PIPING SCHEDULE FOR THIS DRAWING SEE: DWG. M714.
- 7. SUMP REQUIRED IN CONCRETE TO HOUSE GRINDER PUMP, MINIMUM 800 x 800 x 1500 DEEP, HOLDING TANK BY OTHERS.
- 8. CONTRACTOR TO SUPPLY, INSTALL, TEST AND COMMISSION ALL EQUIPMENT AS PER THE DRAWINGS AND SPECIFICATIONS.
- 9. CONTRACTOR TO SUPPLY AND INSTALL ALL SAFETY LABELS AND FLOW DIRECTION WARNINGS FOR PIPING AND EQUIPMENT.



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REVISIONS / ISSUE



PROJECT:
RUTTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:
MECHANICAL
WATER TREATMENT PLANT
SECTIONS

DESIGN BY:	RC/JHH	DATE (YY/MM/DD):	11/12/02
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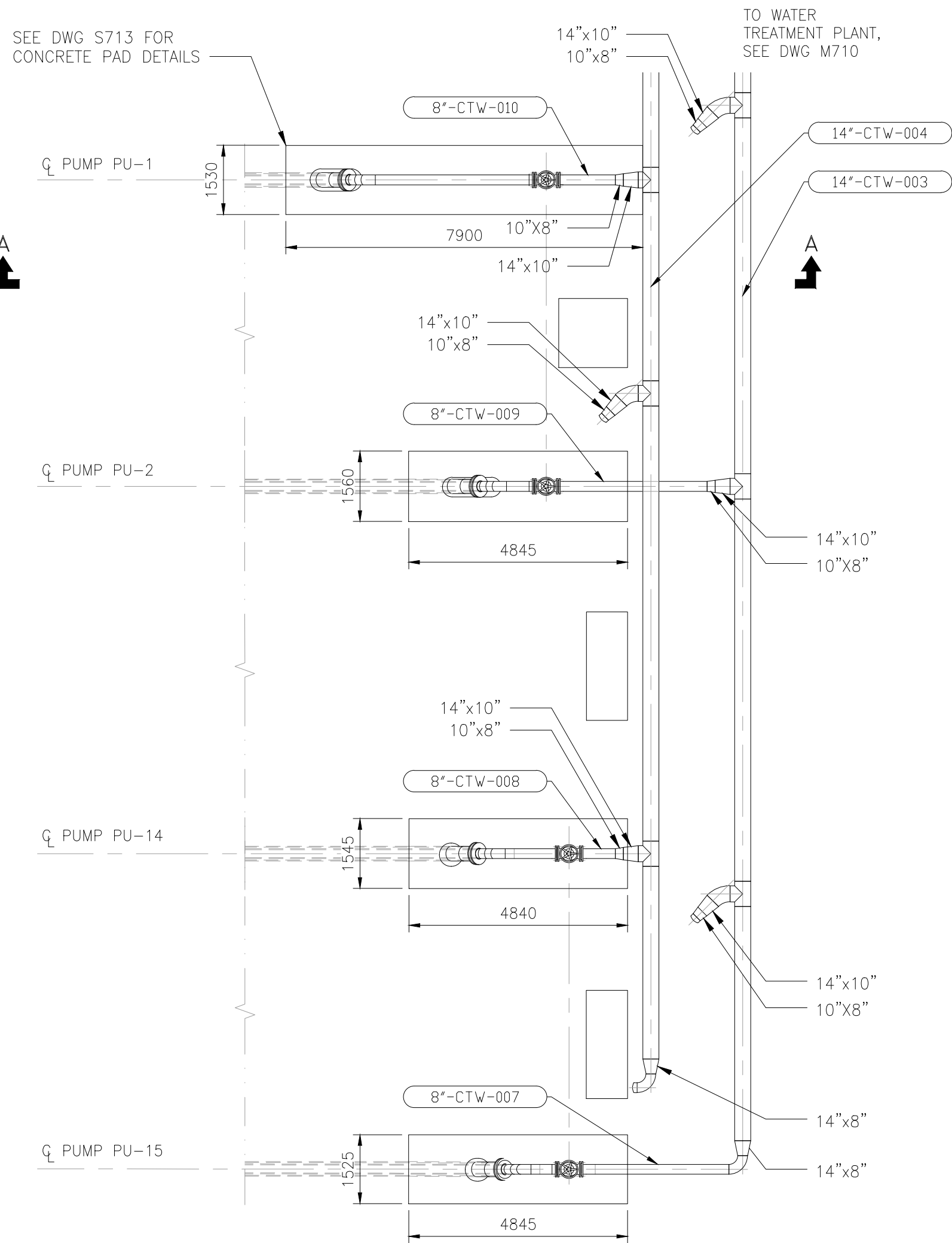
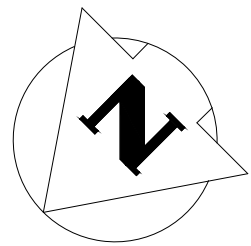
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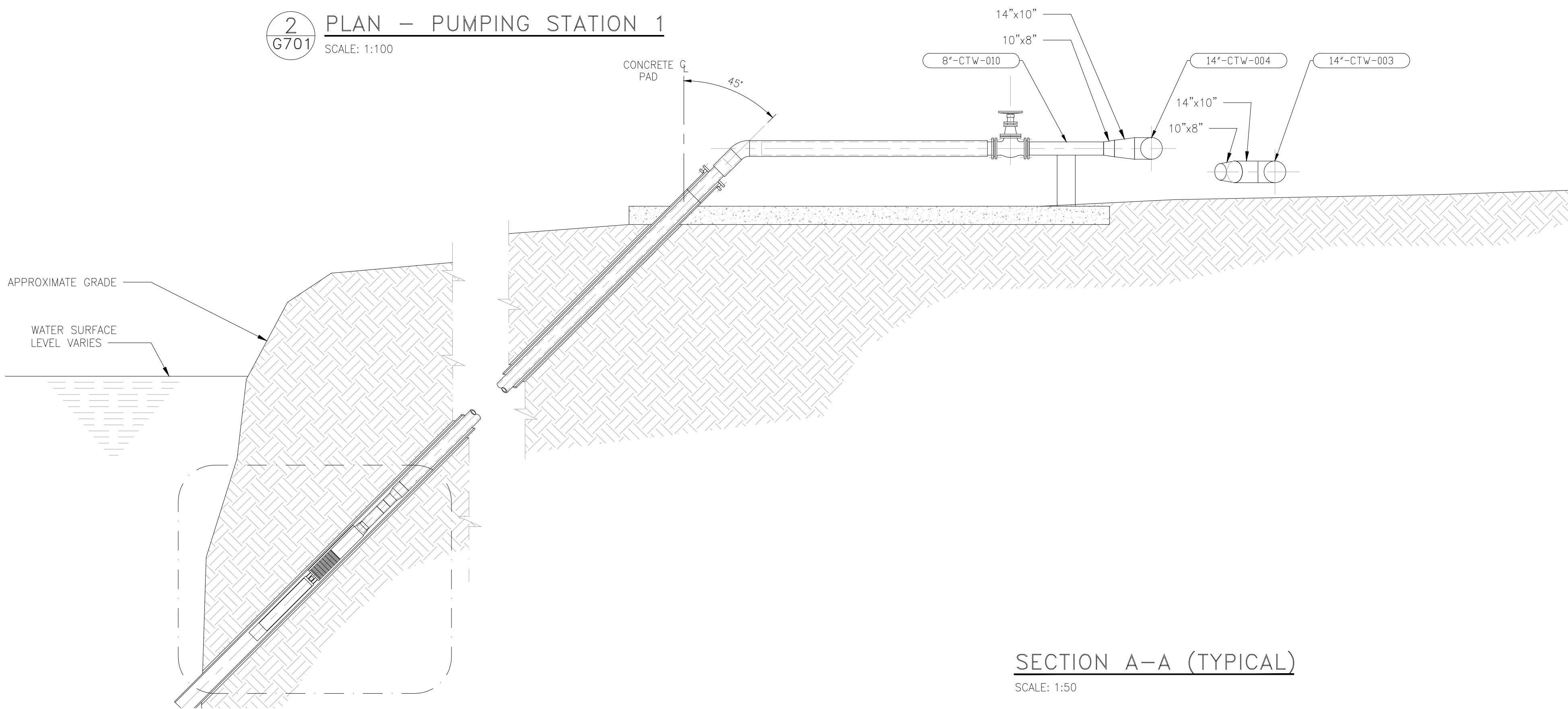
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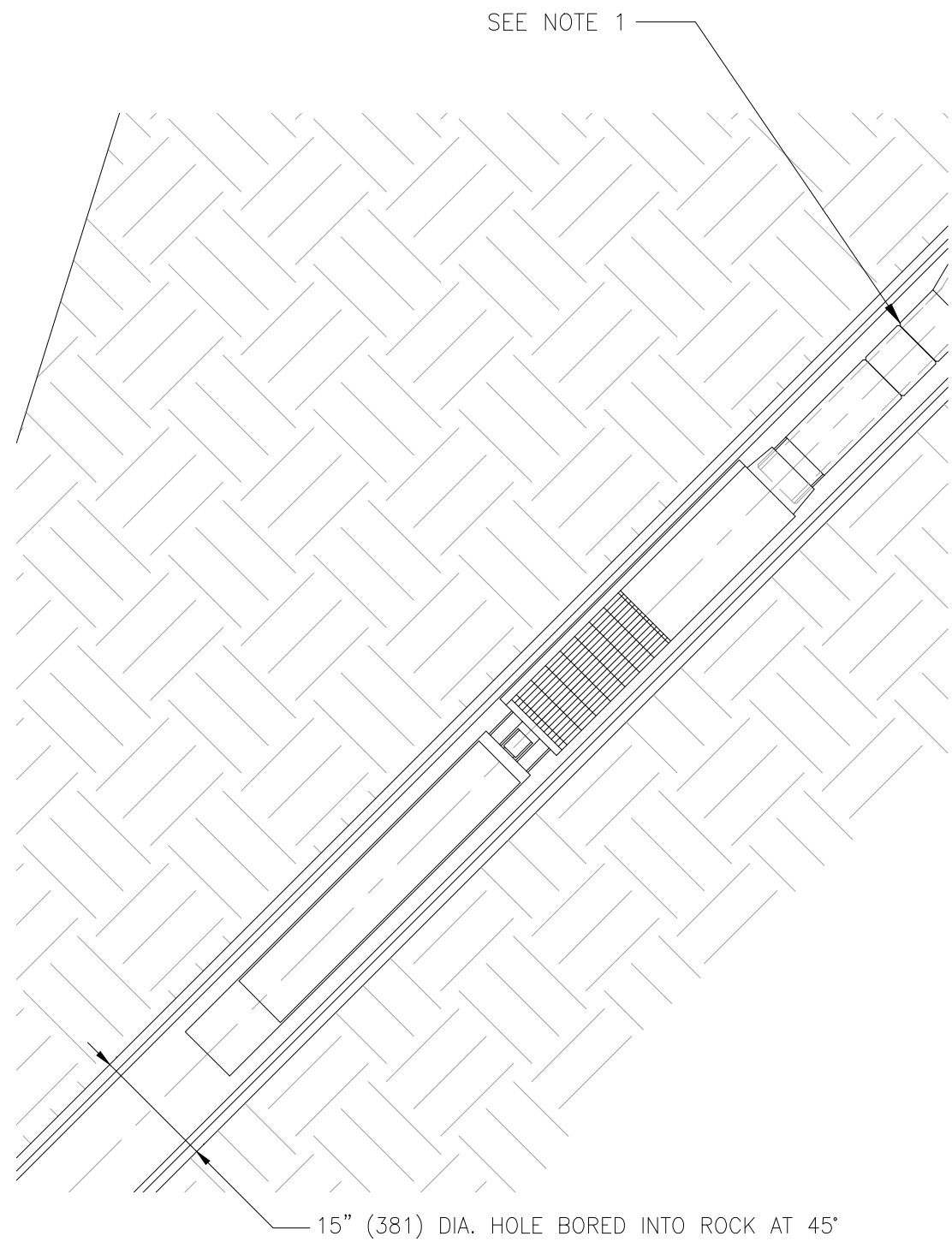
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2
G701 **PLAN — PUMPING STATION 1**
SCALE: 1:100



SECTION A-A (TYPICAL)
SCALE: 1:50



DETAIL 1
SCALE: 1:20

NOTES:

1. PROVIDE 1/4" DIA. WEEP HOLE IN TOP SIDE WALL OF 8" DIA. HDPE PIPE, APPROXIMATELY 12" ABOVE THE DISCHARGE OF EACH SUBMERSIBLE PUMP.
2. ALL DIMENSIONS TO BE CONSIDERED AS APPROXIMATE AS FINAL LOCATIONS WILL DEPEND ON THE ACCURACY OF DRILLING THE FOUR 15" DIA. BORE HOLES.
3. CONTRACTOR TO SUPPLY, INSTALL, TEST AND COMMISSION ALL EQUIPMENT AS PER THE DRAWINGS AND SPECIFICATIONS.
4. CONTRACTOR TO SUPPLY AND INSTALL ALL SAFETY LABELS AND FLOW DIRECTION WARNINGS FOR PIPING AND EQUIPMENT.



SCALE: 1:20mm (24"x36")
1:40mm (11"x17")



SCALE: 1:50mm (24"x36")
1:100mm (11"x17")



SCALE: 1:100mm (24"x36")
1:200mm (11"x17")

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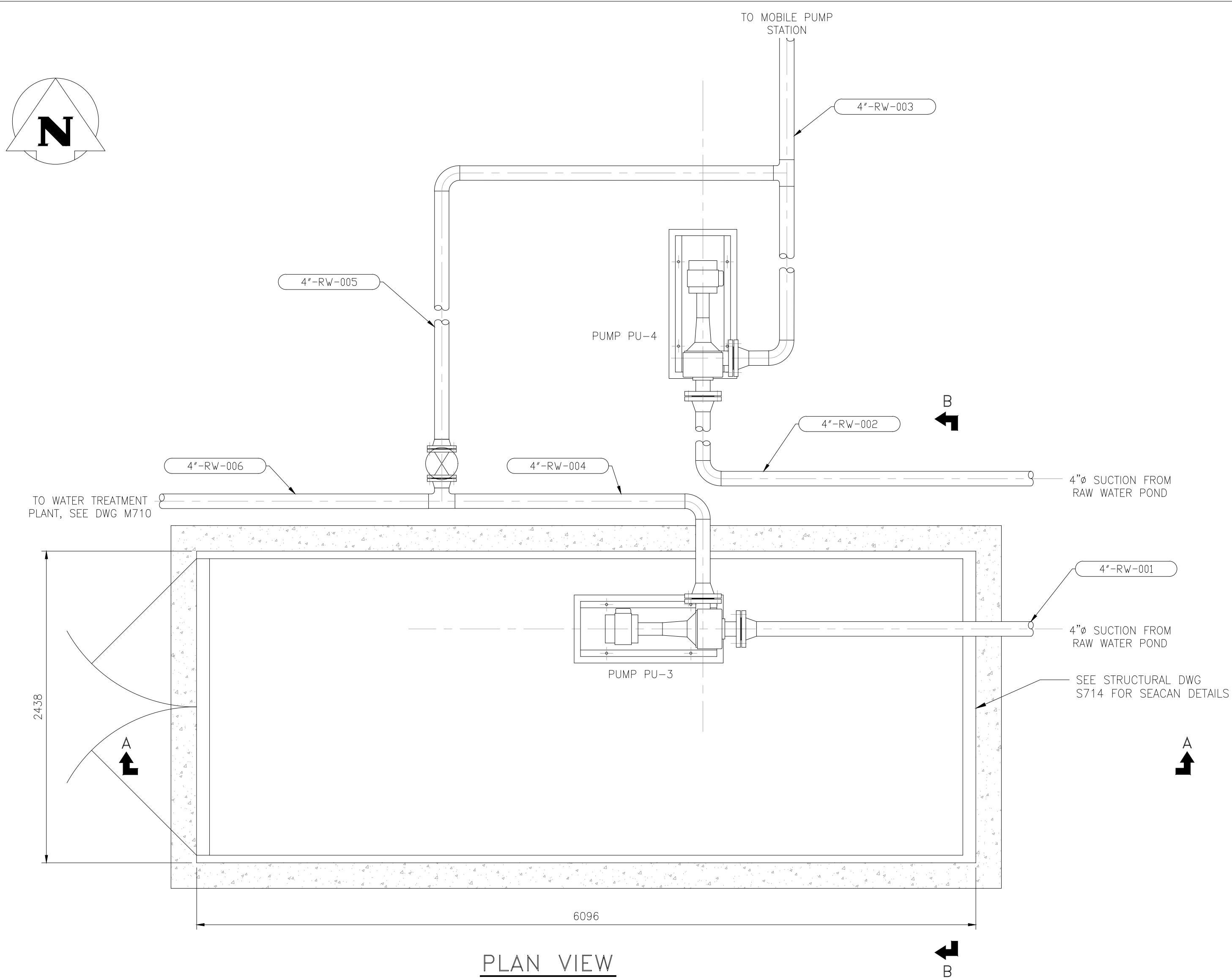
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		DESIGN BY:	RC/JHH
		DESIGN CHECK:	
		DRAWN BY:	JHH
		DWG. CHECK:	
		DATE (YY/MM/DD):	12/06/13
		DATE:	
		DATE:	12/06/13
		DATE:	
DWG. NO.			REV:
21-1357-001 M720			0

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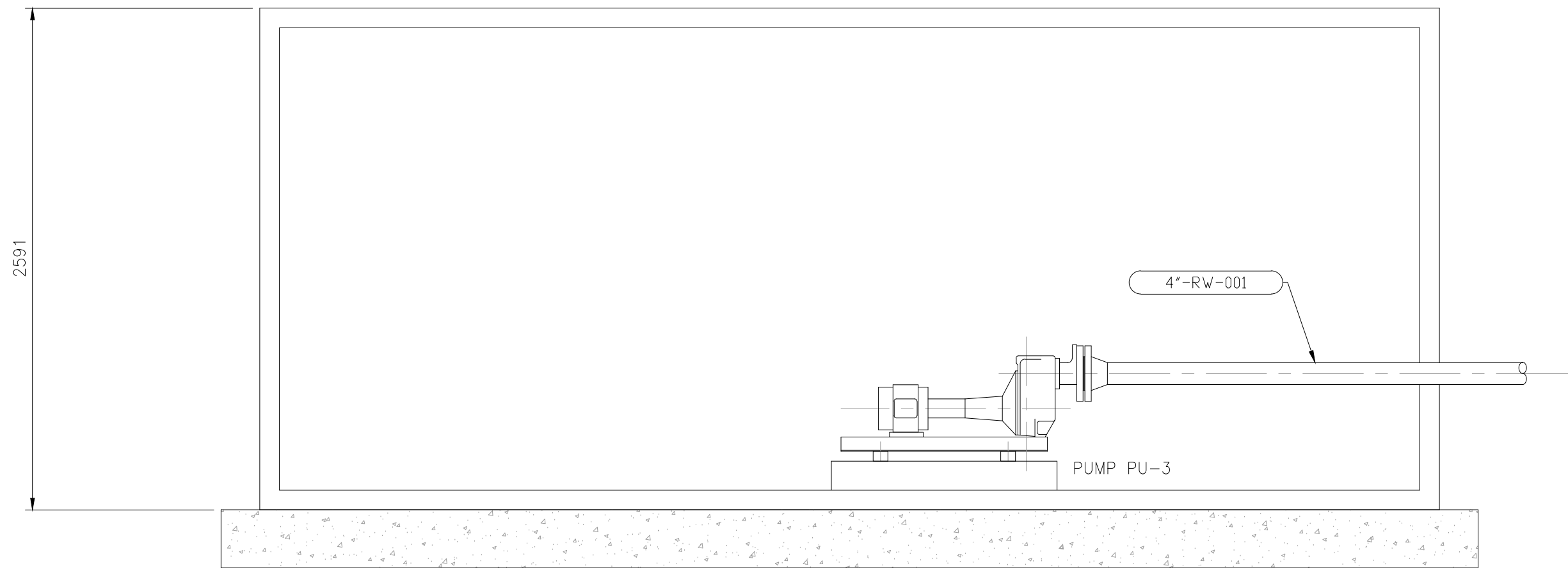
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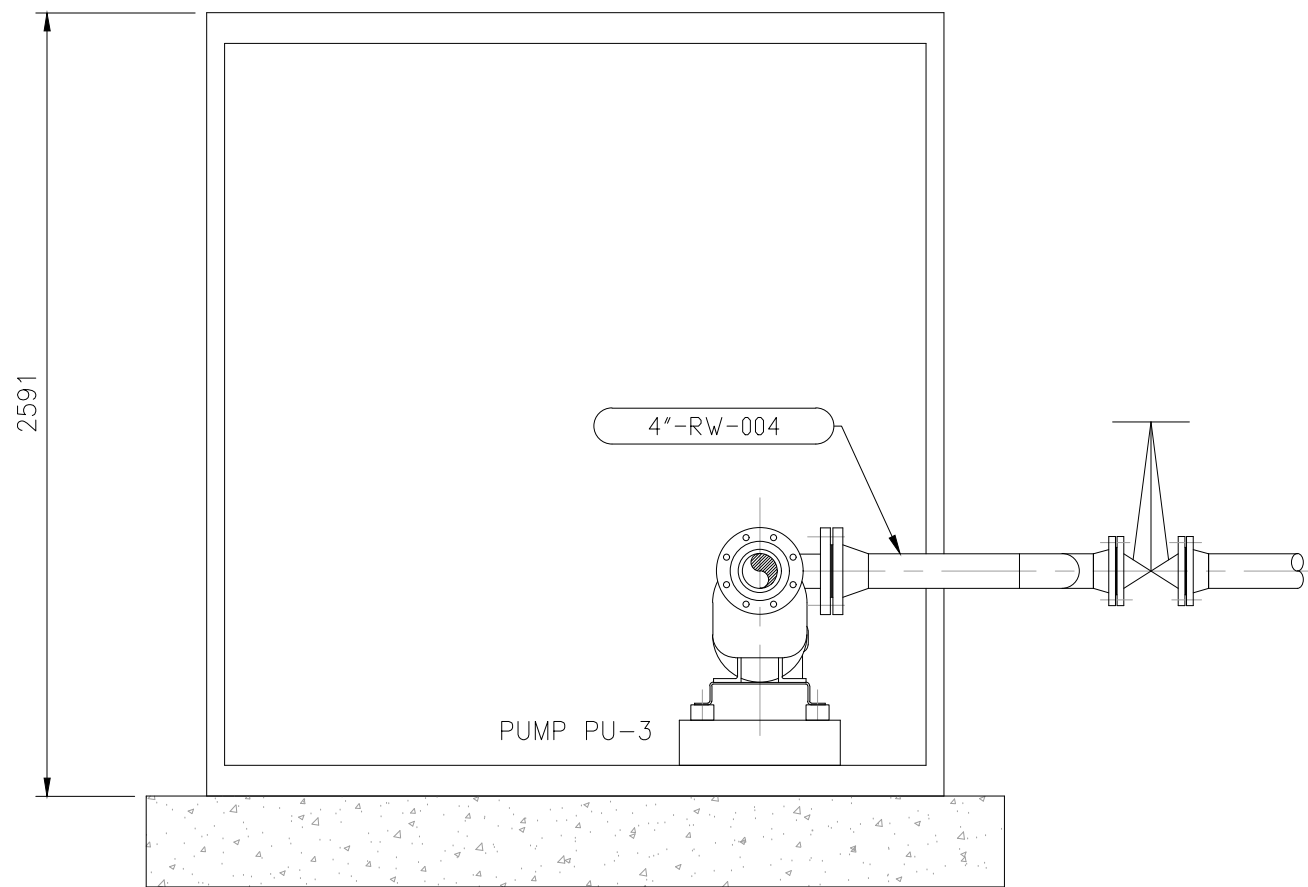
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SCALE: 1:25



SECTION A-A

SCALE: 1:25

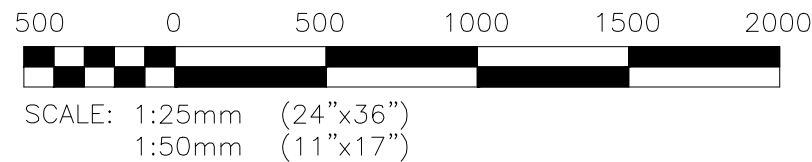


SECTION B-B

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2. CONTRACTOR TO SUPPLY, INSTALL, TEST AND COMMISSION ALL EQUIPMENT AS PER THE DRAWINGS AND SPECIFICATIONS.
3. CONTRACTOR TO SUPPLY AND INSTALL ALL SAFETY LABELS AND FLOW DIRECTION WARNINGS FOR PIPING AND EQUIPMENT.



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CLIENT:



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RUTTAN MINE SITE REMEDIATION
WATER TREATMENT PLANT
CONTRACT C-700

DWG. DESCRIPTION:
MECHANICAL
PUMP STATION 2 EQUIPMENT
GENERAL ARRANGEMENT



DESIGN BY: RC/JHH	DATE (YY/MM/DD): 12/02/24
DESIGN CHECK:	DATE:
DRAWN BY: JHH	DATE: 12/02/24
DWG. CHECK:	DATE:

DWG. NO.
21-1357-001 M730

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ON 2012/10/17

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10-1357-02

3
6701 PUMPING STATION 2
SCALE: 1:25

APPENDIX B

Water Treatment Plant As-Builts

Filename: P:\Projects\2010\10-1357-02\Drawings\Tender Drawings\0770_Operation of Water Treatment Plant\10-1357-02-CS770.dwg - Tab: Layout 01 Plotted By: echubey 12/09/2014 [Tue 3:56pm]

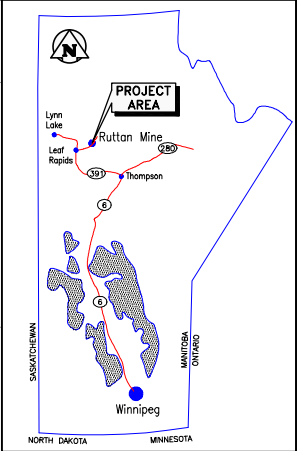
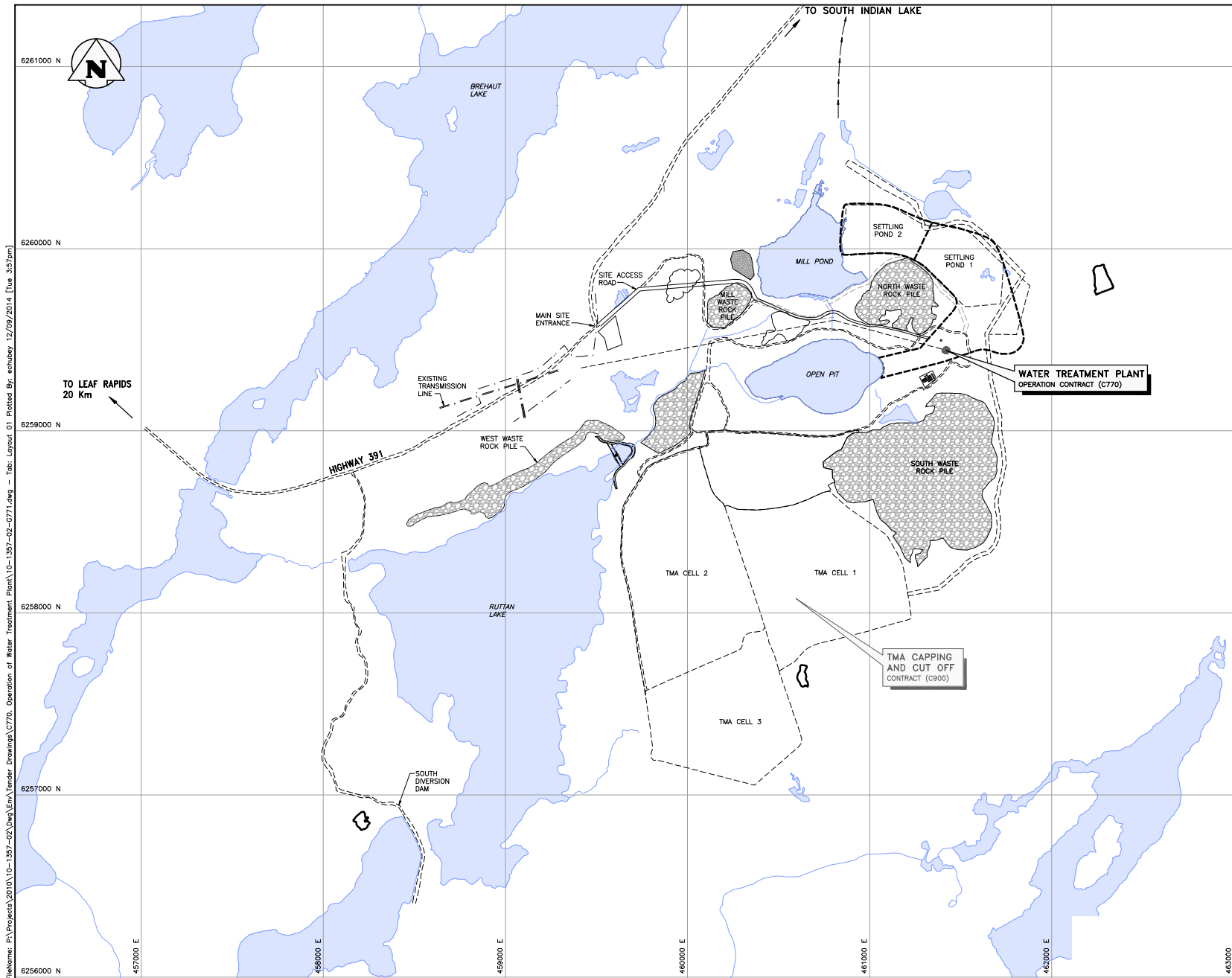
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OPERATION OF WATER TREATMENT PLANT

TENDER CONTRACT No C770

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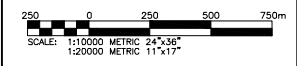
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10-1357-02-G771	OVERALL SITE PLAN AND WORK AREAS
10-1357-02-G772	WATER TREATMENT PLANT SITE PLAN
10-1357-02-M771	MECHANICAL WATER TREATMENT PLANT GENERAL ARRANGEMENT
10-1357-02-M772	P&ID WATER TREATMENT PLAN
10-1357-02-M773	P&ID PUMPING STATIONS



KEY MAP

- LEGEND:**
- == == == == ROADS
 - [Symbol] BUILDINGS
 - [Symbol] LIMITS OF WORK AREA THIS CONTRACT

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A	14/12/10	ISSUED FOR DRAFT TENDER REPORT	EJC

REVISIONS / ISSUE



PROJECT:
RUTAN MINE SITE REMEDIATION
OPERATION OF WATER TREATMENT PLANT
CONTRACT C-770

DWG: 10-1357-02
OVERALL SITE PLAN
AND WORK AREAS

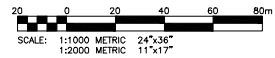
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	DRAWN BY:	PEC	DATE:	14/12/01
	DWG CHECK:		DATE:	

DWG. NO:	10-1357-02	G771	REV:	A
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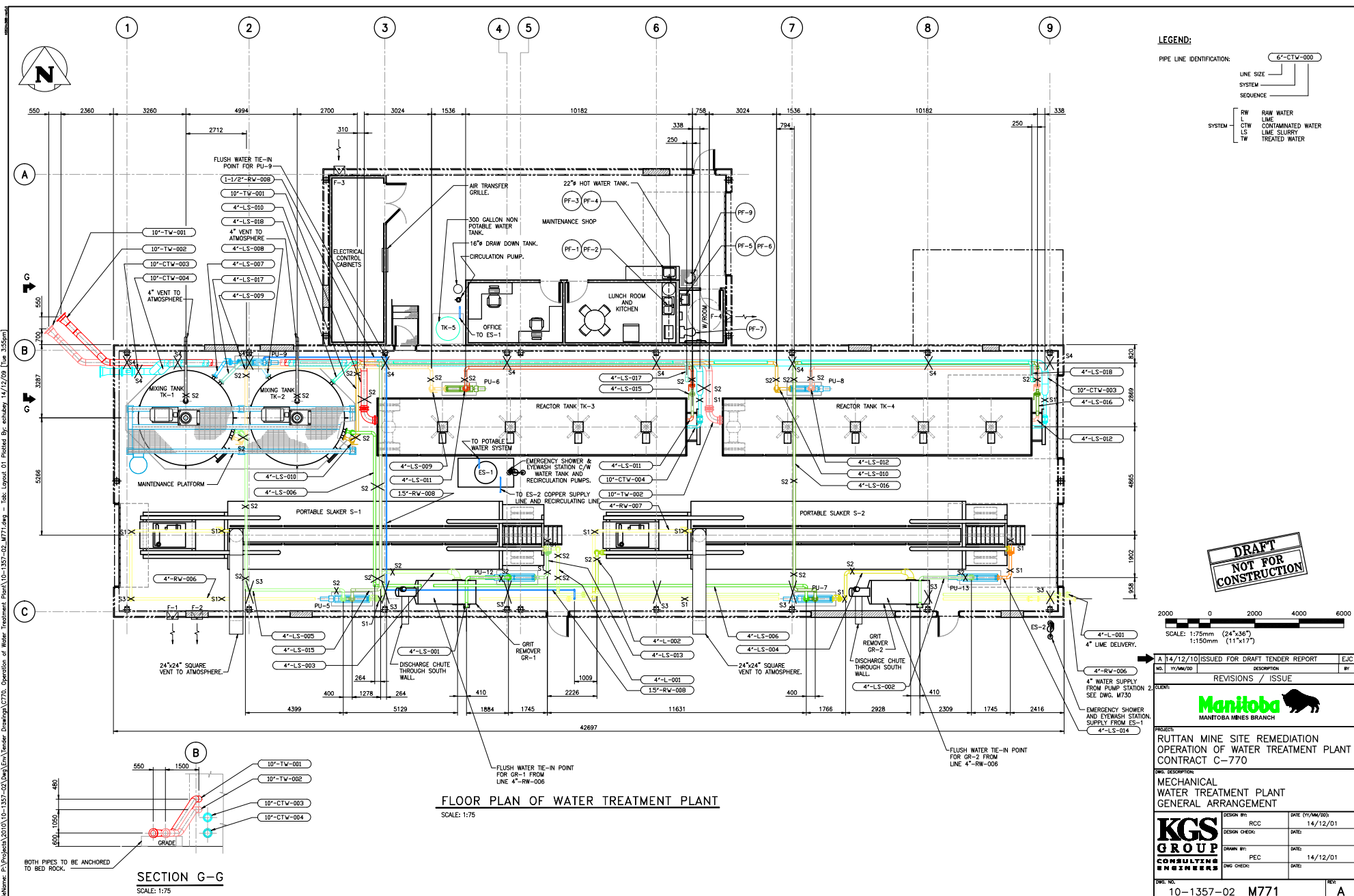
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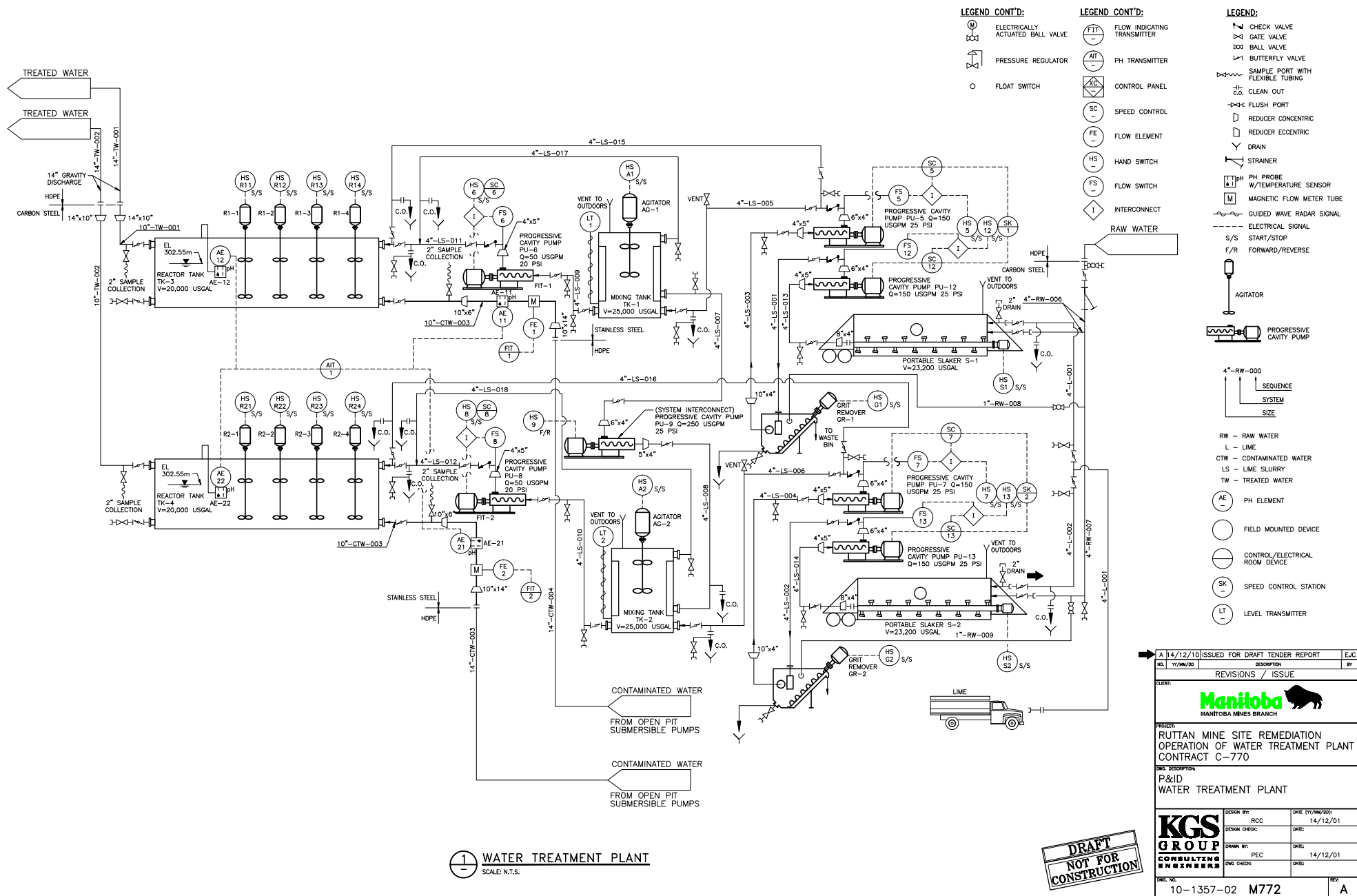
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OPERATION OF WATER TREATMENT PLANT
CONTRACT C-770

DRAWING DESCRIPTION: WATER TREATMENT PLANT
SITE PLAN

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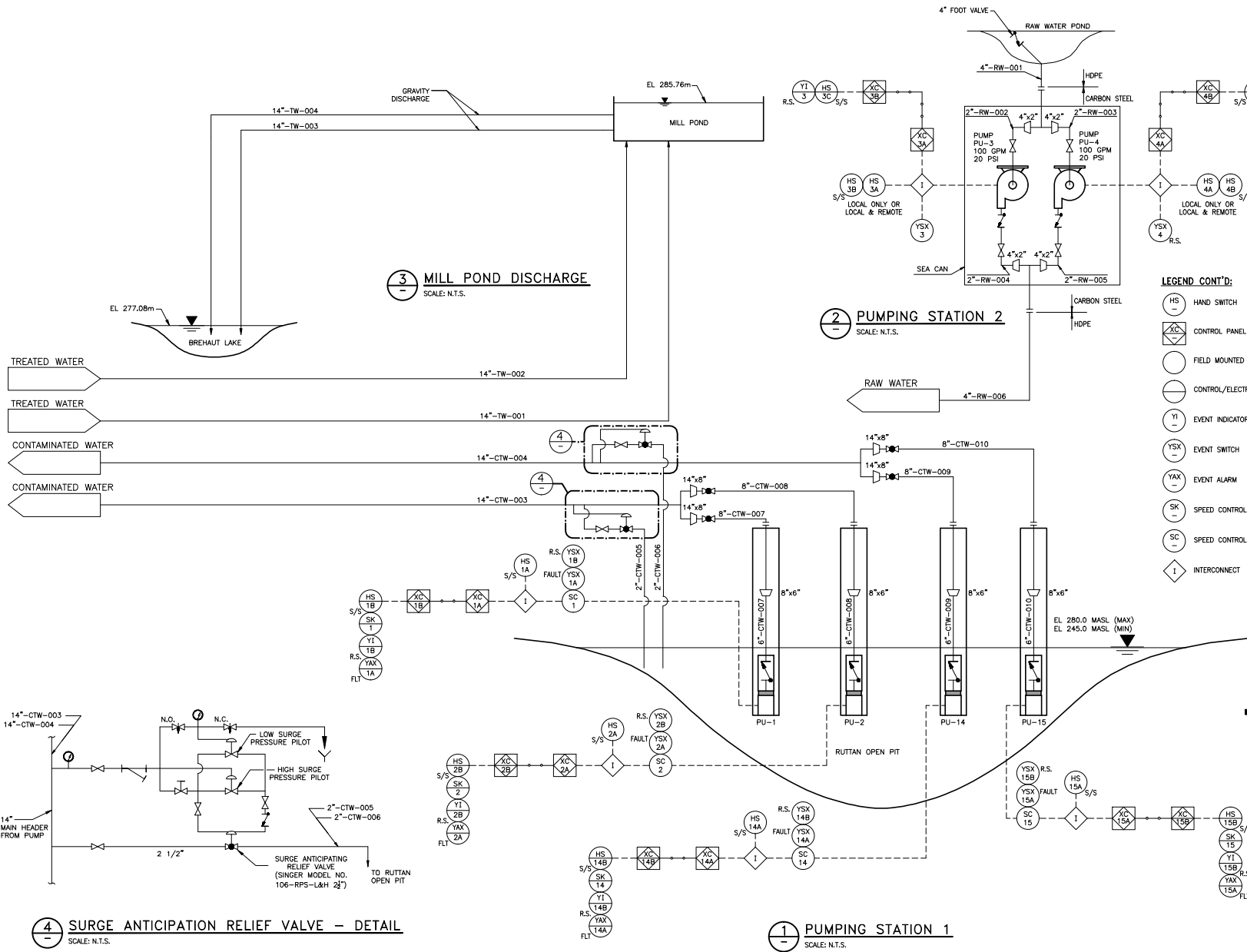
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PROJECT:		
RUTTAN MINE SITE REMEDIATION		
OPERATION OF WATER TREATMENT PLANT		
CONTRACT C-770		
DWG. DESCRIPTION:		
P&ID		
WATER TREATMENT PLANT		
DESIGN BY:		DATE (Y/M/D)
RCC		14/12/01
DESIGN CHECK:		DATE:
DRAWN BY:		DATE:
PEC		14/12/01
DWG. CHECK:		DATE:
DWG. NO.		REV
10-1357-02 M772		A



- LEGEND:**
- CHECK VALVE
 - GATE VALVE
 - NEEDLE VALVE
 - CONTROL VALVE
 - SURGE ANTICIPATING RELIEF GLOBE VALVE
 - PRESSURE PILOT
 - GLOBE VALVE
 - SAMPLE PORT WITH FLEXIBLE TUBING
 - CLEAN OUT
 - FLUSH PORT
 - PRESSURE GAUGE
 - REDUCER
 - DRAIN
 - ELECTRICAL SIGNAL
 - COMMUNICATION LINE
 - S/S START/STOP
 - R.S. RUN STATUS
 - FLT. FAULT

- LEGEND CONT'D:**
- HS HAND SWITCH
 - XC CONTROL PANEL
 - FIELD MOUNTED DEVICE
 - CONTROL/ELECTRICAL ROOM DEVICE
 - YI EVENT INDICATOR
 - YSX EVENT SWITCH
 - YAX EVENT ALARM
 - SK SPEED CONTROL STATION
 - SC SPEED CONTROL
 - I INTERCONNECT

- CENTRIFUGAL PUMP
- SUBMERSIBLE PUMP W/INTEGRAL CHECK VALVE
- 4\"/>

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REVISIONS / ISSUE			
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OPERATION OF WATER TREATMENT PLANT			
CONTRACT C-770			
DRAWN BY			
P&ID PUMPING STATIONS			
DESIGN BY		DATE (Y/M/D)	
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DESIGN CHECK		DATE	
PEC		14/12/01	
DRAWN BY		DATE	
PEC		14/12/01	
DWG CHECK		DATE	
DWG NO.		10-1357-02 M773	
REV			A

APPENDIX C

Water Treatment Plant Operation Manual



RUTTAN MINE WATER TREATMENT PLANT

General Plant Operation

Rev.	Date	Description	Author	Checked by	Approved
A	2014/10/27	Issued for Review	RCC	JAB	EC

1.0 OVERALL PLANT DESCRIPTION

The Ruttan Mine Water Treatment Plant (WTP) is a low density sludge (LDS) system using lime to treat acid mine drainage. The system comprises of a treatment plant inside a metal clad building, two pumping stations (PS-1 and PS-2), a clean water reservoir and 2 settling ponds. The treatment plant is located east of the open pit and includes an electrical room, office, lunchroom, washroom with sewage pump and holding tank, water sample testing station, and plant area that has two water treatment lines. Each treatment line processes lime and clean water into lime slurry to treat the contaminated water from the open pit. The equipment for each line consists of a slaker, grit remover, mixing tank with agitator, and reactor tank. Seven progressive cavity pumps are used to move the lime slurry throughout the plant. Ventilation fans and emergency shower stations are installed throughout the treatment plant. A central domestic water system with storage tank and electric hot water tank supplies water to plumbing fixtures and the emergency shower storage tank. Pumping station PS-1 is located west of the WTP on the east side of the open pit and consists of four submersible pumps and a seacan to store electrical equipment. Pumping station PS-2 is located on the west side of the clean water reservoir and consists of a seacan to store two clean water pumps and electrical equipment.

The WTP is sized to treat 20,000 m³ in a 24 hour period, which is the equivalent of approximately 3,700 US gallons per minute (USGPM). Clean water is pumped from PS-2 to the water treatment plant and is used for process water during the lime slaking process, and for flushing of plant equipment. Quick lime from truck shipments is added to each slaker, which mixes the clean water and lime to create a lime slurry. The lime slurry is then pumped through the grit remover and into the mixing tank. An agitator within the mixing tanks keeps the solids in suspension while a pump transfers lime slurry into the reactor to treat the contaminated water. The contaminated water is pumped from PS-1 into the reactors. The reactors mix the contaminated water and lime slurry, until it reaches a pH of approximately 10 units. The treated water discharges by gravity out of the water treatment plant into an excavated channel, which leads to settling ponds north of the WTP. The two settling ponds operate in series to allow the solids to settle out as sludge. The settled water discharges to Mill Pond, which provides final polishing before the effluent is released to Brehaut Lake through an excavated ditch.

2.0 OPERATION

2.1 START-UP PROCEDURE

The following is a list of start-up items to be completed after winter shutdown of the plant.

- Deliver a load of potable water for the water holding tank and start up the domestic water system.
- Ensure emergency shower stations are operational and check all fire extinguishers are properly charged.
- Inspect all piping and equipment at the water treatment plant and pumping stations (PS-1 and PS-2) for any damage or corrosion.
- Verify motor alignment for equipment.
- Remove all wooden boarding on exterior building vents or hoods (ie: slakers, ventilation fans, mixing tanks, and PS-1 seacan).
- Remove all wooden plywood, insulated tarp, and electric heaters from Active Filter cabinet within PS-1 seacan and store materials within seacan.

- Manually test each valve within the system to ensure proper rotation and full actuation.
- Complete maintenance checks for each piece of equipment as specified in the equipment operating and maintenance manual.
- Bump test all motors to verify rotation.
- Connect clean water intake pipe at pumping station PS-2 to seacan and position piping and floats.
- Flush the water treatment plant piping and equipment with clean water from PS-2. Turn on progressive cavity pumps to move the clean water through the system.
- Check all piping and equipment for leaks.
- Turn on agitators within the slakers, mixing tanks, and reactors to check operation.
- Test level sensors, flow switches, and flow meters.
- Remove pH probes from pH analyzer heated enclosure, install and test. Supply new pH probes as required.
- Transport standby generator to site, immobilize generator by installing on blocks, connect all power and control cables, verify fuel tank is full, test operation of standby generator and automatic transfer switch.

2.2 NORMAL OPERATION

The following is a general description for the normal operation of the water treatment plant. The reactor tanks and mixing tanks are typically run in a continuous operation whereas the slakers and grit removers are run on a batch system depending on the lime truck delivery. Remote controls for the various pumps and motors are located on the perimeter walls of the plant area, with the exception of the slaker and reactor vessels which have their own local control panels.

2.2.1 Lime Slaking Operation

The slaking process starts with clean water from two pumps in pumping station PS-2 being pumped to the water treatment plant. Operators at the water treatment have remote access to turn the clean water pumps ON/OFF via remote controls. The clean water is used to fill the slakers and flush equipment as required. Operators will open or close isolation valves throughout the piping to process the lime slurry and treat the contaminated water.

Slakers are typically pre-filled by the operators prior to each shipment of quick lime. Once the lime truck arrives, the quick lime is blown into each slaker through an exterior cam-lock connection. During lime transfer, the slakers must be operated to ensure mixing of the quick lime with clean water and to avoid the creation of lime clumps. Operators shall utilize the isolation valves to direct the quick lime to the correct slaker. Once loaded, the slakers are isolated from the piping. Operators shall ensure that the slakers remain in operation to ensure that the quick lime and clean water continues slaking. Slaking continues until the pre-determined temperature to ensure the proper amount of reaction between the clean water and the lime is completed.

Once slaking is completed, the lime slurry is transferred to the mixing tank. This involves passing the lime slurry through a grit remover to remove any large clumps or particles

from the slurry. To transfer the lime slurry to the mixing tanks, operators shall do the following:

- Turn ON the motor and open the water supply to the grit remover;
- Open the slaker discharge valve;
- Turn ON the two progressive cavity pumps on both sides of the grit remover. The flow rate of each pump must be adjusted via the dial controls to ensure that the grit remover does not overflow or that the second pump is not run dry.
- The mixing tank agitator motor is turned ON to keep the lime slurry suspended.

Once all the lime slurry is transferred out of the slaker, clean water is added to the slakers and flushed into the mixing tank. Flushing the line ensures that lime does not harden and plug up the piping. Periodic manual cleaning of the slakers, grit removers, mixing tanks, and reactors may be required.

2.2.2 Contaminated Water Treatment Operation

The lime slurry within the mixing tanks is continuously pumped into the reactor tanks to treat the contaminated water. The contaminated water is pumped from pumping station PS-1 into each reactor through the four submersible pumps. These four submersible pumps are controlled via a pump control panel in the electrical room or locally at PS-1. The flow rate of lime slurry transferred into the reactor will be adjusted by the operator via the pump controls on the north wall to meet a discharge pH of 10 for the treated water. This lime slurry flow rate will depend on the concentration of the lime slurry and flow rate of contaminated water. Operators must monitor and control this process at all times while the process line is in operation.

Once the lime slurry and contaminated water are added to the reactor tank, mixers located on top of the reactor must be turned ON. These mixers will continuously mix both fluids to treat the water. The treated water will then gravity discharge out the reactor tank into 14" HDPE lines that flow into the settling ponds and eventually to Mill Pond and Brehaut Lake.

Operators will be monitoring the equipment, piping, flow rates, and controls throughout the process. In addition, the facility will have to be kept clean through continual maintenance. Operators will also complete water sampling and testing to ensure treated water guidelines are met. In addition, periodic inspections of the dikes surrounding the settling ponds and ditches will be required.

2.3 WINTERIZATION

The following is a list of items to be completed prior to shut down of the water treatment plant before winter.

- Perform maintenance storage procedures for all equipment as per Operation and Maintenance manuals.
- Perform pump rotation of all progressive cavity pumps once per month. Rotate fan by ¼ turn.
- Flush and drain all piping and equipment within the water treatment plant and pumping stations.

- Provide portable air compressors to blow out all piping to remove standing water within the piping system.
- Manually operate all valves to the closed position.
- Lock-out main isolation valves at pumping stations (PS-1 and PS-2).
- Disconnect intake piping at pumping station PS-2 and properly anchor the intake piping and floats to the shore line.
- Cover the Active Filter in PS-1 with an insulated tarp. Plug in two space heaters into the dedicated heater receptacles adjacent to the active filter, and place them under the tarp facing the active filter. Use a metal frame or sheet of plywood to hold the tarp away from the heaters. Fasten the tarp to the plywood wall behind the active filter using wood screws.
- Cover all exterior vent openings (ie: slakers, ventilation fans, mixing tanks, and PS-1) with wooden boards to prevent access into the building by wildlife.
- Drain and blow out all domestic water piping and emergency shower piping within the water treatment plant.
- Pump out the sewage holding tank.
- Drain out 14" contaminated water and treated water lines.
- Test the satellite call-out system to ensure that it functions for all call-out scenarios, including unauthorized entry at PS-1, PS-2, the water treatment plant, or the auto-dialer enclosure, or in the event of a power failure.
- Winterization and off-site storage of the backup diesel generator. This includes but is not limited to the following:
 - Perform all manufacturer's procedures for winterization of the diesel motor.
 - Disconnect all cables from the diesel generator to the water treatment plant. The shore power and control cable can be disconnected at both ends, rolled up and stored in the water treatment plant. The main power conductors should be removed from the generator breaker, coiled up and stored underneath the generator termination enclosure;
 - The generator shall be lowered off of its frame jack stands onto its wheels, and all blocking from underneath the generator removed; and
 - Transport generator to a secure location.
- Remove the pH probes from the piping and equipment.
- Store any spare pH probes in a heated enclosure, such as the pH analyzer enclosure or chart recorder enclosure.
- Shut down ventilation fans.

3.0 MAINTENANCE

The following is a list of maintenance activities to be completed throughout the water treatment plant.

- Flush out water treatment plant piping system after every batch of quick lime has been processed.
- Check grease levels on slakers after each batch of quick lime and add additional grease as required.
- Dispose of accumulated grit from grit discharge after each batch of quick lime.

- Flush out clean water strainer after each batch of quick lime.
- Check equipment for leaks on a daily basis.
- Check piping for leaks on a weekly basis.
- Complete maintenance checks for each piece of equipment as specified in the equipment operating and maintenance manual.
- Inspect pumping stations PS-1 and PS-2 on a daily basis to ensure equipment operation and internal seacan temperatures.
- Provide weekly inspections of the surrounding control structures (dikes, weirs, and ditches) and access roads. Inspections should include detailed description of any failures or abnormal operation of any equipment. Maintenance of the access roads and surrounding control structures will be required.